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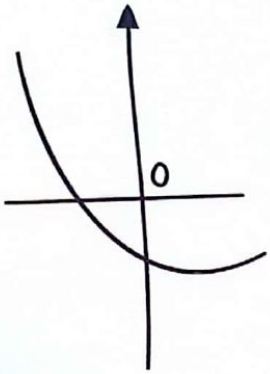
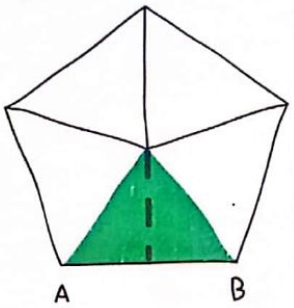
GENERAL

Mathematics

For Grade X

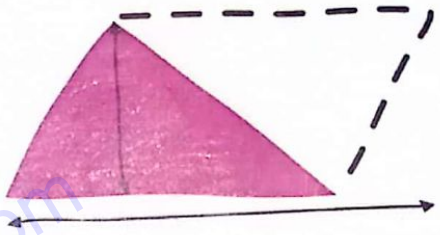


$\{3,4,5,\ldots\}$

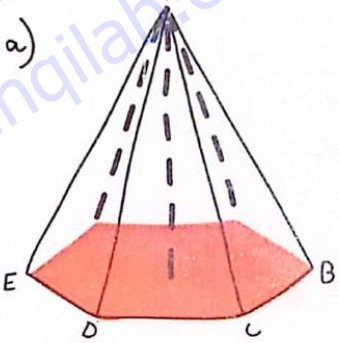


$a=x^2$

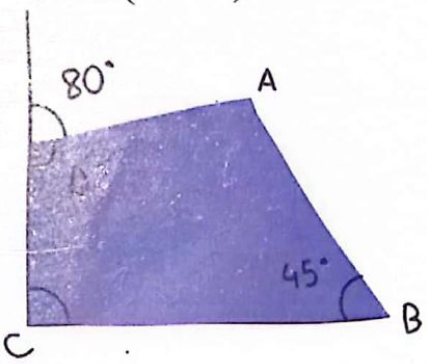
$(\sqrt{5}-\sqrt{3})(8+\sqrt{15})$



$\times\left(\frac{2\pi}{a}-b\right)$ $y(-a)+\left(\frac{\pi}{2}-a\right)$



$2(x-4y)$



$A+B+C+D=360°$

$\left\{6,\frac{-22}{3}\right\}$

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A Textbook of GENERAL MATHEMATICS

Grade X

Khyber Pakhtunkhwa Textbook Board, Peshawar.

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Unit 1

ALGEBRAIC FORMULAS AND APPLICATIONS

In this unit, the students will be able to learn about the:

- Algebraic expression
- Rational expression
- Definition of rational expression
- Examine whether a given algebraic expression is
  - (a) Polynomial or not
  - (b) Rational expression or not
- Define $\frac{p(x)}{q(x)}, q(x) \neq o$ as a rational expression in its lowest terms
- To examine whether a given rational algebraic expression is in lowest form or not
- To reduce a given rational expression to its lowest terms
- Divide a rational expression with another and express the result in its lowest terms.
- Sum, difference and product of rational expressions
- Finding value of algebraic expression at some particular real numbers
- Surds and their applications?
- Rationalization

1.1 Algebraic Expressions

Expressions which join constant and variables with the help of basic mathematical operations such as, additions, division, multiplication and subtraction are called algebraic expressions.

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Examples:

i) $2x+3$ *ii)* $12x$

iii) $2x^2+3x+5$ *iv)* $\frac{16}{x^3-1}$

v) $3x-1$

i) Rational expression behaves like a rational number

The students are already familiar with the real number system. A number of the form of $\frac{p}{q}$ where p, q are integers and $q \neq o$, is called a rational number. In similar way, an expression of the form written as $\frac{p(x)}{q(x)}$ where $p(x)$ and $q(x)$ are polynomials and $q(x) \neq o$ is called the rational expression.

Examples:

i) $\frac{1}{x+1}$, $x \neq -1$

ii) $\frac{x^2+2x+3}{x+2}$, $x \neq -2$

v) $\frac{x+1}{x-1}$, $x \neq 1$

Every polynomial $p(x)$ can be conserved as a rational expression because $p(x)$ can be written as $\frac{p(x)}{1}$, where $q(x)=1$.

ii) Definition of a Rational Expression

An expression of the form $\frac{p(x)}{q(x)}$, $q(x) \neq o$ and both $p(x)$ and $q(x)$ are polynomials is called a rational expression.

Examples:

i) $\frac{3x-2}{2x+1}, x \neq -\frac{1}{2}$ *ii)* $\frac{3x^2-1}{x^2+1}, x \neq 1, -1$

iii) $x+1$

iii) Examine whether a given algebraic expression is

a) Polynomial or not

b) Rational expression or not

(a) To examine whether a given algebraic expression is a polynomial or not, if the exponent of each term in an algebraic expression is zero or a positive integer is called a polynomial.

Example 1:

x^2+2x+3 is a polynomial, of degree 2 in x because the exponent is a positive integer.

Example 2:

3 is a polynomial of degree zero. Thus every constant number is called a constant polynomial.

Example 3:

$x^2+2x^{-1}+3$ is not a polynomial because the exponent of the second term $2x^{-1}$ is -1, not a positive integer.

(b) To examine whether an algebraic expression is rational or not.

If the exponent of any term of an algebraic expression is negative then it is called rational expression.

Example 1:

$\frac{x^2+1}{x^2-1}$, $x^2 \neq 1$, is a rational expression, since $p(x)$ and $q(x)$ are polynomials.

Example 2:

$\frac{x^3+\frac{3}{x^2}+4}{x^2+2x+1}$, is not a rational expression since $x^3+\frac{3}{x^2}+4$ is not a polynomial.

Example 3:

$3x+2=\frac{3x+2}{1}$ is a rational expression, since $p(x)=3x+2$ and $q(x)=1$ are polynomials.

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Exercise 1.1

1. Which expression behaves like a rational number in the following expressions:

i) $x+1$ *ii)* $\frac{x^2+1}{x-1}$ *iii)* $x^2+\frac{3}{x}$

(iv) $\frac{x^2+2x+3}{x^2+2}$ *(v)* $\frac{x^3+2x^2+3x+1}{x^2+4x+5}$

2. Which expressions are polynomial in the following expressions:

i) $\sqrt{x+1}$ *ii)* x^2+2x+3

iii) $x^2-\frac{1}{x}+1$ *iv)* $8x^2y+5xy$

v) $x^3-\frac{1}{2x^2}+\frac{3}{4}$

3. Which of the following are rational expressions?

i) $\frac{x^2+x+3}{x^3+1}$ *ii)* $\frac{x^3+\frac{4}{x}+2}{x^2+1}$

iii) $\sqrt{x^2+3}-1$ *iv)* $\frac{1}{x}$

v) 2 *vi)* $\frac{3}{x^3-1}$

4. Find the degrees of the polynomials:

i) x^3+2x^2+3x+4 *ii)* x^2+2x+3

iii) $\frac{1}{3}x+4$ *iv)* 5

v) $-2x^2+\frac{5}{3}x+4$

iv) Definition of $\frac{p(x)}{q(x)}$ as a rational expression in its lowest form

The expression $\frac{p(x)}{q(x)}$, $q(x)\neq o$, in which the polynomials cannot be factorized with their common factors is called in its lowest form.

Example:

$\frac{2x+1}{x-1}$, is in lowest terms because the factorization of $p(x)=2x+1$, and $q(x)=x-1$ is not possible in term of their common factors.

v) Examine whether a given rational algebraic expression is in lowest form or not

Example 1:

$\frac{x^2+1}{x+1}$ is a rational algebraic expression in its lowest terms, because $p(x)=x^2+1, q(x)=x+1$ cannot be factored in their common factors.

Example 2:

$\frac{x^2-1}{x+1}$, a rational algebraic expression, is not in its lowest terms, because the factorization of $p(x)=x^2-1=(x-1)(x+1)$ is possible in which one factor $(x+1)$ is the same as $q(x)=x+1$.

vi) Reduce a given rational expression to its lowest terms

In order to reduce a rational expression $\frac{p(x)}{q(x)}$ to its lowest terms, both $p(x)$ and $q(x)$ are divided by the common factor and after simplification the remaining expression is written.

Example 1: Reduce $\frac{x^2-1}{x+1}$ to its lowest terms.

Solution: After factorization, the expression becomes:

$$\frac{x^2-1}{x+1}=\frac{(x+1)(x-1)}{x+1}=x-1$$

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Example 2: Reduce the expression $\frac{x^2-5x+6}{x^2-9}$ to its lowest terms.

Solution: Let

$$\frac{p(x)}{q(x)} = \frac{x^2-5x+6}{x^2-9} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$$

Where

$$p(x)=x^2+5x+6 \text{ and } q(x)=x^2-9$$

Factorization of $p(x)$ and $q(x)$ are the following:

$$\begin{aligned} p(x) &= x^2-5x+6 \\ &= x^2-2x-3x+6 \\ &= x(x-2)-3(x-2) \\ &= (x-2)(x-3) \\ q(x) &= x^2-9 \\ &= x^2-3^2 \\ &= (x-3)(x+3) \qquad \therefore a^2-b^2=(a+b)(a-b) \end{aligned}$$

i.e. $q(x)=(x-3)(x+3)$

The factors of $p(x)$ and $q(x)$ are used in the equation (i) to be obtain:

$$\frac{x^2-5x+6}{x^2-9}=\frac{(x-2)(x-3)}{(x+3)(x-3)}=\frac{x-2}{x+3}$$

Exercise 1.2

1. Which of the following expressions are in their lowest term:

i) $\frac{x+1}{2x-1}$ lowest ii) $\frac{3x^2+3}{9x+27}$ not

iii) $\frac{x^2+3x+2}{x+2}$ not iv) $\frac{x^2-4}{x+2}$ not

v) $\frac{4x+8}{2}$ not vi) $\frac{4x^2+3}{x+3}$ lowest

2. Reduce the following rational algebraic expression in their lowest terms:

i) $\frac{x^2-x-6}{x+2}$ ii) $\frac{x^3+1}{x^2-1}$

iii) $\frac{x^2y+xy}{xy^2-xy}$ iv) $\frac{x^4+2x^2+1}{x^4-1}$

v) $\frac{x^4+4}{x^2-2x+2}$ vi) $\frac{z^3-1}{z^2+z+1}$

vii) Sum, Difference and Product of Rational Expressions

a) Sum of rational expressions

The sum of two or more rational expressions is explained in the following examples.

Example 1: Find the sum of the rational expressions:

$$\frac{x+1}{x-1} \text{ and } \frac{2x-1}{x-1}$$

Solution: The sum of given rational expression is:

$$\frac{x+1}{x-1}+\frac{2x-1}{x-1}=\frac{x+1+2x-1}{x-1}=\frac{3x}{x-1}$$

Example 2: Add the rational expression:

$$\frac{3x+1}{x+1} \text{ and } \frac{2x-3}{x^2-1}.$$

Solution: The sum of rational expressions is:

$$\begin{aligned} \frac{3x+1}{x+1}+\frac{2x-3}{x^2-1} &= \frac{3x+1}{x+1}+\frac{2x-3}{(x+1)(x-1)} \\ &= \frac{(3x+1)(x-1)+2x-3}{(x+1)(x-1)} \quad \textit{By taking (L.C.M)} \\ &= \frac{3x^2-3x+x-1+2x-3}{(x+1)(x-1)} \\ &= \frac{3x^2-4}{x^2-1} \end{aligned}$$

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Example 3: Simplify the rational expression:

$$\frac{2}{x+1}+\frac{5}{x-1}+\frac{1}{x^2-1}$$

Solution:

$$\frac{2}{x+1}+\frac{5}{x-1}+\frac{1}{x^2-1}$$

$$=\frac{2}{x+1}+\frac{5}{x-1}+\frac{1}{(x-1)(x+1)}$$

$$=\frac{2(x-1)+5(x+1)+1}{(x-1)(x+1)}, \text{(by taking L.C.M)}$$

$$=\frac{2x-2+5x+5+1}{x^2-1}=\frac{7x+4}{x^2-1}$$

b) Difference of rational expressions

The process of subtraction rational expressions is explained by the following examples.

Example 1: Find the difference of rational expressions:

$$\frac{3}{x+2}-\frac{2}{x^2+3x+2}$$

Solution: The factorization of x^2+3x+2 are:

$$x^2+3x+2=x^2+x+2x+2$$

$$=x(x+1)+2(x+1)=(x+1)(x+2)$$

Now $$\frac{3}{x+2}-\frac{2}{x^2+3x+2}=\frac{3}{x+2}-\frac{2}{(x+1)(x+2)}$$

$$=\frac{3(x+1)-2}{(x+1)(x+2)}, \text{(By taking LCM)}$$

$$=\frac{3x+3-2}{(x+1)(x+2)}=\frac{3x+1}{(x+1)(x+2)}$$

Example 2: Subtract the rational expression $\frac{5}{x-2}$ *from* $\frac{2x+4}{x^2-4}$.

Solution:

$$\frac{2x+4}{x^2-4}-\frac{5}{x-2}=\frac{2(x+2)}{x^2-2^2}-\frac{5}{x-2}$$

$$=\frac{2\cancel{(x+2)}}{(x-2)\cancel{(x+2)}}-\frac{5}{x-2}$$

$$=\frac{2}{x-2}-\frac{5}{x-2}=\frac{2-5}{x-2}=\frac{-3}{x-2}=\frac{3}{2-x}$$

c) Product of Rational Expressions

The product of two or more rational expressions is the product of their numerators and denominators after simplification.

The product of rational expression is explained in the following examples:

Example 1: Find the product of $\frac{x^2-y^2}{2y^2}$ and $\frac{3y}{x+y}$

Solution:

$$\frac{x^2-y^2}{2y^2}\times\frac{3y}{x+y}=\frac{(x-y)\cancel{(x+y)}}{2y\cancel{y}}\times\frac{3\cancel{y}}{\cancel{(x+y)}}=\frac{3(x-y)}{2y}$$

Example 2: Find the product of the rational expressions

$$\frac{8x^2+15x+7}{2x^2+5x+2} \quad \textit{and} \quad \frac{x^2+15x+14}{x^2-1}$$

Solution: First factorize each expression.

$$8x^2+15x+7=8x^2+8x+7x+7$$

$$=8x(x+1)+7(x+1)=(x+1)(8x+7)$$

$$x^2+15x+14=x^2+x+14x+14$$

$$=x(x+1)+14(x+1)$$

$$=(x+1)(x+14)$$

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$2x^2+5x+2 = 2x^2+4x+x+2$
$= 2x(x+2)+1(x+2)$
$= (x+2)(2x+1)$

$x^2-1 = (x+1)(x-1)$

$$\frac{8x^2+15x+7}{2x^2+5x+2} \times \frac{x^2+15x+14}{x^2-1} = \frac{(x+1)(8x+7)}{(x+2)(2x+1)} \times \frac{(x+1)(x+14)}{(x-1)(x+1)}$$

$$= \frac{(x+1)(8x+7)\ \cancel{(x+1)}\ (x+14)}{(x+2)(2x+1)(x-1)\ \cancel{(x+1)}}$$

$$= \frac{(x+1)(8x+7)(x+14)}{(x+2)(2x+1)(x-1)}$$

viii) Division of Rational Expressions

The division of two rational expressions is obtained by dividing the first expression by 2nd expression, after necessary simplification.

Example 1: Divide the rational Expression:

$$\frac{2x+4y}{5x-5} \quad by \quad \frac{2}{25x-25}$$

Solution

$$\frac{2x+4y}{5x-5} \div \frac{2}{25x-25} = \frac{2(x+2y)}{5(x-1)} \div \frac{2}{25(x-1)}$$

$$= \frac{\cancel{2}(x+2y)}{\cancel{5}\ \cancel{(x-1)}} \times \frac{\overset{5}{\cancel{25}}\ \cancel{(x-1)}}{\cancel{2}}$$

$$= 5(x+2y) = 5x+10y$$

Exercise 1.3

1. Find the sum of the following rational expressions

i) $\frac{3x-1}{2x-2}$, $\frac{2}{x-1}$

ii) $\frac{2x+y}{xy}$, $\frac{3y}{x}$

iii) $\frac{1}{y-1}$, $\frac{1}{y+1}$

iv) $\frac{2y}{y^2+1}$, $\frac{4y}{y^2-1}$

v) $\frac{xy^2+y}{x+3}$, $\frac{x^2y+x}{x+3}$

2. Subtract 1st expression from 2nd

i) $\frac{15}{4y-3}$, $\frac{16}{4y+3}$

ii) $\frac{1}{a+b}$, $\frac{1}{a-b}$

iii) $\frac{x+y}{y}$, $\frac{x^2-y^2}{xy}$

iv) $\frac{2x+1}{x+1}$, $\frac{2x-1}{x-1}$

3. Find the product of the following rational expressions

i) $\frac{49x^2-4y^2}{x^2-9}$, $\frac{x-3}{7x-2y}$

ii) $\frac{x^3-y^3}{x^2+2xy+y^2}$, $\frac{x+y}{x^2+xy+y^2}$

iii) $\frac{4x^2+12xy+9y^2}{2x^2+5xy-3y^2}$, $\frac{3x+3}{x^2-1}$

iv) $\frac{x^3+a^3}{x+a}$, $\frac{x+a}{x^3-a^3}$

4. Divide the following rational expressions

i) $\frac{2x^2+13x+15}{4x^2-1}$ by $\frac{2x^2+11x+5}{4x^2-1}$

ii) $a+\frac{2}{x^2}$ by $\frac{a+2}{x^2}$

iii) $\frac{4-x^2}{x^2-9}$ by $\frac{12+6x}{x^2-6x+9}$

iv) $\frac{x^8-1}{4x^2-4}$ by $\frac{x^4-1}{x-2}$

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5. Simplify the following expressions

i) $\frac{1}{a+b}+\frac{1}{a-b}+\frac{2a}{a^2-b^2}$

ii) $\frac{x^2-x-20}{x^2-25}\times\frac{x^2-x-2}{x^2+2x-8}\div\frac{x+1}{x^2+25}$

iii) $\frac{2x^2+9x+9}{3x^2-27}\times\frac{4x^2+8x+3}{x-3}\div\frac{(2x+3)^2}{(3x-9)(x-3)}$

ix) Value of an algebraic expression at some particular real numbers

If the variables in an expression are replaced by real numbers, then the real number so obtained from the expression is called the value of the algebraic expression at those real numbers:

Example 1: When $x=3$, $y=2$ *and* $z=1$. Find the value of $4y+3xz$.

Solution: $4y+3xz=4(2)+3(3)(1)=8+9=17$

Example 2: Find the value of x^2+2x+1, when $x=7$,

Solution: $x^2+2x+1=7^2+2(7)+1=49+14+1=50+14=64$

Example 3: Find $f(2)$, when the expression is $f(x)=3x^2+x+2$.

Solution: $f(x)=3x^2+x+2$

Put $x=2$ in the given expression to obtain:

$f(2)=3(2)^2+2+2=3\times4+4=12+4=16$

Exercise: 1.4

1. Find the values of $5x+3$ at $x=\frac{1}{3}$ and $x=-1$.
2. Find the value of $x^3+y^3+z^3$ at $x=3, y=1$ and $z=4$.
3. Find the value of $H=\frac{2ab}{a+b}$ at $a=4, b=2$.
4. $f(x)=x^3-2x^2+x-1$. Find $f\left(-\frac{1}{2}\right)$.
5. Find the value of expression:

$\frac{x^4}{2}+\frac{y^4}{3}-\frac{z^4}{4}$ at $x=2$, $y=3$ *and* $z=-2$.

6. Find the value of $x^3+2xyz+z^3$ at $x=2$, $z=3$, $y=4$.
7. Compute the value of $\frac{xyz+3xy^2}{x+y}$ at $x=1$, $y=1$ *and* $z=1$.

1.2. Algebraic Formulae

Formulae are rules which make the process of mathematical operations easy.

i) To know and derive some of the algebraic formulae:

Formula 1: $(a+b)^2+(a-b)^2=2(a^2+b^2)$

Proof:

$$L.H.S=(a+b)^2+(a-b)^2=a^2+2ab+b^2+a^2-2ab+b^2$$
$$=2a^2+2b^2=2(a^2+b^2)$$
$$=R.H.S$$

Formula 2: $(a+b)^2-(a-b)^2=4ab$

Proof:

L.H.S $(a+b)^2-(a-b)^2=a^2+2ab+b^2-a^2+2ab-b^2$

$=4ab$

L.H.S $=R.H.S$

Example 1: Simplify the expression:

$(3x+2y)^2+(3x-2y)^2$

Solution: Since $(a+b)^2+(a-b)^2=2(a^2+b^2)$

Put $a=3x$, $b=2y$, we get

$(3x+2y)^2+(3x-2y)^2=2((3x)^2+(2y)^2)$

$=2(9x^2+4y^2)=18x^2+8y^2$

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Example 2: Simplify the expression:

$(4x+y)^2-(4x-y)^2$

Solution: Since $(a+b)^2-(a-b)^2=4ab$.

Put $a=4x$, $b=y$, we get

$$(4x+y)^2-(4x-y)^2=4(4x)(y)$$
$$=16xy$$

Determination of the values of a^2+b^2 and ab when the values of $a+b$ and $a-b$ are known.

Example 1: Find the value of a^2+b^2, when $a+b=4, a-b=2$.

Solution: we know that $(a+b^2)+(a-b)^2=2(a^2+b^2)$

Put $a+b=4$, $a-b=2$, we get

$$2(a^2+b^2)=(4)^2+(2)^2$$
$$=16+4=20$$

Dividing both side by 2, we get

$$a^2+b^2=\frac{20}{2}$$
$$=10$$

Example 2: Find the value of ab when $(a+b)=6,(a-b)=2$

Solution: $a+b=6, a-b=2$

$4ab = (a+b)^2-(a-b)^2$ on putting values

$4ab = 36-4=32$

Dividing both sides by 4,

$ab=8$

Exercise 1.5

1. Simplify with the help of formulae.
   - (i) $(x+y)^2+(x-y)^2$
   - (ii) $(2x^2+3y^2)^2+(2x^2-3y^2)^2$
   - (iii) $(x+4y^2)^2-(x-4y^2)^2$
   - (iv) $(3x+9y^2)^2-(3x-9y^2)^2$
   - (v) $(4x^2+y)^2+(4x^2-y)^2$
   - (vi) $(5x+2y)^2-(5x-2y)^2$

2. Find the value of a^2+b^2 when,
   - *i)* $a+b=2$, $a-b=1$
   - *ii)* $a+b=8$, $a-b=2$

3. Find value of ab when;
   - *i)* $a+b=50$, $a-b=16$
   - *ii)* $a+b=60$, $a-b=20$
   - *iii)* $a+b=40$, $a-b=12$
   - *iv)* $a+b=6$, $a-b=2$

Formula 3: $(a+b+c)^2=a^2+b^2+c^2+2ab+2bc+2ca$

Proof.

LHS $(a+b+c)^2$

$$=(a+b+c)(a+b+c)$$
$$=a^2+ab+ac+ba+b^2+bc+ca+cb+c^2$$
$$=a^2+b^2+c^2+ab+ba+bc+cb+ac+ca$$
$$=a^2+b^2+c^2+2ab+2bc+2ca \quad (\because ac=ca, bc=cb, ca=ac)$$
$$=R.H.S$$

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a) To find the value of $a^2+b^2+c^2$ when the values of $a+b+c$ and $ab+bc+ca$ are given.

Example:

Find value of $a^2+b^2+c^2$ when $a+b+c=3, ab+bc+ca=4$.

Solution:

We know that.

$$(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca$$
$$= a^2+b^2+c^2+2(ab+bc+ca)$$
$$a^2+b^2+c^2 = (a+b+c)^2-2(ab+bc+ca)$$

Putting given values we get,

$$a^2+b^2+c^2 = (3)^2-2(4)$$
$$= 9-8$$
$$= 1$$

b) Find the value of $a+b+c$ when the values of $a^2+b^2+c^2$ and $ab+bc+ca$ are given.

Example 1:

Find values of $a+b+c$ where $a^2+b^2+c^2=6, ab+bc+ca=5$.

Solution:

Given that:

$a^2+b^2+c^2=6$

$ab+bc+ca=5$

$a+b+c=?$

$\because (a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca$

$= 6+2(5)$

$= 6+10$

$= 16$

Taking square root on both sides we get,

$\sqrt{(a+b+c)^2} = \pm\sqrt{16}$

$a+b+c = \pm 4$

c) Find value of $ab+bc+ca$ when values of $a^2+b^2+c^2$ and $a+b+c$ are given.

Example 2:

Find value of $ab+bc+ca$ when $a^2+b^2+c^2=2, a+b+c=4$.

Solution:

Given that

$a^2+b^2+c^2=2$

$a+b+c=4$

$ab+bc+ca=?$

$\therefore \quad (a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ca)$

$\Rightarrow \quad 2(ab+bc+ca) = (a+b+c)^2-(a^2+b^2+c^2)$

$= (4)^2-2$

$= 16-2$

$= 14$

Dividing both sides by 2 we get,

$(ab+bc+ca) = \frac{14}{2}$

$= 7$

Exercise 1.6

1. Find value of $a^2+b^2+c^2$ when:
   - *i)* $ab+bc+ca=2, \quad a+b+c=3$
   - *ii)* $ab+bc+ca=8, \quad a+b+c=4$
   - *iii)* $ab+bc+ca=5, \quad a+b+c=6$
2. Find value of $a+b+c$ when:
   - *i)* $ab+bc+ca=3, \quad a^2+b^2+c^2=6$
   - *ii)* $ab+bc+ca=5, \quad a^2+b^2+c^2=2$
   - *iii)* $ab+bc+ca=7, \quad a^2+b^2+c^2=4$

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3. Find value of $ab+bc+ca$ when:

i) $a+b+c=6,\quad a^2+b^2+c^2=2$

ii) $a+b+c=9,\quad a^2+b^2+c^2=13$

iii) $a+b+c=7,\quad a^2+b^2+c^2=67$

Formula 4: $(a+b)^3=a^3+3ab(a+b)+b^3$

Proof: $L.H.S=(a+b)^3$

$=(a+b)(a+b)^2$

$=(a+b)(a^2+2ab+b^2)$

$=a^3+2a^2b+ab^2+a^2b+2ab^2+b^3$

$=a^3+3a^2b+3ab^2+b^3$ *(combining like terms)*

$=a^3+3ab(a+b)+b^3$

$=R.H.S.$

Formula 5: $(a-b)^3=a^3-3ab(a-b)-b^3$

Proof: $L.H.S=(a-b)^3$

$=(a-b)(a-b)^2$

$=(a-b)(a^2-2ab+b^2)$

$=a^3-2a^2b+ab^2-ba^2+2ab^2-b^3$ *(combining like terms)*

$=a^3-3a^2b+3ab^2-b^3$

$=a^3-3ab(a-b)-b^3$

$=R.H.S.$

Formula 6: $a^3+b^3=(a+b)(a^2-ab+b^2)$

Proof:

$R.H.S=(a+b)(a^2-ab+b^2)$

$=a^3-a^2b+ab^2+a^2b-ab^2+b^3$

$=a^3-\cancel{a^2b}+\cancel{a^2b}+\cancel{ab^2}-\cancel{ab^2}+b^3$

$=a^3+b^3$

$=L.H.S.$

Formula 7: $a^3-b^3=(a-b)(a^2+ab+b^2)$

$R.H.S=(a-b)(a^2+ab+b^2)$

$=a^3+a^2b+ab^2-a^2b-ab^2-b^3$

$=a^3+\cancel{a^2b}+\cancel{a^2b}-\cancel{ab^2}-\cancel{ab^2}-b^3$

$=a^3-b^3$

$=L.H.S$

Find the value of $a^3\pm b^3$ when the values of $a\pm b$ and ab are given.

Example 1:

Find value of a^3-b^3 when $a-b=8$, $ab=2$, $a+b=4$.

Solution:

Given that

$a-b=8$

$ab=2$

$a+b=4$

$a^3-b^3=?$

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$a^3-b^3=(a-b)(a^2+ab+b^2)$

$=8(a^2+b^2+2)$ ..... (i)

Also we know that

$2(a^2+b^2)=(a+b)^2+(a-b)^2$

$=4^2+8^2$

$=16+64$

$=80$

$\Rightarrow a^2+b^2=\frac{80}{2}=40$

$a^2+b^2=40$

Putting in equation (i) we get,

$a^3-b^3=8(40+2)$

$=8(42)$

$=336$

Example 2:

Find value of a^3+b^3 when $a+b=26$, $a-b=14$, $ab=6$.

Solution:

Given that

$a+b=26$

$a-b=14$

$ab=6$

$a^3+b^3=?$

$\because \quad a^3+b^3=(a+b)(a^2-ab+b^2)$

$=(a+b)(a^2+b^2-ab)$

Putting values

$=26(a^2+b^2-6)$ ..... (i)

Now for find (a^2+b^2)

$2(a^2+b^2)=(a+b)^2-(a-b)^2$

Putting given values we get,

$2(a^2+b^2)=(26)^2+(14)^2$

$=676+196$

$=872$

Both sides dividing by "2" we get,

$a^2+b^2=436$

Now putting these values in equation (i) we get,

$a^3+b^3=26(436-6)$

$=26(430)$

$=11180$

Find the continued product of $(x+y)(x-y)(x^2+xy+y^2)(x^2-xy+y^2)$

Example 1: Find continued product of :

$(a+b)(a-b)(a^2+ab+b^2)(a^2-ab+b^2)$

Solution:

$(a+b)(a-b)(a^2+ab+b^2)(a^2-ab+b^2)$

$=(a+b)(a^2-ab+b^2)(a-b)(a^2+ab+b^2)$

$=(a^3+b^3)(a^3-b^3)$

$=(a^3)^2-(b^3)^2$ [using formula $a^2-b^2=(a+b)(a-b)$]

$=a^6-b^6$

Example 2: Find the continued product of:

$(2x+3y)(2x-3y)(4x^2+12xy+9y^2)(4x^2-12xy+9y^2)$

Solution:

$(2x+3y)(2x-3y)(4x^2+12xy+9y^2)(4x^2-12xy+9y^2)$

$=(2x+3y)(4x^2+12xy+9y^2)(2x-3y)(4x^2-12xy+9y^2)$

$=\left((2x)^3+(3y)^3\right)\left((2x)^3-(3y)^3\right)$ (using formulas 6 and 7)

$=(8x^3+27y^3)(8x^3-27y^3)$

$=(8x^3)^2-(27y^3)^2$

$=64x^6-729y^6$

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Exercise 1.7

1. Expand with the help of formulas:
   i) $(x+y)^3$ *ii)* $(2x-3y)^3$
   iii) $\left(\frac{1}{x}+\frac{2}{3}y\right)^3$
2. Reduce each in the form of $(x\pm y)^3$
   i) $x^3+6xy(x+2y)+8y^3$
   ii) $x^3-6x^2y-12xy^2-8y^3$
3. Find the value of a^3+b^3 when $a+b=6$ and $ab=4$
4. Evaluate x^3-y^3 when $x-y=2$ and $xy=12$
5. Using appropriate formulae. Find the continued product of the following.
   i) $(x+2y)(x-2y)(x^2+4y^2+2xy)(x^2+4y^2-2xy)$
   ii) $(2a+3b)(2a-3b)(4a^2+6ab+9b^2)(4a^2-6ab+9b^2)$
6. Find value of a^3+b^3 when
   i) $a+b=3,\quad a-b=5,\quad ab=4$
   ii) $a+b=2,\quad a-b=8,\quad ab=16$
   iii) $a+b=1,\quad a-b=4,\quad ab=12$
7. Find value of a^3-b^3 when
   i) $a+b=7,\quad a-b=9,\quad ab=15$
   ii) $a+b=6,\quad a-b=10,\quad ab=3$
   iii) $a+b=1,\quad a-b=1,\quad ab=3$

1.3 Surds and their application

Surds

Suppose we wish to simplify $\sqrt{\frac{1}{9}}$ we can write it as $\frac{1}{3}$. On the other hand someone numbers involving roots, such as $\sqrt{3}, \sqrt{5}, \sqrt[3]{6}$ etc. can not be expressed exactly in the form of a fraction. Any number of the form $\sqrt[n]{a}$, which cannot be written as a fraction of two integers is called a surd.

Examples:

$$2+\sqrt{3},\ \sqrt{7},\ 5-\sqrt{5},\ 6+\sqrt{5},\ \sqrt{\frac{3}{5}}, \frac{\sqrt{3}}{\sqrt{7}}.$$

Surds of Second Order

Surds of the form $\sqrt{a}$, $\sqrt{b}$ etc where a and b are positive real numbers are called surds or the second order surds, as each can be written as $a^{1/2}$, $b^{1/2}$ where "2" denotes the second order.

Examples

$\sqrt{50}$, $\sqrt{3}$, $\sqrt{5}$ etc.

- **Application of Surds**

There are many applications of surds. For example, by using Pythagoras theorem, the length of the hypotenuse of the triangles with length of base and perpendicular as 1 unit, is given by $\sqrt{2}$.

Solving a quadratic equation by means of a quadratic formula surds arose.

Example

Solve for x, $x^2+3x+1=0$

$$\because\ x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

Here $a=1, b=3, c=1$

So

$$x = \frac{-3\pm\sqrt{9-4}}{2}$$

$$x = \frac{-3\pm\sqrt{5}}{2}$$

Here $\sqrt{5}$ is a surd of second order.

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1.4. Rationalization

The process of rewriting a fraction in such a way that all surds appear in the numerator only, is called rationalization.

For example, rear number of the types $\frac{1}{a+b\sqrt{x}}$, $\frac{1}{\sqrt{x}+\sqrt{y}}$ where a, b are integers and x and y are natural numbers are rationalized by writing simple mathematical operations and algebraic formulae.

Example 1:

Evaluate $\frac{1}{\sqrt{2}+1}$.

Solution:

$$\frac{1}{\sqrt{2}+1}$$

Multiplying numerator and denominator by $\sqrt{2}-1$ then,

$$\frac{1}{\sqrt{2}+1} = \frac{1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1}$$

$$= \frac{\sqrt{2}-1}{(\sqrt{2}+1)(\sqrt{2}-1)}$$

$$= \frac{\sqrt{2}-1}{(\sqrt{2})^2-(1)^2} \qquad \left(\because a^2-b^2=(a+b)(a-b)\right)$$

$$= \frac{\sqrt{2}-1}{2-1}$$

$$= \frac{\sqrt{2}-1}{1} = \sqrt{2}-1$$

Example 2:

Rationalize $\frac{3}{2\sqrt{3}+1}$

Solution:

$$\frac{3}{2\sqrt{3}+1} = \frac{3}{2\sqrt{3}+1} \times \frac{2\sqrt{3}-1}{2\sqrt{3}-1}$$

$$= \frac{3(2\sqrt{3}-1)}{(2\sqrt{3}+1)(2\sqrt{3}-1)}$$

$$= \frac{6\sqrt{3}-3}{(2\sqrt{3})^2-(1)^2}$$

$$= \frac{6\sqrt{3}-3}{4(3)-1}$$

$$= \frac{6\sqrt{3}-3}{12-1}$$

$$= \frac{6\sqrt{3}-3}{11}$$

Example 3:

Rationalize $\frac{1}{2+3\sqrt{5}}$

Solution:

Multiply numerators and denominator by $2-3\sqrt{5}$.

$$\frac{1}{2+3\sqrt{5}} = \frac{1}{2+3\sqrt{5}} \times \frac{2-3\sqrt{5}}{2-3\sqrt{5}}$$

$$= \frac{2-3\sqrt{5}}{(2+3\sqrt{5})(2-3\sqrt{5})}$$

$$= \frac{2-3\sqrt{5}}{2^2-(3\sqrt{5})^2}$$

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$$= \frac{2-3\sqrt{5}}{4-9(5)}$$

$$= \frac{2-3\sqrt{5}}{4-45}$$

$$= \frac{2-3\sqrt{5}}{-41}$$

Example 4:

By means of rationalization, simplify $\frac{1}{\sqrt{5}-\sqrt{3}}$.

Solution:

$$\frac{1}{\sqrt{5}-\sqrt{3}} = \frac{1}{\sqrt{5}-\sqrt{3}} \times \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}}$$

$$= \frac{\sqrt{5}+\sqrt{3}}{(\sqrt{5}-\sqrt{3})(\sqrt{5}+\sqrt{3})}$$

$$= \frac{\sqrt{5}+\sqrt{3}}{(\sqrt{5})^2-(\sqrt{3})^2}$$

$$= \frac{\sqrt{5}+\sqrt{3}}{5-3}$$

$$= \frac{\sqrt{5}+\sqrt{3}}{2}$$

Exercise 1.8

1. Which of the following number are surds.

i) 3 *ii)* $\sqrt{4}$ *iii)* $\sqrt{2}$ *iv)* $3\sqrt{2}$

v) $\sqrt{16}$ *vi)* $\sqrt{\frac{9}{4}}$ *vii)* $\frac{1}{\sqrt{5}}$ *viii)* $2+\sqrt{3}$

ix) $2-\sqrt{3}$ *x)* $\sqrt{\frac{17}{4}}$

2. Separate second order surds from the following.

i) $\sqrt{3}$ *ii)* $3\sqrt{3}$ *iii)* $2^{1/2}$

iv) $\sqrt[4]{5}$ *v)* $\sqrt{7}$

3. Rationalize the following numbers.

i) $\frac{1}{\sqrt{2}+3}$ *ii)* $\frac{1}{\sqrt{3}-1}$ *iii)* $\frac{1}{2\sqrt{3}-1}$

iv) $\frac{1}{\sqrt{5}-\sqrt{3}}$ *v)* $\frac{1}{\sqrt{7}+\sqrt{3}}$

Review Exercise 1.

(Objective Type Questions)

1. Encircle "T" if the statement is true and "F" if the statement is false.

i) x^2+3x+5 is a polynomial. (T/F)

ii) $\frac{x}{x^2+1}$ is a rational expression. (T/F)

iii) $\frac{x-1}{x^2-1}$ is in the lowest term. (T/F)

iv) $\frac{1}{2}$ is not a rational expression. (T/F)

v) In $\frac{p(x)}{q(x)}$, $q(x)$ must be zero. (T/F)

2. Choose the correct option.

i) The degree of x^3-4x^2+3 is:

(a) 2 (b) 3 (c) 1 (d) 4

ii) $(a+b)^2+(a-b)^2 =$

(a) $3(a^2+b^2)$ (b) $2(a^2+b^2)$ (c) (a^2+b^2) (d) $4ab$

iii) $(a+b)^2-(a-b)^2 =$

(a) ab (b) $4ab$ (c) $2ab$ (d) $-4ab$

iv) $a^3+b^3+3a^2b+3ab^2 =$

(a) $(a-b)^3$ (b) $(a+b)^3$ (c) a^3+b^3 (d) a^3

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3. Write a suitable number word in the blank space.

i) The rationalized form of $\frac{1}{\sqrt{2}-1}$ is _____

ii) The process in which surds appear in numerator only is called _____

iii) $(a-b)^3 =$ _____

iv) $(\sqrt{5})^3 - (\sqrt{3})^3 =$ _____

Glossary

- Algebraic Formula: Rule of simplification.
- Algebraic Expression: Combination of constants and variables by basic binary operation.
- Constants: The term which has a fixed value.
- Constant Polynomial: Polynomial with degree zero.
- Difference: The resultant value of subtraction.
- Lowest Term: The most simplified form of a rational expression.
- Mixed Surds: Surds with a rational factor other than unit.
- Polynomial: The algebraic expression in which the exponent of each term is a positive integer.
- Product: The resultant value of multiplication.
- Pure Surd: Surds with rational factor as unity.
- Quotient: The resultant value of division.
- Rational Expression: Expression of the form $\frac{p(x)}{q(x)}, q(x) \neq o.$
- Rationalization: The process in which the surds appear in numerator.
- Sum: The resultant value of addition.
- Surd: The number $\sqrt{a}$ where a is a non-perfect square natural number.
- Second order Surds: Surds of degree 2.
- Variables: Which has no fixed value.

Unit 2

FACTORIZATION

In this unit, the students will be able to learn about the:

- Definition of Factorization
- Factorization of certain forms
  - $kx+ky+kz$
  - $ax+ay+bx+by$
  - $a^2 \pm 2ab+b^2$
  - a^2-b^2
- $(a^2 \pm 2ab+b^2)-c^2$
- $a^4+a^2b^2+b^4$ or a^4+4b^4
- x^2+px+q
- ax^2+bx+c
- $a^3+3a^2b+3ab^2+b^3$
- $a^3-3a^2b+3ab^2-b^3$
- $a^3 \pm b^3$
- Remainder theorem
- Factor theorem
- Factorization of Cubic Polynomial

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2.1 Factorization

Introduction

In lower classes you have learnt that a positive integer can be expressed as a product of two or more positive integers. For example, the possible factorization of 12 are:

$2\times6=12$

$3\times4=12$

$2\times2\times3=12$

Definition

The algebraic expression that can be expressed as the product of two or more algebraic expressions is called the factorization of given algebraic expression.

For example:

$4xy+8x^2y=4xy(1+2x)$ [$4xy$ is common factor to each term]

Here $4xy(1+2x)$ is the factorization of algebraic expression $4xy+8x^2y$. $4xy$ and $1+2x$ are called factors of the given expression.

2.1.2: Factorization of the type: $kx+ky+kz$

The factorization of such type of expression can be found by taking common from the expression $kx+ky+kz$.

Example 1: Factorize the expression $2ax+4ay-8az$.

Solution: Take out 2a as common to obtain:

$2ax+4ay-8az=2a(x+2y-4z)$

Example 2: Factorize the expression $8a^3b^2x+8a^3b^2y$

Solution: Take out $8a^3b^2$ as common to obtain:

$8a^3b^2x+8a^3b^2y=8a^3b^2(x+y)$

Exercise 2.1

Factorize the following algebraic expressions:

1. $2x-8y$
2. $3x^3+6x^2$
3. $4x^3y^3-x^2y^2$
4. $2x+2y-4z$
5. $16x^4-4x^3+2x^2$
6. $28x^2y^4-35xy^3+42x^4y$
7. $7abc-14a^2b^2c^2$
8. $16x+32xy^2$
9. $4x^2+8a^2x^2-4a^3x$
10. $45x^{10}y^4-63x^6y^7+72x^3y^3$

Test:-

2.1.3: Factorization of the type: $ax+ay+bx+by$

In certain expression no factor is common to all terms. In this situation, arrange the common terms and then factorize the expression.

Example 1: Arranging the expression $ax+ay+bx+by$

Solution: Equate common terms to obtain:

$ax+ay+bx+by=(ax+ay)+(bx+by)$

$=a(x+y)+b(x+y)$ (Take a *and* b as common factor)

$=(x+y)(a+b)$ (Take $(x+y)$ as common factor)

Example 2: Factorize the expression $2ax^2+2bx^2+a+3bx+3ax+b$

Solution: Arrange common terms to obtain:

$2ax^2+2bx^2+a+3bx+3ax+b$

$=(2ax^2+2bx^2)+(a+b)+(3ax+3bx)$ (*Arranging the terms*)

$=2x^2(a+b)+1(a+b)+3x(a+b)$

$=(a+b)(2x^2+1+3x)$

$=(a+b)(2x^2+3x+1)$.

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Example 3: Factorize the expression $3x^4y-4x^3y^2-6x^2y^3+8xy^4$

Solution: Take out xy from each term of an expression to obtain:

$$3x^4y-4x^3y^2-6x^2y^3+8xy^4 = xy(3x^3-4x^2y-6xy^2+8y^3)$$
$$= xy\left[x^2(3x-4y)-2y^2(3x-4y)\right]$$
$$= xy(3x-4y)(x^2-2y^2)$$

Exercise 2.2

Test

Factorize the following expressions:

1. $x^2+xy+xz+yz$
2. $a^2+2a+ab+2b$
3. $a^2-2b^3-2ab^2+ab$
4. $ax-a-x+1$
5. $3x^3-2x^2+18x-12$
6. $3x+y+3ax+ay+3bx+b$
7. $2y^5+3y^4+2y^3+3y^2+6y+9$
8. $x^2-xy+xz+3x-3y-yz$

2.14.: Factorization of expression of the type: $a^2 \pm 2ab + b^2$

Such type of expressions can be reduced in the form of perfect square and then take the factors of the resultant expression:

As we know that,

$$a^2+2ab+b^2 = (a+b)^2 = (a+b)(a+b)$$

and $$a^2-2ab+b^2 = (a-b)^2 = (a-b)(a-b)$$

These formulae will be used in solving expressions:

Example 1: Factorize the expression $4x^2+12xy+9y^2$

Solution: Factorize the expression to obtain:

$$4x^2+12xy+9y^2 = (2x)^2+2(2x)(3y)+(3y)^2$$
$$= (2x+3y)^2$$
$$= (2x+3y)(2x+3y)$$

Example 2: Factorize the expression $5x^3-30x^2+45x$

Solution: $5x^3-30x^2+45x = 5x(x^2-6x+9)$
$$= 5x\left[(x)^2-2(x)(3)+(3)^2\right]$$
$$= 5x(x-3)^2 \quad \left[\because a^2-2ab+b^2=(a-b)^2\right]$$
$$= 5x(x-3)(x-3)$$

Exercise 2.3

Test

Factorize the following expressions:

1. x^2+4x+4
2. $y^2-8y+16$
3. $9a^2+12ab+4b^2$
4. $-2x^2+12x-18$
5. $25(a+b)^2-20(a+b)c+4c^2$
6. $49y^2-14yz+z^2$
7. $16a^2+9b^2-24ab$
8. $4x^2+4+\frac{1}{x^2}$

2.1.5: Factorization of expression of the type a^2-b^2

We know that

$$a^2-b^2 = (a+b)(a-b)$$

This rule will be used in solving expressions:

Example 1: Factorize the expression $4x^2-25y^2$

Solution: $4x^2-25y^2 = (2x)^2-(5y)^2$
$$= (2x+5y)(2x-5y)$$

Example 2: Factorize the expression a^4-b^4

Solution:

$$a^4-b^4 = (a^2)^2-(b^2)^2$$
$$= (a^2+b^2)(a^2-b^2)$$
$$= (a^2+b^2)\left[(a+b)(a-b)\right]$$
$$= (a^2+b^2)(a+b)(a-b)$$

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Example 3: Factorize the expression $16x^5 - 9xy^2$

Solution: $16x^5 - 9xy^2 = x(16x^4 - 9y^2)$

$= x\left[(4x^2)^2 - (3y)^2\right]$

$= x(4x^2 + 3y)(4x^2 - 3y)$

Example 4: Factorize the expression $x^2y^2 - x^2 - y^2 + 1$

Solution: $x^2y^2 - x^2 - y^2 + 1 = x^2(y^2 - 1) - 1(y^2 - 1)$

$= (y^2 - 1)(x^2 - 1)$

$= \left[(y)^2 - (1)^2\right]\left[(x)^2 - (1)^2\right]$

$= (y+1)(y-1)(x+1)(x-1)$

Exercise 2.4

Factorize the following expressions:

1. $16x^2 - 9$
2. $x^3y - xy^3$
3. $2x^2 - 288$
4. $2(x-y)a^2 - 18(x-y)$
5. $\frac{x^2}{y^2} - \frac{y^2}{x^2}$
6. $\frac{81}{16}x^4 - 625$
7. $16 - 25x^2$
8. $ab^3 + 2b^2 - ab - 2$
9. $9y^2 - 64x^2$
10. $x^8 - y^8$

2.1.6: Factorization of expression of the type: $(a^2 \pm 2ab + b^2) - c^2$

We know that

$a^2 + 2ab + b^2 - c^2 = (a+b)^2 - (c)^2$

$= (a+b+c)(a+b-c)$ $\left[\because a^2 - b^2 = (a+b)(a-b)\right]$

And $a^2 - 2ab + b^2 - c^2 = (a-b)^2 - (c)^2$

$= (a-b+c)(a-b-c)$

Example 1: Factorize the expression $4a^2 + 4ab + b^2 - c^2$

Solution: $4a^2 + 4ab + b^2 - c^2 = \left[(2a)^2 + 2(2a)(b) + (b)^2\right] - c^2$

$= (2a+b)^2 - (c)^2$

$= (2a+b+c)(2a+b-c)$

Example 2: Factorize the expression $a^2 - b^2 + 2b - 1$

Solution: $a^2 - b^2 + 2b - 1 = a^2 - (b^2 - 2b + 1)$

$= a^2 - \left[(b)^2 - 2(b)(1) + (1)^2\right]$

$= a^2 - (b-1)^2$

$= [a + (b-1)][a - (b-1)]$

$= (a+b-1)(a-b+1)$

Example 3: Factorize the expression $a^2 + b^2 + 2ab - c^2 - d^2 - 2cd$

Solution: $a^2 + b^2 + 2ab - c^2 - d^2 - 2cd = (a^2 + 2ab + b^2) - (c^2 + d^2 + 2cd)$

$= (a+b)^2 - (c+d)^2$

$= \left[(a+b) + (c+d)\right]\left[(a+b) - (c+d)\right]$

$= (a+b+c+d)\ (a+b-c-d)$

Exercise 2.5

Factorize the following expressions.

1. $x^2 - 2xy + y^2 - 4$
2. $9a^2b^2 - 12abc + 4c^2 - 25d^2$
3. $1 - a^2 + 2ab - b^2$
4. $1 + 2xy - y^2 - x^2$
5. $a^2 + 2ab + b^2 - c^2 + 2cd - d^2$
6. $2(xy + pq) - x^2 - y^2 + p^2 + q^2$
7. $-a^2 + b^2 - 2b + 1$
8. $25a^2b^2 - 20abc + 4c^2 - 16d^2$

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2.1.7: Factorization of expression of the type: $a^4+a^2b^2+b^4$ or a^4+4b^4

The algebraic expressions which are neither perfect square nor in the form of the difference of two squares. Then it can be converted into perfect square by adding and subtracting of some suitable terms.

Example 1: Factorize the expression $a^4+a^2b^2+b^4$

Solution: The given expression is not in a perfect square. The addition and subtraction will make it a perfect square:

$$\begin{aligned}a^4+a^2b^2+b^4 &= a^4+a^2b^2+b^4+a^2b^2-a^2b^2\\ &= a^4+2a^2b^2+b^4-a^2b^2\\ &= [(a^2)^2+2(a^2)(b^2)+(b^2)^2]-a^2b^2\\ &= (a^2+b^2)^2-(ab)^2\\ &= (a^2+b^2+ab)(a^2+b^2-ab) \quad [\because\ a^2-b^2=(a+b)(a-b)]\\ &= (a^2+ab+b^2)(a^2-ab+b^2)\end{aligned}$$

Example 2: Factorize the expression a^4+4b^4

Solution: The given expression is not a perfect square. The addition and subtraction will make it a perfect square:

$$\begin{aligned}a^4+4b^4 &= a^4+4b^4+4a^2b^2-4a^2b^2\\ &= \left[(a^2)^2+4(a)^2(b^2)+(2b^2)^2\right]-4a^2b^2\\ &= (a^2+2b^2)^2-(2ab)^2\\ &= (a^2+2b^2+2ab)(a^2+2b^2-2ab) \quad [\because\ a^2-b^2=(a+b)(a-b)]\end{aligned}$$

Example 3: Factorize the expression x^4-7x^2+1

Solution: $x^4-7x^2+1=x^4+1-7x^2$

Add and subtract $2x^2$ to make it a perfect square:

$$\begin{aligned}x^4-7x^2+1 &= x^4+1-7x^2+2x^2-2x^2\\ &= x^4+2x^2+1-7x^2-2x^2\\ &= \left[(x^2)^2+2(x^2)(1)+(1)^2\right]-9x^2\\ &= (x^2+1)^2-(3x)^2\\ &= (x^2+1+3x)(x^2+1-3x)\\ &= (x^2+3x+1)(x^2-3x+1)\end{aligned}$$

Exercise 2.6

Factorize the following expressions:

1. x^4+4
2. $4x^4+81$
3. $4p^4+q^4$
4. x^4+x^2+1
5. x^8+x^4+1
6. $x^4+\frac{1}{x^4}-7$
7. $16x^4+4x^2y^2+y^4$
8. $4a^5b+11a^3b^3+9ab^5$

2.1.8: Factorization of the expression of the type x^2+px+q

The steps involved in the factorization of the above types must be followed.

i. Develop the factors of the constant term q.
ii. Find two numbers whose sum is p and product is q.
iii. Factorize the expression in the light of the factors of the constant term q.

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Example 1: Factorize the expression x^2-6x+8

Solution: Here $p=-6$ and $q=8$, possible factorization of 8 are as under.

factorization of 8	sum of factors of 8
$1\times 8=8$	$1+8=9$
$2\times 4=8$	$2+4=6$
$(-1)\times(-8)=8$	$-1-8=-9$
$(-2)\times(-4)=8$	$-2-4=-6$

Thus, we find two numbers -2 and -4 whose sum is -6 and product is 8.

The factorization of the expression is as under:

$$\begin{aligned} x^2-6x+8 &= x^2-2x-4x+8 \\ &= (x^2-2x)-(4x-8) \\ &= x(x-2)-4(x-2) \\ &= (x-2)(x-4) \end{aligned}$$

Example 2 Factorize the expression y^2+5y+6

Solution: The factorization of expression in the light of constant term 6 is as under:

$$\begin{aligned} y^2+5y+6 &= y^2+3y+2y+6 \\ &= (y^2+3y)+(2y+6) \\ &= y(y+3)+2(y+3) \\ &= (y+3)(y+2) \end{aligned}$$

Exercise 2.7

Factorize the following expressions:

1. x^2+6x+8
2. x^2-2x-8
3. $y^2+6y-55$
4. y^3-8y^2+15y
5. z^2-z-20
6. $-2x^3+48x^2-46x$
7. $-3x^4+6x^3+144x^2$
8. $x^2+10x+21$

2.1.9: Factorization of the expression of the type ax^2+bx+c

The steps involved in the factorization of the expression ax^2+bx+c are the following:

i. Find out the product of ac, where a is the co-efficient of x^2 and c is constant.

ii. Find two numbers whose sum is b and product is ac.

Example 1: Factorize the expression $2x^2-5x+3$

Solution: Compare $2x^2-5x+3$ with ax^2+bx+c, to obtain $a=2, b=-5$, $c=3$ and $ac=2\times 3=6$

Two numbers whose sum is b=–5 and product is ac=6 are –2 and – 3.

$$\begin{aligned} 2x^2-5x+3 &= 2x^2+(-2-3)x+3 \\ &= 2x^2-2x-3x+3 \\ &= (2x^2-2x)-(3x-3) \\ &= 2x(x-1)-3(x-1) \\ &= (x-1)(2x-3) \end{aligned}$$

Example 2: Factorize the expression $10x^2+8x-2$

Solution: Taking 2 as common $2(5x^2+4x-1)$

The two numbers whose sum is 4 and product is –5 are 5 and –1:

$$\begin{aligned} 10x^2+8x-2 &= 2(5x^2+4x-1) \\ &= 2(5x^2+5x-x-1) \\ &= 2[5x(x+1)-1(x+1)] \\ &= 2(x+1)(5x-1) \end{aligned}$$

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Exercise 2.8

Factorize the following expression:

1. $2x^2-7x+3$
2. $4x^2+8x+3$
3. $2y^2-3y+1$
4. $6x^3-15x^2+9x$
5. $2xy^2+8xy-24x$
6. $2+5y-12y^2$
7. $2x^2+13x+20$
8. $4x^2+20x+21$
9. $-16x^2y-20x^2y^2-6xy^3$
10. $8x^2+23x-3$

2.1.10: Factorization of expression of the type $a^3+3a^2b+3ab^2+b^3$ and $a^3-3a^2b+3ab^2-b^3$

We know that

$$(a+b)^3=a^3+3a^2b+3ab^2+b^3$$

and $$(a-b)^3=a^3-3a^2b+3ab^2-b^3$$

These formulae will help us in factorizing the expression of the above type.

Example 1: Factorize the expression $27x^3+27x^2y+9xy^2+y^3$

Solution:

$$27x^3+27x^2y+9xy^2+y^3=(3x)^3+3(3x)^2y+3(3x)y^2+y^3=(3x+y)^3$$

Example 2: Factorize the expression $8a^3-36a^2b+54ab^2-27b^3$

Solution:

$$8a^3-36a^2b+54ab^2-27b^3=(2a)^3-3(2a)^2(3b)+3(2a)(3b)^2-(3b)^3$$
$$=(2a-3b)^3$$

Exercise 2.9

Factorize the following expressions.

1. $x^3+3x^2y+3xy^2+y^3$
2. $64x^3-144x^2y+108xy^2-27y^3$
3. $125a^3+75a^2b+15ab^2+b^3$
4. $27a^3+189a^2b+441ab^2+343b^3$
5. $\frac{a^3}{8}-\frac{1}{4}a^2b+\frac{1}{6}ab^2-\frac{b^3}{27}$
6. $x^3+3x+\frac{3}{x}+\frac{1}{x^3}$
7. $8a^3+48a^2b+96ab^2+64b^3$
8. $8x^3-4x+\frac{2}{3x}-\frac{1}{27x^3}$

2.1.11: Factorization of an expression of the type a^3+b^3 and a^3-b^3

We know that

$$a^3+b^3=(a+b)(a^2-ab+b^2)$$

and $$a^3-b^3=(a-b)(a^2+ab+b^2)$$

These formulae will enable us to factorize different types of expressions.

Example 1: Factorize the expression x^3-64

Solution:

$$x^3-64=(x)^3-(4)^3$$
$$=(x-4)\left[(x)^2+(x)(4)+(4)^2\right]$$
$$=(x-4)(x^2+4x+16)$$

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Example 2: Factorize the expression $128x^3+54y^3$

Solution:

$$128x^3+54y^3=2\left(64x^3+27y^3\right)$$
$$=2\left[(4x)^3+(3y)^3\right]$$
$$=2(4x+3y)\left[(4x)^2-(4x)(3y)+(3y)^2\right]$$
$$=2(4x+3y)\left(16x^2-12xy+9y^2\right)$$

Exercise 2.10

Factorize the following expressions:

1. $2x^3+16$
2. a^3-64b^3
3. x^6-y^6
4. $64y^3+27$
5. $24x^3+3$
6. $a^3+(c+d)^3$
7. $x^{12}-y^{12}$
8. $343a^3-125b^3+7a-5b$
9. $1-27y^3$
10. $18(x-y)^3-144(a-b)^3$

2.2: Remainder Theorem and Factor Theorem

2.2.1: Remainder Theorem

Statement: If a polynomial $P(x)$ of degree $n\geq 1$ is divided by $x-r$, until a constant remainder R is obtained then $R=P(r)$.

Proof: Let $Q(x)$ be the quotient and R is the remainder, when $P(x)$ is divided by $x-r$:

$$P(x)=(x-r)Q(x)+R$$
$$P(r)=(r-r)Q(r)+R, \qquad (r-r=0)$$
$$P(r)=R$$

Example 1.

Divide $p(x)=3x^3-7x^2+6x-3$ by $x-2$ and verify your answer by Remainder theorem.

Solution:

$$\begin{array}{r|l} & 3x^2-x+4 \quad \leftarrow \text{Quotient} \\ \hline x-2 & 3x^3-7x^2+6x-3 \\ & -3x^3 \mp 6x^2 \\ \hline & -x^2+6x \\ & \mp x^2 \pm 2x \\ \hline & 4x-3 \\ & -4x \mp 8 \\ \hline & 5 \quad — \quad \rightarrow \text{Remainder} \end{array}$$

Divisor

Verification:

$p(x)=3x^3-7x^2+6x-3$ ..... (i)

Divisor $=x-2$

Therefore $r=2$

Put r = 2 in equation (i) to obtain:

$$p(r)=p(2)=3(2)^3-7(2)^2+6(2)-3$$
$$=24-28+12-3$$
$$=36-31$$
$$=5$$

$\Rightarrow \quad p(2)=5$

Or $\quad R=p(2)=5$

This is the result previously obtained by direct division.

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Example 2: Without direct division, find the remainder when x^3+4x^2+2x+3 is divided by $x+3$.

Solution: Here $p(x)=x^3+4x^2+2s+3$

and $x-r=x+3=x-(-3)$

Put $x=r=-3$ in the polynomial to obtain R:

$$P(-3)=(-3)^3+4(3)^2+2(-3)+3$$
$$P(-3)=-27+36-6+3$$
$$P(-3)=39-33$$
$$P(-3)=6=R$$

Hence the remainder is 6.

Example 3: Find the value of k, when $3x^3+9x^2-(3k-4)x+2$ is exactly divided by a factor $x-1$.

Solution: If $p(x)=3x^3+9x^2-(3k-4)x+2$ is exactly divided by $x-1$, then the remainder will be zero.

$$R=p(1)=0$$

or $$3(1)^3+9(1)^2-(3k-4)(1)+2=0$$
$$3+9-(3k-4)+2-0$$
$$12-3k+4+2=0$$
$$18-3k=0$$
$$18=3k$$
$$\frac{18}{3}=k$$
$$k=6$$

Exercise 2.11

Find the remainder of the following polynomials by Remainder Theorem.

1. When $x^2-8x+11$ divided by $x-1$.
2. $2x^3+7x^2+4x+2$ divided by $x+3$.
3. $3x^3+x-200$ divided by $x-4$.
4. Without performing division find the value of k, when $2x^3-kx^2-2kx+3x+2$ is exactly divisible by $x+1$.
5. Find the value of 'a' when x^3-4x^2+x-2 is exactly divisible by $x-1$.

2.2.2: Zero of a Polynomial

Let $p(x)$ be a polynomial, A value of x for which $p(x)=0$, is called a zero of the polynomial.

Example Find zero of the polynomial $p(x)=x^2-6x+5$

Solution: The value of $x=1$ reduces the $P(x)$ to zero:

$$p(1)=(1)^2-6(1)+5$$
$$=1-6+5$$
$$=6-6=0$$

Hence 1 is zero of the polynomial x^2-6x+5.

Similarly, $x=5$ is also a zero of the polynomial $P(x)$:

$$p(x)=(5)^2-6(5)+5$$
$$=25-30+5$$
$$=30-30$$
$$=0$$

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Factor Theorem

Statement: If a polynomial $P(x)$ of degree $n \geq 1$ is divided by $x-r$ such that $R=0$. Then $x-r$ is called a factor of $P(x)$.

Proof:

If $(x-r)$ is a factor of the polynomial $p(x)$ then the remainder $R=0$.

i.e. $p(x)=(x-r)Q(x) \quad (\because R=0)$

$p(r)=(r-r)Q(r), x=r$

$\Rightarrow \quad p(r)=0.Q(r)$

$\Rightarrow \quad p(r)=0$

Conversely, if $x=r$ is a root of $P(x)=0$, then $P(r)=0$. But, by remainder theorem, $P(r)=R$

Thus $R=P(r)=0$.

Hence $P(x)=(x-r)Q(x)$.

Therefore

Since $p(x)=(x-r)Q(x)+R$

$\Rightarrow \quad p(x)=(x-r)Q(x)+0$

$\Rightarrow \quad p(x)=(x-r)Q(x)$

$\Rightarrow \quad x-r$ is a factor of $Q(x)$.

This completes the proof of factor theorem.

2.3: Factorization of a Cubic Polynomial

Factorization of the cubic polynomial can be found through factor theorem.

Example 1:

Use factor theorem to, prove that $x-3$ is a factor of x^3-x^2-5x-3. Find also the other factors.

Solution:

$p(x)=x^3-x^2-5x-3$

$(x-3)$ is a factor of $p(x)=x^3-x^2-5x-3$, if and only if $x=r=3$ is a zero of the polynomial.

$p(3)=(3)^2-(3)^2-5(3)-3$

$=27-9-15-3$

$=27-27$

$p(3)=0$

It can also be found through direct division.

$$
\begin{array}{r|l}
x-3 & x^2+2x+1 \\
\hline
 & x^3-x^2-5x-3 \\
 & -x^3 \mp 3x^2 \\
\hline
 & 2x^2-5x-3 \\
 & -2x^2 \mp 6x \\
\hline
 & x-3 \\
 & -x \mp 3 \\
\hline
 & 0
\end{array}
$$

Here $Q(x)=x^2+2x+1$ and $R=0$. The verification of factors theorem is:

Since

$p(x)=(x-r)Q(x)+R$

$\Rightarrow \quad x^3-x^2-5x-3=(x-3)(x^2+2x+1)+0$

$=(x-3)\left[(x)^2+2(x)(1)+(1)^2\right]$

$=(x-3)(x+1)^2$

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Example 2:

Factorize the expression $x^3-17x+26$ by using Factor Theorem.

Solution: The polynomial

$$p(x)=x^3-17x+26$$

Put $x=+1,-1,2,-2,$ in P(x), we see that at $x=2$, the polynomial $p(x)=0$

$$p(2)=(2)^3-17(2)+26$$
$$=8-34+26$$
$$=34-34$$
$$=0$$

Thus $x-2$ is factor of $p(x)$. The remaining factors can be formed through direct division.

Now divide $p(x)$ by $x-2$, we have

$$\begin{array}{r|l} x-2 & x^2+2x-13 \\ \hline & x^3-17x+26 \\ & \mp x^3 \qquad \mp 2x^2 \\ \hline & 2x^2-17x \\ & -2x^2 \mp 4x \\ \hline & -13x+26 \\ & \mp 13x \pm 26 \\ \hline & 0 \end{array}$$

Hence $x^3-17x+26=(x-2)(x^2+2x-13)$

Example 3:

Factorize the expression x^3-7x-6 by Factor Theorem.

Solution: The polynomial

$$p(x)=x^3-7x-6.$$

Put $x=1,-1,+2,-2,$ in P(x) and by inspection, we see that at $x=-2$, $P(x)=0$.

$\therefore$ Put $x=-2$

$$p(-2)=(-2)^3-7(-2)-6$$
$$=-8+14-6$$
$$=14-14$$
$$=0$$

The remaining factors can be found through direct division.

$$\begin{array}{r|l} x+2 & x^2-2x-3 \\ \hline & x^3+0x^2-7x-6 \\ & -x^3 \pm 2x^2 \\ \hline & -2x^2-7x \\ & \mp 2x^2 \mp 4x \\ \hline & -3x-6 \\ & \mp 3x \mp 6 \\ \hline & 0 \end{array}$$

$$P(x)=(x-r)Q(x)$$

Hence $x^3-7x-6 = (x+2)(x^2-2x-3)$

$$=(x+2)[x^2-3x+x-3]$$
$$=(x+2)[x(x-3)+1(x-3)]$$
$$=(x+2)(x-3)(x+1)$$

Exercise 2.12

Factorize the following polynomials, by factor theorem:

1. x^3-2x^2-5x+6
2. x^3-x^2-4x-4
3. $x^3-9x^2+23x-15$
4. $x^3-4x^2-3x+18$
5. $x^3+2x^2-19x-20$
6. $x^3-x^2-14x+24$
7. Prove that $x-1$ is a factor of x^3-7x+6. Find also the other factors.
8. Prove that $x+2$ is a factor of x^3-6x^2+32. Find also the other factors.

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Exercise 2.13

(Objective Type Questions)

1. True and False questions.
Read the following statement carefully and tick "T" in case of true statement and "F" in case of false statement.

(i) $x^2-4=(x+2)(x-2)$ (T/F)

(ii) $2x^2y+8xy=2xy(x+4y)$ (T/F)

(iii) $x^3+8=(x+2)(x^2-2x+4)$ (T/F)

(iv) $x^2-5x+6=(x+6)(x-1)$ (T/F)

(v) $x^4-y^4=(x-y)(x+y)(x^2+y^2)$ (T/F)

2. Fill in the blanks.

(i) $9x^2-16y^2=$ _____

(ii) $x^2-xy-xz+yz=$ _____

(iii) $(2a^2-3b^2)^2=$ _____

(iv) $x^3-27=$ _____

(v) $x^2-11x+30=$ _____

3. Multiple Choices questions.
Write the correct option in the box provided.

(i) Factors of $16x^2-64y^2$ are ☐

a) $(4x-8y)(4x-8y)$ b) $16(x+y)(x-y)$

c) $16(x+2y)(x-2y)$ d) $16(x+2y)(x+2y)$

(ii) Factors of $x^2-2x-24$ ☐

a) $(x+4)(x-6)$ b) $(x-4)(x+6)$

c) $(x+4)(x+6)$ d) $(x-4)(x-6)$

(iii) Factors of a^3-b^3 are ☐

a) $(a+b)(a^2-ab+b^2)$ b) $(a-b)(a^2+ab+b^2)$

c) $(a+b)(a^2+ab-b^2)$ d) $(a+b)(a^2+ab+b^2)$

(iv) $\frac{1}{a+b}+\frac{b}{a^2-b^2}=$ _____ ☐

a) $\frac{b+1}{a^2-b^2}$ b) $\frac{a}{a^2-b^2}$

c) $\frac{b}{a^2-b^2}$ d) $\frac{b+a}{a^2-b^2}$

(v) Factorization of $x^2+2xy+y^2-z^2$ is ☐

a) $(x-y)^2+z^2$ b) $(x-y)^2-z^2$

c) $(x+y+z)(x+y-z)$ d) $(x+y+z)(x+y-z)$

Review Exercise: 2

Factorize the following polynomials.

1. $a^2b+a^2c+a^2d+b^2c+b^2d+b^3$
2. $(a-b)^2-9$
3. $-2x^3+48x^2-46x$
4. x^6+y^6
5. Without direct division, find the remainder when $2x^3-3x^2+x-2$ is divided by $x-3$.

Factorize the following polynomials:

6. $2x^3-128$
7. $y^2-2y-48$
8. x^3+3x^2-9x+5
9. $27x^3-27x^2y+9xy^2-y^3$
10. Use Factor Theorem to prove that $x-3$ is a factor of x^3-x^2-5x-3. Find also the other factors.

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Glossary

Factorization of an algebraic expression

Writing an algebraic expression as the product of two or more algebraic expressions is called factorization of that algebraic expression.

Remainder Theorem

If a polynomial $P(x)$ of degree $n \geq 1$ is divided by a factor until $x-r$ a constant remainder is obtained then $R = P(r)$.

Factor Theorem

Let $p(x)$ be a polynomial. A linear polynomial $x-r$ is a factor of $p(x)$ if and only if r is zero of the polynomial $p(x)$ i.e. $p(r)=0$.

Zero of a Polynomial

Let $P(x)$ be a polynomial. A value of x for which $P(x)=0$, is called zero of the polynomial.

Unit 3

ALGEBRAIC MANIPULATION

In this unit, the students will be able to learn about the:

- Concept of highest common factor (H.C.F)
- Concept of least common multiple (L.C.M)
- Relationship between H.C.F. and L.C.M
- Basic operations on algebraic fractions
- Square root of algebraic expressions.

Introduction:

In previous classes we have studied the concepts of H.C.F, L.C.M and square root of numbers. In this unit these terms are applied to algebraic expression.

3.1: Highest Common Factor and Least Common Multiple

3.1.1: Highest Common Factor (H.C.F)

Definition

The H.C.F of two or more polynomials is a polynomial of highest degree which divides exactly the given polynomials.

The H.C.F can be found by:

i. Factorization
ii. Direct Division.

(i) Factorization

Example 1: Find H.C.F of the expression x^2-y^2 and x^2-xy.

Solution: The common factors of the expressions are:

$$x^2-y^2=(x+y)(x-y)$$

$$x^2-xy=x(x-y)$$

We see that $(x-y)$ is the only common factors to both expressions. Therefore H.C.F = $x-y$

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Example 2: Find H.C.F of the expressions $ax^2+5ax+6a, ax^3+9ax^2+14ax$ and $15ax^2-60a$

Solution: The common factors of the expressions are:

$$ax^2+5ax+6a = a(x^2+5x+6)$$
$$= a(x^2+3x+2x+6)$$
$$= a[x(x+3)+2(x+3)]$$
$$= a(x+3)(x+2) \quad \text{.....(1)}$$

$$ax^3+9ax^2+14ax = ax(x^2+9x+14)$$
$$= ax(x^2+7x+2x+14)$$
$$= ax[x(x+7)+2(x+7)]$$
$$= ax(x+7)(x+2) \quad \text{.....(2)}$$

$$15ax^2-60a = 15a(x^2-4)$$
$$= 15a(x+2)(x-2) \quad \text{.....(3)}$$

From equation (1), (2), (3), we see that $a(x+2)$ is common to given three expression. Therefore the H.C.F = $a(x+2)$

(ii) Direct Division

To find H.C.F by division methods following steps are followed.

1. Write the expression or polynomial in descending order with respect to variable.
2. Divide higher degree polynomial by the polynomial of lower degree.
3. If degrees are the same, then any one can be divided by the other.
4. The 1st divisor is divided by remainder.
5. Repeat the process till the remainder is going to be zero.
6. Last divisor is the required H.C.F of the given polynomial.

In case of three polynomials, find H.C.F of any two of the polynomials and then divide 3rd polynomial by H.C.F of the first two polynomials to obtain the final H.C.F.

Example: Find H.C.F of the two polynomials $2x^3+7x^2+4x-4$ and $2x^3+9x^2+11x+2$.

Solution:

Let $P(x) = 2x^3+7x^2+4x-4$

and $Q(x) = 2x^3+9x^2+11x+2$ are two polynomials. Divide out $Q(x)$ (by $P(x)$ to obtain.

$$\begin{array}{r|l} & 1 \\ \hline 2x^3+7x^2+4x-4 & 2x^3+9x^2+11x+2 \\ & \underline{-\,2x^3+7x^2+4x-4} \\ \hline 2x^2+7x+6 & 2x^3+7x^2+4x-4 \;(\,x \\ & \underline{-\,2x^3+7x^2+6x} \\ \hline & -2 \;\big|\; -2x-4 \\ & x+2 \end{array}$$

Divided by 2

$$\begin{array}{r|l} & 2x^2+3x-2 \\ \hline x+2 & 2x^3+7x^2+4x-4 \\ & \underline{-\,2x^3+4x^2} \\ \hline & 3x^2+4x-4 \\ & \underline{-\,3x^2+6x} \\ \hline & -2x-4 \\ & \underline{+\;\;2x+4} \\ \hline & 0 \end{array}$$

Hence H.C.F of $P(x)$ and $Q(x) = x+2$.

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Example 4: Find H.C.F of $x^3-6x^2+11x-6, 3x^3-5x^2+6x-4$ and $3x^3+5x^2-6x-2$.

Solution:

Let $P(x)=x^3-6x^2+11x-6$

$Q(x)=3x^3-5x^2+6x-4$

and $R(x)=3x^3+5x^2-6x-2$

First we shall find H.C.F of $Q(x)$ and $R(x)$.

Note: You can find the H.C.F of $P(x)$ and $Q(x)$, it depends upon your own choice.

$3x^3-5x^2+6x-4 \;)\; 3x^3+5x^2-6x-2 \;(1$

$-\;3x^3-5x^2+6x-4$

$2 \;|\; 10x^2-12x+2$ (Dividing by 2)

$5x^2-6x+1 \;)\; 3x^3-5x^2+6x-4 \;(3x-7$

$\times 5$ (Multiplying by 5)

$15x^3-25x^2+30x-20$

$-\;15x^3-18x^2+3x$

$-7x^2+27x-20$

$\times 5$ (Multiplying by 5)

$-35x^2+135x-100$

$-35x^2+42x-7$

$93 \;|\; 93x-93$ (Dividing by 93)

$x-1 \;)\; 5x^2-6x+1 \;(5x-1$

$-\;5x^2-5x$

$-x+1$

$-x+1$

0

Hence, H.C.F of $Q(x)$ and $R(x)=x-1$

Now we shall find H.C.F of $x-1$ and $P(x)$.

$x-1 \;)\; x^3-6x^2+11x-6 \;(x^2-5x+6$

$-\;x^3-x^2$

$-5x^2+11x$

$-5x^2+5x$

$6x-6$

$-\;6x-6$

0

Hence required H.C.F of the three polynomials $x^3-6x^2+11x-6$, $3x^3-5x^2+6x-4$ and $3x^3+5x^2-6x-2$ is $(x-1)$.

Exercise 3.1

Find H.C.F of the following expressions by factorization method:

1. $(x-6)^2$ and x^2-36
2. $a^2b^2(a^2+ab+b^2)$ and $a^3b^3(a^3-b^3)$
3. x^4-y^4 and $x^4+2x^2y^2+y^4$
4. $16m^2-16m+4$ and $6m^2+5m-4$
5. $2ax^2-2ax-12a$, $3a^2x^2-7a^2x-6a^2$ and $a^3x^2+4a^3x-21a^3$.

Find H.C.F of the following expressions by direct division method.

6. x^2-x-6 and x^2-2x-3.
7. y^3-3y+2 and y^3-5y^2+7y-3.
8. $2x^5-4x^4-6x$ and $x^5+x^4-3x^3-3x^2$.
9. $2x^3+10x^2+5x+25$ and x^3+5x^2-x-5.
10. x^3-x^2-2x, $2x^3+x^2-10x$ and x^3+2x^2-8x.

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3.1.2: Least Common Multiple (L.C.M)

Definition:

The L.C.M of two or more polynomial is a polynomial of least degree which is divisible by the given polynomials. It is equal to the product of common and non-common factors.

The L.C.M of given expression can be found by:

(a) Factorization
(b) Direct division

(a) L.C.M by Factorization Method

In L.C.M by factorization, the following formula is used for finding LCM.

L. C. M = Common Factors × Non-Common Factors

Example 1: Find L.C.M of the expression $x^2+2x+x+2$ and x^2+4x+4.

Solution: The factorization of the given expression are:

$$x^2+2x+x+2$$
$$=x(x+2)+1(x+2)$$
$$=(x+2)(x+1) \quad \text{.....(1)}$$

$x^2+4x+4=(x)^2+2(x)(2)+(2)^2$, using formula $(a+b)^2=a^2+2ab+b^2$

and
$$=(x+2)^2$$
$$=(x+2)(x+2) \quad \text{.....(2)}$$

In expression (1) and (2), the common factor is $(x+2)$ and non-common factors are $(x+1)$ and $(x+2)$.

L.C.M = Common factors X Non-Common factors
$= (x+2)\times(x+1)(x+2)$

L.C.M $= (x+2)^2(x+1)$

Example 2: Find L.C.M of the expression x^2-4x+3, x^2-3x+2 and x^2-5x+6.

Solution: The factorization of the expression are:

$$x^2-4x+3=x^2-3x-x+3$$
$$=x(x-3)-1(x-3)$$
$$=(x-3)(x-1) \quad \text{.....(1)}$$

$$x^2-3x+2=x^2-2x-x+2$$
$$=x(x-2)-1(x-2)$$
$$=(x-2)(x-1) \quad \text{.....(2)}$$

$$x^2-5x+6=x^2-3x-2x+6$$
$$=x(x-3)-2(x-3)$$
$$=(x-3)(x-2) \quad \text{.....(3)}$$

In expression (1) and (2) $x-1$ is common factor
in expression (2) and (3) $x-2$ is common factor
in expression (1) and (3) $x-3$ is common factor.

The LCM is therefore

L.C.M = common factors × non common factors
$=(x-1)(x-2)(x-3)$

(b) L.C.M by Direct division

L.C.M of two polynomials by division can be found by using the formula:

$$L.C.M=\frac{Product\ of\ two\ polynomials}{H.C.F\ of\ two\ polynomials}$$

This formula is used when factorization of two polynomials is not possible or when we have to find both L.C.M and H.C.F.

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Example 3: Find L.C.M of the expression $x^3-6x^2+11x-6$ and x^3-4x+3.

Solution: First we shall find out H.C.F of given polynomials.

$x^3-4x+3 \overline{)\, x^3-6x^2+11x-6 \;(\,1}$

$-x^3 \pm 4x \mp 3$

$-3 \mid -6x^2+15x-9$ (Dividing by 3)

$2x^2-5x+3 \overline{)\, x^3-4x+3 \;(\,x+5}$

$\times 2$ (Multiplying by 2)

$2x^3-8x+6$

$-2x^3 \mp 3x \pm 5x^2$

$5x^2-11x+6$

$\times 2$ (Multiplying by 2)

$10x^2-22x+12$

$+10x^2 \mp 25x \pm 15$

$3 \mid 3x-3$ (Dividing by 3)

$x-1 \overline{)\, 2x^2-5x+3 \;(\,2x-3}$

$+2x^2 \mp 2x$

$-3x+3$

$+-3x+3$

0

Hence H.C.F of x^3-4x+3 and $x^3-6x^2+11x-6$ is $x-1$.

The L.C.M through given formula is:

$$L.C.M = \frac{\text{Product of two polynomials}}{\text{H.C.F of two polynomials}}$$

$$= \frac{(x^3-4x+3)(x^3-6x^2+11x-6)}{x-1}$$

Divide out $x^3-6x^2+11x-6$ by $x-1$ to obtain:

$x-1 \overline{)\, x^3-6x^2+11x-6 \;(\,x^2-5x+6}$

$-x^3 \pm x^2$

$-5x^2+11x$

$+-5x^2 \pm 5x$

$6x-6$

$-+6x-6$

0

$\therefore$ L.C.M $= (x^3-4x+3)(x^2-5x+6)$

3.1.3: Relationship between H.C.F. and L.C. M

Relationship between H.C.F and L.C.M is shown by the following formula, which we have discussed in section 3.1.2.

Product of two polynomials = H.C.F × L.C.M

Theorem:

If H and L are the H.C.F and L.C.M of the given two polynomials A and B, then $A \times B = H \times L$.

Proof: If H is the common factor of polynomials A and B, then it divides the polynomials A and B:

$$\frac{A}{H} = a \Rightarrow A = Ha \quad \ldots\ldots(1)$$

$$\frac{B}{H} = b \Rightarrow B = Hb \quad \ldots\ldots(2)$$

Clearly a and b have no common factors

The L.C.M formula is:

L.C.M = common factor x non-common factor

$$L = H \times a \times b$$

$$H \times L = H[H \times a \times b]$$

$$H \times L = Ha \times Hb$$

$$= A \times B \quad \text{using eqs. (1) and (2)}$$

or $\boxed{A \times B = H \times L}$

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Example 4: Find H.C.F and L.C.M of the expression $3x^3-2x^2-3x+2$ and $6x^3-7x^2-x+2$

Solution: If the given polynomials are denoted by

$$A=3x^3-2x^2-3x+2,\ B=6x^3-7x^2-x+2$$

Then the H.C.F can be found as follows:

$$3x^3-2x^2-3x+2 \overline{)\ 6x^3-7x^2-x+2\ (\ 2}$$
$$\underline{-\ 6x^3-4x^2-6x+4}$$
$$-1 \mid -3x^2+5x-2 \quad \text{(Dividing by -1)}$$
$$+3x^2-5x+2 \overline{)\ 3x^3-2x^2-3x+2\ (\ x+1}$$
$$\underline{-\ 3x^3-5x^2+2x}$$
$$3x^2-5x+2$$
$$\underline{-\ 3x^2-5x+2}$$
$$0$$

H.C.F of A and B $= 3x^2-5x+2$

To find the L.C.M, we use the formula:

$$A\times B=H\times L$$

$$L=\frac{A\times B}{H}$$

$$=\frac{(3x^3-2x^2-3x+2)(6x^3-7x^2-x+2)}{(3x^2-5x+2)} \quad (x+1)$$

$$L.C.M=(x+1)(6x^3-7x^2-x+2)$$

Example 5: If H.C.F and L.C.M of two polynomials are $x-3$ and $x^3-9x^2+26x-24$ respectively, then find the second polynomial when the first polynomial is x^2-5x+6.

Solution: Given that

$$H=\text{H.C.F.}=x-3$$

$$L=\text{L.C.M.}=x^3-9x^2+26x-24$$

$$A=x^2-5x+6.$$

$$B=?$$

We know that

$$A\times B=H\times L$$

$$\Rightarrow \quad B=\frac{H\times L}{A}$$

$$=\frac{(x-3)(x^3-9x^2+26x-24)}{x^2-5x+6} \quad \ldots\ldots(1)$$

$$x^2-5x+6 \overline{)\ x^3-9x^2+26x-24\ (\ x-4}$$
$$\underline{-\ x^3-5x^2+6x}$$
$$-4x^2+20x-24$$
$$\underline{-4x^2+20x-24}$$
$$0$$

Equation (1) becomes:

$$B=\frac{(x-3)(x^3-9x^2+26x-24)}{(x^2-5x+6)}=(x-3)(x-4)$$

$$B=x^2-7x+12$$

The second polynomial is therefore $x^2-7x+12$.

Example 6: If H.C.F and L.C.M of two polynomials are $x-1$ and x^3+4x^2+x-6. Find the polynomials of degree 2.

Solution: Given that

$$H=x-1$$

$$L=x^3+4x^2+x-6$$

$$A=?$$

$$B=?$$

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Consider $L = x^3 + 4x^2 + x - 6$

$= (x-1)(x+2)(x+3)$

= common factor × non-common factor

Clearly $x-1$ is the common factor of A and B and $(x+2)$ and $(x+3)$ are the non-common factors, then to find the two second degree polynomials use formula $A \times B = H \times L$.

Therefore

$A = (x-1)(x+2) = x^2 + x - 2$

$B = (x-1)(x+3) = x^2 + 2x - 3$

Exercise 3.2

Find L.C.M of the following expressions by factorization:

1. $x^3 - 8$ and $x^2 + x - 6$
2. $x^2 - 5x + 6$ and $x^2 - 4$
3. $x^2 + 3x + 2, x^2 + 5x + 6$ and $x^2 - 2x - 8$
4. $2x^4 + 3x^2 + 1, 5x^3 - 7x^2 + 5x - 7$ and $x^4 - 1$.

Find H.C.F. and L.C.M of the following expressions:

5. $2x^3 - 5x^2 + 14$ and $x^3 - 16x + 24$
6. $x^3 - 2x^2 - 13x - 10$ and $x^3 - x^2 - 10x - 8$
7. $a^4 - a^3 - a + 1$ and $a^4 + a^2 + 1$
8. H.C.F and L.C.M of two polynomials are $x-2$ and $x^3 + 3x^2 - 6x - 8$ respectively. If one polynomial is $x^2 + 2x - 8$, then find out the second polynomial.
9. If product of two polynomials is $x^4 + 6x^3 - 3x^2 - 56x - 48$ and their L.C.M is $x^3 + 2x^2 - 11x - 12$, then find out the H.C.F.
10. H.C.F and L.C.M of two polynomials are $x + 5$ and $2x^3 + 11x^2 + 2x - 15$ respectively. Find the polynomials of degree 2.

3.2: Basic Operations of Algebraic Fractions

If $P(x)$ and $Q(x)$ are algebraic expressions, then $\frac{P(x)}{Q(x)}, Q(x) \neq 0$, is called algebraic fraction. In algebraic fraction, the fundamental operations are $+, -, \times$ and $\div$ acted upon in the same way as in number system. In the following examples, the use of highest common factor and least common multiple is to reduce fractional expressions through operating $+, -, \times, \div$.

3.2.1: Addition and Subtraction of Algebraic Fractions.

Example: Simplify the expressions $\frac{x+y}{3x+2} + \frac{x-y}{3x+2}$

Solution: $\frac{x+y}{3x+2} + \frac{x-y}{3x+2}$

$= \frac{(x+y)+(x-y)}{3x+2} \quad \because L.C.M = 3x+2$

$= \frac{x+y+x-y}{3x+2} = \frac{2x}{3x+2}$

Example 2: Simplify the expression $\frac{x+y}{x-y} + \frac{x^2-2xy}{x^2-y^2}$

Solution: $\frac{x+y}{x-y} + \frac{x^2-2xy}{x^2-y^2}$

$= \frac{x+y}{x-y} + \frac{x^2-2xy}{(x+y)(x-y)} \quad \because x^2 - y^2 = (x+y)(x-y)$

$= \frac{(x+y)(x+y)+(x^2-2xy)}{(x+y)(x-y)} \quad \because L.C.M = (x+y)(x-y)$

$= \frac{x^2+2xy+y^2+x^2-2xy}{(x+y)(x-y)}$

$= \frac{2x^2+y^2}{(x+y)(x-y)} = \frac{2x^2+y^2}{x^2-y^2}$

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Example 3: Simplify the expression $\frac{x^2-xy+y^2}{x^3+y^3}+\frac{x^2+xy+y^2}{x^3-y^3}-\frac{1}{x^2-y^2}$

Solution: $\frac{x^2-xy+y^2}{x^3+y^3}+\frac{x^2+xy+y^2}{x^3-y^3}-\frac{1}{x^2-y^2}$

$$=\frac{\cancel{(x^2-xy+y^2)}}{(x+y)\cancel{(x^2-xy+y^2)}}+\frac{\cancel{(x^2+xy+y^2)}}{(x-y)\cancel{(x^2+xy+y^2)}}-\frac{1}{(x+y)(x-y)}$$

$$=\frac{1}{x+y}+\frac{1}{x-y}-\frac{1}{(x+y)(x-y)}$$

$$=\frac{(x-y)+(x+y)-1}{(x+y)(x-y)}$$

$$=\frac{x-\cancel{y}+x+\cancel{y}-1}{(x+y)(x-y)}=\frac{2x-1}{x^2-y^2}$$

Exercise 3.3

Simplify the following expression:

1. $\frac{2x}{x-y}-\frac{2y}{x-y}$
2. $\frac{x+2}{x+3}-\frac{x-1}{x-3}+\frac{4x}{x^2-9}$
3. $\frac{a}{3x-y}+\frac{a}{3x+y}-\frac{6ax}{9x^2-y^2}$
4. $\frac{1}{a-b}+\frac{1}{a+b}+\frac{2a}{a^2+b^2}+\frac{4a^3}{a^4+b^4}$
5. $\frac{y}{x-y}+\frac{y}{x+y}+\frac{2xy}{x^2+y^2}+\frac{4x^3y}{x^4+y^4}$
6. $\frac{1}{a^2+7a+10}+\frac{1}{a^2+10a+16}$
7. $\frac{4}{x+2}+\frac{8}{x+5}-\frac{8}{x+3}-\frac{4}{x+6}$

NOT FOR SALE

3.2.2: Product of Fractions

Example 1: Simplify the expression $\frac{2x^2}{y^2+3y+2}\times\frac{y^2+6y+5}{4x}$

Solution:

$$\frac{2x^2}{y^2+3y+2}\times\frac{y^2+6y+5}{4x}$$

$$=\frac{2x^2}{\cancel{(y+1)}(y+2)}\times\frac{\cancel{(y+1)}(y+5)}{4x}$$

$$=\frac{x}{(y+2)}\times\frac{y+5}{2}$$

$$=\frac{x(y+5)}{2(y+2)}$$ (Product of numerator and product of denominator)

Example 2: Simplify the expression $\frac{a(a+b)}{a^3-b^3}\times\frac{a^2+ab+b^2}{a^2-b^2}$

Solution:

$$\frac{a(a+b)}{a^3-b^3}\times\frac{a^2+ab+b^2}{a^2-b^2}$$

$$=\frac{a\cancel{(a+b)}}{(a-b)\cancel{(a^2+ab+b^2)}}\times\frac{\cancel{(a^2+ab+b^2)}}{\cancel{(a+b)}(a-b)}$$ (*Factorization*)

$$=\frac{a}{a-b}\times\frac{1}{a-b}$$

$$=\frac{a\times1}{(a-b)(a-b)}=\frac{a}{(a-b)^2}$$

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3.2.3: Division of Fractions

If $p(x), Q(x), R(x)$ and $S(x)$ are the algebraic expressions, then the following rule used for division of fractions is:

$$\frac{P(x)}{Q(x)} \div \frac{R(x)}{S(x)}$$

$$= \frac{P(x)}{Q(x)} \times \frac{S(x)}{R(x)} \qquad \text{where } Q(x), S(x), R(x) \neq 0$$

Example 1: Simplify the expression $\frac{y}{x-4} \div \frac{3y^2}{x^2-x-12}$

Solution:

$$\frac{y}{x-4} \div \frac{3y^2}{x^2-x-12}$$

$$= \frac{y}{x-4} \times \frac{x^2-x-12}{3y^2} \qquad \text{(converting division into multiplication)}$$

$$= \frac{\cancel{y}}{x-\cancel{4}} \times \frac{(x-\cancel{4})(x+3)}{3y^{\cancel{2}}} \qquad \text{(Factorization)}$$

$$= \frac{x+3}{3y} \qquad \text{(Elementary common factors)}$$

Example 2: Simplify the expression $\frac{2x}{3x-12} \div \frac{2x^2-2x}{x^2-6x+8}$

Solution:

$$\frac{2x}{3x-12} \div \frac{2x^2-2x}{x^2-6x+8}$$

$$= \frac{2x}{3x-12} \times \frac{x^2-6x+8}{x^2-2x}$$

$$= \frac{2\cancel{x}}{3\cancel{(x-4)}} \times \frac{\cancel{(x-4)}\,(x-2)}{\cancel{x}\,\cancel{(x-2)}} \qquad \text{(elementary common factors)}$$

$$= \frac{2}{3}$$

Exercise 3.4

Simplify the following expressions:

1. $\frac{x^2-16}{x+4}$
2. $\frac{5}{x-2} \times \frac{x-2}{3-(x-1)}$
3. $\frac{x^3-8}{2x^2} \times \frac{4x}{x^2-5x+6}$
4. $\frac{x^3+y^3}{x^2-y^2} \times \frac{x^2+xy+xz+yz}{x^4+x^2y^2+y^4} \times \frac{x^3-y^3}{x^3+z^3}$
5. $\frac{x+2}{5(x-3)} \div \frac{x+2}{15(x-3)}$
6. $\frac{a^2+b^2}{a^2+ab+b^2} \div \frac{a^4-b^4}{a^3-b^3}$
7. $\frac{x^2-x-6}{x^2+6x+9} \div \frac{x^2-4}{x+3}$
8. $\frac{a^4-8a}{2a^2+5a-3} \times \frac{2a-1}{a^2+2a+4} \div \frac{a^2-2a}{a+3}$

3.3: Square Roots of Algebraic Expressions

Square root of an algebraic expression can be found:

i. By factorization method
ii. By division method

(i) Square Roots by Factorization

In this method, reduce the expression to a perfect square then find out square root of the expression.

Example 1: Find the square roots of $x^2+10xy+25y^2$ by factorization method.

Solution: Reduce the expression to a perfect square:

$$x^2+10xy+25y^2 = (x)^2+2(x)(5y)+(5y)^2$$

$$= (x+5y)^2$$

$$\because \quad \sqrt{x^2+10xy+25y^2} = \sqrt{(x+5y)^2}$$

$$= \pm(x+5y)$$

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Example 2:

Find the square roots of the expression $x^2+\frac{1}{x^2}-10\left(x+\frac{1}{x}\right)+27$ by factorization method.

Solution: Reduce to perfect square:

$$x^2+\frac{1}{x^2}-10\left(x+\frac{1}{x}\right)+27=x^2+\frac{1}{x^2}-10\left(x+\frac{1}{x}\right)+2+25$$

$$=\left(x^2+\frac{1}{x^2}+2\right)-10\left(x+\frac{1}{x}\right)+25$$

$$=\left(x+\frac{1}{x}\right)^2-10\left(x+\frac{1}{x}\right)+(5)^2$$

$$=\left(x+\frac{1}{x}\right)^2-2\left(x+\frac{1}{x}\right)(5)+(5)^2$$

$$=\left(x+\frac{1}{x}-5\right)^2$$

Therefore

$$\sqrt{x^2+\frac{1}{x^2}-10\left(x+\frac{1}{x}\right)+27}=\sqrt{\left(x+\frac{1}{x}-5\right)^2}$$

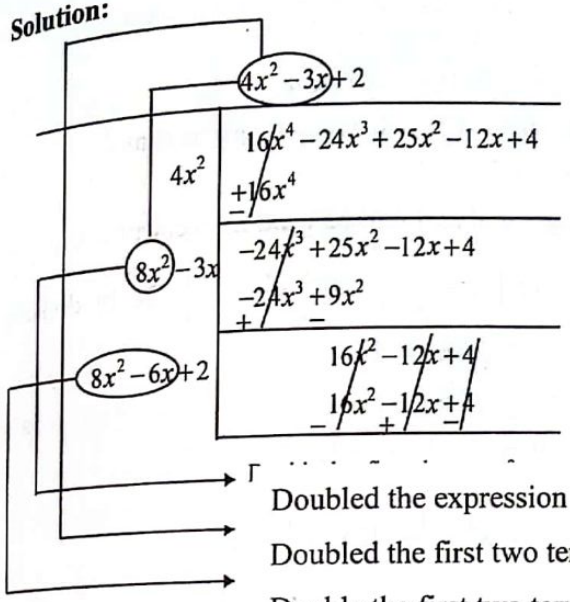
$$=\pm\left(x+\frac{1}{x}-5\right)$$

(ii) Square root by division

The square root by division method is explained in the following examples.

Example 1: Find the square roots of the expression $16x^4-24x^3+25x^2-12x+4$

Solution:



$\sqrt{16x^4}=4x^2$

2nd element of square root $=-\frac{24x^3}{8x^2}=-3x$

3rd element of square root $=\frac{16x^2}{8x^2}=2$

Therefore

$$\sqrt{16x^4-24x^3\,25x^2-12x+4}=\pm(4x^2-3x+2)$$

Rules for finding square root by division method

In the above example, we have followed the following steps:

1. Write the expression in descending order.
2. Take square root of first term of expression i.e. $4x^2$.
3. Subtract square of $4x^2$ from the given expression and get remainder $-24x^3+25x^2-12x+4$.
4. To find the second term of square root divide the first term $-24x^3$ of the remainder by twice $8x^2$ of $4x^2$ to get $-3x$.
5. Add $-3x$ with $4x^2$ on the top.
6. Add $-3x$ with $8x^2$.
7. Multiply $8x^2-3x$ by $-3x$ subtract the product $\left(-24x^3+9x^2\right)$ from the expression $-24x^3+25x^2-12x+4$.
8. Double $4x^2-3x$ and get $8x^2-6x$.

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9. Divide $16x^2$ by $8x^2$ to get 2. Add 2 with $4x^2-3x$ on the top.
10. Add 2 with $8x^2-6x$.
11. Multiply $8x^2-6x+2$ by 2 to get $16x^2-12x+4$.
12. Subtract the product from $16x^2-12x+4$, we get zero as remainder.
13. Required square root $\pm(4x^2-3x+2)$

The above rules are quite general and will be used in every problem.

Example 2: Find the square roots of $\left(x^2-\frac{1}{x^2}\right)^2-12\left(x^2-\frac{1}{x^2}\right)+36$ by division method.

Solution:

$$\left(x^2-\frac{1}{x^2}\right)^2-12\left(x^2-\frac{1}{x^2}\right)+36=x^4-2+\frac{1}{x^4}-12x^2+\frac{12}{x^2}+36$$

$$=x^4+\frac{1}{x^4}-12x^2+\frac{12}{x^2}+34$$

$$=x^4-12x^2+34+\frac{12}{x^2}+\frac{1}{x^4}$$

$\because$ $\sqrt{\left(x^2-\frac{1}{x^2}\right)^2-12\left(x^2-\frac{1}{x^2}\right)+36}=\sqrt{x^4-12x^2+34+\frac{12}{x^2}+\frac{1}{x^4}}$

$$\begin{array}{r|l} & x^2-6-\frac{1}{x^2} \\ \hline x^2 & x^4-12x^2+34+\frac{12}{x^2}+\frac{1}{x^4} \\ & \pm x^4 \\ \hline 2x^2-6 & -12x^2+34+\frac{12}{x^2}+\frac{1}{x^4} \\ & \mp 12x^2 \pm 36 \\ \hline 2x^2-12-\frac{1}{x^2} & -2+\frac{12}{x^2}+\frac{1}{x^4} \\ & \mp 2 \pm \frac{12}{x^2} \pm \frac{1}{x^4} \\ \hline & 0 \end{array}$$

Hence

$$\sqrt{\left(x^2-\frac{1}{x^2}\right)^2-12\left(x^2-\frac{1}{x^2}\right)+36}=\pm\left(x^2-6-\frac{1}{x^2}\right)$$

Which is the required square root.

Exercise: 3.5

Find the square root of the following by factorization method.

1. $4x^2-20xy+25y^2$
2. $16x^2+72xy+81y^2$
3. $(x-y)^2+6(x-y)+9$
4. $\left(x^2+\frac{1}{x^2}\right)^2-4\left(x-\frac{1}{x}\right)^2-4$
5. $x^2+\frac{1}{x^2}-8\left(x+\frac{1}{x}\right)+18$

Find the square root by division method.

6. $x^6-2x^5+3x^4-2x^3+x^2$
7. $\frac{x^4}{4}+4x^2+\frac{ax^2}{3}+\frac{a^2}{9}-2x^3-\frac{4ax}{3}$
8. $x^4+x^3-\frac{31}{4}x^2-4x+16$
9. For what value of k, the expression $4x^4+32x^2+96+\frac{128}{x^2}+\frac{k}{x^4}$ will become a perfect square.

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Exercise: 3.6

(Objective Type Questions)

1. True and False questions.

Read the following sentences carefully and encircle "T" in case of true statement and "F" in case of false statement.

(i) H.C.F of x^2-9 and $x+3$ is $x+3$. (T/F)

(ii) H.C.F. of $a^3-\frac{1}{8}$ and $a^2+\frac{1}{2}a+\frac{1}{4}$ is $a-\frac{1}{2}$. (T/F)

(iii) L.C.M. of a^4-b^4 and a^2-b^2 is a^2+b^2. (T/F)

(iv) L.C.M. x^3+8 and $x+2$ is x^3+8. (T/F)

(v) H.C.F of x^2+3x+2 and x^2+4x+3 is $x+1$. (T/F)

2. Fill in the blanks:

(i) H.C.F of x^2-5x+6 and $x-2$ is _____

(ii) L.C.M of a^3+3a+2 and a^2+4a+3 is _____

(iii) H.C.F of $x^3-\frac{1}{x^3}$ and $x-\frac{1}{x}$ is _____

(iv) L.C.M is the _____ degree polynomial which is divisible by the given polynomials.

(v) H.C.F is the highest degree polynomial which divides _____ the given polynomials.

3. Multiple choices.

Write the correct option against the given box.

(i) H.C.F of a^3-b^3 and a^2+ab+b^2 is [ ]

(a) $a+b$ (b) a^2+ab+b^2

(c) $a-b$ (d) a^2-b^2

(ii) H.C.F of $8a^3-b^3$ and $4a^2-4ab+b^2$ is [ ]

(a) $2a-b$ (b) $a+2b$

(c) $a^2+4ab+b^2$ (d) $2a+b$

(iii) L.C.M of $(a-b)^3$ and $(a-b)^4$ is [ ]

(a) $(a-b)$ (b) $(a-b)^3$

(c) $(a-b)^4$ (d) $(a-b)^7$

(iv) L.C.M of a^2-a+1 and a^3+1 is [ ]

(a) $a+1$ (b) a^2-a+1

(c) a^3+1 (d) a^2-a+1

(v) $\frac{x}{x-y}-\frac{y}{x+y}=$ _____ [ ]

(a) $\frac{x^2+y^2}{x^2-y^2}$ (b) $\frac{x^2-y^2}{x^2+y^2}$

(c) $\frac{x+y}{x^2+y^2}$ (d) $\frac{x-y}{x^2+y^2}$

(vi) L.C.M = _____ [ ]

(a) $\frac{A\times B}{H}$ (b) $\frac{H}{A\times B}$

(c) $\frac{A}{H}$ (d) $\frac{B}{H}$

(vii) $\frac{b}{25a^2-b^2}-\frac{1}{5a-b}=$ _____ [ ]

(a) $\frac{+5a}{25a^2-b^2}$ (b) $\frac{+5a}{5a-b}$

(c) $\frac{-5a}{5a+b}$ (d) $\frac{-5a}{25a^2-b^2}$

(viii) $\frac{ax^3+ay^3}{a^2(x+y)}=$ _____ [ ]

(a) ax^2+ay^2 (b) $\frac{x^2+y^2}{a}$

(c) $\frac{x^2-xy+y^2}{a}$ (d) $\frac{x^2+xy+y^2}{a}$

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Review Exercise: 3.7

1. Find H.C.F of x^3+4x^2+4x+3 and x^3+2x^2-4x-3 by division method.
2. Find L.C.M of $x^3+6x^2+11x+6$ and x^3+3x^2-x-3.
3. If the product of the two polynomials is $x^4+5x^3-6x^2-2x-28$ whose H.C.F is $x-2$, then find out the L.C.M.
4. Simplify the expression $\frac{x-5}{x^2-7x+10}-\frac{2(x-6)}{x^2-9x+18}+\frac{x-7}{x^2-4x-21}$
5. Simplify the expression $\frac{x^3-8}{2x^2}\times\frac{4x}{x^2-5x+6}$
6. Simplify the expression $\frac{5}{x-1}\div\frac{25(x-2)}{x-1}$
7. Find square root of the expression $(x^2+6x+5)(x^2+9x+9)(x^2+13x+40)$ by factorization.

Glossary

Highest Common Factor (H.C.F)

H.C.F of two or more polynomials is a polynomial of highest degree which divides exactly the given polynomial.

Least Common Multiple (L.C.M)

L.C.M of two or more polynomials is a polynomial of least degree which is divisible by the given polynomials.

Relationship between H.C.F and L.C.M

If A and B are the two polynomials and H and L are their H.C.F and L.C.M represented, then

$$A\times B=H\times L$$

Product of two polynomials = $H.C.F\times L.C.M$

Unit 4

LINEAR EQUATIONS AND INEQUALITIES

In this unit, the students will be able to learn about the:

- Recall linear equation in one variable.
- Solution of linear equations with rational coefficients
- Reduce equations, involving radicals, to simple linear form and find their solution.
- Concept of absolute value
- Solution of an equation involving absolute value in one variable
- Concept of inequalities
- Properties of inequalities
- Solution of linear inequalities

4.1: Linear Equations

4.1.1 Introduction:

An equation of the form $ax+b=0$ is called general form of linear equation in one variable with rational coefficients a and b.

Examples:

(*i*) $5x+3=0$

(*ii*) $\frac{3}{x}-4=0$

(*iii*) $3x-2=2x+3$

(*iv*) $\frac{x}{5}=4$

(*v*) $\frac{1}{2}x-\frac{5}{2}=4$

are linear equations in one variable with rational coefficient.

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4.1.2: Solution of Linear Equations

Any value of a variable x that makes the equation a true sentence is called a solution or root of the equation.

The steps involved in solving linear equations are the following:

1. Keep variable term on one side and shift constant terms on the other side of the equation.
2. Simplify both sides of the equation.
3. Divide both sides of the equation by the rational coefficient of the variable.
4. Finally a single numerical value of the variable is obtained.
5. Verify this value by putting in original equation.
6. If the substitution of the value of variable reduces the equation to L.H.S = R.H.S of an equation, then this value of variable will be kept as required solution of the equation.

Example 1: Solve the equation $15x-6=3x+18$

Solution:

$15x-6=3x+18$

Step 1: Subtract $3x$ from both sides of the equation

$15x-6-3x=\cancel{3x}+18-\cancel{3x}$

Step 2: Simplifying both sides

$12x-6=18$

Step 3: Adding 6 with both sides

$12x-6+6=18+6$

Step 4: Simplifying both sides

$12x=24$

Step 5: Divide both sides by 12

$\frac{\cancel{12}x}{\cancel{12}}=\frac{24}{12}$

Step 6: Simplify both sides

$x=2$

Step 7: **Verification:**

Put $x=2$ in original equation, to obtain

$15(2)-6=3(2)+18$

$30-6=6+18$

$24=24$

$L.H.S=R.H.S$

Thus, required solution set is {2}

Note: Write solution set in the form of { } bracket.

Example 2:

Age of brother is twice the age of his sister. Three years back, age of brother was three times the age of his sister. Find their present ages.

Solution:

Let present age of sister $=x$ years

and present age of brother $=2x$ years

Three years back, the

Age of the sister $=(x-3)$ years

and age of brother $=(2x-3)$ years.

According to the given condition

Age of brother $=3$ (age of sister)

$2x-3=3(x-3)$

$\Rightarrow \quad 2x-3=3x-9$

subtract $2x$ both sides

$\Rightarrow \quad 2x-3-2x=3x-9-2x$

$\Rightarrow \quad -3=x-9$

add 9 both sides to obtain

$\Rightarrow \quad 6=x$

or $\quad x=6$

Hence present age of sister = 6 years

and present age of bother = $2\times6=12$ years

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Example 3:

A number consist of two digits. The sum of digits is 8. If the digits of a number are interchanged, then the new number becomes 36 less of the original number. Find the number.

Solution:

Let digit in unit's place $= x$

and digit in ten's place $= 8-x$ [$\because$ sum of digits $= 8$]

the original number $= 10(8-x)+x$

$= 80-10x+x$

$= 80-9x$

When digits are interchanged, then the

new number $= 10(x)+(8-x)$

$= 10x+8-x$

$= 9x+8$

According to given condition

New number = original number $-$ 36

$9x+8 = (80-9x)-36$

$\Rightarrow \quad 9x+8 = 80-9x-36$

$\Rightarrow \quad 9x+8 = 44-9x$

Add $9x$ both sides

$\Rightarrow \quad 9x+8+9x = 44-9x+9x$

$\Rightarrow \quad 18x+8 = 44$

Subtract 8 from both sides

$18x+8-8 = 44-8$

$\Rightarrow \quad 18x = 36$

Divide out both sides by 18

$$\frac{18x}{18} = \frac{36}{18} \Rightarrow x = 2$$

Hence digit in unit's place $= 2$

And digit in ten's place $= 8-2 = 6$

Therefore the original number $= 10\times6+2$

$= 60+2$

$= 62$

Exercise 4.1

Find the solution sets of the following equations and verify their answers.

1. $3x-5=7$
2. $4(x+2)-2(x-2)=18$
3. $6x+3=2x+19$
4. $\frac{1}{x-2}=\frac{2}{x-1}$
5. $6x+3=2x+19$
6. $\frac{2+5}{3}-\left(\frac{2-x}{2}-\frac{1}{4}\right)=\frac{3x+25}{12}$
7. The sum of two consecutive numbers is 23. Find the numbers.
8. The sum of three consecutive odd integers is 159. Find the numbers.
9. The perimeter of a rectangle is 20 m. If length of rectangle is 2 meter more than its breadth. Find the length and breadth of the rectangle.
10. Age of mother is 13 times the age of her daughter. It will be only 5 times after four years. Find their present ages.
11. The sum of two digits is 10. If digits change their places, then the new number is decreased by 18 than the original number. Find the number.
12. Think of a number, if 4 is added to the twice of the number, the result comes to 18. Find the number.

4.1.3: Solution of Equations involving Radicals

Definition (Radical equations)

Equations involving radicals of variables are called radical equations.

For example $\sqrt{x+2}=5$, $\sqrt{x+4}=2$, $\sqrt{2x+3}$, $\sqrt{3x-1}$ are radical equations.

The solution of such equation can be found by taking the square of the given equation. The equation $\sqrt{x+5}=-11$ has no solution, because equation is

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not true for any value of x. On the other hand $\sqrt{x+5}=11$ has a solution set consists of real numbers. To avoid any such situation, we must verify the solution by putting the value of variable in original equation.

Example 1:

Find solution set of $\sqrt{x+5}=3$.

Solution:

$$\sqrt{x+5}=3$$

Squaring both sides to obtain:

$$\Rightarrow \quad \left(\sqrt{x+5}\right)^2=(3)^2$$

$$\Rightarrow \quad x+5=9$$

$$\Rightarrow \quad x+5-5=9-5 \qquad \text{[subtracting 5 both sides]}$$

$$\Rightarrow \quad x=4$$

Substitute $x=4$ in original equation to show:

$$\sqrt{4+5}=3$$

$$\Rightarrow \quad \sqrt{9}=3$$

$$\Rightarrow \quad L.H.S=R.H.S$$

Hence, solution set = $\{4\}$.

Example 2:

Solve $\sqrt{4x-3}=\sqrt{2x+4}$

Solution:

$$\sqrt{4x-3}=\sqrt{2x+4}$$

Squaring both sides of an equation to obtain

$$\left(\sqrt{4x-3}\right)^2=\left(\sqrt{2x+4}\right)^2$$

$$\Rightarrow \quad 4x-3=2x+4$$

Subtracting $2x$ from both sides

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$$4x-3-2x=2\cancel{x}+4-2\cancel{x}$$

$$\Rightarrow \quad 2x-3=4$$

Adding 3 to both sides

$$2x+3-3=4+3$$

$$\Rightarrow \quad 2x=7$$

Dividing both sides by 2

$$\frac{2x}{2}=\frac{7}{2}$$

$$\Rightarrow \quad x=\frac{7}{2}$$

Verification

To verify put $x=\frac{7}{2}$ in given equation to obtain:

$$\sqrt{4+\left(\frac{7}{2}\right)-3}=\sqrt{2\times\left(\frac{7}{2}\right)-3}$$

$$\Rightarrow \quad \sqrt{14-3}=\sqrt{7+4}$$

$$\Rightarrow \quad \sqrt{11}=\sqrt{11}$$

$$L.H.S=R.H.S$$

Hence solution set $=\{\frac{7}{2}\}$

Example 3:

Find solution set of equation $\sqrt{3x+2}+6=2$

Solution:

$$\sqrt{3x+2}+6=2$$

Subtracting 6 from both sides

$$\sqrt{3x+2}+6-6=2-6$$

$$\Rightarrow \quad \sqrt{3x+2}=-4$$

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Squaring both sides

$$\left(\sqrt{3x+2}\right)^2=(-4)^2$$

$\Rightarrow \quad 3x+2=16$

Subtracting 2 from both sides

$$3x+2-2=16-2$$

$\Rightarrow \quad 3x=14$

Dividing both sides by 13

$$x=\frac{14}{3}$$

Verification:

Put $x=\frac{14}{3}$ in original equation.

$$\sqrt{3\left(\frac{14}{3}\right)+2}+6=2$$

$\Rightarrow \quad \sqrt{14+2}+6=2$

$\Rightarrow \quad \sqrt{16}+6=2$

$\Rightarrow \quad 4+6=2$

$\Rightarrow \quad 10\neq 2$

which is impossible.

Thus, given equation has no solution.

Note: When we are taking square of both sides of radical equation, some times the roots will not satisfy the given equation. Such type of roots are called extraneous roots.

Exercise 4.2

1. $\sqrt{x+5}=6$
2. $\sqrt{2x+4}=-6$
3. $\sqrt{x+3}+2=6$
4. $3\sqrt{x}+7=10$
5. $\sqrt{4x-1}=\sqrt{3x+1}$
6. $15-\sqrt{x+2}=10$
7. $\sqrt{3x-1}+7=4$
8. $\sqrt{4(2x-1)}=2\sqrt{x+8}$

4.2: Equations involving absolute value

4.2.1: Absolute Value

Absolute value of any number x is denoted by $|x|$ and is defined as:

$$|x|=x \quad \text{if } x>o$$
$$=o \quad \text{if } x=o$$
$$=-x \quad \text{if } x<o$$

Note: The absolute value of a number is always non-negative.

The following are properties of absolute numbers are the following.

(*i*) $|a+b|\leq|a|+|b|$

(*ii*) $|ab|=|a||b|$

(*iii*) $\left|\frac{a}{b}\right|=\frac{|a|}{|b|}$

These properties are verified by taking particular values of a and b.

For example if $a=2$ and $b=9$ then

$$|2+9|=|2|+|9|$$
$$|11|=2+9$$
$$11=11$$

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4.2.2: Solution of Equations involving absolute value

To solve such equations, we need to consider the possible values of the number. For example if

$$|x| = 2$$

$$\Rightarrow \quad x = 2 \text{ and } x = -2 \text{ because}$$

$$|2| = |-2| = 2$$

To the following example.

Example 1:

Solve $|x-2| = 6$

Solution:

$$|x-2| = 6$$

There are two possibilities

(i) $x-2=6$

(ii) $x-2=-6$

Consider equ. (i)

$$x-2=6$$

Adding 2 to both sides

$$x-2+2=6+2$$

$$x=8$$

Adding 2 to both sides

$$x-2+2=-6+2$$

$$x=-4$$

Therefore

Solution set of given equation = $\{8,-4\}$.

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Example 2:

Find solution set of $|2x+4|-7=3$

Solution:

$$|2x+4|-7=3$$

Adding 7 both sides

$$|2x+4|-7+7=3+7$$

$$\Rightarrow \quad |2x+4|=10$$

There are two possibilities

$2x+4=10$ and $2x+4=-10$.

Subtracting 4 from both sides

$2x+4-4=10-4$	$2x+4-4=-10-4$
$\Rightarrow \quad 2x=6$	$\Rightarrow \quad 2x=-14$

Dividing both sides by 2

$\frac{2x}{2}=\frac{6}{2}$	$\frac{2x}{2}=\frac{-14}{2}$
$\Rightarrow \quad x=3$	$\Rightarrow \quad x=-7$

Therefore,

Solution set $=\{3,-7\}$

Exercise 4.3

Solve the following equations:

1. $|4x|=8$
2. $|x-1|=7$
3. $\left|\frac{x}{5}\right|=6$
4. $|15x-10|=35$
5. $\left|\frac{x+5}{3}\right|=\frac{7}{2}$
6. $|5x-13|+2=14$
7. $|4x-1|=7$
8. $|9x+11|-\frac{2}{5}=\frac{8}{5}$

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4.3: Linear Inequalities

An inequality is a statement that two mathematical expressions are not equal. Usually an inequality indicates that one expression is less than (or greater than) another

The following properties are used to simplify an inequality:

For real numbers a, b and c.

$a<b$ means a is less than b and a lies to the left of b on real number line.

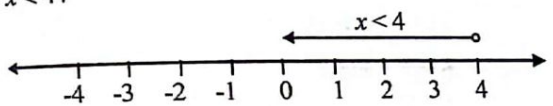
$a>b$ means a is greater than b and a lies to the right of b on real number line.

$a\leq b$ means a is less than or equal to b and a coincides with b or lies to the left of b on real number line.

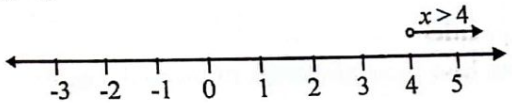
$a\geq b$ means a is greater or equal to b and a coincides with b or lies to the right of b on real number here.

Examples:

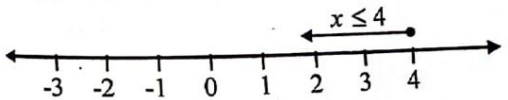
1. $x<4$.



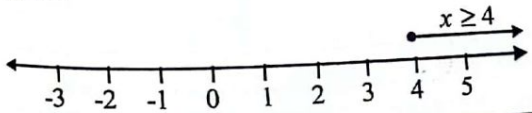
2. $x>4$.



3. $x\leq 4$.



4. $x\geq 4$.



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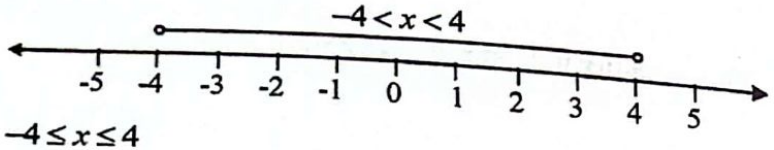
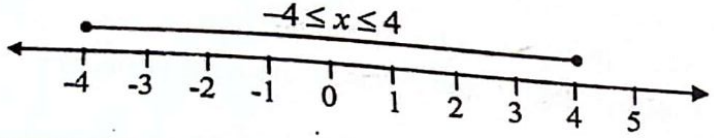
5. $-4<x<4$



6. $-4\leq x\leq 4$



Note: Circle '○' shows that the number is not included and shaded ● circle shows that number is included.

Exercise 4.4

Depict the following inequalities on a real number line:

1. $x>0$
2. $x<0$
3. $x\leq 1$
4. $x\leq -5$
5. $x\geq -3$
6. $x\geq 4$
7. $-5\leq x\leq 6$
8. $3\geq x\geq -2$
9. $4<x<6$
10. $-3<x<0$

4.2.3: Properties of Inequalities

In the following, we have some properties of inequalities.

(i) Trichotomy Property

If x and y are any two real numbers, then one and only one statement is true.

(a) $x>y$

(b) $x<y$

(c) $x=y$

$\forall x, y\in R \quad x>y \quad \vee \quad x=y \quad \vee \quad x<y$

Note: Symbol "$\vee$" means or and "$\wedge$" means "and".

(ii) Transitive Property

If x, y and z are any three real numbers then,

(a) If $x<y$ and $y<z\Rightarrow x<z$

(b) If $x>y$ and $y>z\Rightarrow x>z$.

(iii) Additive Property

If x, y and z are any three real numbers then,

(a) $x<y\Rightarrow x+z<y+z$

(b) $x>y\Rightarrow x+z>y+z$

(iv) Multiplicative Property

If x, y and z are any three real numbers than,

(a) (i) $x<y\Rightarrow xz<yz$

(ii) $x>y\Rightarrow xz<yz$ {where z is a positive number i.e. $z>0$}

(b) (i) $x<y\Rightarrow xz>yz$

(ii) $x>y\Rightarrow xz<yz$ {where z is a negative number i.e. $z<0$}

Note: Sign of inequality must be changed when we multiply or divide its both sides by a negative number.

Examples

Write the names of the properties used in the following statements.

i. $10<21\Rightarrow 20<31$

ii. $20>15\Rightarrow 25>20$

iii. $8>7\Rightarrow 40>35$

iv. $-5>-8\Rightarrow 15<24$

v. If $9>5$ and $5>2$ then $9>2$.

Solution:

(i) $10<21\Rightarrow 10+10<21+10$ (additive property)

Hence $10<21\Rightarrow 20<31$

(ii) $20>15\Rightarrow 20+5>15+5$ (additive property)

Hence $20>15\Rightarrow 25>20$

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(iii) $8<7 \Rightarrow 8\times5<7\times5$ (multiplicative property)

Hence $8<7 \Rightarrow 40<35$

(iv) $-5>-8 \Rightarrow -5\times(-3)<-8\times(-3)$ (multiplicative property)

Hence $-5>-8 \Rightarrow 15<24$

(v) $9>5$ and $5>2$ then $9>2$ (transitive property)

4.4: Solution of Linear Inequalities

Inequalities, like equations have solution sets. Solution sets of inequalities are sets of those number which satisfy the inequalities.

The steps involved in solving linear inequality are the following:

1. Add the same number to both sides of the inequality.
2. Subtract the same number from both sides of the inequality.
3. Multiply both sides of the inequality by the same number. If the number is negative, then the sign of inequality must be changed.
4. Divide both sides of the inequality by the same number. If the number is negative, then the sign of inequality must be changed.

Example 1:

Solve the inequality $2x-5>9$.

Solution:

$2x-5>9$

Add 5 to both sides

$2x-5+5>9+5$

$\Rightarrow \quad 2x>14$

Divide both sides by 2

$\frac{2x}{2}>\frac{14}{2}$

$\Rightarrow \quad x>7$

Therefore, the solution set $=\{8,9,10,...\}=\{x \mid x\in R \text{ and } x>7\}$

Note: An inequality has an infinite number of solutions whereas equation has finite number of solution under some condition.

Example 2:

Find the solution set of the inequality $3x+5\le20$, where x is an integer.

Solution:

$3x+5\le20$

Subtract 5 from both sides

$3x+5-5\le20-5$

$\Rightarrow \quad 3x\le15$

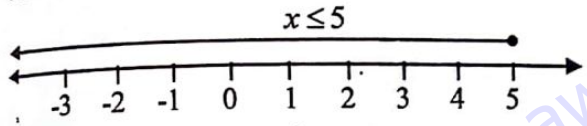
Divide both sides by 3

$\frac{3x}{3}\le\frac{15}{3}$

$\Rightarrow \quad x\le5$

Therefore, the solution set $=\{5,4,3,2,1,0,-1,-2,...\}=\{x \mid x\in R\wedge x\le5\}$

The solution set on the number line is:



Example 3:

Find solution set of the inequality $-3\le x\le4$. where $x\in R$. Depict it on the real number line.

Solution:

$-3\le x\le4$

If can be written as

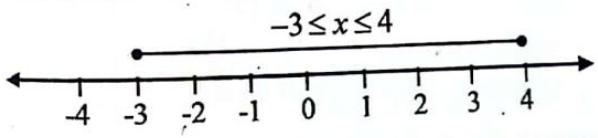
(i) $-3\le x \Rightarrow x\ge-3$ (1)

(ii) $x\le4$ (2)

The solution set of (1) and (2) is: $=\{-3,-2,-1,0,1,2,3,4\}$

$=\{x \mid x\in R\wedge-3\le x\le4\}$

It is represented on number line as



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Exercise 4.5

Find solution sets of the following linear inequalities:

1. $5x+4<34$, $x \in N$
2. $3x-5>1$, $x \in N$
3. $3(x+4)>x+5$, $x \in R$
4. $3x-4>2$, $x \in \{1,2,3,4,5,6\}$
5. $2(x-3)\geq x-2$, $x \in \{-2,-1,0,1,2,3,4\}$
6. $\frac{2x+6}{2} \geq \frac{x-9}{3}$, $x \in R$
7. $\frac{x-1}{3} \leq \frac{1-x}{2}$, $x \in Z$

Find the solution sets of the following inequalities and depict them on the real number line.

8. $-2<x<3$
9. $-4>x>-8$
10. $7 \leq x+4 \leq 4$
11. $5 \geq x \geq -2$
12. $5 \leq x+2 \leq 8$

Exercise 4.6
(Objective Type Questions)

1. True and False questions.

Read the following sentences carefully and encircle "T" in case of true statement and "F" in case of false statement.

(i) $\frac{x}{4}=3$ is linear equation in one variable. (T/F)

(ii) The solution set of $5x+3=8$ is $\{1\}$ (T/F)

(iii) The solution set of $\sqrt{x}+2=4$ is $\{2\}$. (T/F)

(iv) $|x|=-x$ when $x>0$. (T/F)

(v) If $x<y$ and $y<z$ then $x>0$. (T/F)

2. Fill in the blanks:

(i) If $x<y \Rightarrow xz$ ___ yz for $z>0$.

(ii) "$\leq$" means Less than or equal to.

(iii) The solution set of $|x|-2=0$ is $\{2,-2\}$

(iv) The solution set of $\sqrt{x+5}=3$ is $\{\ \}$

(v) The solution set of $0<x+2<3$ when $x \in z$ is $\{-1,0\}$

3. Multiple choices.

Write the correct option against the given box:(i) $|a+b|$ [b]

(a) $=|a|+|b|$ (b) $\leq |a|+|b|$

(c) $>|a|+|b|$ (d) $\geq |a|+|b|$

(ii) The solution set of linear equation has [a]

(a) one element (b) two elements

(c) No element (d) infinite number of elements

(iii) The solution set of $\sqrt{x}=-6$ is: [d]

(a) $\{36\}$ (b) $\{6\}$

(c) $\{-6\}$ (d) $\{\ \}$

(iv) $\sqrt{x+4}+2=7$ is a: [b]

(a) linear equation (b) radical equation

(c) cubic equation (d) quadratic equation

(v) The solution of $5-3x=-4$ is: [c]

(a) $\{-3\}$ (b) $\{1, 3\}$

(c) $\{3\}$ (d) $\{9\}$

(vi) The solution set of $|-x|=0$ is: [c]

(a) $\{1\}$ (b) $\{-1\}$

(c) $\{0\}$ (d) does not exist

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(vii) $x \leq 5$ means:

(a) $x > 5$ (b) $x = 5$
(c) $x > 5 or x = 5$ (d) $x < 5$ *or* $x = 5$

(viii) $-5 < 4$ means:

(a) -5 lies to the right of 4
(b) -5 lies to the left of 4
(c) -5 coincides with 4 and lies to the left of 4
(d) -5 coincides with 4 and lies to the right of 4.

(ix) The solution set of $\left|\frac{4x}{3}\right| = 4$ is:

(a) {3} (b) {4, −4}
(c) 3, −3} (d) {4, −4}

(x) The solution set of $3x - 2 > 7, x \in N$ is:

(a) {1, 2, 3, ...} (b) {3, 4, 5, ...}
(c) {4, 5, 6, ...} (d) {9, 10, 11, ...}

(xi) The solution set of $-1 \leq x \leq 1$, $x \in N$ is:

(a) {−1, 0, 1} (b) {0}
(c) {−1} (d) {1}

(xii) The solution set of $\sqrt{5x+3} + 2 = 4$ is:

(a) $\{\frac{1}{5}\}$ (b) $\{-\frac{1}{5}\}$
(c) {2} (d) 1

(xiii) The symbol ">" means:

(a) less than (b) greater than
(c) less than or equal to
(d) greater than or equal to

(xiv) An inequality has _____ solutions.

(a) finite number of (b) infinite number of
(c) two (d) no

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Absolute value, Absolute value of any number x is denoted by $|x|$ and is defined as:

$$|x| = x \text{ when } x > 0$$
$$= 0 \quad \text{when } x = 0$$
$$= -x \quad \text{when } x < 0$$

Linear Inequalities

An inequality is a statement that two mathematical expressions are not equal. Usually an inequality indicates that one expression is less than or greater than another.

Trichotomy Property

$\forall \quad x, y \in R$

$x < y$ or $x = y$ or $x > y$

Transitive Property

$\forall \quad x, y, z \in R$

$x < y$ and $y < z \Rightarrow x < z$

Additive Property

For any three numbers $x, y, z \in R$

(i) $x < y \Rightarrow x + z < y + z$

(ii) $x > y \Rightarrow x + z > y + z$

Multiplicative Property

For any three numbers $x, y, z \in R$

(i) $x < y \Rightarrow zx < yz$

and $x > y \Rightarrow xz > yz$ {when $z > 0$.}

(ii) $x < y \Rightarrow xz > yz$

and $x > y \Rightarrow xz < yz$ {when $z < 0$}

Unit 5

Quadratic Equations

In this unit, the students will be able to learn about the:

- Quadratic equation.
- Solution of quadratic equations by
  - Factorization.
  - completing square.
- Derivation of quadratic formula.
- Solution of quadratic equations by quadratic formula.
- Real life problems on quadratic equations.

5.1 Definition of a quadratic equation:

An equation with 2 as the highest exponent on the variable is called quadratic.

If a, b and c are real number with $a \neq 0$, then $ax^2 + bx + c = 0$ is called quadratic equation.

Examples of quadratic equations:

$2x^2 - x + 3 = 0$, $a = 2, b = -1, c = 3$

$x^2 + 10x + 1 = 0$, $a = 1, b = 10, c = 1$

$4x^2 - 2x + 25 = 0$, $a = 4, b = -2, c = 25$

$x^2 - 64 = 0$, $a = 1, b = 0, c = -64$

Note:

If $a = 0$, the equation $ax^2 + bx + c = 0$ becomes $bx + c = 0$, which is linear. A quadratic equation written in the form $ax^2 + bx + c = 0$ is said to be in

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standard form. The simplest method of solving a quadratic equation, but one that is not always applied is by factorization. This method depends on the Zero factor property.

Reducible to quadratic equations:

i) $2(p^2-2p)=5$

$\Rightarrow 2p^2-4p=5$

$\Rightarrow 2p^2-4p-5=0$

where $a=2, b=-4, c=-5$ is a quadratic equation in variable p.

ii) $z(z-1)=3$

$\Rightarrow z^2-z=3$

$\Rightarrow z^2-z-3=0$

where $a=1, b=-1, c=-3$ is a quadratic equation in variable z.

iii) $5+\frac{1}{y}-\frac{1}{y^2}=0$

Take L.C.M as y^2

$\frac{5y^2+y-1}{y^2}=0$

$5y^2+y-1=0$ (multiply both side by y^2)

$a=5, b=1, c=-1$ is a quadratic equation in variable y.

Zero-factor property:

If a and b are real numbers, with $ab=0$, then $a=0$ or $b=0$ or both.

Note:

- The letters a, b and c are coefficient (you know those values). They can have any values, except that "a" can't be zero.
- The letter "x" is the variable or unknown you have to find its value.
- The solutions to the quadratic equation are obtained when it is equal to zero.
- There are usually 2 solutions. They are called "roots", or "zeros".

Exercise 5.1

1. Which of the following are the quadratic equations:

(i) $x^2-\frac{1}{x}+3=0$

(ii) $x^4-2x+1=0$

(iii) $x-\frac{6}{x}+5=0$

(iv) $x(2x^2+4)=0$

(v) $\frac{1}{x^2}=1$

(vi) $x=0$

2. Arrange the following equations in the standard form $ax^2+bx+c=0$:

(i) $x^2=3x+1$ → $x-3x-1=0$ Ans.

(ii) $\frac{1}{x^2}-2+\frac{3}{x}=0$ → $\frac{1}{x^2}-2+\frac{3}{x}=0$ $= x^2-\frac{1}{x^2}$

(iii) $x^2-3=0$

(iv) $x^2=5x-1$

(v) $3x^2-x=0$

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5.2 Solution of quadratic equations:

Any real value of the variable that satisfies the given quadratic equation is then called a solution or root of the equation and the set of all solutions is called the solution set of a quadratic equation.

Square-Root property:

If $b>0$, then the solutions of $x^2=b$ are $\sqrt{b}$ and $-\sqrt{b}$.

The quadratic equation

$x^2-9=0$ is true only for

$x=3$ or $x=-3$. Hence $x=3$ and $x=-3$ are the solutions or roots of the quadratic equation and $\{3,-3\}$ is the solution set.

For example, the quadratic equation $x^2=17$ has solutions $\sqrt{17}$ and $-\sqrt{17}$.

The methods for solving quadratic equation are the following:

1. Factorization method
2. Completing the square
3. The quadratic formula.

1) Solution of a quadratic equation by factorization:

If a quadratic equation is in standard form then the simplest method of solving quadratic equation is the factorization. To factor quadratic equation of the form $ax^2+bx+c=0$, $a\neq 0$, then the steps are the following:

1. Find two numbers whose product is ac and whose sum is b.
2. Rewrite the term using the numbers found.

Example 1: Factorize the quadratic equation.

$2x^2+5x+3=0$

$a=2,\ b=5,\ c=3$

We must find two numbers whose product is $a.c=(2)(3)=6$ and whose sum is +5.

$2x^2+2x+3x+3=0$

$2x(x+1)+3(x+1)=0$

$(x+1)(2x+3)=0$ (taking common $(x+1)$)

$(x+1)(2x+3)=0$

The zero-factor property is used to obtain:

$x+1=0$ or $2x+3=0$

$\Rightarrow x=-1$ or $x=\dfrac{-3}{2}$

Solution set $=\left\{-1,\dfrac{-3}{2}\right\}$

Example-2: Solve the quadratic equation $3x^2-6x=-3$

Solution:

Given that

$3x^2-6x=-3$

$\Rightarrow 3x^2-6x+3=0$ (arrange it into standard form)

$3x^2-6x+3=0$

Now here

$a=3, b=-6, c=3$

$ac=-3\times-3=9$

$b=-3+(-3)=-3-3=-6$

So $ac=9,\ b=-6$

$\Rightarrow 3x^2-3x-3x+3=0$

$\Rightarrow 3x(x-1)-3(x-1)=0$

$(x-1)(3x-3)=0$

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Now by zero factor property:

$x-1=0,\ 3x-3=0$

$\Rightarrow x=1,\ 3x=3 \Rightarrow x=1$

The solution set is $\{1\}$

Example 3:

$x^2-5x+4=0$

$a=1,\ b=-5,\ c=4$

$ac=-4\times(-1)=4$

$b=-4-1=-5$

$x-4x-x+4=0$

$x(x-4)-1(x-4)=0$

$(x-4)(x-1)=0$

$\Rightarrow x=4$ and $x=1$

The solution set is $\{4,+1\}$

2) Solution of quadratic equation by completing square:

If the solution of the quadratic equation is not possible by factorization, then the solution can be found by "completing square method".

The steps involved in completing square are the following:

1. Write the given quadratic equation in the standard form $ax^2+bx+c=0$
2. Divide the equation by the co-efficient of x^2 if it is other than 1:

$$x^2+\frac{b}{a}x+\frac{c}{a}=0$$

3. Shift the constant term to the right hand side of the equation:

$$x^2+\frac{b}{a}x=-\frac{c}{a}$$

4. Add the square of the half of the co-efficient of x to both sides of the above equation, thus $x^2+\frac{b}{a}x+\frac{b^2}{4a^2}=-\frac{c}{a}+\frac{b^2}{4a^2}$ and solve this equation to obtain a solution set.

Example: Solve the quadratic equation $2x^2-6x+4=0$

Solution:

$2x^2-6x+4=0$

Divide both sides of equation by "2".

$$x^2-\frac{6}{2}x+\frac{4}{2}=0$$

$\Rightarrow x^2-3x+2=0$

Shift the constant term to right hand side to obtain:

$x^2-3x=-2$

Add $\left(\frac{3}{2}\right)^2$ to both sides of the equation to obtain:

$$x^2-3x+\left(\frac{3}{2}\right)^2=\left(\frac{3}{2}\right)^2-2$$

$$x^2-3x+\frac{9}{4}=\frac{9}{4}-2$$

$$\left(x-\frac{3}{2}\right)^2=\frac{9-8}{4}$$

$$\left(x-\frac{3}{2}\right)^2=\frac{1}{4}$$

Taking square root on both sides.

$$x-\frac{3}{2}=\pm\frac{1}{2}$$

$\Rightarrow x=\frac{3}{2}+\frac{1}{2}$ and $x=\frac{3}{2}-\frac{1}{2}$

$\Rightarrow x=\frac{3+1}{2}$ and $x=\frac{3-1}{2}$

$\Rightarrow x=\frac{4}{2}$ and $x=\frac{2}{2}$

$\Rightarrow x=2$ and $x=1$

So solution set $=\{2,1\}$

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Exercise 5.2

1. Solve of the following quadratic equations by factorization method:

(i) $1+\frac{15}{x^2}=\frac{16}{x},(x\neq 0)$

(ii) $x(3x-5)=(x-6)(x-7)$

(iii) $\frac{x}{x+6}+\frac{3}{x+4}=\frac{6}{(x+6)(x+4)},(x\neq -6,-4)$

(iv) $6x+\frac{1}{x}=7,\ x\neq 0$

(v) $64x^2+16x=-1$

2. Solve the following quadratic equations by completing square:

(i) $x^2-2x-444=0$

(ii) $5x^2-10x-890=0$

(iii) $x^2-4x-921=0$

(iv) $x^2=25x-24$

(v) $x^2-10x+10=0$

(vi) $2x^2=32x+34$

5.3 Solution of quadratic equation by the quadratic formula.

When the solution of the quadratic equation is not possible by factorization and completing square, then the solution can be found by quadratic formula.

The quadratic equation is:

$ax^2+bx+c=0$ where a,b,c are real numbers and $a\neq 0$.

Divide both sides of the equation by a:

$x^2+\frac{b}{a}x+\frac{c}{a}=0$.....(i)

Shift the constant term to the right hand side of the equations (i).

$x^2+\frac{b}{a}x=-\frac{c}{a}$.....(ii)

Add $\left(\frac{b}{2a}\right)^2=\frac{b^2}{4a^2}$ to both sides of the equation (ii)

$x^2+\frac{b}{a}x+\frac{b^2}{4a^2}=\frac{b^2}{4a^2}-\frac{c}{a}$

$\left(x+\frac{b}{2a}\right)^2=\left(\frac{b^2-4ac}{4a^2}\right)$.....(iii)

Take square root of both sides of the equation (iii) to obtain the quadratic formula:

$\left(x+\frac{b}{2a}\right)=\pm\sqrt{\frac{b^2-4ac}{4a^2}}$

$x+\frac{b}{2a}=\pm\frac{\sqrt{b^2-4ac}}{2a}$

$x=-\frac{b}{2a}\pm\frac{\sqrt{b^2-4ac}}{2a}$

$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$.....(iv)

Note: (iv) is called quadratic formula.

Example: Solve the quadratic equation $3x^2+5x-8=0$

Solution: We solve the equation by quadratic formula.

Compare the given equation with the standard form to obtain:

$3x^2+5x-8=0$

$ax^2+bx+c=0$

$a=3,b=5,c=-8$

The quadratic formula is used to obtain the solution set:

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$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$\Rightarrow x=\frac{-5\pm\sqrt{25-4(3)(-8)}}{2(3)}$$

$$\Rightarrow x=\frac{-5\pm\sqrt{25+96}}{6}$$

$$\Rightarrow x=\frac{-5\pm\sqrt{121}}{6}$$

$$\Rightarrow x=\frac{-5\pm11}{6}$$

$$\Rightarrow x=\frac{-5+11}{6} \text{ and } x=\frac{-5-11}{6}$$

$$\Rightarrow x=\frac{6}{6} \text{ and } x=\frac{-16}{6}$$

$$\Rightarrow x=1 \text{ and } x=\frac{-8}{3}$$

So, solution set $=\left\{1,\frac{-8}{3}\right\}$

Exercise 5.3

Solve the equations by quadratic formula:

1. $x^2-8x+15=0$
2. $3x^2-x=2$
3. $4x^2+3x=0$
4. $3x(x-2)+1=0$
5. $x+\frac{1}{6x}=\frac{-5}{6};x\neq0$
6. $(x-1)(x+3)=5(x+2)-3$
7. $\frac{x-2}{x+2}=\frac{2x-3}{x-1}, x\neq2,1$

5.4 Simple real life problems on quadratic equations:-

The following are real life problems related to quadratic equations are the:

Example.1

The cost of a rubber is one rupee more than of a pencil. If the product of their prices is rupees 6. Find the cost of each item.

Solution:

Suppose the cost of a pencil is x.

Then the cost of the rubber is $x+1$.

The product of their prices is rupees 6 that gives the quadratic equation:

$$x(x+1)=6$$

$$x^2+x=6$$

$$x^2+x-6=0$$

$$x^2+x-6=0$$

$$x^2+3x-2x-6=0$$

$$x(x+3)-2(x+3)=0$$

$$(x-2)(x+3)=0$$

$$x-2=0 \text{ and } x+3=0$$

$$x=2 \text{ and } x=-3$$

In this case negative value is not possible. Thus cost of the pencil is Rs.2 and cost of the rubber is Rs. 3.

Example-2

Find two consecutive positive numbers whose product is 42.

Solution:

Let the two consecutive positive numbers be x and $x+1$.

According to the statement it gives:

$$x(x+1)=42$$

$$x^2+x=42$$

$$x^2+x-42=0$$

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$$x^2+x-42=0$$
$$x^2+7x-6x-42=0$$
$$x(x+7)-6(x+7)=0$$
$$(x-6)(x+7)=0$$
$$x-6=0 \text{ or } x+7=0$$
$$x=6 \text{ or } x=-7$$

But we consider $x=6$, The two positive consecutive numbers are 6 and 7.

Exercise 5.4

1. The area of a rectangular plot is 460 square meter. The length of the plot is 12 meter more than its breadth. Find the length and breadth of the plot.
2. The sum of a positive number and its reciprocal is 2. Find the numbers.
3. The area of a board in a classroom is 12 square meters. If the length is 1 meter greater than its width, then find out the dimensions of the board.
4. If the square of the lengths of an equilateral triangle is 4 times the length of its one side. What will be the length of each side?
5. The product of two consecutive integers is 20. Find the integers.
6. Find two integers if their product is -8 and one integer is 6 less than the other.

EXERCISE 5.5

(Objective Type Questions)

1. Encircle (T) if the statement is True and (F) if the statement is False.
   (i) The degree of a quadratic equation is 2. T/F
   (ii) In the quadratic equation $ax^2+bx+c=0$, $a=0$ T/F
   (iii) The square of a number is always negative. T/F
   (iv) If the product of two factors is zero. Then each factor equal to zero. T/F
   (v) The general quadratic equation contains three terms. T/F

2. Write a suitable answer in each blank space.
   (i) A quadratic equation is solved by these methods.
   (ii) A quadratic equation can be factorized into two factors.
   (iii) The formula used for solution of a quadratic equation is called ____.
   (iv) The values of variable which satisfy the quadratic equation are called ____.
3. Three options are given for each question. Choose the best option.
   (i) One root of the equation $x^2-5x+4=0$ is
   (A) 0 (B) 1 (C) -1 (D) -4
   (ii) The degree of the quadratic equation is
   (A) 1 (B) 2 (C) 4 (D) $\frac{1}{4}$
   (iii) Number of roots for the equation $x^2-2x+1=0$
   (A) 1 (B) 2 (C) 0 (D) 3
   (iv) Quad means _____.
   (A) One (B) Square (C) Three (D) four
   (v) One root of $x^2-2x=0$ is _____.
   (A) 1 (B) 3 (C) -3 (D) 2

Glossary

Coefficient: The constant written to the left of a variable.
Degree: The highest power of a variable.
Factor: An expression which can give the original expression by multiplying some other expression
Factorization: The process to change an expression into factors.
Root: The value of the variable.
Quadratic Formula: for $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$
Variable: Which has different values.

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Unit 6

Matrices and Determinants

In this unit, the students will be able to learn about the:

- Definition of a matrix
- Rows and columns of a matrix
- Order of a matrix
- Equality of two matrices
- Types of matrices
  - Row matrix
  - Column matrix
  - Rectangular matrix
  - Square matrix
  - Zero (null) matrix
  - Identity matrix
  - Scalar matrix
  - Diagonal matrix
  - Transpose of a matrix
  - Symmetric and skew-symmetric matrices
- Addition and subtraction of matrices
  - Conformability for addition and subtraction.
  - Multiplication a matrix by a real number.
  - Verification of commutative and associative laws under addition.
  - Definition of addictive identity.
  - Finding additive inverse of a matrix.
- Multiplication of matrices
  - Conformability for multiplication of matrices.

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  - Multiplication of two (or three) matrices.
  - Verification of associative law under multiplication.
  - Verification of distributive laws.
  - Show that commutative law under multiplication does not hold in general.
  - Definition of multiplicative identity of a matrix.
  - Verification of $(AB)^t = B^t A^t$.
- Multiplicative inverse of matrix
  - Definition of determinant of a 2-by-2 matrix.
  - Evaluation of determinant of a 2-by-2 matrix.
  - Singular and non-singular matrices.
  - Adjoint of a 2-by-2 matrix
  - Multiplicative inverse of a 2-by-2 non-singular matrix A and to verify that $AA^{-1} = I = A^{-1}A$
  - Inverse of a 2-by-2 matrix using adjoint method.
  - Verification of the result

    $(AB)^{-1} = B^{-1}A^{-1}$
- Solution of simultaneous linear equations
  - Solution of two linear equations related to real life problems by
    - Matrix inversion method
    - Cramer's rule.

6.1 Introduction to Matrices

Until now we have used matrices only as a convenient shorthand for dealing with systems of equation. However, matrices are also important in the fields of management, natural science, engineering, and social science, because they provide a convenient way to organize data.

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6.6.1 Definition of a matrix:

The arrangement of elements or numbers horizontally and vertically enclosed within a square bracket is called a matrix. Matrices are represented by $A, B, C.....X, Y, Z$ *etc* and elements are represented by $a, b, c...x, y, z$.

Example-1:

The matrix is $A=\begin{bmatrix} 6 & 5 \\ 3 & 4 \\ 5 & -1 \end{bmatrix}$

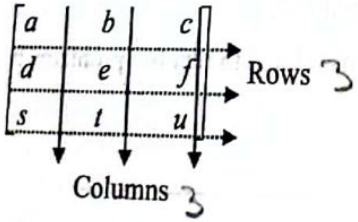
Example-2:

The matrix is $B=\begin{bmatrix} 5 & 8 & 9 \\ 0 & 5 & -3 \\ -4 & 0 & 5 \end{bmatrix}$

Bottle crate, egg-container and glass-packing also represent matrices.

6.1.2: Rows and Columns of a matrix

The horizontal arrangements of numbers are called rows while the vertical arrangements of numbers are called columns.



6.1.3: Order of a Matrix:

If the number of rows in a matrix is m and number of columns in matrix is n then the order of the matrix will be $m \times n$. Remember that order $m \times n$ does not mean a product of m and n. It is read as m by n.

Example:

Orders of the matrices are:

$A=\begin{bmatrix} 2 & 3 & 4 \\ 1 & 0 & 6 \end{bmatrix}$ 2-by-3 or (2×3), $B=\begin{bmatrix} 1 & 1 & 4 \\ 2 & 3 & 1 \\ 1 & -2 & 3 \end{bmatrix}$, 3-by-3 or (3×3)

6.1.4: Equality of two matrices:

If A and B be any two matrices, then A=B when

i) Order of A= Order of B.

ii) The corresponding elements of A equals the corresponding elements of B.

Example-1: The matrices A and B are equal

$$A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad B=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Since, they have same (order) and their corresponding entries are equal.

Example-2:

Check whether the following matrices are equal.

$$A=\begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}, \quad B=\begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}$$

A=B

The matrices A and B are equal, because they have the same and their corresponding elements are equal.

Exercise 6.1

1. Write the Rows and Columns of each of the following matrix.

(i) $A=\begin{bmatrix} 0 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ (ii) $B=\begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$

(iii) $C=\begin{bmatrix} 1 & -2 & 3 \\ 5 & 6 & 7 \end{bmatrix}$ (iv) $D=\begin{bmatrix} 6 & 7 \\ 0 & -2 \end{bmatrix}$

2. Determine the order of each of the matrix given in Q.No.1.

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3. Which of the following pairs of matrices are equal:

(i) $\begin{bmatrix}2 & 3\\ 0 & 4\end{bmatrix}, \begin{bmatrix}2 & 3\\ 0 & 4\end{bmatrix}$ equal

(ii) $\begin{bmatrix}5 & 2\\ 1 & 3\end{bmatrix}, \begin{bmatrix}2 & 1\\ 3 & 5\end{bmatrix}$ equal

(iii) $\begin{bmatrix}6 & 3\\ 1 & 5\end{bmatrix}, \begin{bmatrix}1+5 & 4-1\\ 2-1 & 3+1\end{bmatrix}$

(iv) $[6\ 7\ 8], \begin{bmatrix}6\\ 7\\ 8\end{bmatrix}$ (v) $\begin{bmatrix}8 & 0\\ 4 & 1\end{bmatrix}, \begin{bmatrix}\frac{16}{2} & 1-1\\ \frac{8}{2} & 2-1\end{bmatrix}$

6.2. Type of Matrices

The different types of matrices are:

6.2.1. Row Matrix:

A matrix with only one row is called a row matrix.

Example

The matrix $A=[-3\ -2\ -5]$ is row matrix, having only one row.

6.2.2. Column Matrix

A matrix with only one column is called a column matrix.

Example:

$\begin{bmatrix}1\\ 2\\ 3\\ 4\\ 5\end{bmatrix}$ is a column matrix, having only one column.

6.2.3. Square matrix:

If the number of rows equals the number of columns then the matrix is said to be a square matrix.

Example:

$$\begin{bmatrix}1 & 0 & 3\\ 5 & 3 & 1\\ 7 & 9 & 2\end{bmatrix}$$

Add

$$\begin{bmatrix}a & b & c\\ d & e & f\\ g & h & I\end{bmatrix}$$

are square matrices of order 3.

6.2.4. Rectangular Matrix:

A matrix is said to be rectangular matrix, if the number of its rows is not equal to the number of its columns.

Example:

$\begin{bmatrix}2 & 5 & 3\\ 7 & 6 & 1\end{bmatrix}$ and $\begin{bmatrix}1 & 3 & 0 & 2\\ 2 & 5 & 4 & 3\\ 8 & 1 & 7 & 4\end{bmatrix}$

are rectangular matrices.

6.2.5 Zero or null matrix:

If each element of a matrix is zero, then it is called a zero or null matrix.

Example: The matrices

$[0], [0\ 0\ 0], \begin{bmatrix}0\\ 0\\ 0\end{bmatrix}, \begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$ zero matrices.

6.2.6. Identity matrix

A square matrix of order n having diagonal elements one (1), is called an identity matrix or unit matrix. The elements above and below the diagonal "1" are zero. It is denoted by I.

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$I=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, $I=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$, $I=\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$ are the unit matrices of order 2, 3, and 4.

6.2.7. Scalar Matrix:

A square matrix having the same diagonal entries is called a scalar matrix.

Example:

The matrices:

(1) $\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}, \begin{bmatrix}3 & 0 & 0\\ 0 & 3 & 0\\ 0 & 0 & 3\end{bmatrix}, \begin{bmatrix}4 & 0 & 0 & 0\\ 0 & 4 & 0 & 0\\ 0 & 0 & 4 & 0\\ 0 & 0 & 0 & 4\end{bmatrix}$

Scalar matrices of order 2,3 and 4 respectively.

6.2.8. Diagonal matrix:

A square matrix A of order n with zero elements everywhere except at least one (non zero) element in the principle diagonal is called a diagonal matrix.

Examples:

The matrices

$[3], \begin{bmatrix}4 & 0\\ 0 & 3\end{bmatrix}, \begin{bmatrix}0 & 0 & 0\\ 0 & 5 & 0\\ 0 & 0 & 3\end{bmatrix}, \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 6 & 0 & 0\\ 0 & 0 & 3 & 0\\ 0 & 0 & 0 & 8\end{bmatrix}$ are diagonal matrices of order 1, 2, 3 and 4.

6.2.9. Transpose of a Matrix:

If the rows and column of a matrix A of order m-by-n (m×n) are interchanged then the new matrix of order n-by-m is (n×m) called the transpose of a matrix and is denoted by A^t.

Example:

$$A=\begin{bmatrix}7 & 1\\ 0 & 5\\ 3 & 4\end{bmatrix}$$

$$A^t=\begin{bmatrix}7 & 0 & 3\\ 1 & 5 & 4\end{bmatrix}$$

A^t is obtained by interchanging its rows by columns.

6.2.10 Symmetric matrix

A matrix A is said to be symmetric if

$$A^t = A$$

Example-1: The matrix A is symmetric, since $A^t = A$:

$$A=\begin{bmatrix}7 & 1 & 5\\ 1 & 3 & 0\\ 5 & 0 & 7\end{bmatrix}$$

$$A^t=\begin{bmatrix}7 & 1 & 5\\ 1 & 3 & 0\\ 5 & 0 & 7\end{bmatrix}$$

6.2.11. Skew-Symmetric matrix:

A matrix 'A' is said to be skew symmetric, if $A^t = -A$.

Example: The matrix A is skew symmetric if $A^t = -A$:

$$A=\begin{bmatrix}0 & 1 & -5\\ -1 & 0 & 0\\ 5 & 0 & 0\end{bmatrix}$$

$$A^t=\begin{bmatrix}0 & -1 & 5\\ 1 & 0 & 0\\ -5 & 0 & 0\end{bmatrix}=-A$$

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Exercise 6.2

1. Identify row matrix in the following matrices:

(i) $\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 2 & 4 \end{bmatrix}$

(iii) $\begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$ (iv) $\begin{bmatrix} 1 & 2 & 8 \\ 0 & 1 & 3 \end{bmatrix}$

(v) $\begin{bmatrix} 0 & 4 & 9 \end{bmatrix}$

2. Identify column matrices in the following matrices:

(i) $\begin{bmatrix} 1 & 3 & 0 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 4 \end{bmatrix}$

(iii) $\begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix}$ (iv) $\begin{bmatrix} 1 & 0 \\ 2 & 3 \\ 8 & 7 \end{bmatrix}$

(v) $\begin{bmatrix} 0 \\ 7 \\ 1 \end{bmatrix}$

3. Identify the rectangular and square matrices in the following matrices:

(i) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 4 & 0 & 5 \end{bmatrix}$

(iii) $\begin{bmatrix} 1 & 0 & 4 \\ 3 & 2 & 0 \end{bmatrix}$ (iv) $\begin{bmatrix} 1 & 2 \\ 3 & 0 \\ 0 & 0 \end{bmatrix}$

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(v) $\begin{bmatrix} 1 & 2 & 7 \\ 7 & 0 & 1 \\ 1 & 2 & 8 \end{bmatrix}$ (vi) $\begin{bmatrix} 0 & 3 \end{bmatrix}$

(vii) $\begin{bmatrix} 3 \end{bmatrix}$ viii) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(ix) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ (x) $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 5 \end{bmatrix}$

4. Which of the following are zero/null matrices:

(i) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ (ii) $\begin{bmatrix} 0 \end{bmatrix}$

(iii) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ (iv) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(v) $\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$ (vi) $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

(vii) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

5. Separate the scalar, unit and diagonal matrices in the following matrices:

(i) $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

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(iii) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (iv) $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

(v) $\begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}$ (vi) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$

(vii) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (viii) $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

6. Find the transpose of the following matrices:

(i) $\begin{bmatrix} 1 & 2 \\ 5 & 7 \end{bmatrix}$ (ii) $\begin{bmatrix} 0 & 1 & 3 \\ 3 & 3 & 0 \\ 4 & 5 & 0 \end{bmatrix}$

(iii) $\begin{bmatrix} 3 & 2 & 1 \\ 1 & 0 & 4 \\ 5 & 1 & 7 \end{bmatrix}$ (iv) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

7. Which of the following are the symmetric matrices:

(i) $\begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 3 & 7 \\ 3 & 0 & 2 \\ 7 & 2 & 4 \end{bmatrix}$

(iii) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (iv) $\begin{bmatrix} 0 & 1 & 4 \\ 1 & 7 & 9 \\ 4 & 9 & 2 \end{bmatrix}$

8. Which of the following matrices are skew-symmetric matrices:

(i) $\begin{bmatrix} 0 & 1 & 7 \\ -1 & 0 & 3 \\ -7 & -3 & 0 \end{bmatrix}$ (ii) $\begin{bmatrix} 3 & 1 & 0 \\ -1 & 2 & 4 \\ 7 & -4 & -7 \end{bmatrix}$

(iii) $\begin{bmatrix} 0 & 9 & 6 \\ -9 & 0 & 3 \\ -7 & -3 & 0 \end{bmatrix}$ (iv) $\begin{bmatrix} 0 & 3 & 5 \\ -3 & 0 & 1 \\ -5 & -1 & 0 \end{bmatrix}$

(v) $\begin{bmatrix} 0 & 2 & 7 \\ -2 & 0 & 3 \\ -7 & -3 & 0 \end{bmatrix}$ (vi) $\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 9 \\ -2 & -9 & 0 \end{bmatrix}$

6.3 Operation of Addition, Subtraction and Scalar multiplication of matrices

6.3.1 Rule for addition or subtraction of matrices

If A and B be any two matrices, then the sum A+B can be obtained, if the order of A equals to the order of B. similarly the difference A-B of two matrices can be obtained, if order of A equals to order of B.

6.3.2. Addition of matrices

If A and B are two matrices of the same order *m*-by-*n*, then their sum A+B is a matrix of the same order obtained by taking the sum of their corresponding elements.

Example 1: Add the matrices

$$A = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}, \text{ and } B = \begin{bmatrix} 6 & 7 \\ 9 & 1 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 2+6 & 3+7 \\ 0+9 & 1+1 \end{bmatrix} = \begin{bmatrix} 8 & 10 \\ 9 & 2 \end{bmatrix}$$

Example 2: Add the matrices

$$A = \begin{bmatrix} 4 & 4 & 2 \\ 7 & 5 & 9 \end{bmatrix}, \text{ and } B = \begin{bmatrix} 4 & 5 & 0 \\ 9 & 1 & 6 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 4 & 4 & 2 \\ 7 & 5 & 9 \end{bmatrix} + \begin{bmatrix} 4 & 5 & 0 \\ 9 & 1 & 6 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 4+4 & 4+5 & 2+0 \\ 7+9 & 5+1 & 9+6 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 8 & 9 & 2 \\ 16 & 6 & 15 \end{bmatrix}$$

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6.3.3. Scalar Multiplication

If a matrix A is multiplied by a scalar quantity K, then the resultant matrix KA is called a scalar multiplication of A by K.

Example:

If $A=\begin{bmatrix}1 & 3 & 2\\ 4 & 2 & 1\\ 3 & 7 & 5\end{bmatrix}$ is a matrix K is any scalar then the matrix.

$$KA=K\begin{bmatrix}1 & 3 & 2\\ 4 & 2 & 1\\ 3 & 7 & 5\end{bmatrix}$$

$$=\begin{bmatrix}K & 3K & 2K\\ 4K & 2K & K\\ 3K & 7K & 5K\end{bmatrix}$$

is called a scalar multiple of A.

6.3.4. Subtraction of matrices.

If A and B are two matrices of the same order m-by-n, then their difference A-B is a matrix of the same order obtained by taking the difference of then corresponding elements.

Example:

Subtract the matrix B from matrix A.

$$A=\begin{bmatrix}1 & 4 & 0\\ 3 & 0 & 1\\ 3 & 2 & 5\end{bmatrix} \text{ and } B=\begin{bmatrix}2 & 1 & 0\\ 3 & -2 & 4\\ 7 & -5 & 8\end{bmatrix}$$

$$A-B=\begin{bmatrix}1 & 4 & 0\\ 3 & 0 & 1\\ 3 & 2 & 5\end{bmatrix}-\begin{bmatrix}2 & 1 & 0\\ 3 & -2 & 4\\ 7 & -5 & 8\end{bmatrix}$$

$$=\begin{bmatrix}1 & 4 & 0\\ 3 & 0 & 1\\ 3 & 2 & 5\end{bmatrix}-\begin{bmatrix}2 & 1 & 0\\ 3 & -2 & 4\\ 7 & -5 & 8\end{bmatrix}$$

$$=\begin{bmatrix}1-2 & 4-1 & 0+0\\ 3-3 & 0+2 & 1-4\\ 3-7 & 2+5 & 5-8\end{bmatrix}$$

$$A-B=\begin{bmatrix}-1 & 3 & 0\\ 0 & 2 & -3\\ -4 & +7 & -3\end{bmatrix}$$

6.3.5. Commutative and Associative Laws under addition.

i) Commutative Law under addition:

If A and B are two matrices of the same order then

$A + B = B + A$.

This property is called commutative property of addition of matrices.

Example:

If $A=\begin{bmatrix}2 & 0 & 2\\ 3 & 5 & 1\\ 7 & 4 & 6\end{bmatrix}$

and

$B=\begin{bmatrix}1 & 0 & 3\\ 2 & 1 & 5\\ 4 & 3 & 6\end{bmatrix}$, then prove that $A+B=B+A$

L.H.S.

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$$A+B=\begin{bmatrix}2 & 0 & 2\\ 3 & 5 & 1\\ 7 & 4 & 6\end{bmatrix}+\begin{bmatrix}1 & 0 & 3\\ 2 & 1 & 5\\ 4 & 3 & 6\end{bmatrix}$$

$$=\begin{bmatrix}2+1 & 0+0 & 2+3\\ 3+2 & 5+1 & 1+5\\ 7+4 & 4+3 & 6+6\end{bmatrix}$$

$$A+B=\begin{bmatrix}3 & 0 & 5\\ 5 & 6 & 6\\ 11 & 7 & 12\end{bmatrix} \quad \ldots\ldots\ldots\ldots \text{(i)}$$

R.H.S

$$B+A=\begin{bmatrix}1 & 0 & 3\\ 2 & 1 & 5\\ 4 & 3 & 6\end{bmatrix}+\begin{bmatrix}2 & 0 & 2\\ 3 & 5 & 1\\ 7 & 4 & 6\end{bmatrix}$$

$$=\begin{bmatrix}1+2 & 0+0 & 3+2\\ 2+3 & 1+5 & 5+1\\ 4+7 & 3+4 & 6+6\end{bmatrix}$$

$$B+A=\begin{bmatrix}3 & 0 & 5\\ 5 & 6 & 6\\ 11 & 7 & 12\end{bmatrix} \quad \ldots\ldots\ldots\ldots \text{(ii)}$$

From (i) and (ii) we prove that

$A + B = B + A$

Hence commutative law under addition is hold in matrices addition.

6.3.6 Associative Law under addition:

If A, B and C are three matrices of the same order,

$(A+B)+C = A+(B+C)$.

This property is called associative property of addition of matrices.

Example:

If $A=\begin{bmatrix}0 & 1 & 3\\ 2 & 4 & 1\\ 5 & 0 & 2\end{bmatrix}$, $B=\begin{bmatrix}3 & 2 & 2\\ 1 & 0 & 4\\ 0 & 5 & 1\end{bmatrix}$

and $C=\begin{bmatrix}1 & 2 & 1\\ 3 & 1 & 0\\ 2 & 0 & 4\end{bmatrix}$, then prove that $(A+B)+C=A+(B+C)$

$$(A+B)=\begin{bmatrix}0 & 1 & 3\\ 2 & 4 & 1\\ 5 & 0 & 2\end{bmatrix}+\begin{bmatrix}3 & 2 & 2\\ 1 & 0 & 4\\ 0 & 5 & 1\end{bmatrix}$$

$$=\begin{bmatrix}0+3 & 1+2 & 3+2\\ 2+1 & 4+0 & 1+4\\ 5+0 & 0+5 & 2+1\end{bmatrix}$$

$$=\begin{bmatrix}3 & 3 & 5\\ 3 & 4 & 5\\ 5 & 5 & 3\end{bmatrix}$$

Now

$$(A+B)+C=\begin{bmatrix}3 & 3 & 5\\ 3 & 4 & 5\\ 5 & 5 & 3\end{bmatrix}+\begin{bmatrix}1 & 2 & 1\\ 3 & 1 & 0\\ 2 & 0 & 4\end{bmatrix}$$

$$=\begin{bmatrix}3+1 & 3+2 & 5+1\\ 3+3 & 4+1 & 5+0\\ 5+2 & 5+0 & 3+4\end{bmatrix}$$

$$=\begin{bmatrix}4 & 5 & 6\\ 6 & 5 & 5\\ 7 & 5 & 7\end{bmatrix} = \text{L.H.S} \ldots\ldots\ldots\text{(i)}$$

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$$B+C=\begin{bmatrix}3 & 2 & 2\\ 1 & 0 & 4\\ 0 & 5 & 1\end{bmatrix}+\begin{bmatrix}1 & 2 & 1\\ 3 & 1 & 0\\ 2 & 0 & 4\end{bmatrix}$$

$$=\begin{bmatrix}3+1 & 2+2 & 2+1\\ 1+3 & 0+1 & 4+0\\ 0+2 & 5+0 & 4+1\end{bmatrix}$$

$$B+C=\begin{bmatrix}4 & 4 & 3\\ 4 & 1 & 4\\ 2 & 5 & 5\end{bmatrix}$$

Now

$$A+(B+C)=\begin{bmatrix}0 & 1 & 3\\ 2 & 4 & 1\\ 5 & 0 & 2\end{bmatrix}+\begin{bmatrix}4 & 4 & 3\\ 4 & 1 & 4\\ 2 & 5 & 5\end{bmatrix}$$

$$=\begin{bmatrix}0+4 & 1+4 & 3+3\\ 2+4 & 4+1 & 1+4\\ 5+2 & 0+5 & 2+5\end{bmatrix}$$

$$=\begin{bmatrix}4 & 5 & 6\\ 6 & 5 & 5\\ 7 & 5 & 7\end{bmatrix}=\text{R. H. S}\ldots\ldots\ldots\ldots\ldots\text{(ii)}$$

From (i) and (ii), we prove that

$$(A+B)+C=A+(B+C)$$

6.3.7 Additive Identity of a Matrix:

It is known that "O" (zero) is the additive identity for real number, since the addition of zero to any real number does not alter that real number. Exactly, if Null matrix is added to a matrix of the same order, and the matrix does not change its value. Hence, Null matrix is called the additive identity.

Example-1: Show that "O" is the additive identity matrix.

$$A+O=O+A=A$$

Let $$A=\begin{bmatrix}2 & 4 & 0\\ 1 & 2 & 3\\ 3 & 0 & 1\end{bmatrix}$$

$$O=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

$$A+O=\begin{bmatrix}2 & 4 & 0\\ 1 & 2 & 3\\ 3 & 0 & 1\end{bmatrix}+\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

$$=\begin{bmatrix}2+0 & 4+0 & 0+0\\ 1+0 & 2+0 & 3+0\\ 3+0 & 0+0 & 1+0\end{bmatrix}$$

$$=\begin{bmatrix}2 & 4 & 0\\ 1 & 2 & 3\\ 3 & 0 & 1\end{bmatrix}=\text{A}$$

Now

$$O+A=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}+\begin{bmatrix}2 & 4 & 0\\ 1 & 2 & 3\\ 3 & 0 & 1\end{bmatrix}$$

$$=\begin{bmatrix}0+2 & 0+4 & 0+0\\ 0+1 & 0+2 & 0+3\\ 0+3 & 0+0 & 0+1\end{bmatrix}$$

$$=\begin{bmatrix}2 & 4 & 0\\ 1 & 2 & 3\\ 3 & 0 & 1\end{bmatrix}=A$$

Hence $$A+O=O+A=A$$

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6.3.8. Additive inverse of a matrix:

If A is a matrix, then the additive inverse of A is –A, under condition:

$$A+(-A)=0$$

Let

$$A=\begin{bmatrix}2 & 1 & 5\\ 7 & 0 & 3\\ 4 & 6 & 9\end{bmatrix}$$

$$A+(-A)=\begin{bmatrix}2 & 1 & 5\\ 7 & 0 & 3\\ 4 & 6 & 9\end{bmatrix}+(-1)\begin{bmatrix}2 & 1 & 5\\ 7 & 0 & 3\\ 4 & 6 & 9\end{bmatrix}$$

$$=\begin{bmatrix}2 & 1 & 5\\ 7 & 0 & 3\\ 4 & 6 & 9\end{bmatrix}+\begin{bmatrix}-2 & -1 & -5\\ -7 & 0 & -3\\ -4 & -6 & -9\end{bmatrix}$$

$$A+(-A)=\begin{bmatrix}2 & 1 & 5\\ 7 & 0 & 3\\ 4 & 6 & 9\end{bmatrix}+\begin{bmatrix}-2 & -1 & -5\\ -7 & 0 & -3\\ -4 & -6 & -9\end{bmatrix}$$

$$=\begin{bmatrix}2-2 & 1-1 & 5-5\\ 7-7 & 0-0 & 3-3\\ 4-4 & 6-6 & 9-9\end{bmatrix}$$

$$=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

$$A+(-A)=O$$

Exercise 6.3

1. Add following matrices:

(i) $\begin{bmatrix}5 & 2 & 0\\ -1 & 1 & -2\end{bmatrix}, \begin{bmatrix}0 & -2 & 4\\ 2 & 1 & 1\end{bmatrix}$

(ii) $\begin{bmatrix}1\\0\\5\end{bmatrix}, \begin{bmatrix}7\\9\\2\end{bmatrix}$

(iii) $[2], [5]$

(iv) $[2 \quad 2 \quad 2], [3 \quad 1 \quad 4]$

(v) $\begin{bmatrix}0 & 1 & 4\\ 4 & 0 & 2\\ 3 & 5 & 9\end{bmatrix}, \begin{bmatrix}0 & 2 & 3\\ 3 & 1 & 7\\ 6 & 4 & 1\end{bmatrix}$

(vi) $\begin{bmatrix}2 & 4\\ 6 & 8\end{bmatrix}, \begin{bmatrix}5 & 3\\ 1 & 1\end{bmatrix}$

2. Subtract 2nd matrix from 1st:

(i) $\begin{bmatrix}3\\4\\5\end{bmatrix}, \begin{bmatrix}3\\-2\\-1\end{bmatrix}$

(ii) $[1 \quad 0 \quad 3], [-4 \quad -5 \quad -6]$

(iii) $\begin{bmatrix}5 & 4 & 3\\ 2 & -7 & 6\end{bmatrix}, \begin{bmatrix}1 & 3 & 5\\ 2 & 4 & 6\end{bmatrix}$

(iv) $\begin{bmatrix}0 & 1 & 2\\ 4 & 5 & 7\\ 6 & 2 & 7\end{bmatrix}, \begin{bmatrix}3 & 0 & 2\\ 4 & 1 & 5\\ 7 & 8 & 6\end{bmatrix}$

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3. If

$$A=\begin{bmatrix}3 & 2 & 6\\ 4 & 5 & 0\\ 1 & 3 & 4\end{bmatrix}, B=\begin{bmatrix}5 & 0 & 4\\ -2 & 6 & 1\\ -3 & 2 & 1\end{bmatrix}, C=\begin{bmatrix}6 & 1 & 1\\ 4 & 3 & 2\\ 0 & 1 & -5\end{bmatrix}$$

are the matrices of the same order then show that:

i) $A+B=B+A$

ii) $A+(B+C)=(A+B)+C$

iii) $A+O=O+A=A$

iv) $A+(-A)=(-A)+A=O$

4. Find the additive inverse of the following matrices:

(i) $[8]$ (ii) $\begin{bmatrix}2 & 1\\ 3 & 3\end{bmatrix}$

(iii) $\begin{bmatrix}9 & 0 & 0\\ 0 & 9 & 0\\ 0 & 0 & 9\end{bmatrix}$ (iv) $\begin{bmatrix}1 & 3 & 2\\ 5 & 8 & 9\end{bmatrix}$

5. Verify the additive identity property for tie following matrices:

(i) $[2 \quad 4]$ (ii) $\begin{bmatrix}1\\ 0\end{bmatrix}$

(iii) $\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$ (iv) $\begin{bmatrix}7 & 0\\ 0 & 7\end{bmatrix}$

6.4 Multiplication of Matrices

Two matrices A and B are said to be confo:nable for multiplication giving the product AB if the number of columns in matrix A equals to the number of rows of matrix B.

The product matrix of A and B is denoted by AB.

Example:

Find the product of the matrices.

$$A=\begin{bmatrix}3 & 2\\ 2 & 1\end{bmatrix}, B=\begin{bmatrix}1 & 2\\ 3 & 1\end{bmatrix}$$

If $A=\begin{bmatrix}3 & 2\\ 2 & 1\end{bmatrix}, B=\begin{bmatrix}1 & 2\\ 3 & 1\end{bmatrix}$, the product of A and B matrices is:

$$AB=\begin{bmatrix}3 & 2\\ 2 & 1\end{bmatrix}_{2\times2}\begin{bmatrix}1 & 2\\ 3 & 1\end{bmatrix}_{2\times2}$$

$$=\begin{bmatrix}3\times1+2\times3 & 3\times2+2\times1\\ 2\times1+1\times3 & 2\times2+1\times1\end{bmatrix}_{2\times2}$$

$$=\begin{bmatrix}3+6 & 6+2\\ 2+3 & 4+1\end{bmatrix}$$

$$=\begin{bmatrix}9 & 8\\ 5 & 5\end{bmatrix}$$

6.4.1 Multiplication of two or three matrices

Example-1:

Find product of the matrices

$$A=\begin{bmatrix}0 & 1\\ 3 & 2\end{bmatrix}, B=\begin{bmatrix}4 & 1\\ 2 & 0\end{bmatrix}$$

$$AB=\begin{bmatrix}0 & 1\\ 3 & 2\end{bmatrix}\begin{bmatrix}4 & 1\\ 2 & 0\end{bmatrix}$$

$$=\begin{bmatrix}0\times4+1\times2 & 0\times1+1\times0\\ 3\times4+2\times2 & 3\times1+2\times0\end{bmatrix}=\begin{bmatrix}0+2 & 0+0\\ 12+4 & 4+0\end{bmatrix}=\begin{bmatrix}2 & 0\\ 16 & 4\end{bmatrix}$$

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Example-2: Find the product of the matrices

$$A=\begin{bmatrix}0 & 2\\ 1 & 0\end{bmatrix}, B=\begin{bmatrix}2 & 2\\ 0 & 0\end{bmatrix}, C=\begin{bmatrix}3 & 0\\ 0 & 3\end{bmatrix}, \text{ find } ABC=?$$

To find the product of $(AB)C$. Since we need to find out product AB:

$$AB=\begin{bmatrix}0 & 2\\ 1 & 0\end{bmatrix}\begin{bmatrix}2 & 2\\ 0 & 0\end{bmatrix}$$

$$=\begin{bmatrix}0\times2+2\times0 & 0\times2+2\times0\\ 1\times2+0\times0 & 1\times2+0\times0\end{bmatrix}$$

$$=\begin{bmatrix}0+0 & 0+0\\ 2+0 & 2+0\end{bmatrix}=\begin{bmatrix}0 & 0\\ 2 & 2\end{bmatrix}$$

Then AB and C is therefore:

$$(AB)C=\begin{bmatrix}0 & 0\\ 2 & 2\end{bmatrix}\begin{bmatrix}3 & 0\\ 0 & 3\end{bmatrix}$$

$$=\begin{bmatrix}0\times3+0\times0 & 0\times0+0\times3\\ 2\times3+2\times0 & 2\times0+2\times3\end{bmatrix}$$

$$=\begin{bmatrix}0+0 & 0+0\\ 6+0 & 0+6\end{bmatrix}$$

$$ABC=\begin{bmatrix}0 & 0\\ 6 & 6\end{bmatrix}$$

Associative Law under multiplication

The associative law of multiplication for the matrices A, B and C is

$$(AB)C=A(BC)$$

For proof, if

$$A=\begin{bmatrix}2 & 4\\ 0 & 0\end{bmatrix}, B=\begin{bmatrix}0 & 3\\ 2 & 1\end{bmatrix}, C=\begin{bmatrix}0 & 6\\ 5 & 0\end{bmatrix}$$ are the matrices, then

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$$AB=\begin{bmatrix}2 & 4\\ 0 & 0\end{bmatrix}\begin{bmatrix}0 & 3\\ 2 & 1\end{bmatrix}$$

$$=\begin{bmatrix}2\times0+4\times2 & 2\times3+4\times1\\ 0\times0+0\times2 & 0\times3+0\times1\end{bmatrix}$$

$$=\begin{bmatrix}0+8 & 6+4\\ 0+0 & 0+0\end{bmatrix}$$

$$AB=\begin{bmatrix}8 & 10\\ 0 & 0\end{bmatrix}$$

Now, the L.H.S is:

$$(AB)C=\begin{bmatrix}8 & 10\\ 0 & 0\end{bmatrix}\begin{bmatrix}0 & 6\\ 5 & 0\end{bmatrix}$$

$$=\begin{bmatrix}8\times0+10\times5 & 8\times6+10\times0\\ 0\times0+0\times5 & 0\times6+0\times0\end{bmatrix}$$

$$(AB)C=\begin{bmatrix}0+50 & 48+0\\ 0+0 & 0+0\end{bmatrix}$$

$$\text{L.H.S}=\begin{bmatrix}50 & 48\\ 0 & 0\end{bmatrix}\ldots\ldots\ldots\text{(i)}$$

For R.H.S, the product of B and C is:

$$BC=\begin{bmatrix}0 & 3\\ 2 & 1\end{bmatrix}\begin{bmatrix}0 & 6\\ 5 & 0\end{bmatrix}$$

$$=\begin{bmatrix}0\times0+3\times5 & 0\times6+3\times0\\ 2\times0+1\times5 & 2\times6+1\times0\end{bmatrix}$$

$$BC=\begin{bmatrix}0+15 & 0+0\\ 0+5 & 12+0\end{bmatrix}=\begin{bmatrix}15 & 0\\ 5 & 12\end{bmatrix}$$

Now, the R.H.S is:

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$$A(BC)=\begin{bmatrix}2 & 4\\ 0 & 0\end{bmatrix}\begin{bmatrix}15 & 0\\ 5 & 12\end{bmatrix}$$

$$=\begin{bmatrix}30+20 & 0+48\\ 0+0 & 0+0\end{bmatrix}$$

R.H.S $=\begin{bmatrix}50 & 48\\ 0 & 0\end{bmatrix}$ .....(ii)

From (i) and (ii)

L.H.S = R.H.S

The associative law holds for matrix multiplication.

6.4.2. Left Distributive Law of multiplication over addition

The left distributive law for the matrices A, B and C is:

$A(B+C)=AB+AC$

If $A=\begin{bmatrix}2 & 4\\ 0 & 1\end{bmatrix}$, $B=\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}$, $C=\begin{bmatrix}1 & 4\\ 0 & 2\end{bmatrix}$ are the matrices, then

L.H.S. $=A(B+C)=\begin{bmatrix}2 & 4\\ 0 & 1\end{bmatrix}\left[\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}+\begin{bmatrix}1 & 4\\ 0 & 2\end{bmatrix}\right]$

$$=\begin{bmatrix}2 & 4\\ 0 & 1\end{bmatrix}\begin{bmatrix}1+1 & 3+4\\ 0+0 & 1+2\end{bmatrix}$$

$$=\begin{bmatrix}2 & 4\\ 0 & 1\end{bmatrix}\begin{bmatrix}2 & 7\\ 0 & 3\end{bmatrix}$$

$$=\begin{bmatrix}2 & 4\\ 0 & 1\end{bmatrix}\begin{bmatrix}2 & 7\\ 0 & 3\end{bmatrix}$$

$R.H.S=AC+BC$

$$=\begin{bmatrix}2\times2+4\times0 & 2\times7+4\times3\\ 0\times2+1\times0 & 0\times7+1\times3\end{bmatrix}$$

$$=\begin{bmatrix}4+0 & 14+12\\ 0+0 & 0+3\end{bmatrix}$$

$$=\begin{bmatrix}4 & 26\\ 0 & 3\end{bmatrix}$$ .....(i)

Now for the R.H.S, we have.

$$AB=\begin{bmatrix}2 & 4\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}$$

$$=\begin{bmatrix}2\times1+4\times0 & 2\times3+4\times1\\ 0\times1+1\times0 & 0\times3+1\times1\end{bmatrix}$$

$$=\begin{bmatrix}2+0 & 6+4\\ 0+0 & 0+1\end{bmatrix}$$

$$=\begin{bmatrix}2 & 10\\ 0 & 1\end{bmatrix}$$

And $AC=\begin{bmatrix}2 & 4\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 4\\ 0 & 2\end{bmatrix}$

$$=\begin{bmatrix}2\times1+4\times0 & 2\times4+4\times2\\ 0\times1+1\times0 & 0\times4+1\times2\end{bmatrix}$$

$$AC=\begin{bmatrix}2+0 & 8+8\\ 0+0 & 0+2\end{bmatrix}$$

$$=\begin{bmatrix}2 & 16\\ 0 & 2\end{bmatrix}$$

The R.H.S is therefore:

R.H.S $=AB+BC=\begin{bmatrix}2 & 10\\ 0 & 1\end{bmatrix}+\begin{bmatrix}2 & 16\\ 0 & 2\end{bmatrix}$

$$=\begin{bmatrix}2+2 & 10+16\\ 0+0 & 1+2\end{bmatrix}$$

$$=\begin{bmatrix}4 & 26\\ 0 & 3\end{bmatrix}$$ .....(ii)

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from (i) and (ii), we have

$$A(B+C)=AB+BC$$

Hence the law is proved.

6.4.3. Right Distributive law of multiplication over additions

If A, B, C are any three matrices, then the right distributive law is:

$$(A+B)C=AC+BC$$

Example:

If $A=\begin{bmatrix}1 & 0\\ 2 & 3\end{bmatrix}, B=\begin{bmatrix}-1 & 3\\ 4 & 5\end{bmatrix}, C=\begin{bmatrix}2 & 5\\ 0 & 3\end{bmatrix}$ are the matrices,

Then verify that

$(A+B)C=AC+BC$

Solution:

L.H.S. $=(A+B)C$

$$\text{L.H.S.}=\left(\begin{bmatrix}1 & 0\\ 2 & 3\end{bmatrix}+\begin{bmatrix}-1 & 3\\ 4 & 5\end{bmatrix}\right)\begin{bmatrix}2 & 5\\ 0 & 3\end{bmatrix}$$

$$=\begin{bmatrix}0 & 3\\ 6 & 8\end{bmatrix}\begin{bmatrix}2 & 5\\ 0 & 3\end{bmatrix}$$

$$=\begin{bmatrix}0\times2+3\times0 & 0\times5+3\times3\\ 6\times2+8\times0 & 6\times5+8\times3\end{bmatrix}$$

$$=\begin{bmatrix}0+0 & 0+9\\ 12+0 & 30+24\end{bmatrix}$$

$$=\begin{bmatrix}0 & 9\\ 12 & 54\end{bmatrix}$$

$R.H.S=AC+BC$

$$=\begin{bmatrix}1 & 0\\ 2 & 3\end{bmatrix}\begin{bmatrix}2 & 5\\ 0 & 3\end{bmatrix}+\begin{bmatrix}-1 & 3\\ 4 & 5\end{bmatrix}\begin{bmatrix}2 & 5\\ 0 & 3\end{bmatrix}$$

$$=\begin{bmatrix}2+0 & 5+0\\ 4+0 & 10+9\end{bmatrix}+\begin{bmatrix}-2+0 & -5+9\\ 8+0 & 20+15\end{bmatrix}$$

$$=\begin{bmatrix}2 & 5\\ 4 & 19\end{bmatrix}+\begin{bmatrix}-2 & 4\\ 8 & 35\end{bmatrix}$$

$$=\begin{bmatrix}2-2 & 5+4\\ 4+8 & 19+35\end{bmatrix}$$

$$=\begin{bmatrix}0 & 9\\ 12 & 54\end{bmatrix}$$

Hence $(A+B)C=AC+BC$

6.6.4. Commutative law under multiplication

The commutative law of multiplication for the matrices A and B does not hold; $AB\neq BA$ but $AI=IA$ where I is identity matrix.

Examples

Let $A=\begin{bmatrix}5 & 2\\ 0 & 1\end{bmatrix}, B=\begin{bmatrix}0 & 3\\ 2 & 1\end{bmatrix}$

Solution:

To verify $AB\neq BA$

L.H.S $=AB$

$$=\begin{bmatrix}5 & 2\\ 0 & 1\end{bmatrix}\begin{bmatrix}0 & 3\\ 2 & 1\end{bmatrix}$$

$$=\begin{bmatrix}5\times0+2\times2 & 5\times3+2\times1\\ 0\times0+1\times2 & 0\times3+1\times1\end{bmatrix}$$

$$=\begin{bmatrix}0+4 & 15+2\\ 0+2 & 0+1\end{bmatrix}$$

$$=\begin{bmatrix}4 & 17\\ 2 & 1\end{bmatrix}\quad\text{.....(i)}$$

Now

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R.H.S $= BA$

$$= \begin{bmatrix} 0 & 3 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 5 & 2 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0\times5+3\times0 & 0\times2+3\times1 \\ 2\times5+1\times0 & 2\times2+1\times1 \end{bmatrix}$$

$$= \begin{bmatrix} 0+0 & 0+3 \\ 10+0 & 4+1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 3 \\ 10 & 5 \end{bmatrix} \ldots\ldots\ldots\ldots\ldots\text{(ii)}$$

From (i) and (ii)

$L.H.S \neq R.H.S$

$\Rightarrow AB \neq BA$

This means that the matrix multiplication is not commutative.

6.4.5. Multiplicative Identity:

In matrix multiplication the unit matrix (identity matrix) is called the multiplicative identity. If A is any matrix, then

$A.I = I.A = A$

Example:

If $A = \begin{bmatrix} 8 & 5 \\ 7 & 3 \end{bmatrix}$ *and* $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then

A.I $= \begin{bmatrix} 8 & 5 \\ 7 & 3 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$= \begin{bmatrix} 8\times1+5\times0 & 8\times0+5\times1 \\ 7\times1+3\times0 & 7\times0+3\times1 \end{bmatrix}$$

$$= \begin{bmatrix} 8+0 & 0+5 \\ 7+0 & 0+3 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 5 \\ 7 & 3 \end{bmatrix} = A \ldots\ldots\ldots\ldots\text{(i)}$$

I.A $= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 8 & 5 \\ 7 & 3 \end{bmatrix}$

$$= \begin{bmatrix} 1\times8+0\times7 & 1\times5+0\times3 \\ 0\times8+1\times7 & 0\times5+1\times3 \end{bmatrix}$$

$$= \begin{bmatrix} 8+0 & 5+0 \\ 0+7 & 0+3 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 5 \\ 7 & 3 \end{bmatrix} = A \ldots\ldots\ldots\ldots\text{(ii)}$$

So from (i) and (ii)

$A.I = I.A = A$

Example:

If $A = \begin{bmatrix} 2 & 1 \\ 4 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 4 \\ 4 & 0 \end{bmatrix}$, then

Show that $(AB)^t = B^t A^t$

Solution:

Given that

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 4 \\ 4 & 0 \end{bmatrix}$$

The product of A and B is:

$$AB = \begin{bmatrix} 2 & 1 \\ 4 & 0 \end{bmatrix}\begin{bmatrix} 0 & 4 \\ 4 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2\times0+1\times4 & 2\times4+1\times0 \\ 4\times0+0\times4 & 4\times4+0\times0 \end{bmatrix} = \begin{bmatrix} 0+4 & 8+0 \\ 0+0 & 16+0 \end{bmatrix}$$

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$$= \begin{bmatrix} 4 & 8 \\ 0 & 16 \end{bmatrix}$$

$$(AB)' = \begin{bmatrix} 4 & 0 \\ 8 & 16 \end{bmatrix}$$

Now

$$A' = \begin{bmatrix} 2 & 4 \\ 1 & 0 \end{bmatrix}$$

$$B' = \begin{bmatrix} 0 & 4 \\ 4 & 0 \end{bmatrix}$$

$$B'A' = \begin{bmatrix} 0 & 4 \\ 4 & 0 \end{bmatrix}\begin{bmatrix} 2 & 4 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0\times2+4\times1 & 0\times4+0\times0 \\ 4\times2+0\times1 & 4\times4+0\times0 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 8 & 16 \end{bmatrix}$$

Hence, prove that $(AB)' = B'A'$

Exercise 6.4

1. Multiply the following matrices:

(i) $\begin{bmatrix} 2 & 3 \end{bmatrix}\begin{bmatrix} 4 \\ 1 \end{bmatrix}$ (ii) $\begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix}$

(iii) $\begin{bmatrix} 4 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 5 & 3 \\ 0 & 1 \end{bmatrix}$ (iv) $\begin{bmatrix} 3 & 2 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix}$

(v) $\begin{bmatrix} 3 & 3 \\ 2 & 2 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (vi) $\begin{bmatrix} 1 & -2 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

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2. For the matrices $A = \begin{bmatrix} 0 & 1 \end{bmatrix}$ [illegible]

show that

(i) $(A+B)' = A' + B'$

(ii) $(AB)' = B'A'$

3. If $A = \begin{bmatrix} 3 & 0 \\ 2 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -5 \\ 3 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 1 & -2 \\ -1 & -1 \end{bmatrix}$

are the matrices, then show that:

(i) $(A+B)C = AC + BC$

(ii) $A(B+C) = AB + AC$

4. If $A = \begin{bmatrix} -5 & 1 \\ 0 & -4 \end{bmatrix}$, $B = \begin{bmatrix} -2 & -3 \\ -1 & 5 \end{bmatrix}$, $C = \begin{bmatrix} 1 & -2 \\ -1 & -1 \end{bmatrix}$

are the matrices, then find: $2A+3B-4C$

5. If $A = \begin{bmatrix} 2 & 0 \\ -1 & 4 \end{bmatrix}$ is a matrix, then

Show that

(i) $(A')' = A$ (ii) $AA' \neq A'A$

6.5 Multiplicative inverse of matrix.

If A and B are two non-zero square matrices. Such that their product is equal to the multiplicative identity matrix I i.e. AB = BA = I, then A and B are called the multiplicative inverses of each other

i.e. $A = B^{-1}$ and $B = A^{-1}$

Then B is called inverse of A or A is called inverse of B if AB=BA=I. for the matrix A, its inverse is denoted by A^{-1}.

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6.5.1 Determinant of a matrix.

Let $A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is any square matrix, then

determinant of A is denoted by

$$|A|=\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

6.5.2 Evaluation of the determinant of a matrix

If a square matrix is

$$A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Then the determinant of A is.

$$|A|=\begin{vmatrix} a & b \\ c & d \end{vmatrix}=ad-bc$$

The resultant value is ad-bc a real number.

Example:

Find $|A|$ where

$$A=\begin{bmatrix} 3 & -1 \\ 2 & -1 \end{bmatrix}$$

Solution:

$$|A|=\begin{vmatrix} 3 & -1 \\ 2 & -1 \end{vmatrix}$$

$=(3)(-1)-(-1)(2)$ (using definition)

$=-3+2=-1$

6.5.3 Singular matrix

If the determinant of a square matrix A is equal to zero, then the matrix A is called singular matrix.

Example: Show that $A=\begin{bmatrix} 2 & 6 \\ 1 & 3 \end{bmatrix}$ is singular.

Solution:

The determinant of A is

$$|A|=\begin{vmatrix} 2 & 6 \\ 1 & 3 \end{vmatrix}$$

$=(2)(3)-(6)(1)$

$=6-6$

$|A|=0$

Thus, the matrix A is singular matrix.

6.5.4 Non-singular matrix:

If the determinant of a square matrix A is not equal to zero, then A is called a non-singular matrix.

Example-1: Show that $A=\begin{bmatrix} 2 & 5 \\ 0 & 1 \end{bmatrix}$ is non-singular matrix

Solution:

$$|A|=\begin{vmatrix} 2 & 5 \\ 0 & 1 \end{vmatrix}$$

$=2\times1-5\times0$

$=2-0$

$=2$

$|A|\neq0$

Thus the matrix A is non-singular matrix.

6.5.5 Adjoint of A matrix:

If A is a square matrix, of order 2,

$$A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Then Adjoint of A obtained by interchanging the position of a and d, and multiplying c and b by negative sign.

It is denoted by: Adj $A=\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Examples:

Find the adjoint of the following matrices:

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(i) $A=\begin{bmatrix}4 & 5\\ 6 & 7\end{bmatrix}$

$\text{Adj } A=\begin{bmatrix}7 & -5\\ -6 & 4\end{bmatrix}$

(ii) $C=\begin{bmatrix}3 & 1\\ 5 & 8\end{bmatrix}$

$\text{Adj } C=\begin{bmatrix}8 & -1\\ -5 & 3\end{bmatrix}$

6.5.6. Inverse of a matrix

If A is a non-singular matrix of order 2, then the inverse of a matrix A is

$$A^{-1}=\frac{1}{|A|}adj\ A$$

Consider a non-singular matrix, $A=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$

Verification:

i.e. $AA^{-1}=I=A^{-1}A$

$$AA^{-1}=\begin{bmatrix}a & b\\ c & d\end{bmatrix}.\frac{1}{(ad-bc)}\begin{bmatrix}d & -b\\ -c & a\end{bmatrix}$$

$$=\frac{1}{(ad-bc)}\begin{bmatrix}a & b\\ c & d\end{bmatrix}\begin{bmatrix}d & -b\\ -c & a\end{bmatrix}$$

$$=\frac{1}{(ad-bc)}\begin{bmatrix}ad-bc & -ab+ab\\ cd-cd & -bc+ad\end{bmatrix}$$

$$=\frac{1}{(ad-bc)}\begin{bmatrix}ad-bc & 0\\ 0 & ad-bc\end{bmatrix}$$

$$AA^{-1}=\frac{1}{(ad-bc)}\begin{bmatrix}ad-bc & 0\\ 0 & ad-bc\end{bmatrix}$$

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$$=\begin{bmatrix}\frac{ad-bc}{ad-bc} & \frac{0}{ad-bc}\\ \frac{0}{ad-bc} & \frac{ad-bc}{ad-bc}\end{bmatrix}$$

$$=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}=I$$

Similarly it can be shown that $A^{-1}A=I$

So $AA^{-1}=I=A^{-1}A$

Example 1:

Verify that B is the inverse of A.

$$A=\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}, B=\begin{bmatrix}\frac{1}{2} & 0\\ 0 & \frac{1}{2}\end{bmatrix}$$

Solution:

If the product of AB = I, then B is the inverse of A:

$$AB=\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}\begin{bmatrix}\frac{1}{2} & 0\\ 0 & \frac{1}{2}\end{bmatrix}=\begin{bmatrix}1+0 & 0+0\\ 0+0 & 0+1\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$AB=I$

Similarly BA=I

So AB=BA=I

Therefore B is the inverse of A.

Example 2:

Use the adjoint method to find out the inverse.

$$A=\begin{bmatrix}2 & 0\\ 6 & 3\end{bmatrix}$$

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Solution:

The determinant of A is:

$$|A| = \begin{bmatrix} 2 & 0 \\ 6 & 3 \end{bmatrix}$$

$$= 6 - 0$$

$$|A| = 6$$

$$A \neq 0$$

So inverse exists.

The Adj A is:

$$\text{Adj}\,(A) = \begin{bmatrix} 3 & 0 \\ -6 & 2 \end{bmatrix}$$

The inverse of A is:

$$A^{-1} = \frac{1}{|A|}\begin{bmatrix} 3 & 0 \\ -6 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{6}\begin{bmatrix} 3 & 0 \\ -6 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3}{6} & \frac{0}{6} \\ -\frac{6}{6} & \frac{2}{6} \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ -1 & \frac{1}{3} \end{bmatrix}.$$

Exercise 6.5

1. Find singular and on singular matrices in the following.

(i) $\begin{bmatrix} 2 & 6 \\ 1 & 3 \end{bmatrix}$ (ii) $\begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix}$

(iii) $\begin{bmatrix} \frac{5}{2} & 6 \\ \frac{5}{6} & 2 \end{bmatrix}$ (iv) $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$

(v) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (vi) $\begin{bmatrix} 7 & 0 \\ 40 & 1 \end{bmatrix}$

(vii) $\begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix}$ (viii) $\begin{bmatrix} 3 & 4 \\ 2 & -1 \end{bmatrix}$

(ix) $\begin{bmatrix} -2 & 1 \\ -5 & -3 \end{bmatrix}$ (x) $\begin{bmatrix} -1 & 4 \\ -3 & 12 \end{bmatrix}$

2. Find the adjoint of the following matrices.

(i) $\begin{bmatrix} -3 & 2 \\ -1 & 0 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & -4 \\ 0 & 2 \end{bmatrix}$

(iii) $\begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}$ (iv) $\begin{bmatrix} 1 & -1 \\ -5 & 3 \end{bmatrix}$

(v) $\begin{bmatrix} 3 & 0 \\ -1 & 0 \end{bmatrix}$ (vi) $\begin{bmatrix} 1 & 0 \\ -1 & 3 \end{bmatrix}$

3. Find the inverse of a matrix:

$$A = \begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix}$$

4. Verify $(AB)^{-1} = B^{-1}A^{-1}$ for the matrices:

$$A = \begin{bmatrix} -1 & 1 \\ 2 & 3 \end{bmatrix}, \; B = \begin{bmatrix} 4 & -3 \\ 2 & 1 \end{bmatrix}$$

5. Find inverse matrix of the following matrices:

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(i) $\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$ (ii) $\begin{bmatrix} 6 & 3 \\ 1 & \frac{1}{2} \end{bmatrix}$

(iii) $\begin{bmatrix} 4 & 1 \\ 1 & 0 \end{bmatrix}$ (iv) $\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$

(v) $\begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}$ (vi) $\begin{bmatrix} 5 & 0 \\ 1 & 2 \end{bmatrix}$

6.6. Solution of simultaneous linear equations:

Two linear equations in variables x and y are given below.

$ax + by = m$ --------- (i)

$cx + dy = n$ -------- (ii) where $a, b, c, d, m, n \in R$

This is also called simultaneous linear equations.

6.6.1. Solution of simultaneous linear equations by using Matrices:-

Consider the following system of linear equations in two variables (unknown)

Example

$$\left.\begin{aligned} a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2 \end{aligned}\right] \quad (1)$$

Where a_1, b_1, c_1, a_2, b_2 *and* c_2 are constant real numbers.

System (1) in matrix form can be written as

$$\begin{bmatrix} a_1x + b_1y \\ a_2x + b_2y \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

or $$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \quad (2)$$

or $AX = B$ (3)

where $A = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} X = \begin{bmatrix} x \\ y \end{bmatrix}$ *and* $B = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$

If A is non-singular, then A^{-1} exists. Multiply both sides of (3) by A^{-1}.

So $A^{-1}(AX) = A^{-1}B$

Or $(A^{-1}A)X = A^{-1}B$ (4)

(using associative property of matrix multiplications)

Since $A^{-1}A = I$, *So* (4) becomes

$IX = A^{-1}B$

but $IX = X$

$X = A^{-1}B$ (5)

As A^{-1} is a matrix of order 2-by-2.

So $A^{-1}B$ is also a matrix of order 2-by-1.

Hence for each x and y a unique value exists in $A^{-1}B$.

Example:

Solve the following system of linear equations by matrix method.

$$\left.\begin{aligned} 2x + y &= 1 \\ x - 2y &= 2 \end{aligned}\right\} \quad (1)$$

The matrix form of the system (1) is

$$\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad (2)$$

or $AX = B$ (3)

where $A = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}, B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

To solve (3) we first find A^{-1}.

but $A^{-1} = \frac{1}{|A|}\ adj\ A$

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So $|A| = \begin{vmatrix} 2 & 1 \\ 1 & -2 \end{vmatrix}$

$= (2)(-2) - (1)(1)$

$= -4 - 1$

$= -5$

adj A $= \begin{bmatrix} -2 & -1 \\ -1 & 2 \end{bmatrix}$

$\therefore A^{-1} = \frac{1}{-5}\begin{bmatrix} -2 & -1 \\ -1 & 2 \end{bmatrix}$.

Multiply both sides of (3) by A^{-1}.

So $A^{-1}(AX) = A^{-1}B$

or $(A^{-1}A)X = A^{-1}B$

or $IX = A^{-1}B$

or $X = A^{-1}B$

$= \frac{1}{-5}\begin{bmatrix} -2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$= \frac{-1}{5}\begin{bmatrix} -2 & -2 \\ -1 & +4 \end{bmatrix}$

$= \frac{1}{-5}\begin{bmatrix} -4 \\ 3 \end{bmatrix}$

$= \begin{bmatrix} 4/5 \\ -3/5 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4/5 \\ -3/5 \end{bmatrix}$

$\therefore \quad x = \frac{4}{5}$ *and* $y = \frac{-3}{5}$ Solution set $= \left\{\left(\frac{4}{5}, -\frac{3}{5}\right)\right\}$

> Note: The system has no solution if $|A| = 0$ i.e. if A is singular.

6.6.2. Cramer's Rule:

Cramer was a famous mathematition who introduced a method for finding the solution of a system of linear equations. This method is named as Cramer's Rule, which is explained as below:-

Consider the system of linear equations:

$$\left.\begin{aligned} a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2 \end{aligned}\right\} \qquad (1)$$

In matrix from (1) is written as

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

or $AX = B$

where $A = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$ *and* $B = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$

we define two matrices A_1 and A_2 as

$A_1 = \begin{bmatrix} c_1 & b_1 \\ c_2 & b_2 \end{bmatrix}$ *and* $A_2 = \begin{bmatrix} a_1 & c_1 \\ a_2 & c_2 \end{bmatrix}$

Both A_1 and A_2 are derived from A.

After numerical solution $|A|, |A_1|$ *and* $|A_2|$ is obtained.

So $x = \frac{|A_1|}{|A|}$ *and* $y = \frac{|A_2|}{|A|}$

Where $|A| \neq 0$ i.e. A must not be singular.

Example:

Solve the following system of equations by Cramer's Rule.

$x - 2y = 1$

$2x + y = 2$

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So The matrix form of the given system is

$$\begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

Or $AX = B$

Where $A = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}, B = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$

Let $A_1 = \begin{bmatrix} 0 & -2 \\ 2 & 1 \end{bmatrix}$ *and* $A_2 = \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix}$

Now we evaluate $|A|, |A_1|, |A_2|$.

So $|A| = \begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} = 1+4 = 5$

$$|A_1| = \begin{vmatrix} 0 & -2 \\ 2 & 1 \end{vmatrix} = 0+4 = 4$$

$$|A_2| = \begin{vmatrix} 1 & 0 \\ 2 & 2 \end{vmatrix} = 2-0 = 2$$

$$\therefore \quad x = \frac{|A_1|}{|A|} = \frac{4}{5} \quad \textit{and} \quad y = \frac{|A_2|}{|A|} = \frac{2}{5}$$

Exercise 6.6

1. Write each of the following system of equations in matrix form.

(i) $x + 2y = 0$

$2x - 3y = 1$

(ii) $\frac{1}{2}x - 3y = 4$

$x + \frac{3}{2}y = \frac{1}{2}$

(iii) $\sqrt{2}x + y = \sqrt{3}$

$x - \sqrt{3}y = 1$

(iv) $\frac{4}{5}x - y = 6$

$\frac{1}{3}x + 2y = 0$

2. Solve each of the following system of linear equations by matrix method.

(i) $x - y = 2$

$x + 2y = 1$

(ii) $2x + 3y = 0$

$x - 4y = 1$

(iii) $4x + 5y = \frac{1}{2}$

$x - \frac{1}{2}y = 3$

(iv) $x - y = 2$

$x + y = 4$

3. **Solve Q2 by Cramer's Rule.**

4. Cost of 4 dozens eggs and 2 dozen oranges is Rs.200 and cost of 3 dozens eggs and 2 dozen oranges is Rs.180. Find cost per dozen of each item?

5. The sum of the ages of a sister and her brother is 26 years. Three years ago sister's age was 4 times the age of her brother. Find their present ages?

6. The sum of the digits of a (two-digited) number is 13. If the digits are inter -changed, the new number so formed is increased by 27. Find the number?

EXERCISE 6

(Objective Type Questions)

1. True False questions:

Encircle (T) of the statement is true and (F) if the statement is False.

(i) For addition of two matrices their order must not be same. T/F

(ii) A square matrix "B" is called singular if $|B| = 0$ T/F

(iii) If the order of matrix A is m-by-n then the order of A^{-1} is also m-by-n. T/F

(iv) In scalar multiplication order of a matrix changes. T/F

2. Fill in items:

(i) If A is non-singular, then A^{-1} _____.

(ii) A square matrix "C" is symmetric if C^t = _____

(iii) Determinant of a square matrix is a _____ number.

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(iv) The product matrix IX=_____

3. Multiple choice questions (MCQs)

Choose the correct option in each.

(i) For the product matrix XY, number of columns in X= no of _____

(A) Columns in Y (B) Rows in Y (C) Rows in X (D) some of rows and columns

(ii) If at least one element in the principle diagonal of a square matrix is non-zero and the remaining elements are zero, it is called

(A) Diagonal matrix (B) Rectangular matrix
(C) Scalar matrix (D) None of these

(iii) In scalar matrix each diagonal element is a constant number equal to

(A) 1 (B) 0 (C) Any real number other than 1.

(iv) Multiplicative inverse of matrix A is

(A) A^{-1} (B) $\frac{1}{A}$ (C) I (D) $|A|^{-1}$

Glossary

Additive identity:	Zero or Null matrix
Additive inverse:	Matrix of the same order with opposite sign
Columns:	Vertical arrangement of elements in a matrix.
Determinant:	The scalar number
Identity matrix:	A square matrix in which each diagonal element is 1.
Inverse matrix:	For a square matrix A, inverse is A^{-1} as $A.A^{-1} = I$
Linear equation:	An equation in which the power of each variable is 1.
Order of a matrix:	Number of rows by number of columns of a matrix.
Row:	The horizontal arrangement of the elements in a matrix.
Transpose:	The process of interchanging rows into columns or columns to rows.

Unit 7

FUNDAMENTALS OF GEOMETRY

In this unit, the students will be able to learn about the:

i. Define adjacent complementary, supplementary and vertically opposite angles.
ii. Calculate unknown angles involving adjacent angles, complementary angles, supplementary angles and vertically opposite angles.
iii. Calculate unknown angle of a triangle.
iv. Define parallel lines and demonstrate through figures the following properties of parallel lines:
   a. Two lines which are parallel to the same given line are parallel to each other.
   b. If three parallel lines are intersected by two transversals in such a way that the two intercepts on one transversal are equal to each other, the two intercepts on, the second transversal are also equal.
   c. A line through the midpoint of a a side of a triangle parallel to another side bisects the third side (an application of above property)
v. Draw a transversal to intersect two parallel lines and demonstrate corresponding angles, alternate-interior angles, vertically-opposite angles and interior angles on the same side of transversal.
vi. Describe and demonstrate through figures the following transversal intersects two parallel lines:
   a. Pairs of corresponding angles are equal
   b. Pairs of alternate interior angles are equal
   c. Pairs of interior angles on the same side of transversal is supplementary and demonstrate them through figures.
vii. Identify congruent and similar figures and recognize the symbol of congruency.
viii. Apply the properties for two figures to be congruent or similar.
ix. Apply the properties of SSS ≅ SSS, SAS ≅ SAS, ASA ≅ ASA and RHS ≅ RHS for congruency between two triangles.

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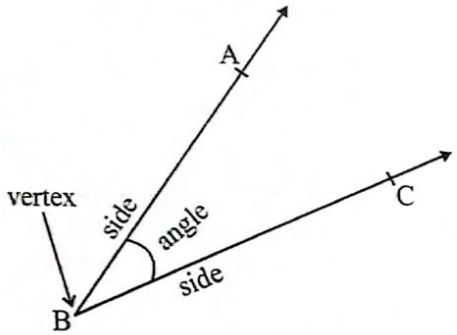
x. Demonstrate the properties of a square, a rectangle and a parallelogram.
xi. Describe a circle, its centre, radius, diameter, chord, arc, major and minor areas, semi circle and segment of the circle.
xii. Describe the terms: sector and secant of on circle, concyclic points, tangent to a circle and concentric circles.
xiii. Demonstrate and apply the following properties in different geometrical figures:
   a. The angle in a semi circle is a right angle.
   b. The angles in the same segment of a circle are equal.
   c. The central angle of a minor arc of a circle is double that of the angle subtended by the corresponding major arc.

Introduction

We have studied in previous classes that two rays having a common end point form an angle. The point where the rays intersect is called the 'vertex' of the angle. The two rays are called the sides or arms of the angle.



We can specify an angle by using a point on each ray and the vertex.

For example two rays $\overrightarrow{AB}$ and $\overrightarrow{BC}$ have a common end point B, therefore they form an angles ABC or CBA. We can write these angles as $\angle ABC$ or $\angle CBA$.

(Note that vertex point is always given in the middle.

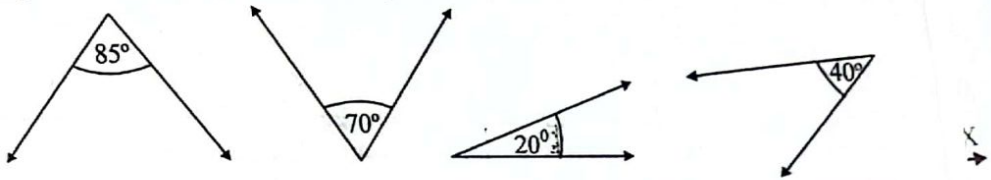
According to their measurements, angles can be classified as:

i. Acute angle

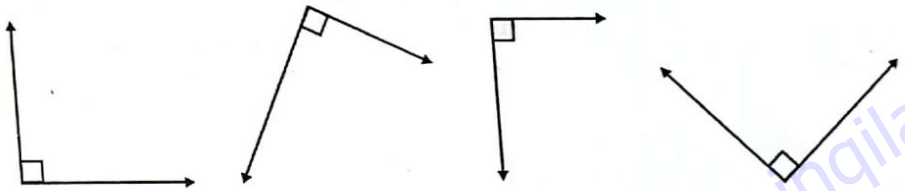
ii. Right angle, and

iii. Obtuse angle

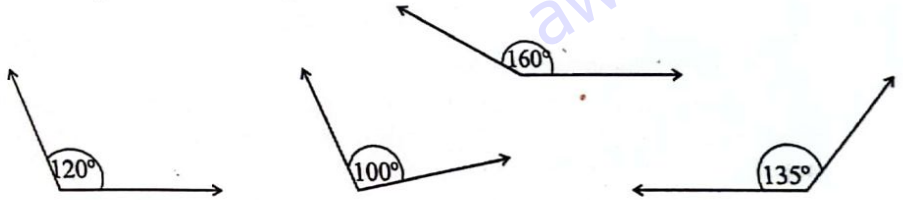
An acute angle is an angle measuring less than 90°. The following angles are the examples of acute angle.



A right angle is an angle measuring 90°. The following angles are the examples of right angle.



An obtuse angle is an angle measuring greater than 90°. Following are the examples of obtuse angles.

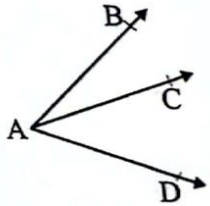


7.1.1 Adjacent angles

Two angles are said to be adjacent angles, if:

i) they have a common vertex and a common arm; and

ii) the intersection of their interiors is an empty set.



In the figure $\angle BAC$ and $\angle CAD$ are adjacent angle because A is their common vertex, $\overrightarrow{AC}$ is their common arm and intersection of their interiors is a null set.

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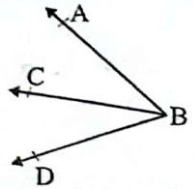
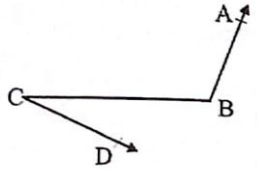
Note that:

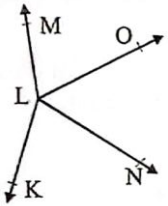


i) ∠ABC and ∠ABDare not adjacent angles because intersection of their interiors is ∠ABC which is not an empty set;

ii) ∠ABC and ∠BCD are not adjacent angles because they do not have a common vertex; similarly



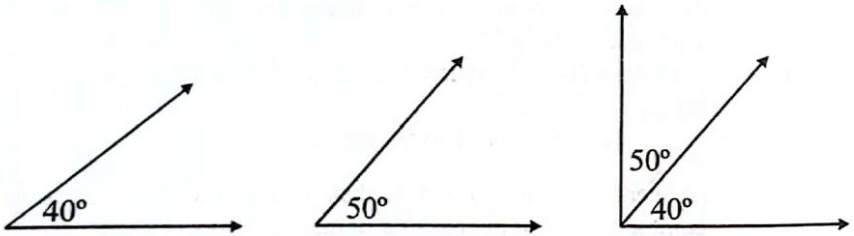
iii) ∠KLMand ∠OLN are not adjacent angles because they do not have a common arm.



7.1.2 Complementary angles

Two angles are said to be complementary angles if the sum of their measurements equals 90°. Each of the complementary angles is called complement of the other.

In the figure two angles of measures 40° and 50° are given. These two angles can be pasted together to form a right angle. The angles are complementary angles.



Each one of the two angles is called complement of the other.

7.1.3 Supplementary angles

Two angles are said to be supplementary angles if the sum of their measurements equals 180°. Each of the supplementary angles is called supplement of the other.

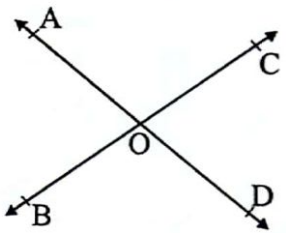
In the figure two angles of measure 130° and 50° are given. These angles can be pasted together to form a straight line. The angles are supplementary angles.

Each one of the two angles is called supplement of the other.

7.1.4 Vertically opposite angles

If two lines intersect each other, then the non-adjacent angles so formed are called vertically opposite angles.

These angles have the same vertex and be on the opposite side of each other. Arms of these angles consist of opposite rays.



In the adjoining figure, $\overleftrightarrow{AD}$ and $\overleftrightarrow{BC}$ intersect each other at O. (∠AOB, ∠COD) and (∠AOC, ∠BOD) are two pairs of vertically opposite angles.

Vertically opposite angles are equal in measure i.e.

m∠AOB= m∠CODand m∠AOC= m∠BOD

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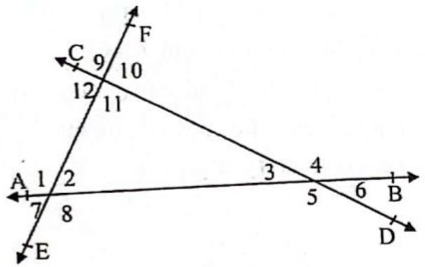
7.1.5 Calculating unknown angles

Consider the following example

Example 1:

In the adjoining figure $m\angle 1 = 120°$, $m\angle 6 = 30°$ and $m\angle 9 = 90°$. Find the measures of unknown angles.



Solution:

$\because m\angle 1 = 120°$

$\because m\angle 8 = 120°$ (vertically opposite angles)

Also $m\angle 1 + m\angle 2 = 180°$ (supplementary angles)

$\because m\angle 2 = 180° - m\angle 1$

$= 180° - 120°$

$= 60°$

Hence $m\angle 7 = 60°$ (vertically opposite angle)

Again $m\angle 6 = 30°$

$\because m\angle 3 = 30°$ (vertically opposite angles)

Also $m\angle 5 + m\angle 6 = 180°$ (supplementary angles)

$\because m\angle 5 = 180° - m\angle 6$

$= 180° - 30°$

$= 150°$

Hence $m\angle 4 = 150°$ (vertically opposite angle)

Because $m\angle 9 = 90°$

$\because m\angle 11 = 90°$ (vertically opposite angle)

Also $m\angle 9 + m\angle 10 = 180°$ (supplementary angles)

$\because m\angle 10 = 180° - m\angle 9$

$= 180° - 90°$

$= 90°$

Hence $m\angle 12 = 90°$ (vertically opposite angle)

Example 2: Find the complement of the following angles:

i) $78°$ ii) $60°$

Solution: i) Let the measure of the complement by $x°$

$x° + 78° = 90°$

$x° = 90° - 78°$

$= 12°$

Therefore complement of $78°$ is $12°$.

ii) Let the measure of the complement by $x°$

$x° + 63° = 90°$

$x° = 90° - 63°$

$= 27°$

Therefore complement of $63°$ is $27°$.

Example 3: Find the supplement of the following angles:

i) $167°$ ii) $57°$

Solution: i) Let the measure of the supplement be $x°$

$x° + 167° = 180°$

$x° = 180° - 167°$

$= 13°$

$\therefore$ supplement of $167°$ is $13°$.

ii) Let the measure of the supplement be $x°$

$\therefore x° + 57° = 180°$

$x° = 180° - 57°$

$= 123°$

Hence supplement of $57°$ is $123°$.

Example 4: Find the measures of the following angles:

a) $(x + 12), (3x - 2)$ when they are complementary angles.

b) $(2x + 5), (3x - 10)$ when they are supplementary angles.

a) As the angle are complementary

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$\therefore x + 12 + 3x - 2 = 90°$

$4x + 10 = 90°$

or $4x = 90° - 10°$

or $= 80°$

or $x = \frac{80°}{4}$

$= 20°.$

Hence 1st angle is $x + 12 = 20 + 12 = 32°$
and 2nd angle is $3x - 2 = 60 - 2 = 58°$
So the required angles are 32° and 58°.

b) As the angles are supplementary

$\therefore 2x + 5 + 3x - 10 = 180°$

or $5x - 5 = 180°$

or $5x = 180° + 5°$

$= 185°$

or $x = \frac{185°}{5}$

$= 37°.$

Hence 1st angle $= 2x + 5 = 2 \times 37 + 5$

$= 74 + 5 = 79°$

And 2nd angle $= 3x - 10 = 3 \times 37 - 10$

$= 111 - 10$

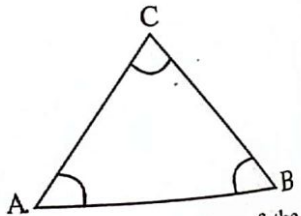
$= 101°$

So the required angles are 79° and 101°.

7.1.5 Interior and Exterior angles of a triangle

Angles related to a triangle can be classified as Interior and Exterior angles.

Interior angles are inside a triangle and are formed at the point where two sides meet. The sum of the measures of three interior angles of a triangle is always equal to 180°.



In the figure $\angle ABC, \angle BCA$ and $\angle CAB$ are the interior angles of the triangle ABC and

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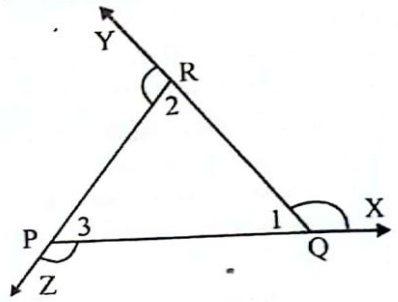
$m\angle A + m\angle B + m\angle C = 180°.$

An exterior angle of a triangle is formed when one of its side is extended beyond one of its vertices. Sum of the exterior angles of a triangle is equal to 360°. Therefore, measure of an exterior angle of a triangle is equal to the sum of the measures of its non-adjacent interior angles. In the figure, $\angle QPZ$, $\angle RQX$ and $\angle YRP$ are the exterior angles of ΔABC.



Here $m\angle QPZ + m\angle RQX + m\angle YRP = 360°.$

and $m\angle QPZ = m\angle 1 + m\angle 2;$

$m\angle RQX = m\angle 2 + m\angle 3;$ and

$m\angle YRP = m\angle 3 + m\angle 1.$

Also, every exterior angle of a triangle is the supplement of its adjacent interior angle.

Now consider the following examples:

Example-5:

In an isosceles triangle, the measure of the vertex (angle included between congruent sides) angle is 80°. Find the measures of base angles.

Solution:

Sum of interior angles of triangle = 180°

Measure of the vertex angle = 80°

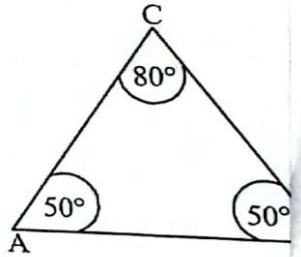
∴ Sum of the remaining two angles = 180°-80°

As the triangle is isosceles

Therefore measure of the base angles will be the same.

Sum of the base angles = 100°

∴ Measure of each base angle $= \frac{100}{2} = 50°$



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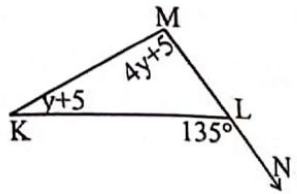
Example-6:
Measure of an exterior angle of a triangle is 135°. If the measure sof its non-adjacent interior angles are $(y+5)$ and $(4y+5)$ degrees, find the measures of all interior angle of the triangle.

Solution:



As the measure of exterior angle is 135°

$\therefore$ measure of its adjacent interior angle = $180° - 135°$
$= 45°$

Measure of non-adjacent interior angles = $(y+5),(4y+5)$

$\therefore \quad y+5+4y+5 = 135°$

or $\quad 5y + 10 = 135°$

or $\quad 5y = 135° - 10$
$= 125°$

Or $\quad y = \frac{125}{5} = 25°$

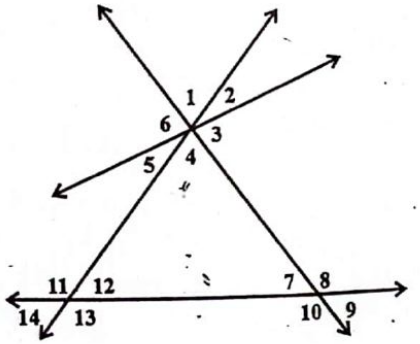
Hence one angle $= 25+5=30°$

2^{nd} angle $= 4\times25+5=105°$

And all interior angles are 45°, 30°, 105°.

Exercise7.1

1. From the adjoining figure:
   (i) Write any two pairs of vertically opposite angles.
   (ii) Write any four pairs of adjacent angles.
   (iii) Write any one pair of complementary angles.
   (iv) Write any four pairs of supplementary angles.
   (v) Write any two pairs of adjacent angles that are neither complementary nor supplementary.



2. Find the supplement of the following angles:-
   (i) 28° (ii) 86° (iii) 110° (iv) 154°
3. Find the complement of the following angles:-
   (i) 14° (ii) 33° (iii) 56° (iv) 78°
4. Find the measures of the following angles, if they are supplement of each other:-
   (i) $5x+8, 7x+4$ (ii) $3x+9, 5x-5$
   (iii) $4x+1, x+9$ (iv) $x+3, 8x+15$
5. Find the measures of the following angles, if they are complement of each other:-
   i) $2x+5, 3x-10$ (ii) $4x+5, 2x-11$
   (iii) $x+9, 2x-15$ (iv) $8x+15, x+3$
6. Find the measure of the third angle of a triangle if the measures of its two angles are 37° and 56°.
7. A triangle is a right-angled isosceles triangle. Find the measure of each of its base angle.
8. Angles of a triangle are in the ratio of 2:3:4. Find their measures.
   (Hints: $2x+3x+4x=180°$
9. In a right angled triangle, the acute angles are in the ratio of 5:13. Fin the measure of these angles.
10. Measures of exterior angle of a triangle is 85°. measures of its non-adjacent interior angles are $(7x+2)$ and $(9x+3)$ degrees. Find the measures of the interior angles of the triangle.

7.2.1 Co-planer lines

A plane is a flat surface having only two dimensions. Lines lying in the same plane are called co-planer lines.

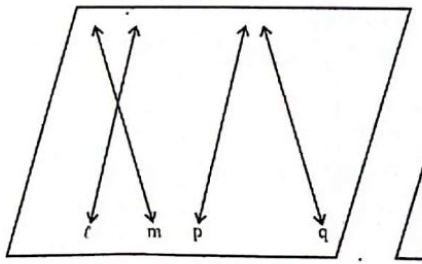


Fig (i)

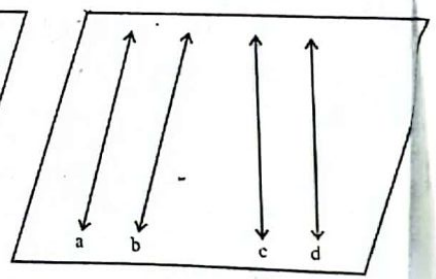


Fig (i)

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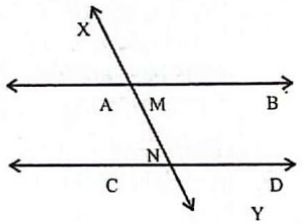
Two co-planer lines either intersect each other, or do not intersect. In figure (i) lines l and m intersect at A. point A is common in both the lines. A is called the intersecting point of l and m. in the same figure p and q are two lines. Apparently they don't have any common point. But as we know, a line can be extended indefinitely, it is evident that the lines p and q, if extended, will meet at a point i.e. both the lines will have a point of intersection. Therefore we can easily say that p and q are intersecting lines. In figure (ii), lines a and b, even if extended in any direction, do not intersect each other. Same is the case of lines c and d. they remain at the same distance from each other and do not intersect each other.

7.2.2 Parallel lines

Co-planer lines that remain at the same distance from each other and never intersect, are called parallel lines. Symbol II is used to denote parallel lines. In figure (ii), a || b and c || d.

7.2.3 Transversal

A line that cuts two or more co-planer line is called transversal. In the figure

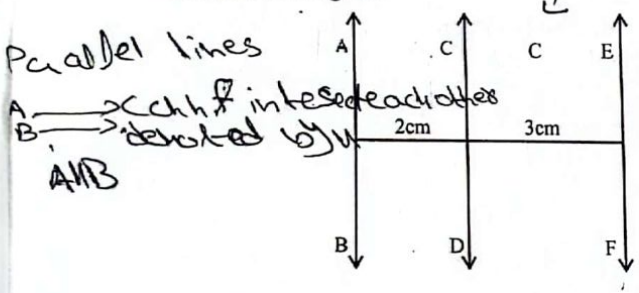


$\overline{AB}$ and $\overline{CD}$ are two coplaner lines. $\overline{XY}$ cuts them at M and N. $\overline{XY}$ is the transversal.

7.2.4 Properties of parallel lines

(i) If two lines are parallel to the same given line, then they are parallel to each other.

Consider the figure:



In this figure, $\overline{AB} \parallel \overline{CD}$

and the distance between these two lines is 2cm.

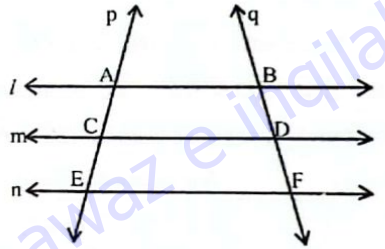
Similarly $\overline{EF} \parallel \overline{CD}$

and the distance between these two lines is 3cm.

Now if we measure the distance between $\overrightarrow{AB}$ And $\overrightarrow{EF}$, it comes to be 5cm i.e. $(2+3)cm$, and the distance remains the same at all points. Therefore we can say that $\overrightarrow{AB}$ And $\overrightarrow{EF}$, are parallel.

(ii) If three parallel lines are intersected by two transversals in such a way that the two intercepts on one transversal are equal to each other, the two intercepts on the second transversal are also equal.

Let us take any transversal p and take three points A, C and E on it, such that $m\overline{AC} = m\overline{CE}$. We draw three lines l, m and n through A, C and E respectively such that they are parallel to each other.



Then the intercepts made by the parallel lines l, m and n on the transversal p are $\overline{AC}$ and $\overline{CE}$, which are equal.

We now draw another transversal q intersecting the lines l, m and n at B, D and F respectively.

If we measure the lengths $\overline{BD}$ and $\overline{DF}$, we find that the intercepts made on transversal q are also equal i.e. $\overline{BD} = \overline{DF}$.

If we repeat this activity, we get the same result. Therefore we come to the conclusion that:

> If three parallel lines are intersected by two transversals in such a way that the two intercepts on one transversal are equal to each other, then the two intercepts on the second transversal are also equal.

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Two co-planer lines either intersect each other, or do not intersect. In figure (i) lines l and m intersect at A. point A is common in both the lines. A is called the intersecting point of l and m. in the same figure p and q are two lines. Apparently they don't have any common point. But as we know, a line can be extended indefinitely, it is evident that the lines p and q, if extended, will meet at a point i.e. both the lines will have a point of intersection. Therefore we can easily say that p and q are intersecting lines. In figure (ii), lines a and b, even if extended in any direction, do not intersect each other. Same is the case of lines c and d. they remain at the same distance from each other and do not intersect each other.

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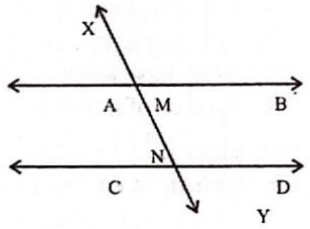

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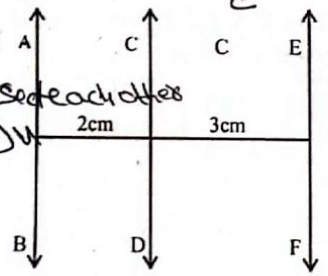

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Similarly $\overline{EF} \parallel \overline{CD}$

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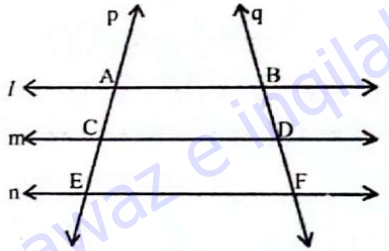
Now if we measure the distance between $\overrightarrow{AB}$ And $\overrightarrow{EF}$, it comes to be 5cm i.e. $(2+3)cm$, and the distance remains the same at all points. Therefore we can say that $\overrightarrow{AB}$ And $\overrightarrow{EF}$, are parallel.

(ii) If three parallel lines are intersected by two transversals in such a way that the two intercepts on one transversal are equal to each other, the two intercepts on the second transversal are also equal.

Let us take any transversal p and take three points A, C and E on it, such that $m\overline{AC} = m\overline{CE}$. We draw three lines l, m and n through A, C and E respectively such that they are parallel to each other.



Then the intercepts made by the parallel lines l, m and n on the transversal p are $\overline{AC}$ and $\overline{CE}$, which are equal.

We now draw another transversal q intersecting the lines l, m and n at B, D and F respectively.

If we measure the lengths $\overline{BD}$ and $\overline{DF}$, we find that the intercepts made on transversal q are also equal i.e. $\overline{BD} = \overline{DF}$.

If we repeat this activity, we get the same result. Therefore we come to the conclusion that:

> If three parallel lines are intersected by two transversals in such a way that the two intercepts on one transversal are equal to each other, then the two intercepts on the second transversal are also equal.

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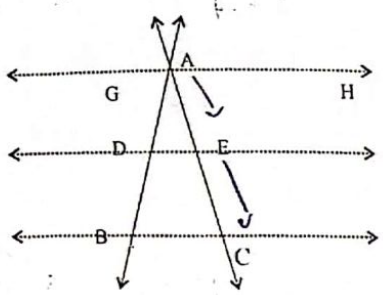
(iii) A line through the midpoint of a side of a triangle parallel to another side bisects the third side.

Let us take a triangle ABC, in which D is the midpoint of $\overline{AB}$ and $\overline{DE} \parallel \overline{BC}$.



We draw $\overline{BC}, \overline{DE}, \overline{AB}$ and $\overline{AC}$. We also draw through A, $\overline{GAH} \parallel \overline{BC}$. Now as $\overline{DE} \parallel \overline{BC}$ and $\overline{GH} \parallel \overline{BC}$ we can say that $\overline{GH} \parallel \overline{DE} \parallel \overline{BC}$. The figure shows that three parallel lines $\overline{GH}, \overline{DE}$ and $\overline{BC}$ are cut by a transversal $\overline{AB}$ in such a way that $\overline{AD} = \overline{DB}$. $\overline{AC}$ is another transversal which cuts the same parallel lines at A, E and C. Hence $\overline{AE} = \overline{EC}$ or E is the midpoints of $\overline{AC}$.

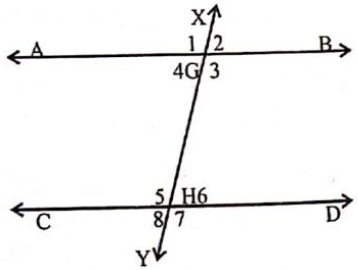
We conclude that:

> If a line, through the midpoint of one side of a triangle, is drawn parallel to the second side then it bisects the third side of the triangle.

7.2.5 Angles formed by transversal

When a transversal intersects two parallel lines, we see that, eight angles are formed.

The figure shows that a transversal $\overline{XY}$ intersects two parallel lines $\overline{AB}$ and $\overline{CD}$ at G and H.



Four angles $\angle 1, \angle 2, \angle 3$ and $\angle 4$ are formed when $\overline{XY}$ intersects $\overline{AB}$ at G and four angles $\angle 5, \angle 6, \angle 7$ and $\angle 8$ are formed when $\overline{XY}$ intersects $\overline{CD}$ at H.

We know that $(\angle 1, \angle 3), (\angle 2, \angle 4), (\angle 5, \angle 7)$ and $(\angle 6, \angle 8)$ are four pairs of vertically opposite angles.

$\angle 2, \angle 3, \angle 6$ *and* $\angle 7$ are on the right side of the transversal and $\angle 1, \angle 4, \angle 5$ *and* $\angle 8$ are on the left side of the transversal. Also $\angle 3, \angle 4, \angle 5$ *and* $\angle 6$ are the interior angles whereas $\angle 1, \angle 2, \angle 7$ *and* $\angle 8$ are the exterior angles.

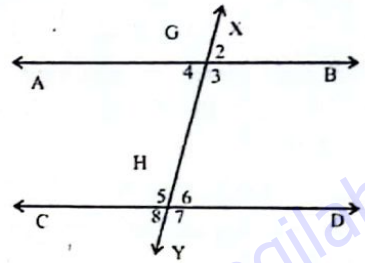
7.2.6 Corresponding angles

Consider angles $\angle 1$, *and* $\angle 5$.

(i) Both of the angles are on the same side of the transversal $\overline{XY}$.

(ii) $\angle 1$ is exterior whereas $\angle 5$ is an interior angle.

(iii) These are non-adjacent angles. $\angle 1$ *and* $\angle 5$ are called corresponding angles.



$(\angle 2, \angle 6), (\angle 3, \angle 7)$ *and* $(\angle 4, \angle 8)$ are also pairs of corresponding angles.

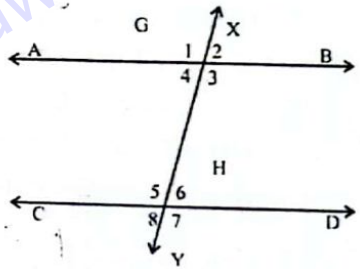
7.2.7 Alternate interior angles

Consider $\angle 3$, *and* $\angle 5$.

(i) Both of the angles are interior angles

(ii) These are non-adjacent angles.

(iii) Both of the angles are not on the same side of the transversal. $\angle 3$ *and* $\angle 5$ are called alternate interior angles.



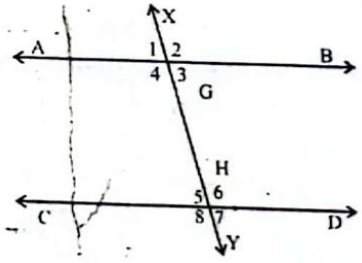
Similarly $\angle 4$ *and* $\angle 6$ is another pair of alternate interior angles.

7.2.8 Vertically opposite angles

$\overline{AB}$ and $\overline{XY}$ intersect at G. Consider $\angle 1$, *and* $\angle 3$.

(i) These are non-adjacent angles.

(ii) Both of the angles have the same vertex.



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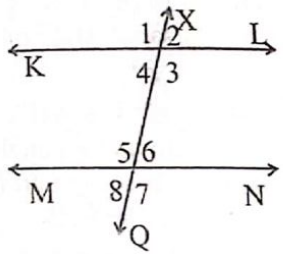
These are vertically opposite angles.

$(\angle 2, \angle 4), (\angle 5, \angle 7)$ *and* $(\angle 6, \angle 8)$ are also pairs of vertically opposite angles.

7.2.9 Interior angles

When a transversal intersects two parallel lines, four interior angles are formed.

$\angle 3, \angle 4, \angle 5$ *and* $\angle 6$ are interior angles. $\angle 3$ and $\angle 6$ are on one side of the transversal while $\angle 4$ and $\angle 5$ are on the other side of the transversal.



$(\angle 3, \angle 6)$ *and* $(\angle 4, \angle 5)$ are called interior angles on the same side of the transversal.

7.2.10 Relation between angles when two parallel lines are intersected by a transversal

Draw two parallel lines l and m.

Draw a transversal n to intersect l and m at P and Q. $m\angle 1 = 60^\circ$.

$\therefore$ $\angle 1$ and $\angle 4$ are vertically opposite angles.

$\therefore$ $m\angle 4 = 60^\circ$

Also $\angle 1$ and $\angle 2$ are supplementary angles

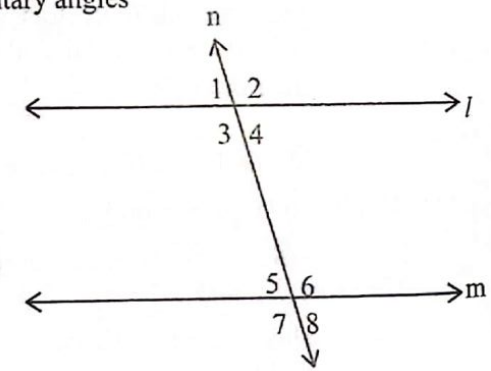
i.e. $m\angle 1 + m\angle 2 = 180^\circ$

$\therefore$

$m\angle 2 = 180^\circ - m\angle 1$

$= 180^\circ - 60^\circ$

$= 120^\circ$



As $\angle 2$ and $\angle 3$ are vertically opposite angles

$\therefore$

$m\angle 2 = m\angle 3 = 120^\circ$

Now measure $\angle 5$ with the help of protractor, which comes to be 60°.

Hence $m\angle 8 = m\angle 5 = 60^\circ$ (vertically opposite angles)

and $m\angle 6 = 180^\circ - m\angle 5$ (supplementary angles)

$= 180^\circ - 60^\circ$

$= 120^\circ$

$\therefore$ $m\angle 7 = m\angle 6 = 120^\circ$

There are four pairs of corresponding angles i.e.

$(\angle 1, \angle 5), (\angle 2, \angle 6), (\angle 3, \angle 7)$ *and* $(\angle 4, \angle 8)$

We have seen that:

$m\angle 1 = m\angle 5 = 60^\circ$

$m\angle 2 = m\angle 6 = 120^\circ$

$m\angle 3 = m\angle 7 = 120^\circ$

and $m\angle 4 = m\angle 8 = 60^\circ$

We conclude that:

> If a transversal intersects two parallel lines then pairs of corresponding angles are equal

We know that there are two pairs of alternate interior angles i.e. $(\angle 3, \angle 6)$ *and* $(\angle 4, \angle 5)$.

We have seen that

$m\angle 3 = m\angle 6 = 120^\circ$

and $m\angle 4 = m\angle 5 = 60^\circ$

This shows that:

> If a transversal intersects two parallel lines then pairs of alternate interior angles are equal

From the results given above, we note that:

$m\angle 3 = m\angle 5 = 120^\circ + 60^\circ = 180^\circ$

and $m\angle 4 = m\angle 6 = 60^\circ + 120^\circ = 180^\circ$

Where $(\angle 3, \angle 5)$ and $(\angle 4, \angle 6)$ are pairs of interior angles on the same side of the transversal.

We conclude that:

> If a transversal intersects two parallel lines then pairs of interior angles on the same side of the transversal are supplementary.

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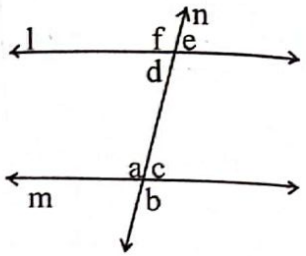
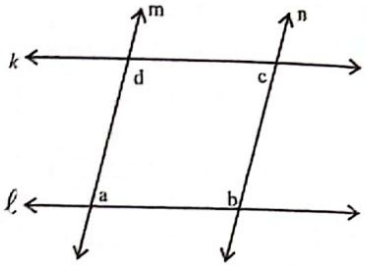
Exercise7.2

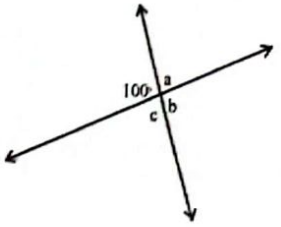
1. In the given figure, $l \parallel m$ and $a = 110°$. Find the measures of unknown angles.



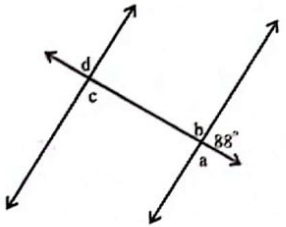
2. In the given figure, $k \parallel l$ and $m \parallel n$. If $a = 65°$, find the measures of unknown angles.



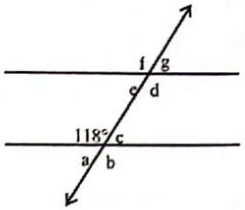
3. Find the measures of unknown angles in the following figures:



(i)

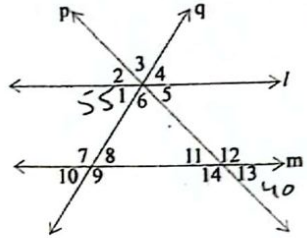


(ii)

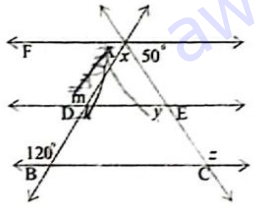


(iii)

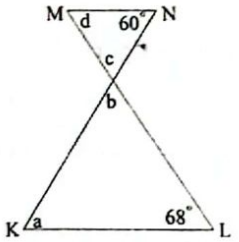
4. In the given figure $l \parallel m$, $m\angle 1 = 55°$ and $m\angle 13 = 40°$. Find the measures of all unknown angles.



5. In the figure $\overleftrightarrow{BC} \parallel \overleftrightarrow{DE} \parallel \overleftrightarrow{FA}$. Find the measures of unknown angles k, m, x, y and z. If $\overline{AD}$ and $\overline{DB}$ are equal then find another pair of equal segments.



6. In the given figure $\overline{KL} \parallel \overline{MN}$. Find the measures of angles a, b, c and d.



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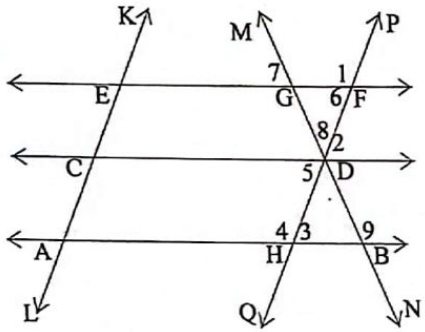
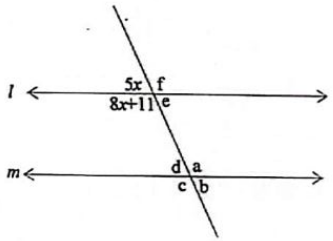
7. In the figure $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD} \parallel \overleftrightarrow{EF}$. If $\overline{AC} = \overline{CE}$ then name two other pairs of congruent line segments. Also find the measures of unknown angles, when $m\angle 1 = 100^\circ$ *and* $m\angle 7 = 70^\circ$



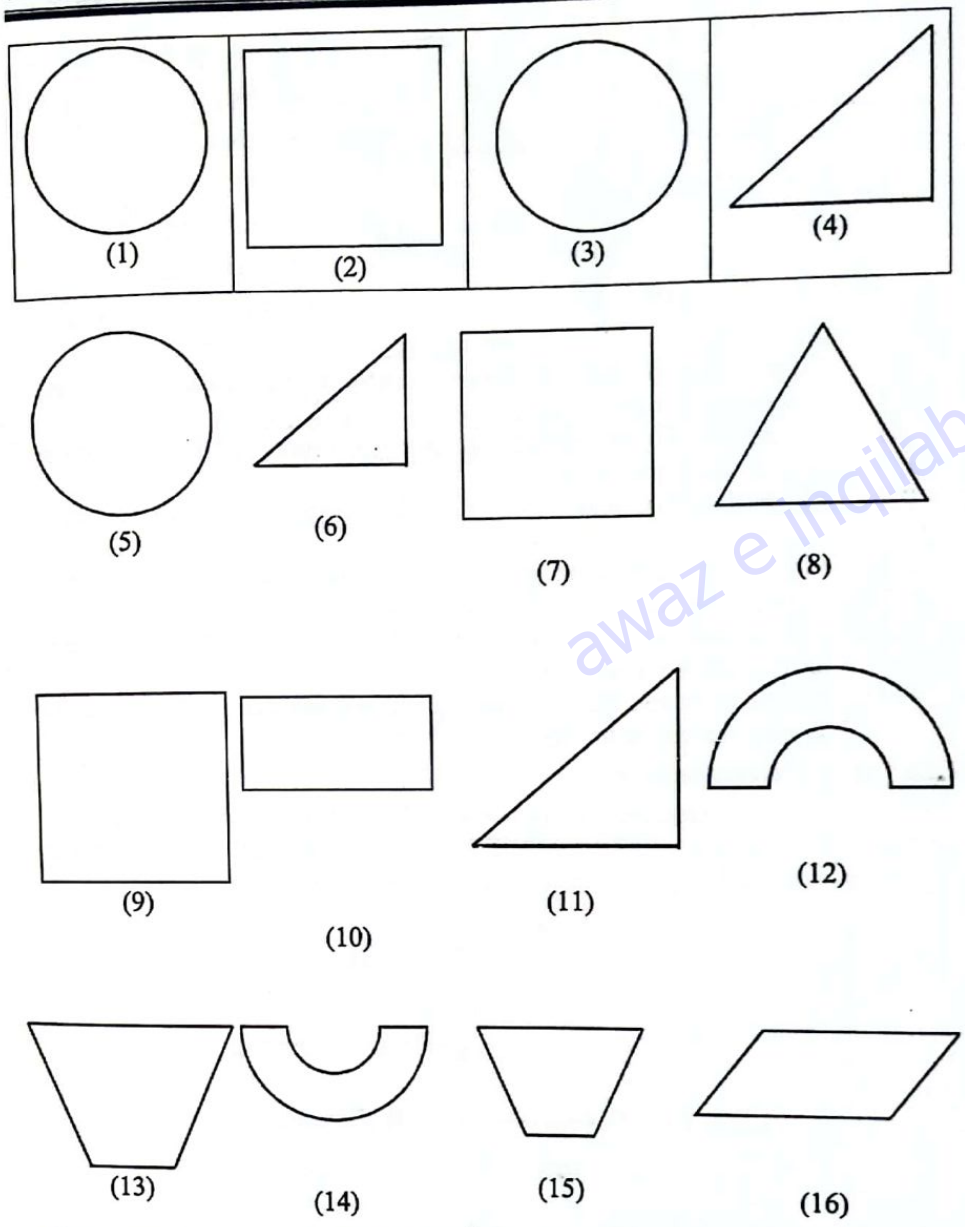
8. Find the measures of unknown angles a, b, c, d, e and f when $l \| m$.



7.3.1 Congruent and similar figures

Figures having the same shape and same size are called congruent figures; whereas similar figures have the same shape but they may or may not have the same size.

If we look at the figures given below, we find that:



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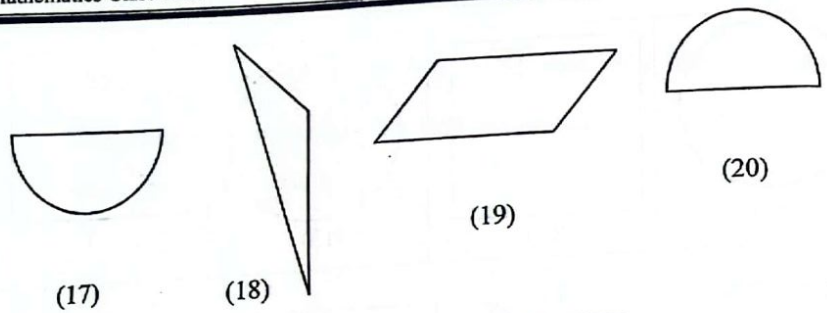


(i) Figure (1), (3) and (5) are congruent because they have the same shapes and the same sizes.

(ii) Figure (2), (7) and (9) are congruent because they have the same shapes and same sizes.

(iii) Figure (4), (6) and (11) are similar because they have the same shapes.

(iv) Figure (12) and (14) are similar because they have the same shapes.

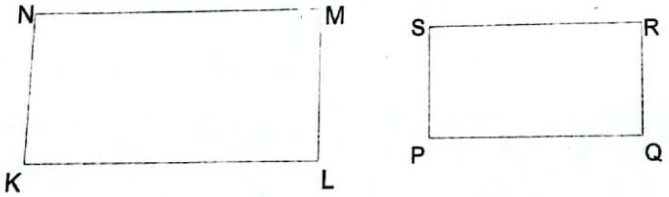
(v) Figure (13) and (15) are similar because they have the same shapes.

(vi) Figure (16) and (19) are congruent because they have the same shapes and the same sizes.

(vii) Figure (17) and (20) are congruent because they have the same shapes and the same sizes.

7.3.2 (1-1) Correspondence

For determining congruence and similarity between two figures, we have to compare CORRESPONDING sides and CORRESPONDING angles, i.e. sides and angles taken in a sequence, e.g. in the figure given below:-



Figures KLMN ⟷ figure PQRS means

$\overline{KL}$ is corresponding to $\overline{PQ}$,

$\overline{LM}$ is corresponding to $\overline{QR}$,

$\overline{MN}$ is corresponding to $\overline{RS}$,

$\overline{NK}$ is corresponding to $\overline{SP}$,

$\angle K$ is corresponding to $\angle P$,

$\angle L$ is corresponding to $\angle Q$,

$\angle M$ is corresponding to $\angle R$, and

$\angle N$ is corresponding to $\angle S$.

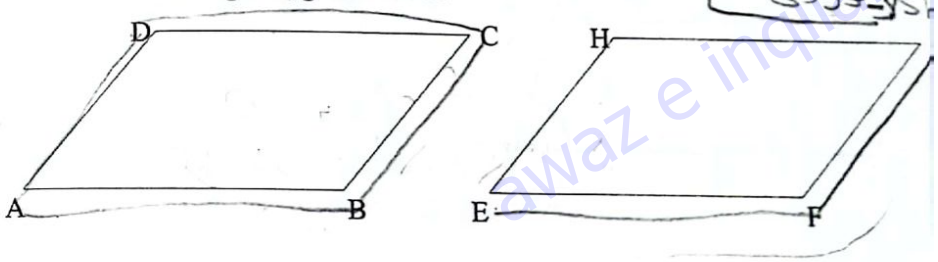
7.3.3 Congruent figures

When two figures are congruent,

- Their corresponding angles are congruent; and
- Their corresponding sides are congruent.

Congruency is shown by symbol ≅

In figures, given below:



$\overline{AB} \cong \overline{EF}$,

$\overline{BC} \cong \overline{FG}$,

$\overline{CD} \cong \overline{GH}$,

$\overline{DA} \cong \overline{HE}$,

Also $\angle A \cong \angle E$,

$\angle B \cong \angle F$,

$\angle C \cong \angle G$

$\angle D \cong \angle H$

Therefore Fig. $ABCD \cong Fig.EFGH$

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7.3.4 Similar figure

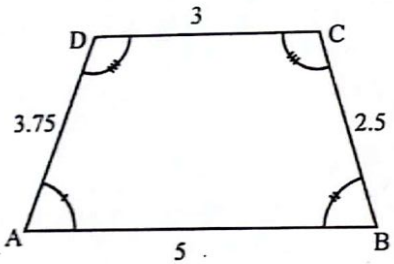
When two figures are similar,

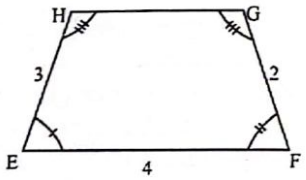
- Their corresponding angles are congruent.
- Their corresponding sides are proportional i.e.

Corresponding sides have the same ratio.

Similarity is shown by the symbol ~

In the figures given below:





$\angle A \cong \angle E$,

$\angle B \cong \angle F$,

$\angle C \cong \angle G$

$\angle D \cong \angle H$

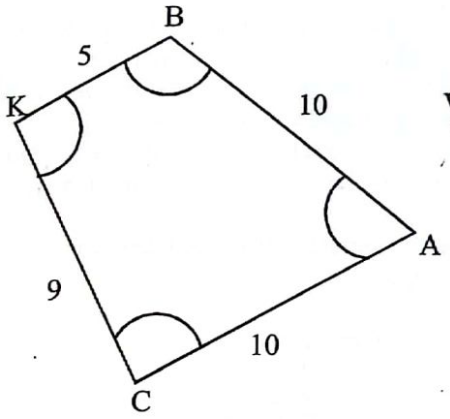
Also $\frac{m\overline{EF}}{m\overline{AB}} = \frac{m\overline{FG}}{m\overline{BC}} = \frac{m\overline{GH}}{m\overline{CD}} = \frac{m\overline{HE}}{m\overline{DA}} = \frac{4}{5}$

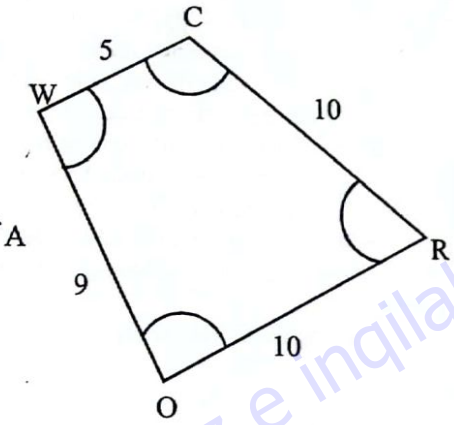
$\therefore$ *Fig.ABCD* ~ *Fig.EFGH*

The concept of similar and congruent is further elaborated with the help of examples:

Example-1:

Are the following figures similar or congruent?





Solution: In the given correspondence of

Polygon BACK ↔ polygon CROW

$m\overline{AB} = m\overline{RC} = 10$

$m\overline{AC} = m\overline{RO} = 10$

$m\overline{CK} = m\overline{OW} = 9$

$m\overline{KB} = m\overline{WC} = 5$

Also $\angle B \cong \angle C$, $\angle K \cong \angle W$

$\angle C \cong \angle O$, $\angle A \cong \angle R$

$\therefore$ *Figrue BACK* $\cong$ *Figure CROW*

7.3.5 Similar triangles

Two triangles are said to be similar if:

i) Corresponding angles are of the same measure.

ii) Lengths of the corresponding sides have the same ratio.

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However, to determine that two triangles are similar, it is not necessary that each and every element is measured and then the conditions given above are applied. ONE of the following conditions is sufficient to determine that the two triangles are similar:

i) TWO ANGLES of a triangle are congruent to the two corresponding angles of another triangle. (AA similarity).

ii) ALL the three sides of a triangle are proportional to the corresponding sides of another triangle. (SSS similarity).

iii) TWO sides of triangle are proportional to two corresponding sides of another triangle AND their included angles are congruent. (SAS similarity).

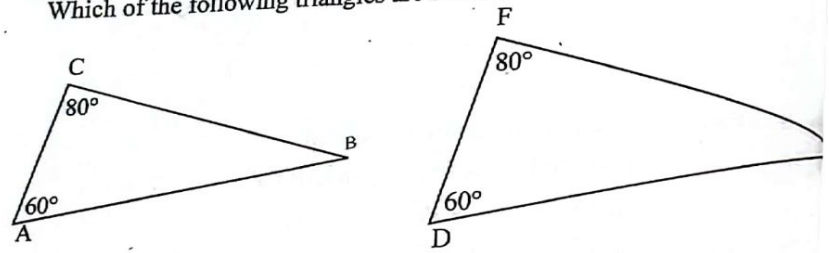
These help in finding the unknown elements of similar triangles.

Now consider the following examples:

Example

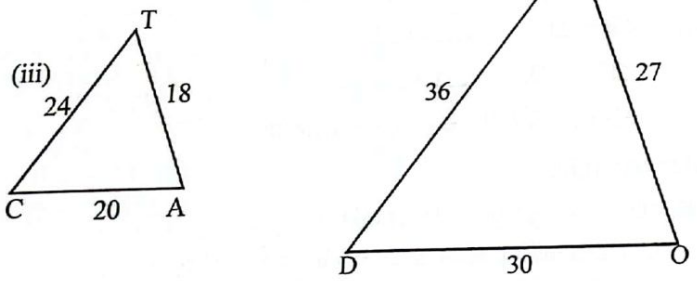
Which of the following triangles are similar?

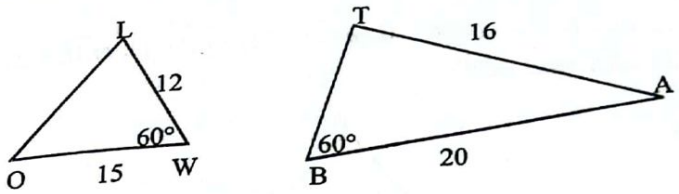
(i)



(ii)

(iii)





(iii)

Solution:

(i) In $\Delta ABC \leftrightarrow \Delta DEF$

$\angle A \cong \angle D$

$\angle C \cong \angle F$

$\therefore \Delta ABC \sim \Delta DEF$ (AA similarity)

(ii) In $\Delta CAT \leftrightarrow \Delta DOG$

$\frac{m\overline{CA}}{m\overline{DO}} = \frac{20}{30} = \frac{2}{3}$

$\frac{m\overline{AT}}{m\overline{OG}} = \frac{18}{27} = \frac{2}{3}$ and

$\frac{m\overline{CT}}{m\overline{DG}} = \frac{24}{36} = \frac{2}{3}$

$\therefore \Delta CAT \sim \Delta DOG$ (SSS similarity)

(iii) In $\Delta OWL \leftrightarrow \Delta BAT$

$\frac{m\overline{OW}}{m\overline{BA}} = \frac{15}{20} = \frac{3}{4}$

$\frac{m\overline{WL}}{m\overline{AT}} = \frac{12}{16} = \frac{3}{4}$

But $\angle W \cong \angle B$ which are not the included angles of proportional sides.

$\therefore$ ΔOWL *and* ΔBAT are not similar.

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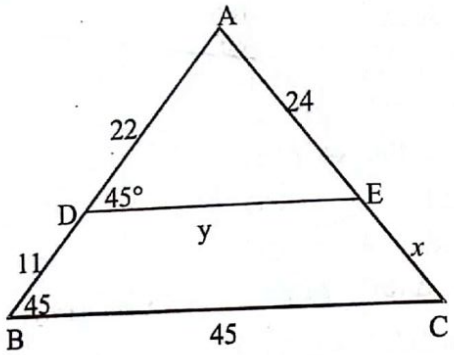
Example

Show that $\Delta ADE \sim \Delta ABC$. If $m\overline{AD}=22$, $m\overline{DB}=11$, $m\overline{AE}=24$, and $m\overline{BC}=45$, find $m\overline{EC}$ and $m\overline{DE}$.



Solution: In $\Delta ADE \leftrightarrow \Delta ABC$

$\angle D \cong \angle B$

$\angle A \cong \angle A$

$\therefore \Delta ADE \sim \Delta ABC$

We have $m\overline{AD}=22$, $m\overline{DB}=11$

i.e. $m\overline{AB}=22+11=33$

and $m\overline{AE}=24$. Also $m\overline{BC}=45$

let $m\overline{EC}=x$ and $m\overline{DE}=y$

$\therefore \Delta ADE \sim \Delta ABC$

$$\therefore \frac{m\overline{AD}}{m\overline{AB}}=\frac{m\overline{AE}}{m\overline{AC}}=\frac{m\overline{DE}}{m\overline{BC}}$$

or $$\frac{22}{33}=\frac{24}{24+x}=\frac{y}{45}$$

or $$\frac{24}{24+x}=\frac{y}{45}=\frac{2}{3}$$

Now $\frac{24}{24+x}=\frac{2}{3}$ and $\frac{y}{45}=\frac{2}{3}$

or, $2(24+x)=3\times24$

or, $48+2x=72$

or, $2x=72-48$

or, $=24$

or, $\therefore x=\frac{24}{2}=12$

or, $3y=2\times45$

or, $3y=90$

or, $y=\frac{90}{3}$

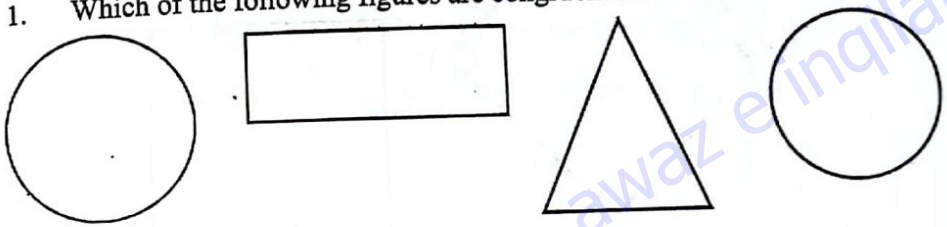
$=30$

Hence unknown sides are:

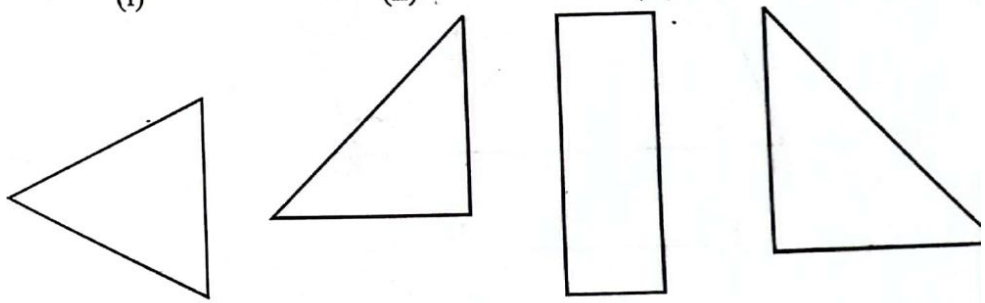
$m\overline{EC}=12$ and $m\overline{DE}=30$

Exercise 7.3

1. Which of the following figures are congruent and which are similar?



(i) (ii) (iii) (iv)



(v) (vi) (vii) (viii)

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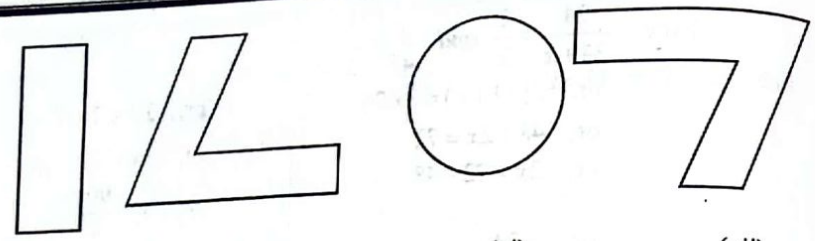
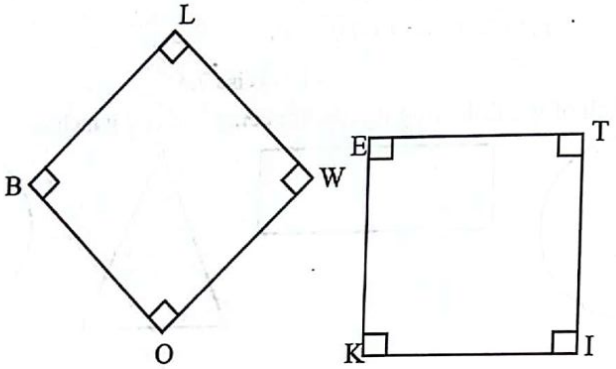


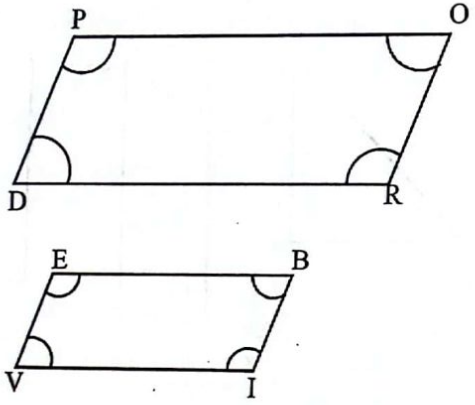
(ix) (x) (xi) (xii)

2. Which of the following figures are similar and which are congruent?

(i)

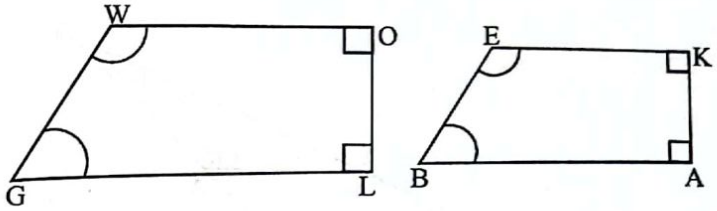


(ii)

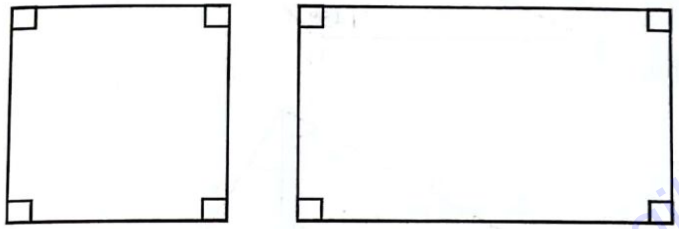


NOT FOR SALE

(iii)

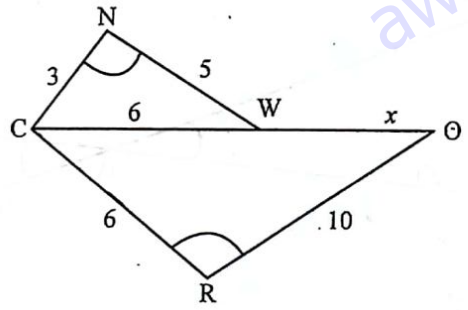


(iv)

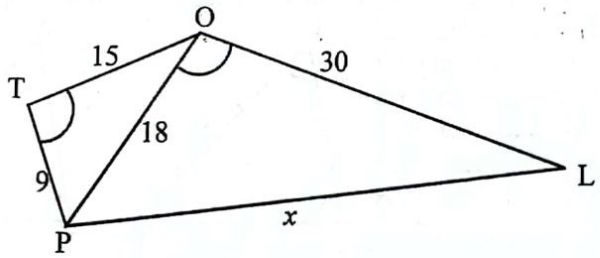


3. Are the triangles given in each figure similar? If yes, find the unknown elements.

(i)



(ii)



NOT FOR SALE

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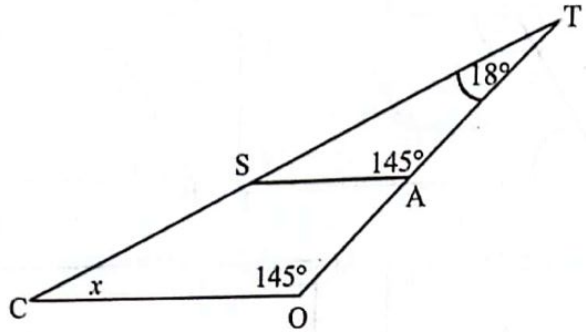
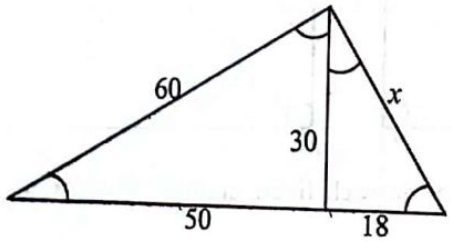
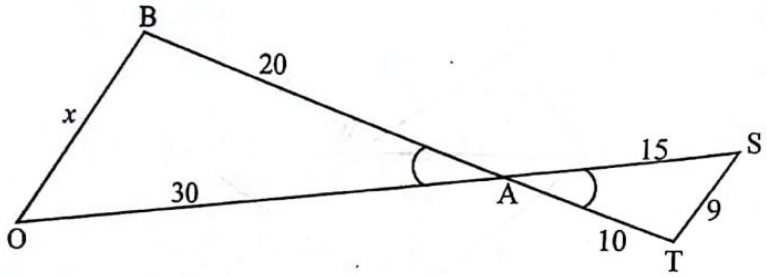
(iii)



(iv)

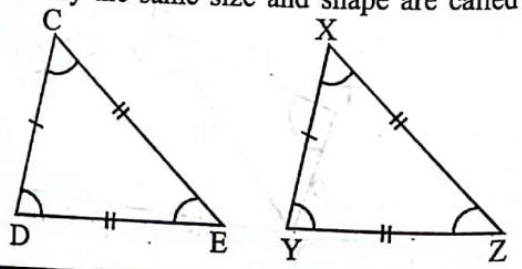


(v)



7.4 Congruent triangles

Triangles that have exactly the same size and shape are called congruent triangles. Sides and angles of a triangle are called their elements. Each triangle has six elements i.e. three sides and three angles. Two triangles are congruent when all the six elements of one triangle have the same measurement as the corresponding elements of another triangle. Congruent triangles are named by writing their vertices in corresponding order. In figure:-



$$\Delta CDE \cong \Delta XYZ$$

i.e. $\overline{CD} \cong \overline{XY}$

$\overline{DE} \cong \overline{YZ}$

$\overline{EC} \cong \overline{ZX}$

and $\angle C \cong \angle X$

$\angle D \cong \angle Y$

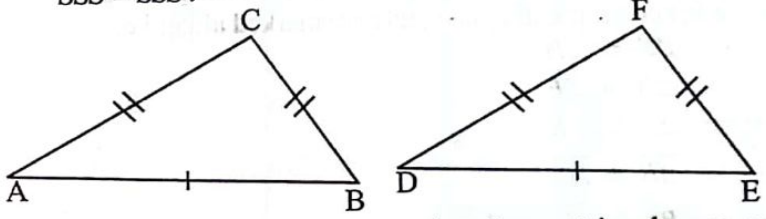
$\angle E \cong \angle Z$

7.4.2 Test for congruence of two triangles

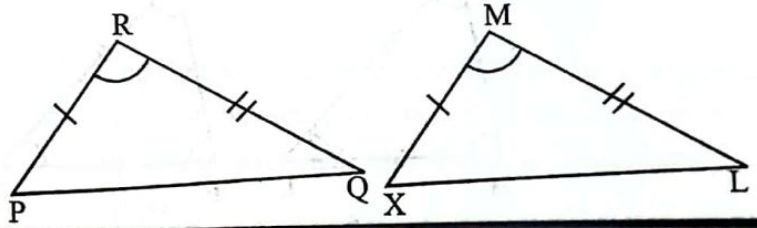
To show that triangles are congruent, it is not necessary that all six pairs of corresponding elements are equal. If three pairs of corresponding elements including atleast one pair of corresponding sides are equal, the triangles are said to be congruent, i.e.

Two triangles are congrueent, if:

(i) Three sides of one triangle are congruent respectively to the corresponding three sides of another triangle. This is referred to as $SSS \cong SSS$.



(ii) Two sides and included angle of one triangle are congruent respectively to the corresponding two sides and the included angle of another triangle. This is referred to as $SAS \cong SAS$.



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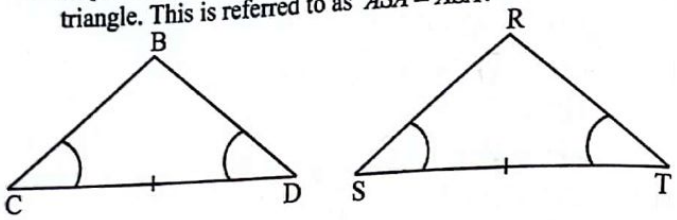
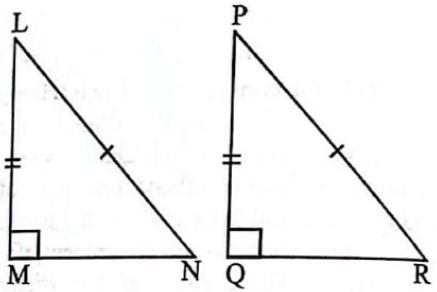
(iii) If two angles and the included side of one triangle are congruent respectively to two angles and the included side of another triangle. This is referred to as $ASA \cong ASA$.



(iv) If the hypotenuse and any one side of a right angled triangle are congruent respectively to the corresponding hypotenuse and side of another right angled triangle. This is referred to as $RH \cong RH$



Example-1:

Write the congruence statement for the following triangles:

Here the corresponding elements are marked alike: i.e.

$\angle A \cong \angle H$

$\angle B \cong \angle E$

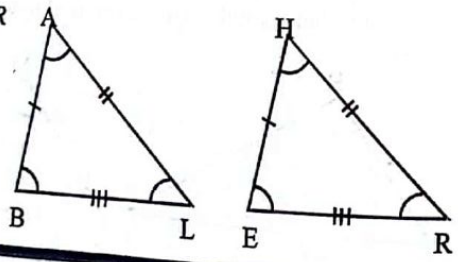
$\angle L \cong \angle R$

$\overline{AB} \cong \overline{HE}$

$\overline{BL} \cong \overline{ER}$ and

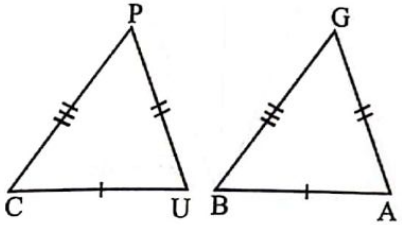
$\overline{LA} \cong \overline{RH}$

$\therefore \Delta ABL \cong \Delta HER$

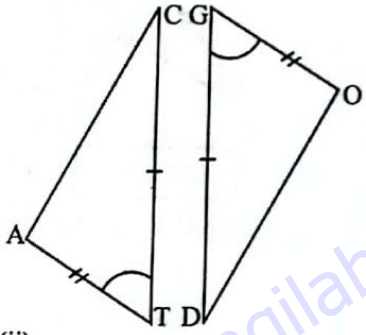


Exercise 7.4

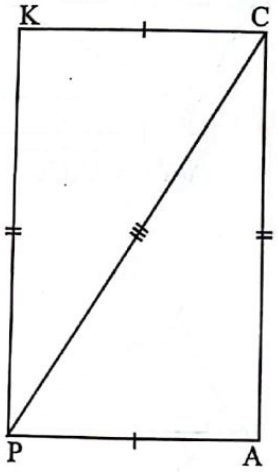
1. Consider the marked elements of the triangles and show how are the triangles congruent?



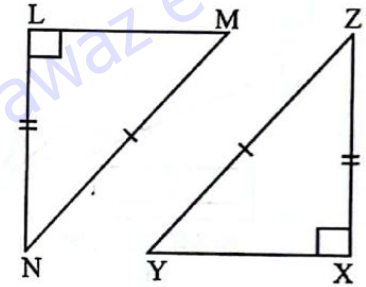
(i)



(ii)



(iii)



(iv)

2. In the following figures, some of the elements of the triangles have been marked to show that they are congruent. Write at least one corresponding element which when combined with the given information, triangles congruent.

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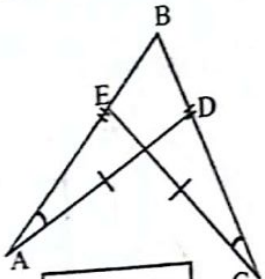
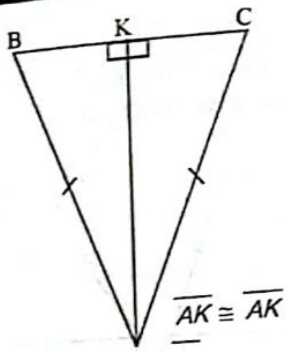


$\angle B \cong \angle B$

$\Delta ADB \cong \Delta CBE$

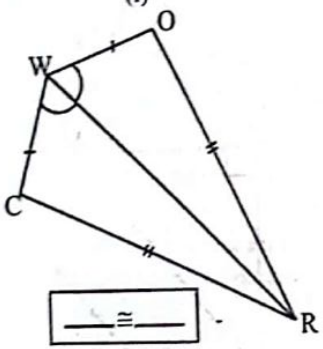
(i)



$\overline{AK} \cong \overline{AK}$

$\Delta ABK \cong \Delta ACK$

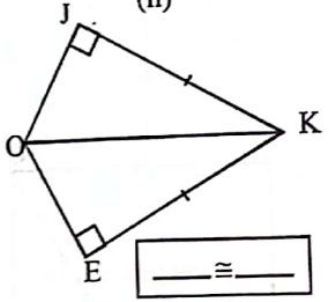
(ii)



$___ \cong ___$

$\Delta CRW \cong \Delta ORW$

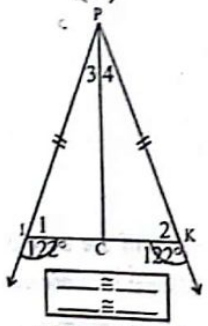
(iii)



$___ \cong ___$

$\Delta KOJ \cong \Delta KOE$

(iv)

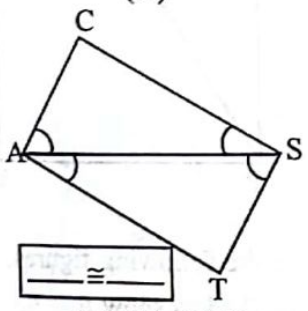


$___ \cong ___$

$___ \cong ___$

$\Delta PCI \cong \Delta PKC$

(v)



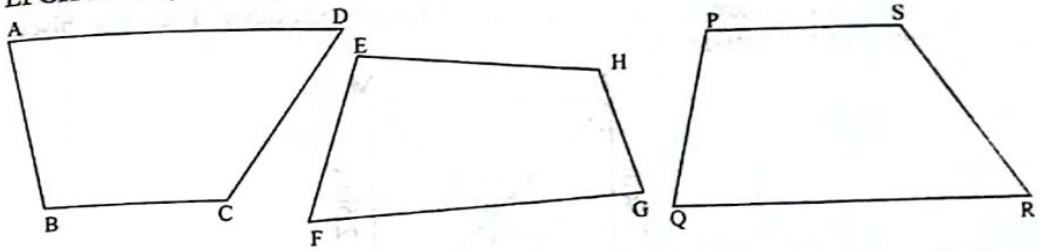
$___ \cong ___$

$\Delta ACS \cong \Delta STA$

(vi)

7.5 Quadrilaterals

A closed figure having four sides is called a quadrilateral. ABCD, EFGH and PQRS are quadrilaterals.

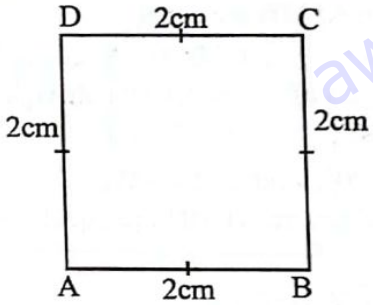


All quadrilaterals, have four angles and the sum of the measures of all those angles is $360°$.

Below are given the properties of some special kind of quadrilaterals.

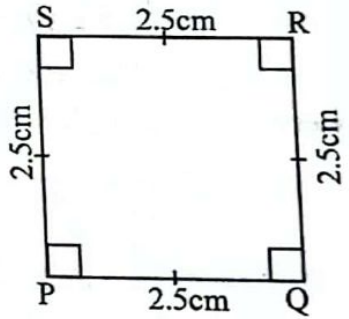
7.5.1 Square

A square is a quadrilateral in which all the sides are of the same length.



ABCD is a square. Length of each side of this square is 2cm.

All the four angles of a square are right angles i.e. measure of each angle $= 90°$.



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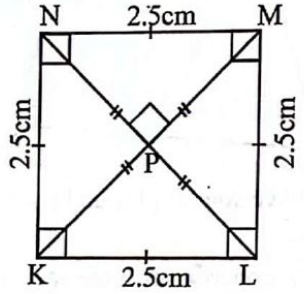
In the square PQRS, $m\angle P = m\angle Q = m\angle R = m\angle S = 90^\circ$ i.e. all angles of a square are right angles.

Line segment joining opposite vertices of a square is called its diagonal diagonals of a square are always equal and bisect each other. They also bisect the angles of the square.



KLMN is a square. It has the following properties:

(i) Four sides of KLMN are equal

$i.e.\ \overline{KL} \cong \overline{LM} \cong \overline{MN} \cong \overline{KN}$

(ii) The four angles of square KLMN are right angles.

$i.e.\ m\angle k = 90^\circ, \quad m\angle L = 90^\circ$

$m\angle M = 90^\circ$, *and* $m\angle N = 90^\circ$

(iii) Diagonals of square KLMN are equal and intersect each other at right angle.

$i.e.\ \overline{KM} \cong \overline{LN}$ *and*

$\overline{KP} \cong \overline{MP}, \overline{PN} \cong \overline{PL}$

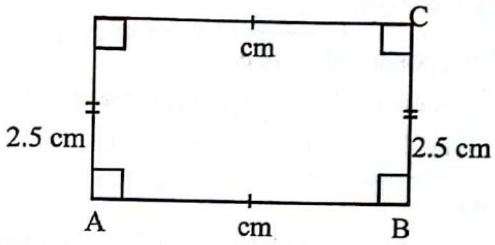
Also $\angle KPL$, $\angle LMP$, $\angle MPN$ *and* $\angle NPK$ are right angles i.e. measure of each angle is 90°.

So in a square:

- All the four sides are equal;
- All the four angles are right angles; and
- Both the diagonals equal and bisect each other.

7.5.2 Rectangle

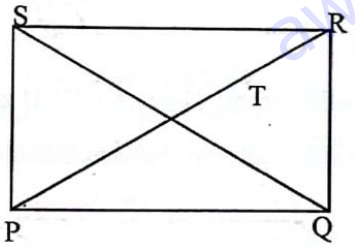
A rectangle is a quadrilateral in which opposite sides are equal and all the angles are right angles.



ABCD is a rectangle. Length of its opposite sides is equal

$i.e. m\overline{AB} = m\overline{DC} = 4cm,\ and\ \ m\overline{AD} = m\overline{BC} = 2.5cm$

Also all of its angles are right angles i.e. measure of each angle is equal to 90°.



PQRS is a rectangle. Its diagonals $\overline{PR}$ and $\overline{SQ}$ are equal i.e. $m\overline{PR} = m\overline{SQ}$.

Both the diagonals intersect each other at T and bisect each other i.e.

$PT \cong \overline{TR}$ *and* $\overline{ST} \cong \overline{TQ}$

The difference between a square and a rectangle is that:

(i) In a square the adjacent sides are equal whereas in a rectangle the adjacent sides are not equal.

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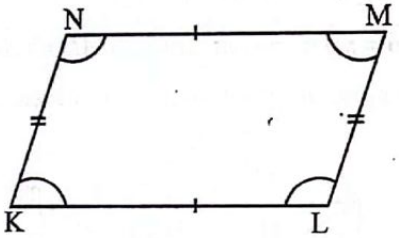
(ii) Diagonals of a square intersect each other at right angle, whereas the diagonals of a rectangle do not form a right angle at the point of their intersection.

So in a rectangle:

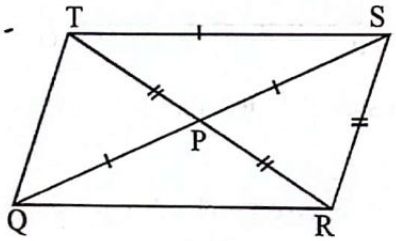
- Opposite sides are equal;
- All the angles are right angles; and
- The diagonals bisect each other.

7.5.3 Parallelogram

A parallelogram is a quadrilateral in which opposite sides are parallel and equal.



KLMN is a parallelogram. Here $\overline{KL} \parallel \overline{NM}$ and $m\overline{KL} = m\overline{NM}$. Also $\overline{KN} \parallel \overline{LM}$ and $m\overline{KN} = m\overline{LM}$. Again $m\angle K = m\angle M$ and $m\angle N = m\angle L$.



QRST is a parallelogram. Its diagonals $\overline{QS}$ and $\overline{TR}$ intersect each other at P such that $m\overline{QP} = m\overline{PS}$ and $m\overline{TP} = m\overline{PR}$.

The difference between a parallelogram and a rectangle is that:

(i) All the angles of a rectangle are right angles whereas in a parallelogram, the angles are not right angles.

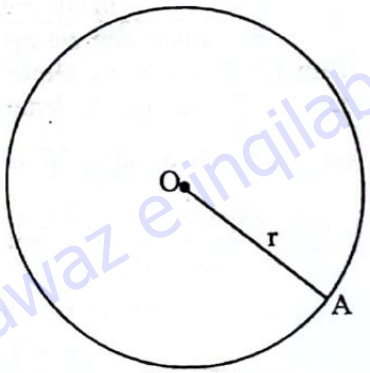
(ii) Diagonals of a rectangle are equal whereas the diagonals of a parallelogram are not equal.

So in a parallelogram:

- Opposite sides are equal;
- Opposite angles are equal; and
- The diagonals bisect each other.

7.6 Circle

Let us recall that a circle is a set of all such points in a plane, which are equidistant from a fix point of the plane. The fixed point is called the centre of the circle.



7.6.1 Radial segment and radius

Line segment joining the centre of the circle to any point on the circle is called its radial segment. Measure of the radial segment is called radius. It means that radius is a number and not a line segment.

In the given figure O is the centre, $\overline{OA}$ is a radial segment and radius i.e. $m\overline{OA} = r$.

Please note that:

i) There is only one centre of a circle.

ii) Centre is always inside the circle.

iii) Measure of all radial segment of a circle is the same.

7.6.2 Circumference of a circle

Measure of the line joining all the points on a circle which are equidistance from its centre is called circumference of the circle.

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7.6.3 Chord and diameter

A line segment which joins two points on the circumference of a circle is called is chord.

If a chord passes through the centre of the circle, then it is called a diameter of the circle.

In the adjoining figure $\overline{AB}$ and $\overline{DE}$ are chords whereas $\overline{AC}$ and $\overline{DF}$ are diameters.

Please remember that:

i) Diameter of a circle is always twice its radius.

ii) All diameters are chords but all chords are not necessarily the diameters of the circle.

Ratio between the circumference and the diameter of the same circle is constant and is denoted by a greek letter π (pi) which is approximately equal to $\frac{22}{7}$ or 3.14.

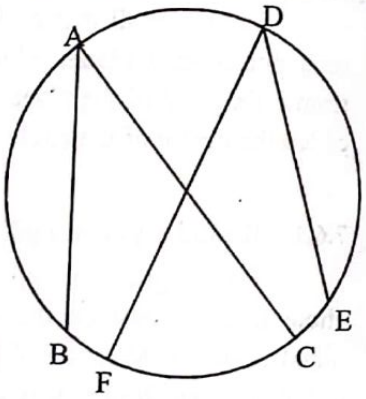
$$\frac{circumfernece(c)}{diameter(d)} = \pi$$

or $C = \pi d$

$C = 2\pi r \quad (\because d = 2r)$

Circumference of a circle $= 2\pi r$

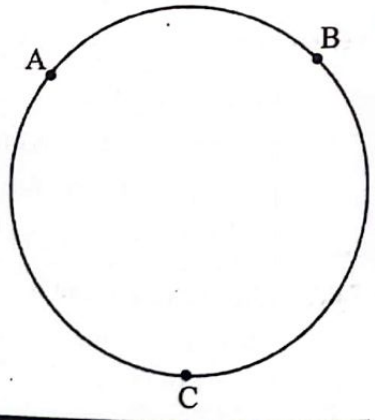
$C = \pi d$



7.6.4 Arc, minor are and major are

A part or a portion of a circle joining any points A and B of the circle is called an arc, and is denoted by $\overset{\frown}{AB}$ two points on the circumference of the circle define two arcs. The shorter one is called minor arc whereas the bigger one is called major arc.

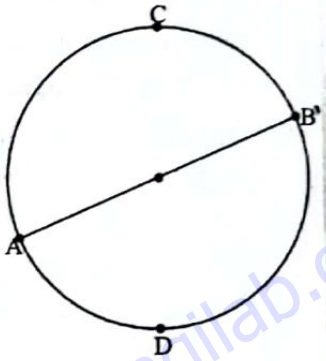
In the figure $\overset{\frown}{AB}$ is the minor arc and $\overset{\frown}{ABC}$ is the major arc. (In $\overset{\frown}{ACB}$, C is included to



avoid confusion because $\overset{\frown}{AB}$ define both the arcs: minor as well as major). If we draw chord $\overline{AB}$, we can say that both the arcs are intercepted by chord $\overline{AB}$.

7.6.5 Semi-circle

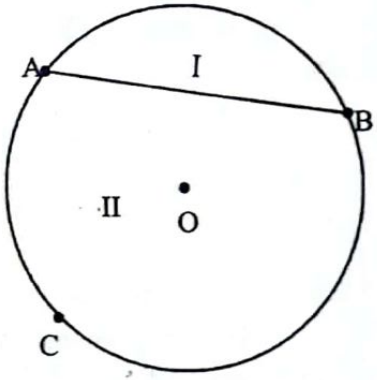
A diameter of a circle divides the circle into two equal portions. Each of these portions is called a semi-circle or half circle. We can also say that if the lengths of major and minor arcs of a circle are equal then they divide the circle into two semi-circle regions. Each of these regions is a semi-circle or half circle. In this case, none of these arcs is called major or minor arc.



In the given figure, diameter $\overline{AB}$ or $\overline{AOB}$ divides the circle into two semi-circle. Neither of arcs $\overset{\frown}{ACB}$ and $\overset{\frown}{ADB}$ is a major or minor arc.

7.6.6 Segment of the circle

A chord of a circle divides the circle into two parts. Each part is called segment of the circle. In the figure chord $\overline{AB}$ divides the circle into two segments.



Segment I is the region bounded by chord $\overline{AB}$ and minor arc $\overset{\frown}{AB}$. Therefore it is called minor segment similarly segment II is the region bounded by chord $\overline{AB}$ and major arc $\overset{\frown}{ACB}$. It is therefore called major segment.

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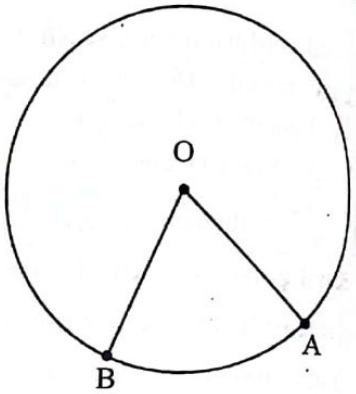
7.6.7 Sector of a circle

A circular region bounded by two radial segment and the arc intercepted by them is called a sector of the circle.

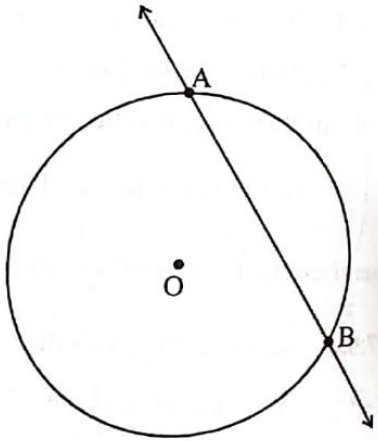
In the figure the region bounded by $\overline{AO}, \overline{BO}$ and $\overset{\frown}{AB}$ represents a sector of the circle.



7.6.8 Secant of the circle

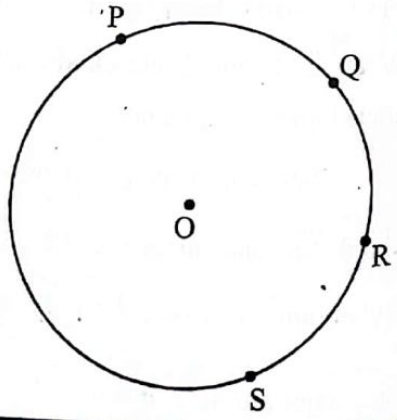
A line which intersects a circle at exactly two points is called a secant of the circle.

In the figure, $\overline{AB}$ is a secant of the circle.



7.6.9 Concyclic points

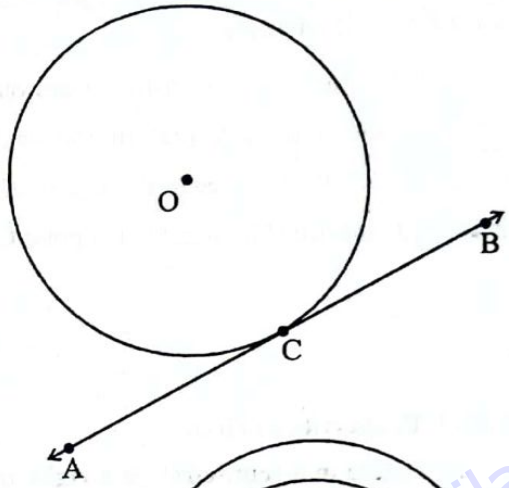
Four or more points which lie on the circumference of a circle are called concyclic points. In the figure P, Q, R, S are concyclic points.



7.6.10 Tangent to a circle

A line which touches a circle at just one point is called a tangent to the circle. It must be noted that a tangent touches a circle at a single point only, as if it becomes closer to the centre it will intersect the circle at two points and thus it will become a secant.

In the figure $\overline{AB}$ is a tangent which touches the circle at C only.

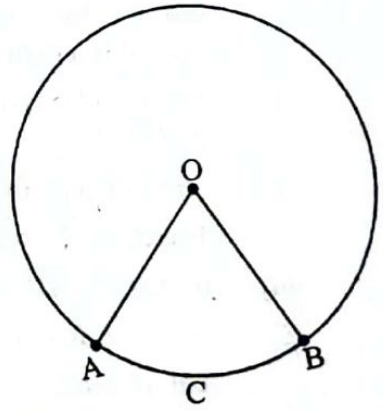


7.6.11 Concentric circle

Circles of different radii having the same centre arc called concentric circles. In the figure, all the circles arc of different size but have a common centre O.

7.6.12 Central angle

The angle made at the centre of a circle by the radii at the end points of an arc is called the central angle. In the figure $\angle AOB$ is the central angle subtended by arc ACB.



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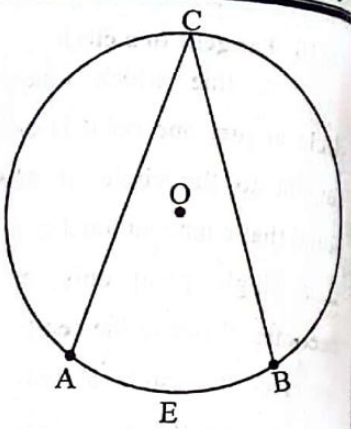
7.6.13 Inscribed angle

The angle subtended by an arc on any point on the remaining part of the circle is called an inscribed angle. In the figure $\angle ACB$ is an angle inscribed by arc AEB at point C.



7.6.14 Properties of circle

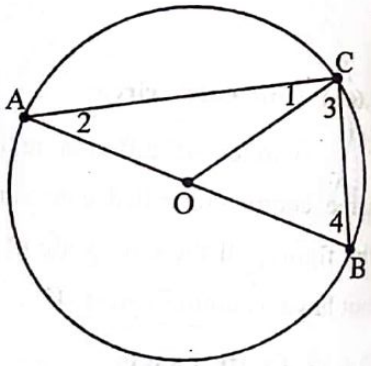
(a) Angle in a semi-circle is a right angle

i) Draw a circle with centre O.

ii) Draw its diameter $\overline{AOB}$ this divides the circle into two semi-circles.

iii) Take a point C on the circle and join it with the end points of the diameter i.e. A and B.

iv) In this way ΔABC is formed and we have to show that $\angle ACB$ is a right angle.



v) Join C to O. ΔABC is converted into two triangles ΔAOC and ΔBOC.

vi) $\overline{AO}, \overline{BO}$ and $\overline{CO}$ are the radial segments of the same circle. Therefore $\overline{AO} \cong \overline{BO} \cong \overline{CO}$.

vii) In ΔAOC, $\overline{AO} \cong \overline{CO}$

$\therefore$ ΔAOC is an isosceles triangle and $\angle 1 \cong \angle 2$

viii) Similarly ΔBOC is an isosceles triangle

$\therefore$ $\angle 3 \cong \angle 4$

ix) Because the sum of the interior angles of a triangle is equal to 180°.

$\therefore$ for ΔABC

$$m\angle A + m\angle ACB + m\angle B = 180°$$

or $$m\angle 2 + (m\angle 1 + m\angle 3) + m\angle 4 = 180°$$

or $$m\angle 2 + (m\angle 2 + m\angle 4) + m\angle 4 = 180°$$

or $$2m\angle 2 + 2m\angle 4 = 180°$$ $(\because \angle 1 \cong \angle 2 \text{ and } \angle 3 \cong \angle 4)$

or $$2(m\angle 2 + m\angle 4) = 180°$$

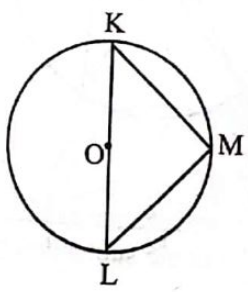
or $$m\angle 2 + m\angle 4 = \frac{180°}{2} = 90°$$

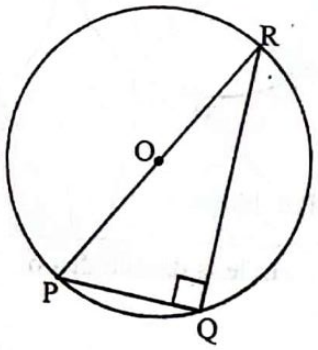
or $$m\angle ACB = 90$$

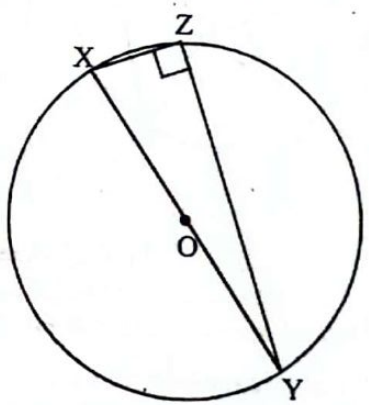
This shows that:

Angle in a semi-circle is a right angle.

Draw some circle and verify that the angle in a semi-circle is always a right angle.







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(b) Central angle of a minor arc of a circle is double that of the angle subtended by the corresponding major arc.

Activity

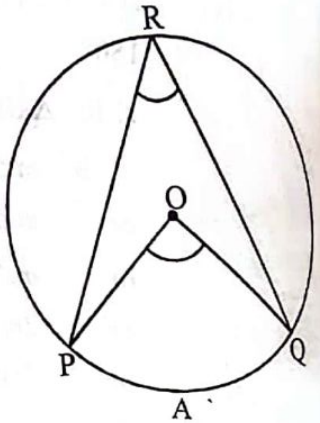
i) Draw a circle with centre O.

ii) take a minor arc PAQ and a point R at the corresponding major arc so that PRQ is an inseribed angle of arc PAQ.

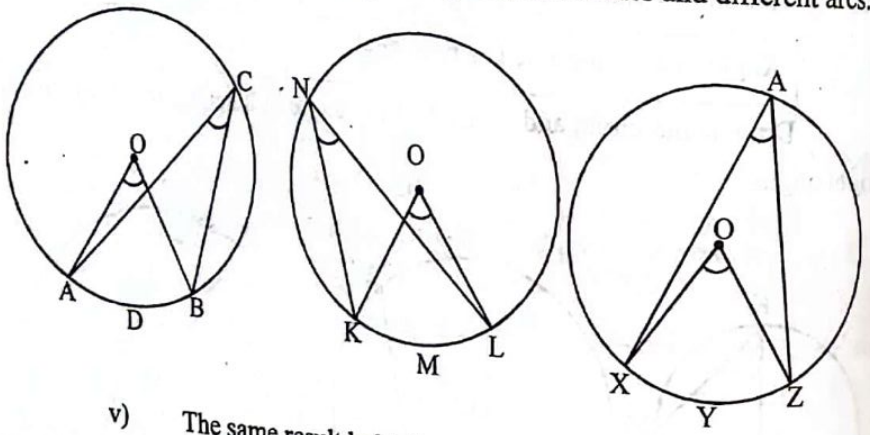
iii) Measure $\angle POQ$ and $\angle PRQ$.

iv) We observe that

$m\angle POQ = 2m\angle PRQ$



(iii) Repeat this activity by taking different circles and different arcs.



v) The same result holds i.e.

Central angle of a minor arc of a circle is double that of the angle subtended by the corresponding major arc.

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(c) Angles in the same segment of a circle are equal

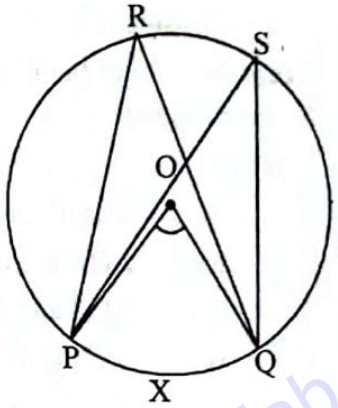
i) Draw a circle with centre O.

ii) Draw $\angle PRQ$ and $\angle PSQ$ in the same segment of the circle subtended by arc PXQ.

iii) Join P and Q to O. $\angle POQ$ is the central angle subtended by are PXQ.

iv) Because $\angle POQ$ is the central angle of minor are and $\angle PRQ$ is the angle subtended by the corresponding to major arc.

$m\angle POQ = 2m\angle PRQ \longrightarrow (i)$



v) Similarly $\angle POQ$ is the central angle of minor arc and $\angle PSQ$ is the angle subtended by the corresponding to major arc.

$\therefore m\angle POQ = 2m\angle PSQ \longrightarrow (ii)$ ($\because$ angle subtended by the corresponding major arc)

vi) From (i) and (ii)

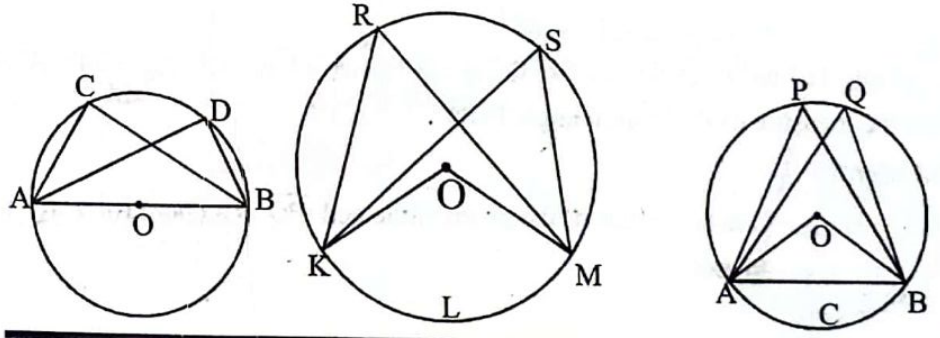
$2m\angle PRQ = 2m\angle PSQ$

or $m\angle PRQ = m\angle PSQ$

i.e.

Angles in the same segment of a circle are congruent

Repeat this activity to verify the result.



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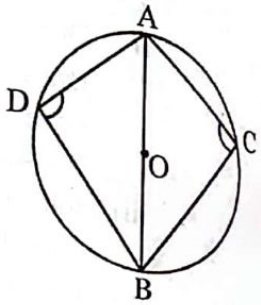
7.6.15 Application of the properties of a circle

Consider the following examples.



Example 1: Find the measures of $\angle C$ and $\angle D$ from the given figure.

Solution:

i) O is the centre of the circle.

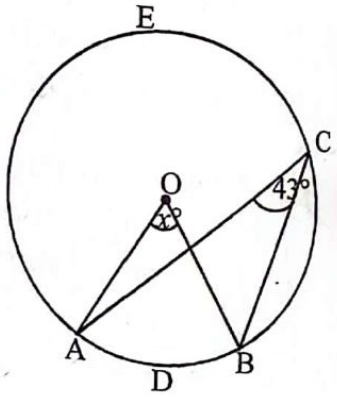
ii) AOB is a diameter

iii) ACB and ADB are two semi-circle.

iv) $\angle C$ is formed in one semi-circle and $\angle D$ in another semi-circle.

v) $\therefore m\angle C = 90°$ and $m\angle D = 90°$.

Example 2: Find the value of x in the given figure $m\angle ACB$ is given to be equal to 43°.



Solution:

i) In the given circle $\angle AOB$ is the central angle of arc ADB.

ii) $\angle ACB$ is the angle subtended by the corresponding major are AEB.

iii) $m\angle AOB = 2m\angle ACB$

or $x° = 2(43°)$

$= 86°$

hence value of x is 86°.

Example 3: Find the value of x in the given figure, when $m\angle PRQ = 54°$. Also find the measure of the central angle POQ.

Solution:

i) O is the centre of the given circle and $\overline{PQ}$ is a chord (or PAQ is an arc)

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ii) $\angle PRQ$ and $\angle PSQ$ arc in the same segment of the circle i.e. PBQ.

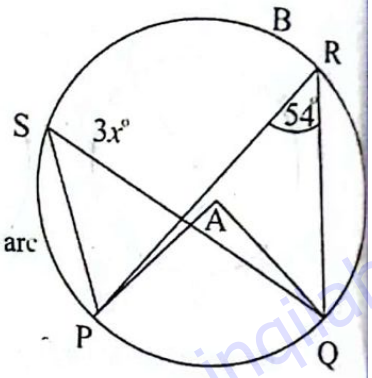
iii) $m\angle PSQ = m\angle PRQ$

Or, $3x° = 54°$

Or, $x° = \frac{1}{3}(54°)$

$= 18°$

Hence the value of x is 18°.



iv) $\angle POQ$ is the central angle of arc PAQ

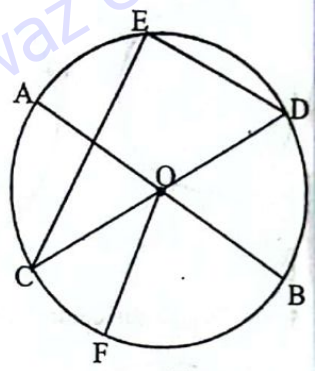
$\therefore m\angle POQ = 2m\angle PRQ$

$= 2(54°)$

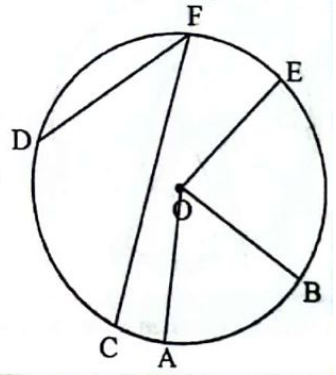
$= 108°$

Exercise 7.5

1. Name the centre of the given circle. Name 2 chords and 2 diameters of the circle. Also name a radial segment shown in the circle.



2. Name the radii and chords in this figure. Also name the concyclic points.



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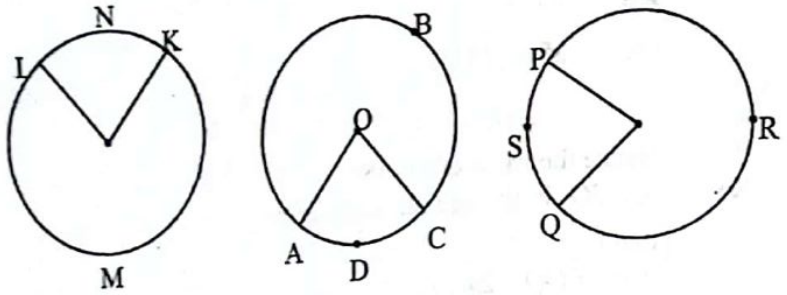
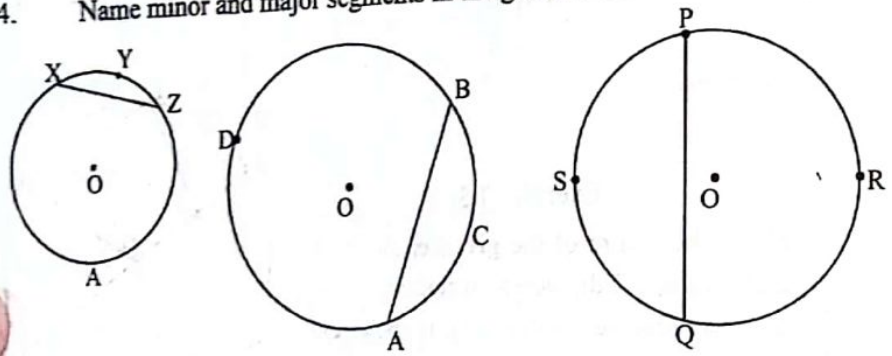
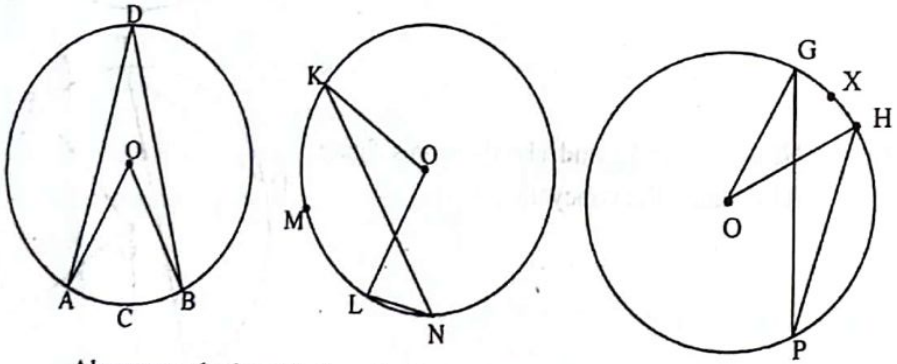
3. Name major arc and minor arc in the given circles. Also name the sectors in the given figures.



4. Name minor and major segments in the given circle.



5. Name the central angles of arcs $\overset{\frown}{ABC}$, $\overset{\frown}{KLM}$ and $\overset{\frown}{GXH}$.



Also name the inscribed angles in the corresponding major arcs

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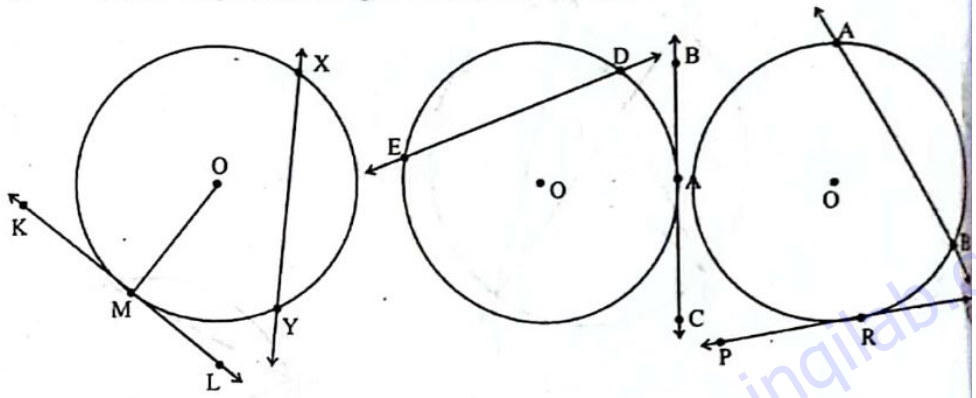
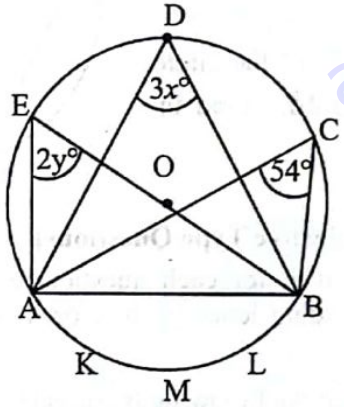
6. Draw concentric circles of radii 2cm, 2.5 cm and 3cm. What is the diameter of each circle?

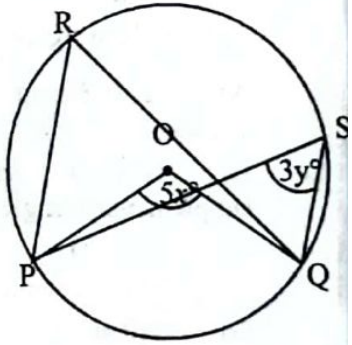
7. Name tangents to the given circles. Also name the secants.



8. In the given figure, $m\angle ACB = 54°$. Find the value of x and y.



9. In the given figure, $m\angle PRQ = 45°$, find x and y, when O is the centre of the circle.



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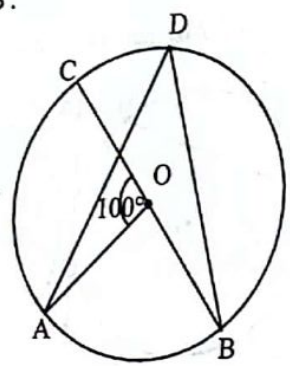
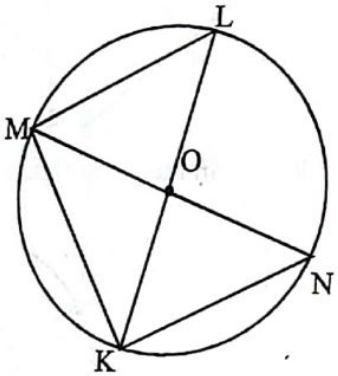
10. O is the centre of the circle. $\overline{BC}$ is a diameter. $m\angle AOC = 100°$. Find $m\angle ADB$.



11. In the figure, O is the centre of the circle. $\overline{MN}$ is the bisector of $\angle KML$. Find the measure of $\angle KNM$.



Exercise 7 (Objective Type Questions)

Four suggested answers are given for each question. Select the correct answer and write the corresponding letter (*a*, b, c or d) in the box given against each equation.

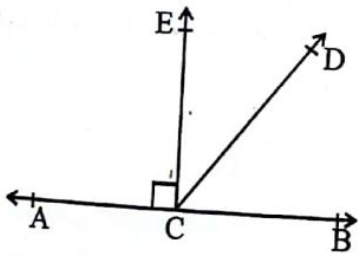
1. For the given figure which of the following is correct? ☐

A) $\angle ACE$ and $\angle ECD$ are adjacent angles.

B) $\angle ACD$ and $\angle BCD$ are supplementary angles.

C) $\angle BCD$ and $\angle ECD$ are complementary angles.

D) All are correct.



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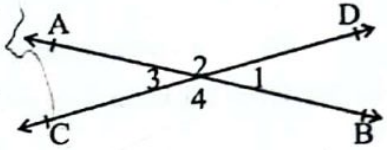
2. For the given figure which of the following is correct? ☐

A) $\angle 1$ and $\angle 2$ are vertically opposite angles.

B) $\angle 1$ and $\angle 3$ are complementary angles.

C) $\angle 2$ and $\angle 4$ are vertically opposite angles.

D) $\angle 2$ and $\angle 4$ are supplementary angles.



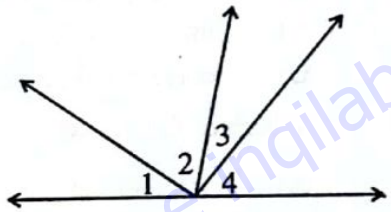
3. For the given figure which of the following are adjacent angles? ☐

A) $\angle 1, \angle 3$

B) $\angle 2, \angle 3$

C) $\angle 4, \angle 2$

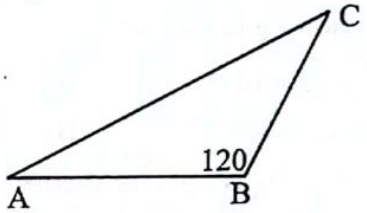
D) $\angle 1, \angle 4$



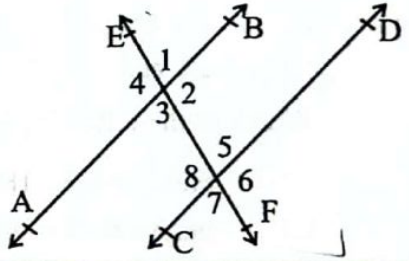
4. In ΔABC, $m\angle B = 120°$, Ratio between $m\angle A$ and $m\angle C$ is 5 : 7, $m\angle C =$ _____. ☐

A) 35° D) 25°

C) 60° D) 30°



In the given figure $\overline{AB} \parallel \overline{CD}$. Answer Q. 5 to Q.13 with reference to this figure.



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5. $\angle 3 \cong \angle 5$ because these are: ☐

A) vertically opposite angles B) corresponding angles
C) alternate angles D) complementary angles

6. $\angle 6 \cong \angle 8$ because these are: ☐

A) corresponding angles B) vertically opposite angles
C) alternate angles D) supplementary angles

7. $\angle 2 \cong \angle 6$ because these are: ☐

A) alternate angles
B) interior angles of the transversal
C) corresponding angles D) vertically opposite angles

8. Which of the following is not a pair of corresponding angles. ☐

A) $\angle 1$ and $\angle 5$ B) $\angle 4$ and $\angle 8$
C) $\angle 7$ and $\angle 3$ D) $\angle 5$ and $\angle 4$

9. Which of the following is not a pair of vertically opposite angles? ☐

A) $\angle 2$ and $\angle 4$ B) $\angle 6$ and $\angle 8$
C) $\angle 1$ and $\angle 3$ D) $\angle 5$ and $\angle 3$

10. Which of the following is not correct? ☐

A) $m\angle 2 + m\angle 5 = 180°$ B) $m\angle 3 + m\angle 8 = 180°$
C) $m\angle 1 + m\angle 3 = 180°$ D) $m\angle 1 + m\angle 4 = 180°$

11. Which of the following is a pair of alternate angles? ☐

A) $\angle 2$ and $\angle 4$ B) $\angle 3$ and $\angle 5$
C) $\angle 1$ and $\angle 5$ D) $\angle 4$ and $\angle 8$

12. Which of the following is not correct? ☐

A) $m\angle 6 + m\angle 8 = 180°$ B) $m\angle 2 + m\angle 5 = 180°$
C) $m\angle 3 + m\angle 8 = 180°$ D) $m\angle 6 + m\angle 7 = 180°$

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13. If $m\angle 2 = 120°$, then which of the following is not correct? ☐

A) $m\angle 7 = 120°$ B) $m\angle 8 = 120°$
C) $m\angle 6 = 120°$ D) $m\angle 4 = 120°$

14. If a transversal intersects two parallel lines, how many pains of corresponding angles are formed? ☐

A) 2 B) 4
C) 6 D) 8

15. Two angles are called adjacent angels, if: ☐

A) they have a common vertex
B) they have a common arm
C) intersection of their interiors is a null set
D) all of the above

16. Diagonals of a square ☐

A) are equal in length
B) bisect each other
C) make a right angle at the point of intersection
D) all of the above

17. Which of the following is not true for a rectangle? ☐

A) its diagonals are equal in length
B) it opposite sides are parallel
C) its sides do not intersect at right angle
D) its adjacent sides are not equal in length

18. Which of the following is not a sufficient condition for congruency of two triangles? ☐

A) $SSS \cong SSS$ B) $AAA \cong AAA$
C) $SAS \cong SAS$ D) $ASA \cong ASA$

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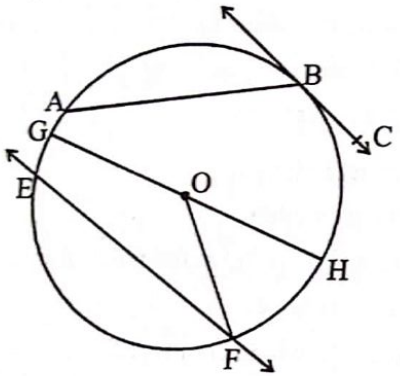
19. Which of the following statements is ALWAYS true when parallel lines are cut by a transversal? ☐

A) sum of the measures of corresponding angles is 180°
B) vertically opposite angles are acute
C) corresponding angles are complementary
D) the interior angles on the same side of a transversal are supplementary

20. Measure of a line segment joining the mid points of two sides of a triangle is 4.5 cm. Measure of its third side is: ☐

A) 9 cm B) 6 cm
C) 11 cm D) 2.25 cm



Note: Answer Q. 21 to Q. 26 with reference to given figure.

21. $\overline{AB}$ is called a: ☐

A) chord B) secant
C) diameter D) radial segment

22. $\overline{GH}$ is called a: ☐

A) chord B) secant
C) diameter D) radial segment

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23. $\overline{EF}$ is called a: ☐

A) chord B) secant
C) tangent D) radial segment

24. $\overline{OF}$ is called a: ☐

A) chord B) tangent
C) diameter D) radial segment

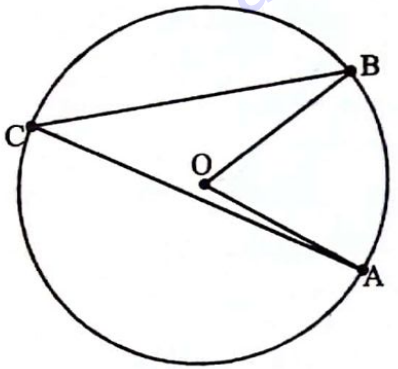
25. $\overline{BC}$ is called a: ☐

A) secant B) tangent
C) radial segment D) diameter

26. which of following points are called conayelic points? ☐

A) O, F, G, H B) A, B, C, E
C) G, H, E, F D) E, B, O, C

Note: Answer Q. 27 to Q. 29 with reference to given figure.



27. _____ is a central angle. ☐

A) $\angle ACB$ B) $\angle AOB$
C) $\angle BOC$ D) $\angle OAB$

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28. _____ is the inscribed angle.

A) $\angle ACB$ B) $\angle AOB$

C) $\angle BOC$ D) $\angle OAB$

29. Which of the following is correct?.

A) $m\angle ACB = m\angle CBA$

B) $m\angle ACB = 2m\angle AOB$

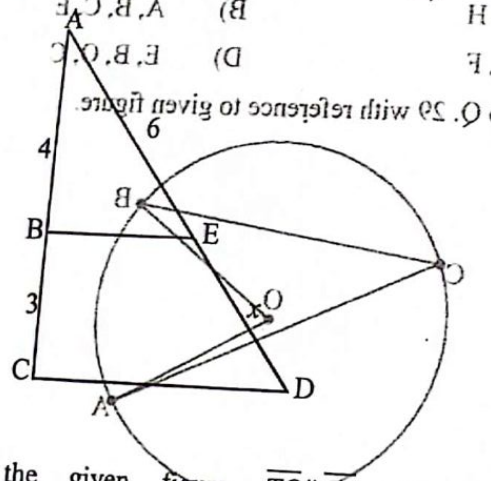
C) $m\angle AOB = m\angle AOC + m\angle BOC$

D) $m\angle AOB = 2m\angle ACB$

30. In the given figure $\overline{BE} \parallel \overline{CD}$. The length of $\overline{AD}$ is:

A) 7 B) 10

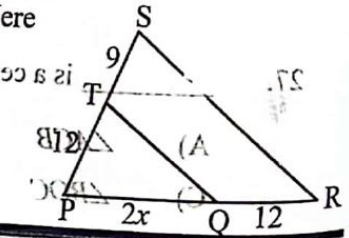
C) 10.5 D) 12



31. In the given figure $\overline{TQ} \parallel \overline{SR}$. Here x= _____.

A) 8 B) 16

C) 28 D) 9



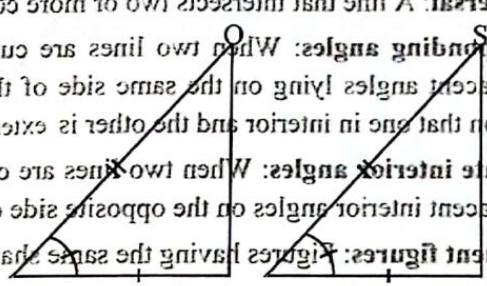
32. In the given figure $\Delta ONE \cong \Delta STA$. Because

A) $SSS \cong SSS$

B) $ASA \cong ASA$

C) $SAS \cong SAS$

D) $RHS \cong RHS$



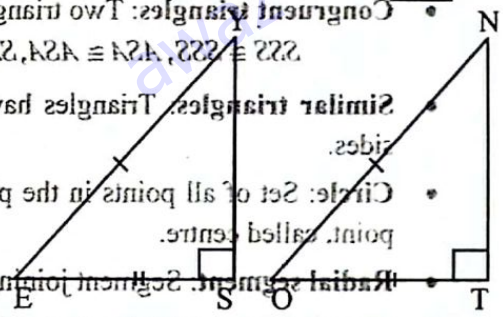
33. In the given figure $\Delta\ YES \cong \Delta NOT$. Because

A) $RH \cong RH$

B) $SAS \cong SAS$

C) $SSS \cong SSS$

D) $ASA \cong ASA$



Glossary

- **Adjacent Angles:** Two angles having a common vertex, a common arm and an empty set as the intersection of their interiors.
- **Complementary angles:** two angles having a sum of their measurement as 90°.
- **Supplementary angles:** Two angles having a sum of their measurement as 180°.

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- **Vertically opposite angles:** Non adjacent angles formed by intersectio of two lines.
- **Co-planer lines**: Lines lying in the same plane.
- **Parallel lines**: Coplaner lines that remain at the same distance from each other.
- **Transversal**: A line that intersects two or more coplaner lines.
- **Corresponding angles**: When two lines are cut by a transversal, the non-adjacent angles lying on the same side of the transversal with the condition that one in interior and the other is exterior.
- **Alternate interior angles**: When two lines are cut by a transversal, the non-adjacent interior angles on the opposite side of the transversal.
- **Congruent figures**: Figures having the same shape and same size.
- **Similar figures**: Figures having the same shape but not necessarily the same shape.
- **Congruent triangles**: Two triangles are congruent if:

  $SSS \cong SSS, ASA \cong ASA, SAS \cong SAS, RH \cong RH$
- **Similar triangles**: Triangles having the same angles and proportional sides.
- **Circle**: Set of all points in the plane those are equidistant from a fixed point, called centre.
- **Radial segment**: Segment joining the centre to any point on the circle.
- **Chord**: Segment joining two points of circle.
- **Diameter**: A chord passing through the centre of the circle.
- **Minor arc**: The shorter covered distance between two points taken on the circle.
- **Major arc**: The longer covered distance between two points taken arc on the circle.
- **Minor segment**: The region bounded by a chord and the minor arc.

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Unit 8

PRACTICAL GEOMETRY

In this unit, the students will be able to learn about the:

- Construct a triangle when:
  - Two sides and the included angle are given
  - One side and two of the angles are given
  - Two of its sides and the angle opposite to one of them (with all three possibilities
- Draw:
  - Angle bisectors
  - Altitudes
  - Perpendicular bisectors
  - Medians of given triangle and verify their concurrency
- Construct a rectangle when:
  - Two sides are given
  - Diagonal and one side are given
- Construct a square when its diagonal is given.
- Construct a parallelogram when two adjacent sides and the angle included between them is given.
- Locate the centre of a given circle
- Draw a circle passing through three give non-collinear points
- Draw a tangent to a given circle from a point p when p lies:
  - On the circumference
  - Outside the circle
- Draw:
  - Direct common tangent or external tangent
  - Transverse common tangent or internal tangent to two equal circles and to two unequal circles.
- Draw a tangent to:
  - Two unequal touching circles
  - Two interesting circles

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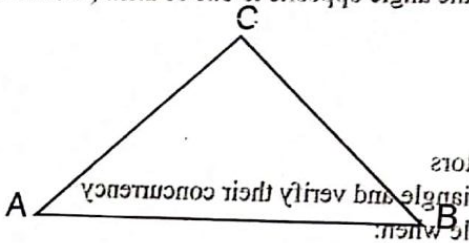
8.1 Introduction

Practical geometry helps us in understanding the graphical views of standard theorems. In lower classes the students have learnt about geometrical figures with the help of compass and ruler that are the constructor of some angles of specific measures bisection of line segment and angle, right bisector of line segment.

8.1.1 Construction of triangle

The steps related to construct a triangle are the following:



A triangle consists of three sides and three angles : In the given figure, the sides are $\overline{AB}$, $\overline{BC}$, $\overline{CA}$ and the angles are $\angle A$, $\angle B$ and $\angle C$.

iii) A, B and C are called the vertices of a ΔABC.

iv) The sum of the measures of the three angles of a triangle is always equal to 180°.

v) The sum of the length of two sides of a triangle is always greater than the length of its third side.

Following examples will elaborate the process of construction.

A triangle can be constructed, if

i) The two sides and the included angles are available.

ii) One side and two angles are available.

iii) Two sides and the angle opposite to one of them are available.

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Case 1: When two sides and the included angle are given

Example 1

Construct a triangle ABC when $m\overline{AB} = 5$cm, $m\overline{BC} = 3.5$cm and $\angle B = 60°$.

Steps of construction

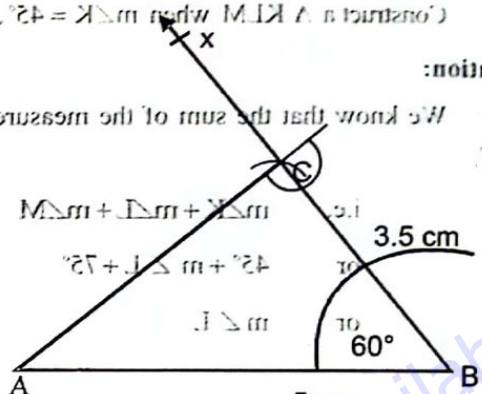
i. Draw $\overline{AB}$ measuring 5 cm.

ii. At vertex B, construct an angle of 60°.

iii. With B as centre draw an arc of radius 3.5 cm to cut $\overline{BX}$ at C.

iv. Join C to A.

v. The resultant figure is the required triangle ΔABC.



Case 2: When one side and two of the angles are given

Example 2

Construct a ΔXYZ when $m\angle X = 30°$, $m\angle Y = 45°$ and $m\overline{XY} = 4.5$cm.

Steps of construction

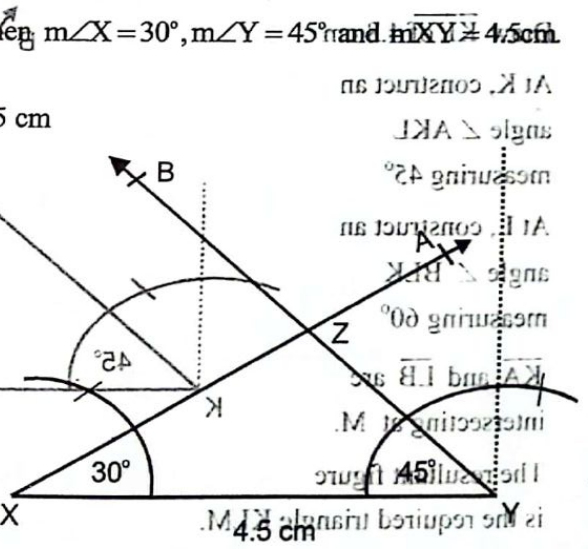
i. Draw $\overline{XY}$, which is 4.5 cm

ii. At X, construct an angle $\angle AXY$ of 30°

iii. At Y, construct an angle $\angle BYX$ of 45°.

iv. $\overline{XA}$ and $\overline{YB}$ are intersecting at Z.

v. The resultant figure is the required triangle XYZ.



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Example 3

Construct a Δ KLM when $m\angle K = 45°, m\angle M = 75°$ and $m\overline{KL} = 4.8cm$.

Solution:

We know that the sum of the measures of three angles of a triangle is 180°.

i.e. $m\angle K + m\angle L + m\angle M = 180°$

or $45° + m\angle L + 75° = 180°$

or $m\angle L = 180° - (45° + 75°)$

$= 180° - 120°$

$= 60°$

We have to construct $\angle KLM$ when $m\angle K = 45°$ and $m\angle L = 60°$.

Steps of construction

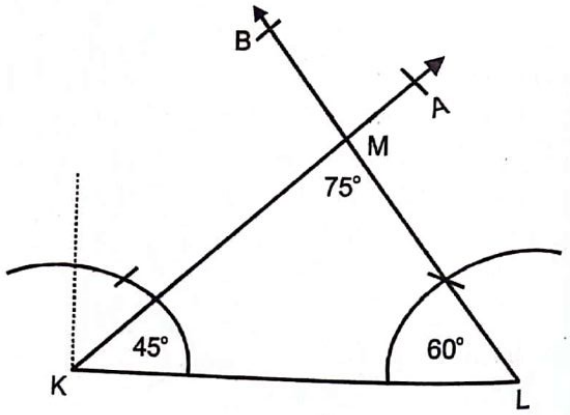
i. Draw $\overline{KL}$ of 4.8 cm

ii. At K, construct an angle $\angle AKL$ measuring 45°

iii. At L, construct an angle $\angle BLK$ measuring 60°

iv. $\overline{KA}$ and $\overline{LB}$ are intersecting at M.

v. The resultant figure is the required triangle KLM.



4.

Case 3: When two sides and the angle opposite to one of them are given

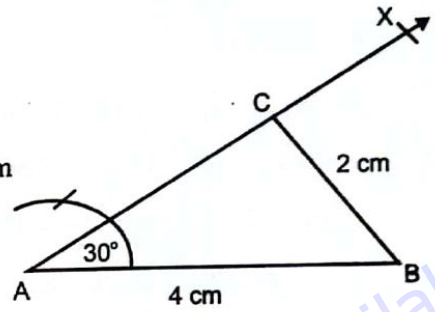
Example 4

Construct a Δ ABC when $m\overline{BC} = 2cm$, $m\overline{AB} = 4cm$ and $m\angle A = 30°$.

i. Draw $\overline{AB}$ of 4 cm.

ii. At A, construct an angle BAX of 30°.

iii. With B as centre draw an arc of 2cm that equal the length of $\overline{BC}$; $\overline{BC}$ intersects $\overline{AX}$ at only one point C.



iv. Join C to B.

v. The resultant figure is the required triangle ABC.

Example 5

Construct Δ KLM when;

$m\overline{KL} = 5.3cm$, $m\overline{LM} = 6cm$ and $m\angle M = 45°$.

Steps of construction:

i. Draw $\overline{LM}$ of 6 cm.

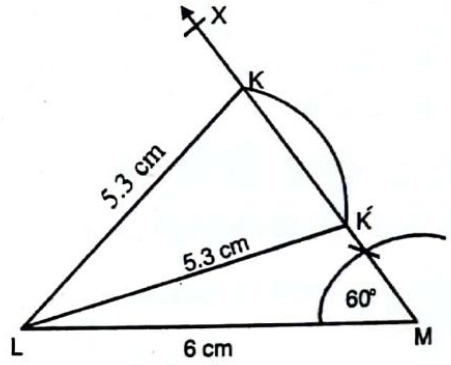
ii. Construct $\angle M$ of 45°.

iii. With L as centre, draw an arc of radius 5.3 cm which intersects $\overline{MX}$ at two points K and K′

iv. Join K and K′ to L.

v. Two triangles KLM and K′LM are obtained, which satisfy the given conditions.

vi. The resultant figure Δ KLM and ΔK′LM are the required triangles.



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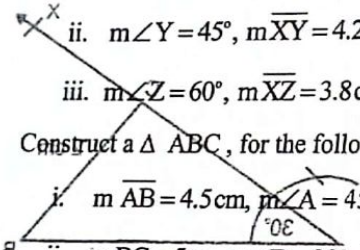
Exercise 8.1

1. Construct a Δ XYZ, for the following assumptions:
   i. $m\angle X = 30°$, $m\overline{XY} = 3.5$ cm, and $m\overline{XZ} = 4$ cm
   ii. $m\angle Y = 45°$, $m\overline{XY} = 4.2$ cm, and $m\overline{YZ} = 4.5$ cm
   iii. $m\angle Z = 60°$, $m\overline{XZ} = 3.8$ cm, and $m\overline{YZ} = 4.4$ cm



2. Construct a Δ ABC, for the following assumptions:
   i. $m\overline{AB} = 4.5$ cm, $m\angle A = 45°$, and $m\angle B = 60°$
   ii. $m\overline{BC} = 5$ cm, $m\angle B = 30°$, and $m\angle C = 75°$
   iii. $m\overline{AC} = 4.8$ cm, $m\angle A = 120°$ cm, and $m\angle C = 30°$

3. Construct a Δ KLM, for the following assumptions:
   i. $m\overline{KL} = 4.8$ cm, $m\angle K = 45°$, and $m\angle M = 60°$
   ii. $m\overline{LM} = 3.8$ cm, $m\angle K = 30°$, and $m\angle M = 75°$
   iii. $m\overline{KM} = 5$ cm, $m\angle K = 105°$, and $m\angle L = 45°$

4. Construct a ΔABC (whenever possible), for the following assumptions:
   i. $m\angle A = 30°$, $m\overline{BC} = 2.5$ cm, and $m\overline{AC} = 5$ cm
   ii. $m\angle B = 30°$, $m\overline{AB} = 8$ cm, and $m\overline{AC} = 4$ cm
   iii. $m\overline{AB} = 7$ cm, $m\overline{AC} = 5.5$ cm, and $m\angle B = 45°$
   iv. $m\overline{AB} = 6$ cm, $m\overline{AC} = 5.6$ cm, and $m\angle B = 60°$
   v. $m\overline{CA} = 4.5$ cm, $m\overline{BC} = 7.5$ cm, and $m\angle B = 45°$
   vi. $m\overline{AB} = 6$ cm, $m\overline{AC} = 2.5$ cm, and $m\angle A = 60°$

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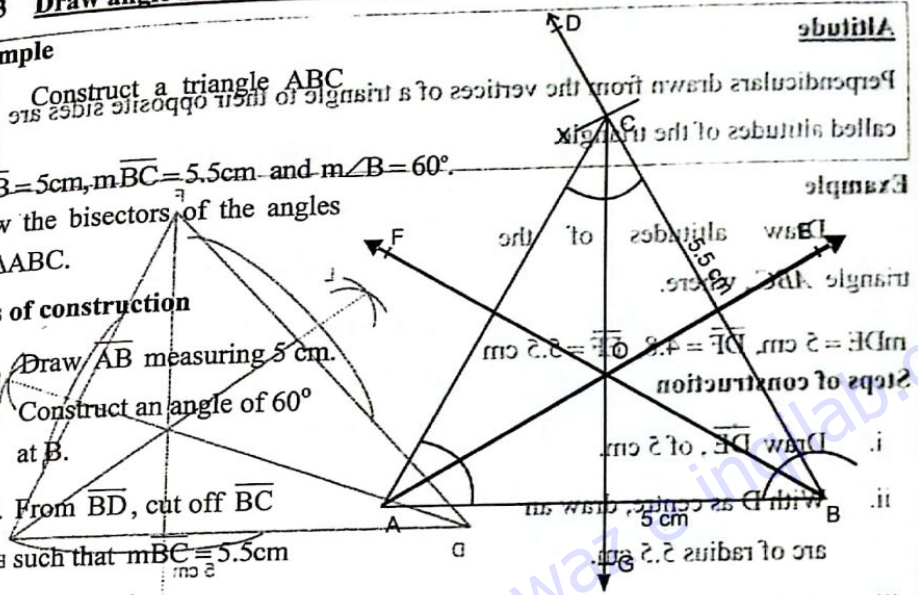
8.1.3 Draw angle bisectors of a triangle

Example

Construct a triangle ABC with $m\overline{AB} = 5$ cm, $m\overline{BC} = 5.5$ cm and $m\angle B = 60°$. Draw the bisectors of the angles of a ΔABC.



Steps of construction

i. Draw $\overline{AB}$ measuring 5 cm.
ii. Construct an angle of 60° at B.
iii. From $\overline{BD}$, cut off $\overline{BC}$ such that $m\overline{BC} = 5.5$ cm
iv. Join C to A.
v. We get Δ ABC.
vi. Draw the bisector $\overline{AE}, \overline{BF}$ and $\overline{CG}$ of ∠A, ∠B and ∠C respectively (i.e. the three angles of the triangle)
vii. The bisectors intersect each other at O.

We see that the bisectors of the angles of Δ ABC pass through one point i.e. O. in other words we can say that the bisectors of the angles of a triangle are concurrent.

Note: point of intersection of the angle bisectors is equidistant from the sides of the triangle.

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8.1.4 To draw the altitudes of a triangle

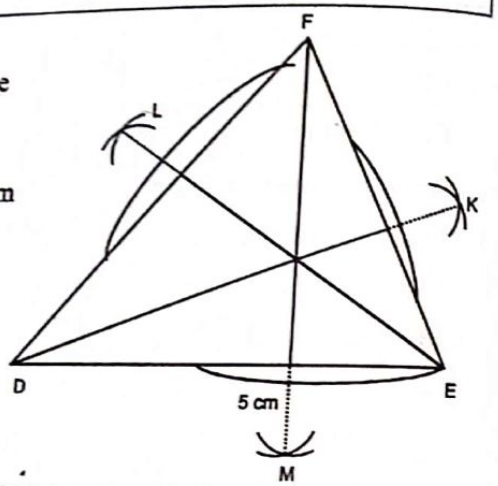
Altitude

Perpendiculars drawn from the vertices of a triangle to their opposite sides are called altitudes of the triangle.

Example

Draw altitudes of the triangle ABC, where.

$m\overline{DE} = 5$ cm, $\overline{DF} = 4.8$, $\overline{EF} = 5.5$ cm



Steps of construction

i. Draw $\overline{DE}$, of 5 cm.

ii. With D as centre, draw an arc of radius 5.5 cm.

iii. With E as centre, draw another arc of radius 4.8 cm which intersects the first arc at F.

iv. Join F to D and E.

v. We get Δ DEF.

vi. From D, E and F draw perpendiculars on the opposite sides.

vii. Here $\overline{DK} \perp \overline{EF}$, $\overline{EL} \perp \overline{DF}$ and $\overline{FM} \perp \overline{DE}$.

viii. Hence $\overline{DK}, \overline{EL}$ and $\overline{FM}$ are the required altitudes.

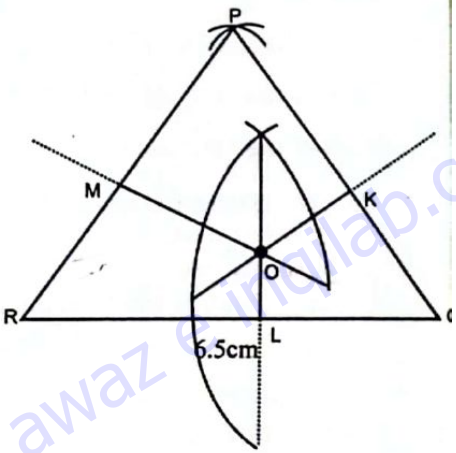
We observe that the three altitudes intersect each other at O, which shows that the altitudes of a triangle are concurrent.

8.1.5 To draw perpendicular bisectors of a triangle

Example

Construct a triangle PQR in which $m\overline{PQ} = 6$cm, $m\overline{QR} = 6.5$cm and $m\overline{PR} = 5.5$cm. Draw its perpendicular bisectors and verify their concurrency.

Steps of construction



i. Draw QR of 6.5 cm.

ii. With Q and R as centre draw arcs of radius 6 cm and 5.5 cm respectively that intersect each other at P.

iii. Join P to Q and R to get a Δ PQR.

iv. Draw the perpendicular bisectors of sides $\overline{PQ}$, $\overline{QR}$ and $\overline{PR}$ which meet at O.

v. $\overline{OK}$, $\overline{OL}$ and $\overline{OM}$ are the required perpendicular bisectors of a ΔPQR.

We observe that the perpendicular bisectors of the sides of a triangle are concurrent.

Note:

Point of intersection of the perpendicular bisectors is equidistant from the vertices of the triangle.

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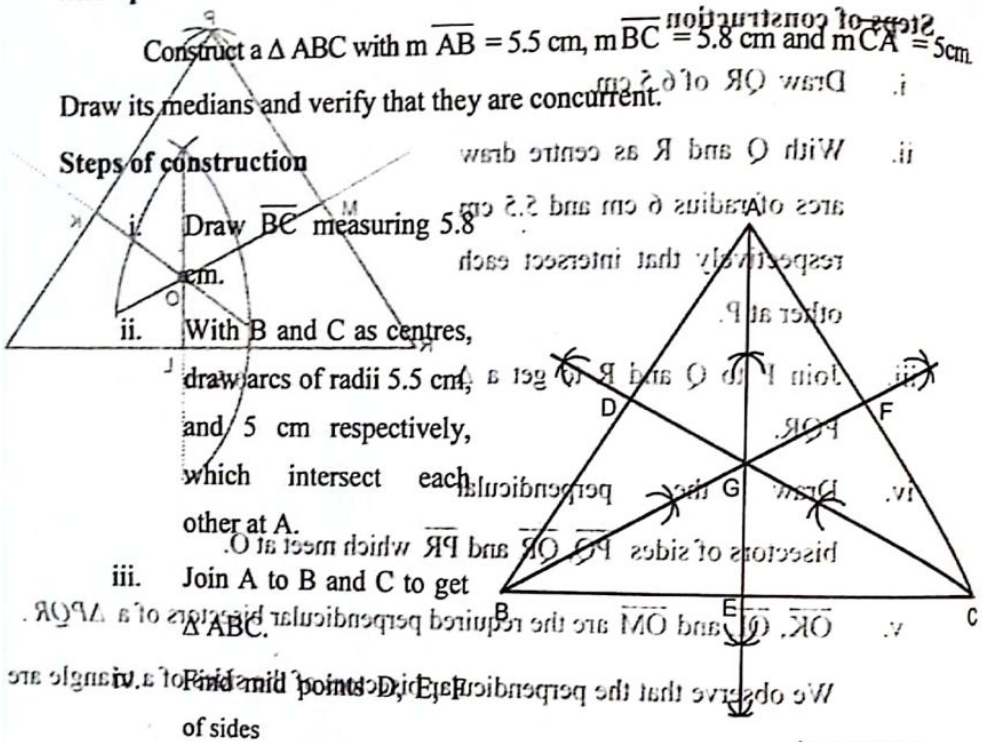
8.2.4 To draw medians of a triangle

Median

Line segments joining the mid points of the sides of a triangle to their opposite vertices are called Medians of the triangle.

Example

Construct a Δ ABC with m $\overline{AB}$ = 5.5 cm, m $\overline{BC}$ = 5.8 cm and m $\overline{CA}$ = 5cm. Draw its medians and verify that they are concurrent.



Steps of construction

i. Draw $\overline{BC}$ measuring 5.8 cm.

ii. With B and C as centres, draw arcs of radii 5.5 cm, and 5 cm respectively, which intersect each other at A.

iii. Join A to B and C to get Δ ABC.

iv. Find mid points D, E, F of sides $\overline{AB}, \overline{BC}$ and $\overline{CA}$ respectively.

v. Join E to A, F to B and D to C.

vi. $\overline{AE}, \overline{BF}$ and $\overline{CD}$ are the required medians.

We observe that all the three medians pass through a point G which verifies the concurrency of the medians of a triangle.

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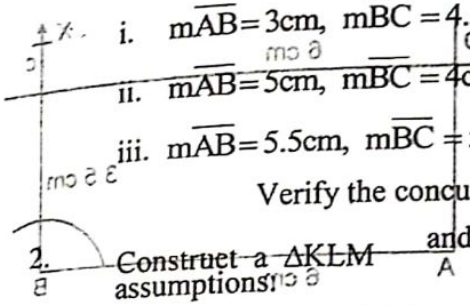
Exercise 8.2

1. Construct a ΔABC and draw their angle bisectors, for the following assumptions:



i. m$\overline{AB}$ = 3cm, m$\overline{BC}$ = 4.5cm and m$\overline{AC}$ = 4cm

ii. m$\overline{AB}$ = 5cm, m$\overline{BC}$ = 4cm and m∠B = 60°

iii. m$\overline{AB}$ = 5.5cm, m$\overline{BC}$ = 5cm and m$\overline{AC}$ = 6cm

Verify the concurrency of these angle bisectors.

2. Construct a ΔKLM and draw their altitudes, for the following assumptions:

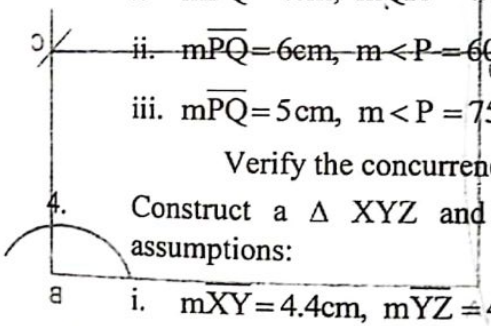
i. m$\overline{KL}$ = 4cm, m$\overline{LM}$ = 3.8cm and m$\overline{KM}$ = 4.4cm

ii. m$\overline{KL}$ = 5cm, m∠K = 60° and m∠L = 75°

iii. m$\overline{KL}$ = 4.8cm, m∠K = 75° and m$\overline{KM}$ = 5cm

Verify the concurrency of these altitudes.

3. Construct a Δ PQR and draw their perpendicular bisectors, for the following assumptions:



i. m$\overline{PQ}$ = 4cm, m$\overline{QR}$ = 4.6cm and m$\overline{RP}$ = 5cm

ii. m$\overline{PQ}$ = 6cm, m∠P = 60° and m∠Q = 45°

iii. m$\overline{PQ}$ = 5cm, m∠P = 75° and m$\overline{PR}$ = 4.6cm

Verify the concurrency of these perpendicular bisectors.

4. Construct a Δ XYZ and draw their medians, for the following assumptions:

i. m$\overline{XY}$ = 4.4cm, m$\overline{YZ}$ = 4cm and m$\overline{XZ}$ = 5.5cm

ii. m$\overline{XY}$ = 5cm, m∠X = 60° and m∠Y = 75°

iii. m$\overline{XY}$ = 6cm, m∠X = 45° and m$\overline{XZ}$ = 5.5cm

Verify the concurrency of these medians.

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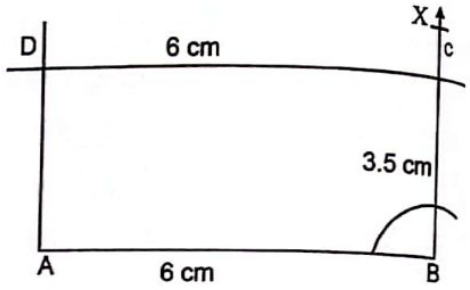
8.2 Construction of quadrilateral

Example 1

Construct a rectangle ABCD, when $m\overline{AB}=6cm$ and $m\overline{BC}=3.5cm$

Steps of construction



i. Draw $\overline{AB}$ measuring 6 cm.

ii. At B draw $\overline{XB} \perp \overline{AB}$

iii. Cut off $\overline{BC}$, such that $m\overline{BC}=3.5cm$.

iv. With A and C as centers draw arcs of radii 3.5 cm and 6 cm respectively to meet each other at D.

v. Join D to A and C.

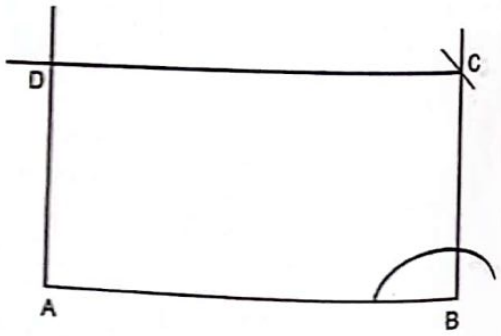
vi. ABCD is the required rectangle

8.2.1 To construct a rectangle when its diagonal and one side are given

Example

Construct a rectangle ABCD when $m\overline{AB}=5m$ and $m\overline{AC}=6cm$.

Steps of construction



i. Draw $\overline{AB}$ measuring 5 cm.

ii. At B, draw an angle of 90°.

iii. With A as centre draw an arc of 6 cm to cut $\overline{BX}$ at C.

iv. From C draw an arc of radius 5cm.

v. With A as centre draw another arc of radius = $m\overline{BC}$ to cut the first arc at D.

vi. Join D to A and C.

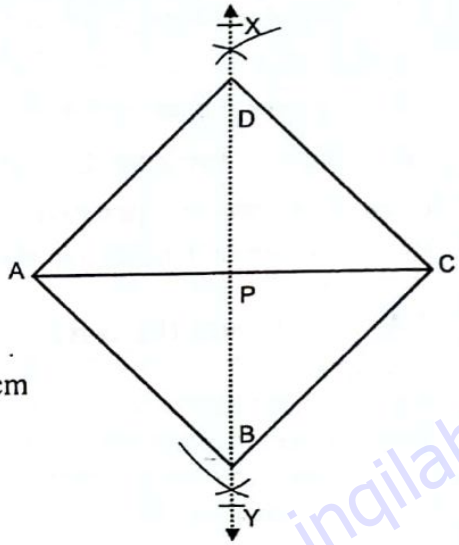
vii. ABCD is the required rectangle.

8.2.2 To construct a square when its diagonal is given

Example

Construct a square ABCD when $m\overline{AC}=5cm$.

Steps of construction



i. Draw $\overline{AC}$ measuring 5 cm.

ii. Draw the perpendicular bisector $\overline{XY}$.

iii. Cut off $\overline{PD}$ and $\overline{PB}$ measuring 2.5 cm i.e. half of $\overline{AC}$.

iv. Join D to A and C.

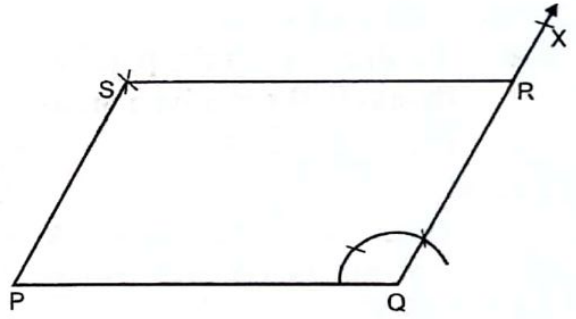
v. Also join B to A and C.

vi. ABCD is the required square.

8.2.3 To construct a parallelogram when two adjacent sides and the angle included between them are given.

Example

Construct a parallelogram PQRS when $m\overline{PQ}=4cm$, $m\overline{QR}=2.5cm$ and $m\angle Q$



i. Draw $\overline{PQ}$ measuring 4 cm.

ii. At Q, draw $\angle PQX$ measuring 120°.

iii. From $\overline{QX}$, cut off $\overline{QR}$ such that $m\overline{QR}=2.5cm$.

iv. With P and R as centers, draw arc of radii 2.5 cm and 4 cm respectively to cut each other at S.

v. Join S to P and R.

vi. PQRS is the required parallelogram.

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Exercise 8.3

1. Construct a rectangle ABCD for the following assumptions:
   i. m $\overline{AB} = 3.8cm$ and $m\overline{BC} = 2cm$
   ii. $m\overline{AB} = 4.5cm$ and $m\overline{AD} = 2.5cm$
   iii. $m\overline{BC} = 3cm$ and $m\overline{CD} = 5cm$
2. Construct a square ABCD with $m\overline{AB} = 4cm$ $m\angle B = 90°$:
3. Construct a parallelogram ABCD with $m\overline{AB} = 4cm$ and m BC = 3cm $m\angle B = 60°$:

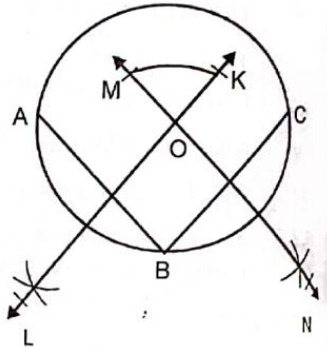
8.3 To locate the centre of a given circle

Example

Locate the centre of a circle.

Steps of construction (Method – 2)

i. Take any three points A, B and C on the given circle.
ii. Draw the chords $\overline{AB}$ and $\overline{BC}$.
iii. Draw the perpendicular bisectors $\overline{KL}$ and $\overline{MN}$ of $\overline{AB}$ and $\overline{BC}$ respectively to meet each other at O.
iv. O is the required centre of the given circle.



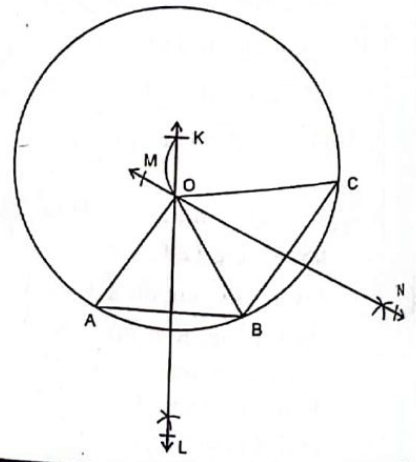
8.4 To draw a circle passing through three given non-collinear points

Example

Three non-collinear points A, B, C are given. Draw a circle passing through the given non-collinear points A, B and C.

Steps of construction

i. Join B to A and C.
ii. Draw the perpendicular bisectors $\overline{KL}$ and $\overline{MN}$ of $\overline{AB}$ and $\overline{BC}$ respectively to meet each other at O.
iii. O is the centre of the required circle.
iv. With O as centre, draw circle with radius equal to $m\overline{OA}$ or $m\overline{OB}$ or $\overline{OC}$.
v. This is the required circle passing through the given points A, B and C.



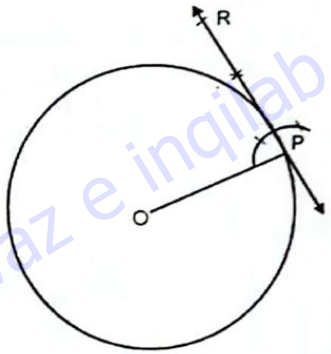
8.5.1 (i) To draw a tangent to a given circle at a point which lies on the circumference.

Example

The radius of a circle is 2.5 cm. Take a point P on the circumference of the circle. Draw a tangent to the circle at P.

Steps of construction

i. Draw a circle with centre O, and a radius of 2.5 cm.
ii. Take a point P on its circumference and join it to O.
iii. At the other end of $\overline{OP}$ draw $\overline{RP} \perp \overline{OP}$
iv. $\overline{RP}$ is the required tangent to the circle.



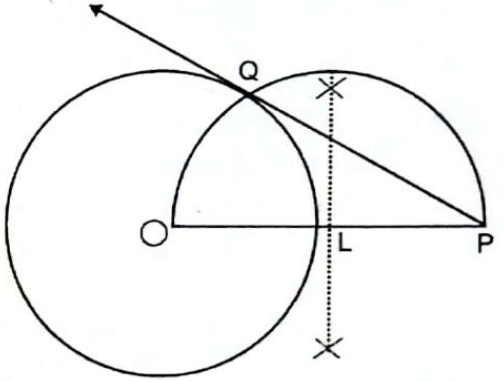
8.5.2 (ii) To draw a tangent to a given circle from a point which lies outside the circle

Example

The radius of a circle is 2 cm. Draw a tangent to the circle from a point P which is at a distance of 4.5 cm from the centre of the circle.

Steps of construction

i. Draw a circle with centre O and a radius of 2 cm.
ii. Take a point P outside the circle such that $m\overline{OP} = 4.5cm$.



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iii. Bisect $\overline{OP}$ at L.

iv. With L as center draw, a semi-circle with radius $m\overline{OL}, or\ m\overline{LP}$ to intersect the circle at Q.

v. Draw $\overline{PQ}$.

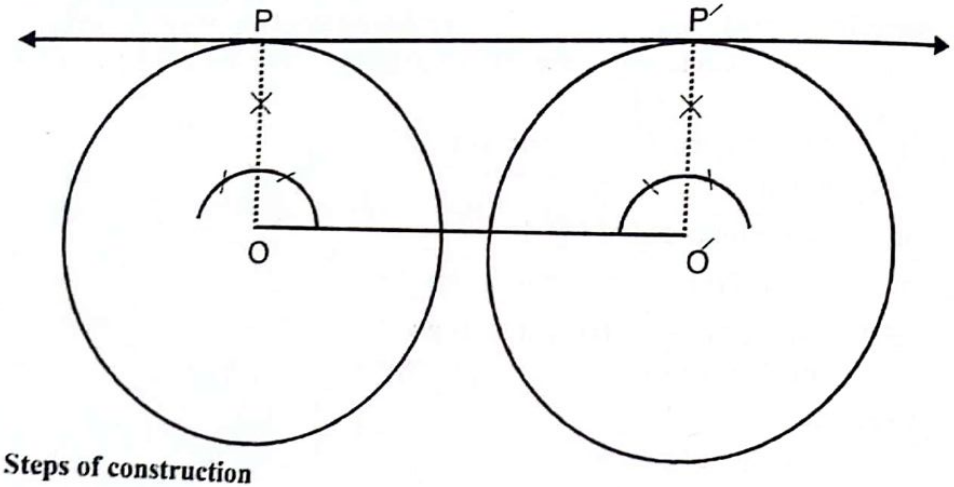
vi. $\overline{PQ}$ is the required tangent.

Note: Two tangents can be drawn to a circle from a point outside the circle.

8.6.1 To draw direct common tangent to two equal circles

Example

Draw direct common tangent to two circles having radii of 3 cm each and the distance between their centers is 7 cm.



Steps of construction

i. Draw $\overline{OO'}$ measuring 7 cm.

ii. With O and O′ as centres draw circles of radius 3 cm each.

iii. Draw $\overline{OP} \perp \overline{OO'}$ and $\overline{O'P'} \perp \overline{OO'}$ that intersect the circles at P and P' respectively.

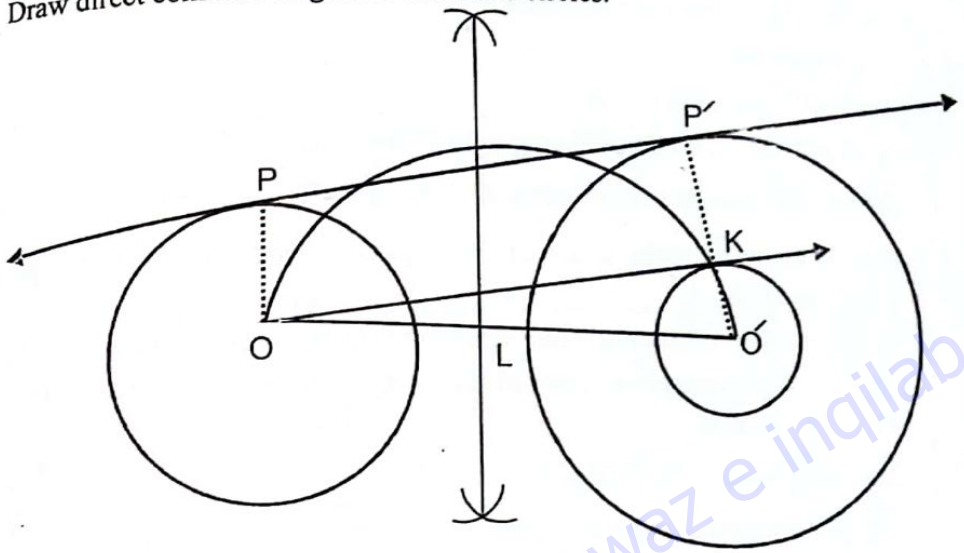
iv. Draw $\overline{PP'}$, which represents direct common tangent to the given circles.

Note: the same process can be extended to other side to draw second direct common tangent.

8.6.2 To draw direct common tangent to two unequal circles

Example

Two circles have their radii 2 cm and 3 cm. their centers are 6 cm apart. Draw direct common tangent to these two circles.



Steps of construction

i. Draw $\overline{OO'}$ such that $m\overline{OO'} = 6cm.$

ii. Draw circles of radii 2 cm and 3 cm with centers O and O′ respectively.

iii. With centre O′, draw a third circle of radius equal to the difference of the radii of the given circles i.e. $3-2=1$ cm.

iv. Take L is the midpoint of $\overline{OO'}$.

v. With L as centre draw a semicircle of radius = $m\overline{LO}\ or\ m\overline{LO'}$ that intersect the third circle at K.

vi. Draw $\overline{O'K}$ to cut the bigger circle at P'.

vii. Through O, draw $\overline{OP} \,||\, \overline{O'P'}$.

viii. Draw $\overline{PP'}$, which represents the direct common tangent to the given circles.

Note: (i) direct common tangent is also called external tangent.

(ii) Two direct common tangents can be drawn by extending the process to the other side of $\overline{OO'}$.

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8.6.3 To draw transverse common tangent to two equal circles

Example

Centers of two circles having radii 2.5 cm and 7 cm apart. Draw transverse common tangent to these circles.

Steps of construction

i. Draw $\overline{OO'}$ measuring 7 cm.

ii. At O and O' draw circles having radii 2.5 cm each.

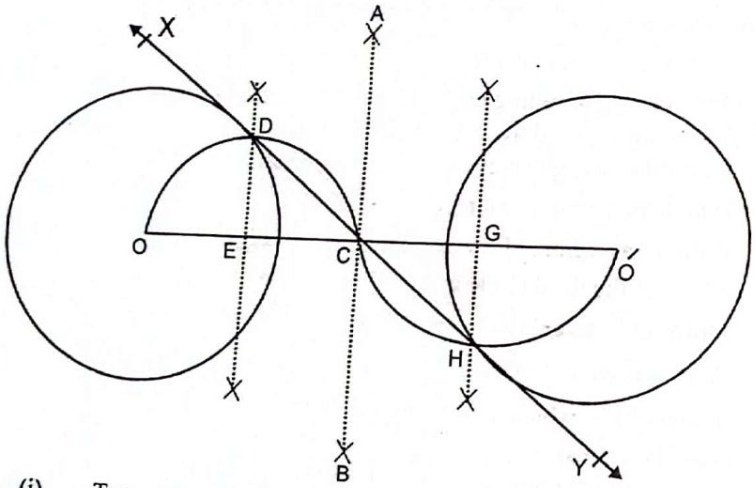
iii. Divide $\overline{OO'}$ into four equal segments, $\overline{OE}, \overline{EC}, \overline{CG}$ *and* $\overline{GO'}$.

iv. With E as centre, draw a semicircle of radius $= m\overline{EO}$ *or* $m\overline{EC}$ to cut one circle at D.

v. With G as centre, draw another semicircle (in the opposite direction of the semi circle already drawn) of radius = $m\overline{GC}$ *or* $m\overline{GO'}$ to cut the second circle at H.

vi. Draw $\overline{XY}$ through D, C and H.

vii. $\overline{XY}$ is the required transverse common tangent.



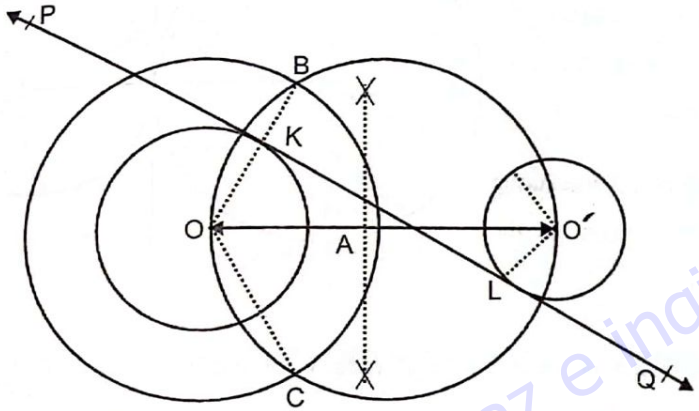
Note: (i) Transverse common tangent is also called internal tangent.

(ii) Two transverse common tangents can be drawn to two equal circle.

8.6.4 Draw transverse common tangent to two unequal circles

Example

Centers of two circles having radii 2 cm and 1 are 5.5 cm apart. Draw their transverse common tangent.



Steps of construction

i. Draw $\overline{OO'}$ measuring 5.5 cm.

ii. Draw circles with O and O' as centers and radius 2 cm and 1 cm respectively.

iii. Locate the midpoint of $\overline{OO'}$ at A.

iv. With O as centre draw the third circle having radius equal to the sum of radii of two given circle i.e. $2+1=3cm$.

v. With A as centre and radius $= m\overline{AO}$ *or* $m\overline{AO'}$, draw a circle which cuts the third circle at B and C.

vi. Draw $\overline{OB}$ to cut the first circle at K.

vii. Draw $\overline{O'L} \parallel \overline{OK}$.

viii. Through K and L, draw $\overline{PQ}$.

ix. $\overline{PQ}$ is the required transverse common tangent to the given circles.

Note: Two transverse common tangents can be drawn to two unequal circles.

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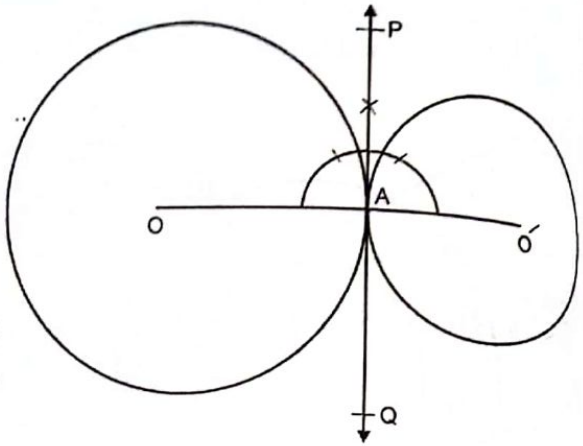
8.7.1 To draw a tangent to two unequal touching circles

Example

Two circles of radii 3cm and 2cm are touching each other. Draw a tangent to the circles.



i) Draw $\overline{OO'}$ measuring 5cm (= sum of the radii).

ii. At O and O' draw circles of radii 3 cm and 2 cm respectively, which touch each other at A.

iii. At A, draw $\overline{PA} \perp \overline{OO'}$.

iv. $\overline{PQ}$ is the required tangent.

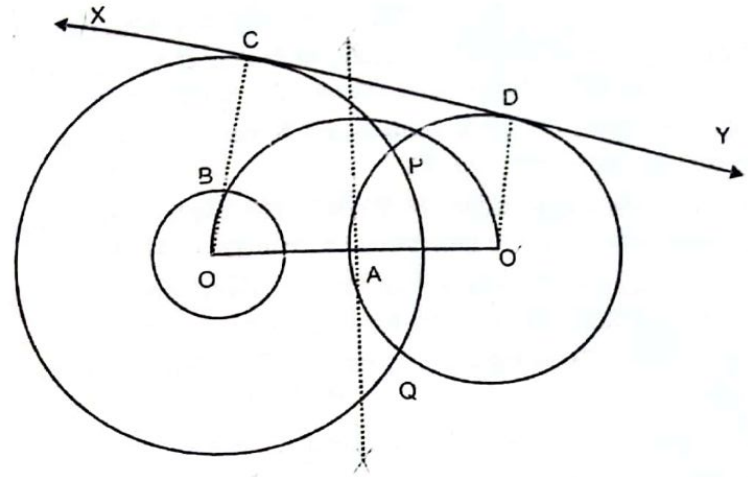
8.7.2 To draw tangent to two intersecting circles

Example

Two circles of radii 3 cm and 2 cm intersect each other at P and Q. Draw a tangent to these circles.

Steps of construction

i. Draw $\overline{OO}$ of measure 3.6 cm and draw two circles with centers O and O', of radii 3 cm and 2 cm respectively which intersect each other at P and Q.



ii. With centre O, draw third circle of radius equal to the difference of the radii of the given circles i.e. $3-2=1$cm.

iii. Bisect $\overline{OO'}$ at A and draw a semicircle with centre A and radius equal to $m\overline{AO}$ or $m\overline{AO'}$ to cut the third circle at B.

iv. Extend $\overrightarrow{OB}$ to cut the first circle at C.

v. Draw $\overline{O'D} \parallel \overline{OC}$.

vi. Draw $\overline{CD}$.

vii. $\overline{XY}$ is the required tangent to the given circles.

Note: Two tangents can be drawn in this case by extending the process to the other side of $\overline{OO'}$.

Exercise 8.4

1. Draw a circle with the help of a round object. Locate its centre.
2. Draw a circle which passes through non-collinear points A, B and C.
3. Draw a circle of radius 2 cm. Take a point on the circumference of the circle and draw a tangent to the circle at that point.

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4. Draw a circle of radius 2.5 cm. Take a point P which is at a distance of 3 cm from the centre of the circle. Draw a tangent to the circle from point P.
5. Draw two circles of radii of 2.5 cm each, such that the distance between their centers is 8 cm. Draw direct common tangent to the circles.
6. Centers of two circles of radii 2.5 cm and 1.5 cm are 6 cm apart. Draw direct common tangent to these circles.
7. Two circles have the same radius of 2 cm each. Their centers are 6 cm apart. Draw transverse common tangent to these circles.
8. Draw transverse common tangent to two circles having radii 2.5 cm and 1.5 cm respectively and their centers are 7.5 cm apart.
9. Draw a tangent to the circles radii 1.5 cm and 2.5 cm that are touching each other externally.
10. Two circles of radii 2.5 cm and 2 cm intersect each other at two points. Draw a tangent to these circles.

Revision Exercise 8

1. Which of the following statements are true and which are false. Encircle T in case the statement in true and F for being false.
   i. The point of intersection of the altitudes of any triangle is equidistant from its vertices. /F
   ii. If measures of three line segments are given, it is not necessary that a triangle be constructed from them. /F
   iii. If measures of two angles and one side are given, we can draw a triangle. /F
   iv. Only one tangent can be drawn to circle from a point which lies on its circumference. /F
2. Fill in the blanks.
   i. Line segment joining the vertex of a triangle to the midpoint of its opposite side is called a _____.
   ii. A line which passes through the midpoint of a line segment and is also perpendicular to it, is called its _____
   iii. Vertices of a triangle are _____ from the point of intersection of its perpendicular bisectors.

   iv. The point of intersection of the bisectors of three angles of a triangle is equidistant from its _____
   v. Perpendicular drawn from a vertex of a triangle to its opposite side is called _____.
   vi. If three or more lines pass through the same point, the lines are said to be the ____.
   vii. Point of intersection of the medians of a triangle divides them in the ratio_____.
   viii. Bisectors of the angles of a triangle are _____.
3. Four suggested answers are given for each question. Selection the correct answer and write the corresponding letter (a, b, c, or d) in the box provided for each question.
   (i) The point of intersection of the _____ of a triangle divides them in the ratio 2:1.
   a) perpendicular bisectors b) altitudes
   c) Medians d) all are possible
   (ii) If the measures of sides $\overline{AB}$, $\overline{BC}$ *and* $\angle B$ are given, then how many triangles can be constructed.
   a) no triangle b) only one triangle
   c) two triangles d) all are possible
   (iii) Point of intersection of the perpendicular bisectors of the sides of a triangle is equidistant from the _____ of the triangle.
   a) three vertices b) three sides
   c) both (a) and (b) d) none of the above
   (iv) Point of intersection of the bisectors of angles of a triangle is equidistant from the _____ of the triangle.
   a) three sides b) three vertices
   c) both (a) and (b) d) none of the above
   (v) The line segment joining the vertex of a triangle to the midpoint of its opposite side is called _____
   a) altitude b) bisector of angle
   c) perpendicular bisector d) medians

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(vi) perpendicular drawn from vertex of a triangle to its opposite side is called ____

a) altitude b) perpendicular bisector
c) median d) bisector of angle

(vii) a rectangle can be constructed if the measure(s) of at least _____ is/are given.

a) two opposite sides b) two non adjacent sides
c) any two sides d) one side and a diagonal

(viii) _____ tangents can be drawn to a circle from a point outside the circle

a) one b) two
c) three d) infinite

Glossary

- Only one triangle can be constructed when:
- Two sides and their included angle ; or
- One side and two of the angles are given.
- When two sides and the angle opposite to one of the sides are given, there exist three possibilities.
- Only one triangle is formed; or
- Two triangles may be formed; or
- No triangle is formed
- Angle bisectors, altitudes, perpendicular bisectors, and medians of a triangle are concurrent.
- Point of intersection of the angle bisectors of a triangle is equidistant from the sides of the triangle.
- Point of intersection of the perpendicular bisectors of a triangle is equidistant from the vertices of the triangle.
- Point of intersection of the medians of a triangle divides the medians in the ratio of 2 :1.
- Only one tangent can be drawn to a circle from a point which lies on its circumference.
- Two tangents can be drawn to a circle from a point which lies outside the circle.

Unit 9

AREAS AND VOLUME

In this unit, the students will be able to learn about the:

- State Pythagoras theorem
- Solve right angled triangle using Pythagoras theorem.
- Find the area of :
  - A triangle when three sides are given, by applying Hero's formula.
  - A triangle whose base and altitude are given.
  - An equilateral triangle when its side is given.
  - A rectangle when its two sides are given.
  - A parallelogram when base and altitude are given
  - A square when its side is given.
  - Boundary walls of a room when its length, width and height are given.
- Find the cost of turfing a square/rectangular field.
- Find the number of tiles of given dimensions, required to pave the footpath of given width carried around the outside of a rectangular plot.
- Find the area of a circle and a semi-circle when its radii are given.
- Find the area embossed by two concentric circles whose radii are given.
- Solve real life problems related to areas of triangle, rectangle, square, parallelogram and circle.
- Find the volume of:
  - A cube when its edge is given
  - A cuboid when its length, breadth and height are given
  - A right circular cylinder whose base radius and height are given
  - A right circular cone whose radius and height are given
  - A sphere and a hemisphere when radii are given.
- Solve real life problems relating to volume of cube, cuboids, cylinder, cone and sphere.

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9.1 Pythagorus Theorem

We know that a triangle having one right angle is called Right-angled triangle. The sides of right angled triangle have a special relation amongst them. This was first discovered by a Greek mathematician Pythagoras more than 2500 years ago. The theorem is, therefore, named after him and is called 'Pythagoras theorem'.

> Square of the hypotenuse in any right angled triangle equals to the sum of squares of base and altitude of the triangle, i.e.
>
> $(\text{hypotenuse})^2 = (\text{base})^2 + (\text{altitude})^2$.

Remember that the side opposite to the right angle in a right-angled triangle is called a 'hypotenuse'. The adjacent horizontal side of the right angle is called the base and vertical side is called the altitude.

Example 1

A right-angled triangle ABC with right angle at B has m $\overline{BC} = 5$ and m $\overline{AB} = 12$. Find $m\ \overline{AC}$.

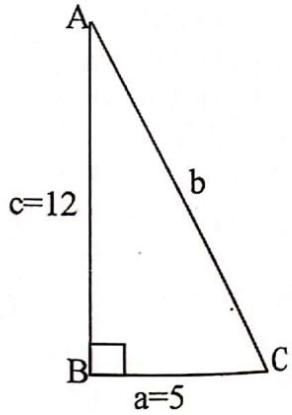
Solution:

In ΔABC, let m $\overline{BC}$ =a=5,

m $\overline{AB}$ =c=12 and m $\overline{AC}$ =b.

Pythagoras theorem gives us:

$$b^2 = a^2 + c^2$$
$$= (5)^2 + (12)^2$$
$$= 25 + 144$$
$$= 169 = (13)^2$$
$$\therefore b = 13$$



Example 2

In a right-angled triangle ABC, $\angle A$ is right a angle, m $\overline{BC}$ = 10 and $m\overline{AB} = 6$. Find m $\overline{AC}$.

Solution:

Let m $\overline{AB}$ =c=6, m $\overline{BC}$ =a=10

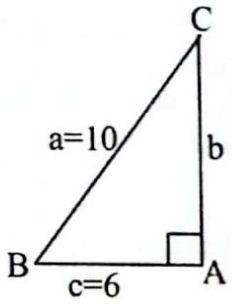
and m $\overline{AC}$ =b.

So we have $a = 10$ and $c = 6$

Pythagoras theorem gives us:

$$a^2 = b^2 + c^2$$

or, $$b^2 = a^2 - c^2$$
$$= (10)^2 - (6)^2$$
$$= 100 - 36$$
$$= 64$$
$$b = \sqrt{64}$$
$$b = 8$$
$$\therefore m\ \overline{AC} = 8$$

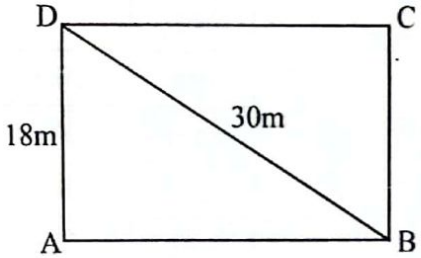


Example 3

The length of a diagonal of rectangular field is 30m. If the width of the field is 18m, then find out its length.

Solution:

We know that all the angles of a rectangle are right angles and a diagonal divides a rectangle into two right-angled triangles, having equal areas.



Let ABCD be the rectangle with

m $\overline{AB} = x$

In the right-angled triangle ABD

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$m\angle A = 90^\circ$, so the hypotenuse is $\overline{BD}$,

$m\overline{AD} = 18m$

and $m\overline{BD} = 30m$

By Pythagoras theorem

$$\left(m\,\overline{BD}\right)^2 = \left(m\,\overline{AB}\right)^2 + \left(m\,\overline{AD}\right)^2$$

or $$\left(m\,\overline{AB}\right)^2 = \left(m\,\overline{BD}\right)^2 - \left(m\,\overline{AD}\right)^2$$

$$= (30)^2 - (18)^2$$

$$= 900 - 324$$

$$= 576$$

$$\therefore\ m\overline{AB} = \sqrt{576}$$

$$= 24m$$

Hence the length of the rectangle is 24m.

Exercise 9.1

1. In ΔXYZ, $m\angle Z = 90^0$. Fill in the blanks in the following table:

	$m\overline{XY}$	$m\overline{YZ}$	$m\overline{XZ}$
(i)		28	21
(ii)		2	4.8
(iii)	6.5		3.9
(iv)	37.5		10.5
(v)	9.7	6.5	
(vi)	22.5	21.6	

2. Asad travelled a distance of 3.6 kms on his bicycle towards West. He then travelled distance of a 7.7kms towards South. Find the total distance covered from the starting point to the end point.

3. Diagonal of a rectangle is 3.25cm long. Find the width of the rectangle if its length is 3.12 cm.
4. Hypotenuse of an isosceles right-angled triangle is $7\sqrt{2}$ cm. Find the length of the congruent sides of the triangle.
5. Side of a square is 13cm long. Find the length of its diagonal.
6. A ladder 13m long rests against a wall and reaches a height of 12m on the wall. Find the distance from the foot of the ladder to the wall.
7. A right triangular shaped piece of land has a hypotenuse of 52m. Find its altitude if its base is 48m.
8. Top of a light house is 224m from the surface of the water and 280m from a boat sailing in the sea. Find the distance between the boat and the light house.

9.2.1 To find the area of a triangle by applying Hero's formula

The area of a triangle when the measures of three sides are given can be found by the Hero's formula. Area of ΔABC by Hero's formula is:

Area of a $\Delta ABC = \sqrt{s(s-a)(s-b)(s-c)}$

Where a, b, c are the lengths of the sides of the triangle and s is the semi perimeter, i.e. $s = \dfrac{a+b+c}{2}$.

Example: Find the area of a triangular region ABC when the measures of its sides are $a = 4cm,\quad b = 5cm \text{ and } c = 7cm$

Solution:

Here $a = 4cm,\quad b = 5cm,\quad c = 7cm$

Since $s = \dfrac{a+b+c}{2} = \dfrac{4+5+7}{2} = 8$

Hero's formula gives us:

Area of triangular region ABC $= \sqrt{s(s-a)(s-b)(s-c)}$

$$= \sqrt{8(8-4)(8-5)(8-7)}$$

$$= \sqrt{8\times4\times3\times1}$$

$$= \sqrt{96}$$

$$= \sqrt{16\times6} = 4\sqrt{6}\ cm^2$$

Hence the area of a triangular region ABC is $4\sqrt{6}\ cm^2$.

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9.2.2 To find the area of a triangle when base and altitude are given

The area of a triangular region when the measure of its base and the perpendicular or altitude are given, can be found with the help of formula given by:-

Area of a triangle $= \frac{1}{2}$ (base × altitude).

Example 1:

Find the area of a triangular region having base 65m and altitude 18m.

Solution:

measure of base $= 65$m, Measure of altitude $= 18$m

$\therefore$ Area of a triangular region $= \frac{1}{2}$ (base × altitude)

$= \frac{1}{2} \times 65 \times 18 = 585 m^2$

Example 2:

Area of a triangular region is $364 m^2$. Find its base when the measure of its altitude is $52 m$.

Solution:

Suppose the measure of the base is x m.

The area of triangular region is:

Area of triangular region $= \frac{1}{2}$ (base × altitude).

$$364 = \frac{1}{2}(x \times 52)$$

$$364 = \frac{52x}{2}$$

$$x = 364 \times \frac{2}{52}$$

$$x = \frac{728}{52} = x = 14$$

Hence, the base is 14m.

9.2.3 To find the area of an equilateral triangle when measure of its side is given

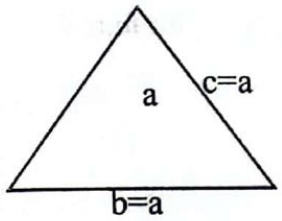
If all the three sides of a triangle are known, then the area can be found by Hero's formula we use Hero's formula in case of an equilateral triangle, all the sides are of the same length,

$$a = b = c$$

$$\therefore \quad s = \frac{a+b+c}{2}$$

$$= \frac{a+a+a}{2}$$

$$= \frac{3a}{2}$$



($\because$ all the three sides are equal)

Area of a triangle $= \sqrt{s(s-a)(s-b)(s-c)}$

$\therefore$ Area of an equilateral triangle $= \sqrt{\frac{3a}{2}\left(\frac{3a}{2}-a\right)\left(\frac{3a}{2}-a\right)\left(\frac{3a}{2}-a\right)}$

$$= \sqrt{\frac{3a}{2} \cdot \frac{a}{2} \cdot \frac{a}{2} \cdot \frac{a}{2}}$$

$$= \sqrt{\frac{3a^4}{16}}$$

$$= \frac{\sqrt{3}a^2}{4}$$

Area of an equilateral triangle $= \frac{\sqrt{3}a^2}{4}$,

Where 'a' is the length of any one side of the triangle

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Example 1

Find the area of an equilateral triangle, when the length of whose one side is 6cm.

Solution:

Here $a = 6cm$

Applying the formula

$$A = \frac{\sqrt{3}}{4}a^2$$

$$= \frac{\sqrt{3}}{4}a^2$$

$$= \frac{\sqrt{3}}{4}(6)^2 = \frac{\sqrt{3}}{4} \times 36$$

$$= \frac{\sqrt{3}}{\cancel{4}_1} \times \cancel{36}^9$$

$$= 9\sqrt{3}\ cm^2$$

Example 2

Area of an equilateral triangle is $16\sqrt{3}\ cm^2$. Find the length of its one side.

Solution:

If the

Area of an equilateral triangle is $\frac{\sqrt{3}}{4}a^2$, then:

$$16\sqrt{3} = \frac{\sqrt{3}}{4}a^2$$

or $$\frac{\sqrt{3}}{4}a^2 = 16\sqrt{3}$$

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or $$a^2 = 16\sqrt{3} \times \frac{"4}{\sqrt{3}}$$

or $$a^2 = 64$$

or $$(a)^2 = (8)^2$$

or $$a = 8$$

Hence, the length of one side of the triangle is 8cm.

Exercise 9.2

1. Find the area of the triangle whose sides are the following:
   - i) 21cm, 17cm and 10cm
   - ii) 20m, 13m and 11m
   - iii) 13m, 14m and 15m
   - iv) 56cm, 39cm and 25cm
2. Find the area of the triangles, for the assumptions:
   - i) base = 16cm , altitude = 7cm
   - ii) base = 5m , altitude = 12m
   - iii) base = 3.4m , altitude = 5.5m
   - iv) base = $3\frac{1}{7}$m , altitude = $8\frac{3}{4}$m
3. Area of a triangular region is $72m^2$. Find the altitude if base is of 9m.
4. Area of a triangular region is $9.9m^2$. Find its base, when the altitude is of 5.5m.
5. Find the area of equilateral triangles when the length of one of its sides is the following:
   (i) 4cm (ii) $5\sqrt{2}cm$
6. Find the length of a side of an equilateral triangle, when the areas are the following:
   (i) $9\sqrt{3}cm^2$ (ii) $25\sqrt{3}cm^2$

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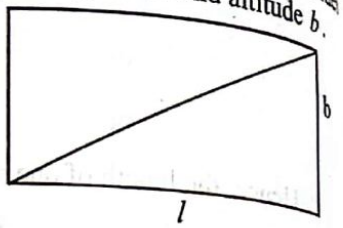
9.2.4 To find the area of a rectangle when its two sides are given

We know that rectangle is a quadrilateral whose opposite sides are congruent and all the angles are right angles. If we draw a diagonal, it divides the rectangle into two congruent right triangles, having base l and altitude b.



If the area of one of these right angled triangles is $\frac{1}{2}(l \times b)$

Then the area of both the right angled triangles is.

The area of the rectangle $= 2 \times \frac{1}{2}(l \times b) = l \times b$

Area of a rectangle = length × breadth

Example 1:

Find the area of a rectangle, whose adjacents sides are 5.4 cm and 2.5 cm.

Solution:

Length of the rectangle = 5.4 cm

Breadth = 2.5 cm

As Area = length × breadth

$\therefore$ Area $= (5.4 \times 2.5)cm^2$

$= 13.5cm^2$

Example 2:

Sides of a rectangle are in the ratio 3:5. If its area is $1215m^2$, find out its length and breadth.

Solution:

In light of ratio between the sides=3:5, the sides of the rectangle are $3x$ and $5x$.

Since area of the rectangle = length x breadth

1215 $= 5x \times 3x$

$= 15x^2$

The area of the rectangle is given by:

$15x^2 = 1215$

The length and breadth can be found:

or $x^2 = \frac{1215}{15}$

or $x^2 = 81$

or $x = \sqrt{81}$

or $x = 9.$

The length of the rectangle is $5x = 5 \times 9$

$= 45m$

And breadth=$3x$ $= 3 \times 9$

$= 27m.$

Example3:

Breadth of a rectangle is 3.9m and its diagonal is of 8.9m. Find the area of the rectangle.

Solution:

We know that diagonal of a rectangle divides it into two congruent right angled triangles. So diagonal of the rectangle is the hypotenuse of both the right angled triangles. First we will find the length by using Pythagoras thereon. Let the length of the rectangle be x

$\therefore \quad x^2 + (3.9)^2 = (8.9)^2$

or $x^2 + 15.21 = 79.21$

or $x^2 = 79.21 - 15.21$

$x^2 = 64$

or $x = \sqrt{64}$

$x = 8.$

The area of a rectangle whose length is 8m and breadth 3.9m is:

So Area = Length × Breadth

$= 8 \times 3.9$

$= 31.2m^2.$

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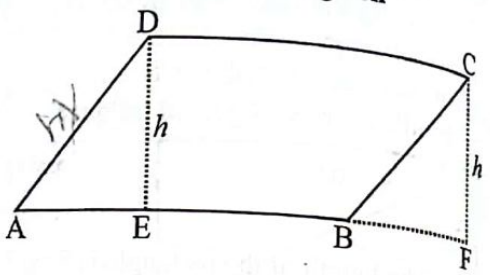
9.2.5 Area of a parallelogram when base and altitude are given

We know that a parallelogram is a quadrilateral having parallel opposite sides. A parallelogram differs from a rectangle to the extent that its angles are not right angles.



Look at the figure. ABCD is a parallelogram. $\overline{DE}$ is its altitude. If we shift triangle ADE and place it along-with $\overline{CD}$ to form a triangle CBF, then we have a parallelogram ABCD that is converted into rectangle CDEF, having same base and same altitude.

As the area of parallelogram ABCD is $= \ell \times b$

$$= m\overline{CD} \times m\overline{CF}$$

$\therefore$ Area of rectangle CDEF is also $= m\overline{CD} \times m\overline{CF}$

$$= m\overline{AB} \times m\overline{DE} \qquad \left(\because m\overline{CD} = m\overline{AB},\ m\overline{CF} = m\overline{DE}\right)$$

= base × altitude.

Hence

Area of a parallelogram = Base x Height = Area of rectangle

Example.1:

Find the area of a parallelogram whose base is 5.8cm and altitude 2.5cm.

Solution:

Here base = 5.8cm

altitude = 2.5cm

Area of parallelogram = base × altitude

= 5.8 x 2.5

= $14.5cm^2$

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Example2:

The area of a parallelogram is $102m^2$. If its altitude is 8.5m, find its base.

Solution:

$\therefore$ Area of a parallelogram = Base x Altitude

$\therefore$ $Base = \dfrac{Area}{Altitude}$

Using Area $= 102m^2$ and altitude = 8.5m in the above equation:

$\therefore$ $Base = \dfrac{102}{8.5}$

$$= \frac{^{6}\cancel{102} \times \cancel{10}^{2}}{\cancel{85}_{\cancel{17}_{1}}}$$

=12m

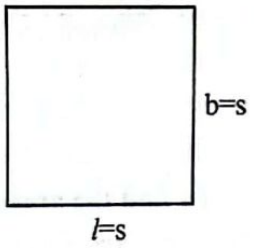
9.2.6 Area of a square when its side is given:

A square is a quadrilateral in which all sides are of equal length and all angles are right angles.

It differs from a rectangle to the extent that its length= its breadth.

If the length of one side of a square is s,

Then the length = breadth $\Rightarrow$ $l = b = s$



$\therefore$ Area of a square $= l \times b$

$$= s \times s$$

$$= s^2$$

Area of a square $= (length\ of\ side)^2$

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Example 1:

Find the area of a square, when one of its sides is of 8.5 cm.

Solution:

Here length of one side of the square= 8.5cm

Area of a square $= (lengh\ of\ side)^2$

$\therefore$ Area of a square $= (8.5)^2$

$= 72.25cm^2$

Example 2:

Area of a square is $11881m^2$. Find its perimeter.

Solution:

Area of a square $= (lengh\ of\ side)^2$

$\therefore$ length of the side $= \sqrt{Area\ of\ a\ square}$

$= \sqrt{11881}$

$= 109m$

Perimeter of square $= 4 \times side$

$= 4 \times 109$

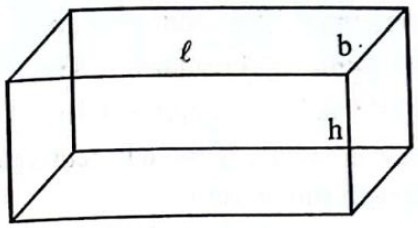
$= 436m.$

9.2.7 To find the area of four walls of a room when its length, width and height are given:

A room is normally a combination of four rectangular regions, called its walls. To find the area of its four walls, First find out the area of each wall and then add them to obtain the area of a room.

Let us explore if we can find a formula for finding the area of four walls of a room.

Take an empty shoe box. Remove its lid and bottom.



The remaining four walls can be considered as four rectangle

area one large rectangle (wall) = Length of the room × Height

$= \ell \times h$

Area of rectangle (wall) $= 2(\ell \times h)$

Area of one small rectangle (wall) = Breadth of room × Height

$= b \times h$

Area of two small rectangles (walls) $= 2(b \times h)$

Area of four wall or rectangles $= 2(\ell \times h) + 2(b \times h)$

$= 2(\ell + b) \times h$

Exercise 9.3

1. Find the area of rectangles having the following assumptions:

i)	Length	=	18m	:	breadth	=	6.5m
ii)	Length	=	19.6m	:	breadth	=	9.5m
iii)	Length	=	27.5m	:	breadth	=	17.8m

2. Find the area of parallelograms having the following assumptions:

i)	Base	=	27cm	:	altitude	=	18.5cm
ii)	Base	=	19.2cm:		altitude	=	8.5cm
iii)	Base	=	46.5cm:		altitude	=	19.2cm

3. Find the area of a square having following side:

(i) 17cm (ii) 8.9cm (iii) 12.6m

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4. Find the area of four walls of a room having the following assumptions:
   i) Length = 6.9m; breadth = 4.7m; Height = 4.5m
   ii) Length = 5.4m; breadth = 4.6m; Height = 3.5m
   iii) Length = 4.9m; breadth = 4.1m; Height = 4.2m
5. Area of a rectangle is $315m^2$. If the adjacent sides are in the ratio 5:7, then find out its length and breadth.
6. There is a ratio of 7:13 in between the sides of a rectangle. Find the measure of its sides, when its area is $1456m^2$.
7. Length of a rectangle is 55cm. if its diagonal is of 73cm, find its area.
8. Width of a rectangle is 12m. Find its area, if the measure of its diagonal is of 37m.
9. Area of a parallelogram is $13.5m^2$. Find the altitude, if its base is 9m.
10. Altitude of a parallelogram is 6.5cm. Find its base, if the area of the parallelogram is $55.25cm^2$.
11. Area of a square is $272.25m^2$. Find its perimeter.
12. Perimeter of a square is 54m. Find its area.

9.2.8 Find the cost of turfing a square/rectangular field:

This is explained in the following examples:

Example 1:

A rectangular field is 140m long and 95m wide. Find the cost of turfing the field at the rate of Rs. 1.5 per meter square.

Solution:

Length of the field = 140m
Breadth of the field = 95m
Area of rectangular field = Length x Breadth
= 140 x 95
= $13300m^2$
Cost of turfing per m^2 = Rs. 1.5
$\therefore$ Total cost of turfing = Rs. 1.5 x 13300
= Rs. 19950

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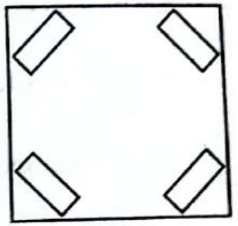
Example 2:

Side of a square garden is 16m. find the cost of turfing the garden at Rs. 1.10/m^2, if there are 4 flower beds of 0.75m × 0.5m in the garden.



Solution:

Side of the square = 16m
Area of square = $(side)^2$
= $(16m)^2$
= $256m^2$
Area of 4 flower beds = $4(0.75\times0.5)$
= $1.5m^2$
Remaining area = 256 - 1.5
= $254.5m^2$
Cost of turfing per m^2 = Rs. 1.10
$\therefore$ Total cost of turfing = 254.5×1.1
= Rs. 279.95

9.2.9 To find the cost of paving the footpath around a rectangular plot:

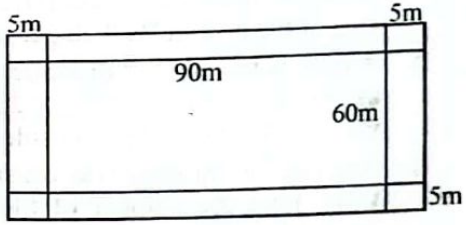
Consider the following examples:

Example 1:

A rectangular park is 90m long and 60m wide. What will be the number of tiles required to pave a footpath 5m wide carried around the oneside of the park, if one tile is 25cm x 25cm.

Solution:

Length of the park = 90m
Breadth of the park= 60m
Area of the park = 90 x 60
= $5400m^2$



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After adding the breadth of the pavement

the length of the outer rectangle	=	90 + 5 + 5
	=	100m
and breadth of the outer rectangle	=	60 + 5 + 5
	=	70m
Area of the outer rectangle	=	100 x 70
	=	$7000m^2$
Area of pavement	=	7000 - 5400
	=	$1600m^2$.
Area of one tile	=	25cm x 25cm
	=	0.25m x 0.25 m
	=	$0.0625m^2$.
So number of tiles required	=	$1600 \div 0.0625$
	=	$\frac{1600 \times \cancel{10000}^{400^{16}}}{\cancel{625}_{25_1}}$
	=	25600

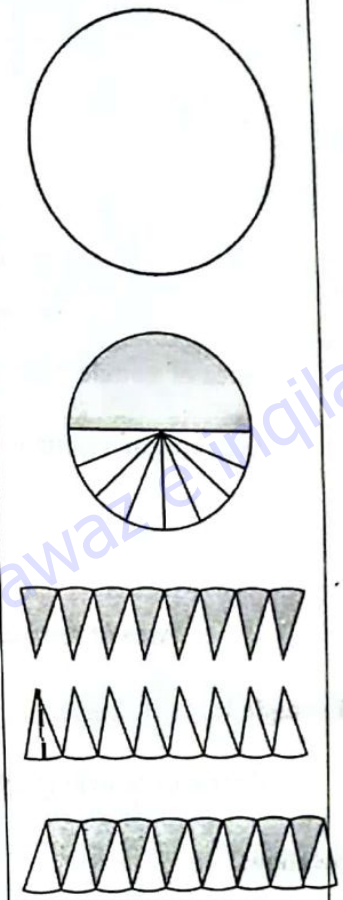
Exercise 9.4

1. Find the cost of turfing a square shaped garden having one side 19.5m at the rate of Rs.4 per square meter.
2. Find the cost of turfing a rectangular plot 120m long and 90m wide at the rate of Rs.5.50 per square meter.
3. A rectangular plot is 46m long and 34.5m wide. What will be cost of bricking the plot if 23cm x 11.5 cm sized bricks cost Rs.3500 per 1000 bricks.
4. A park is 250m long and 200 m wide. There are 12 flower beds in the park each measuring 2.5m x 1.5m. Find the cost of turfing the remaining area of the park at Rs.0.50 per m^2.
5. A park is rectangular in shape, and is 180m long and 120m wide. Find the number of tiles 25cm x 25cm required to pave the footpath 2.5m wide carried around the outside of the park.
6. A verandah 2m wide was constructed around a hall 46m long and 26m wide. Find the number of tiles 16cm x 16cm required for tiling the verandah.

9.2.10 To find the area of a circle and a semi circle when radius is given:

Let us find the area of a circle with the help of an activity:

1. Draw a circle of suitable radius on a piece of paper.
2. Cut out the circle with the help of a pair of scissors.
3. Fold it to get two equal parts of the circle.
4. Simultaneous folding will convert it into 4, 8 and 16 equal section.
5. Shade half of the circle dark.
6. With the help of a pair of scissors cut out 16 congruent sectors of cutting the folding signs.
7. Divide one of the white sectors into two equal parts.
8. Arrange the sectors according to the given figure.
9. Place one of the half white sector on one end and the other on the next end.
10. In this way, the figure has nearly been converted into a rectangle, whose length is ½ of the circumference of the circle and breadth is equal to the radius of the circle.
11. Area of this rectangle is equal to the area of the circle.



equal length of rectangle is $\frac{1}{2}$ of circumference

Breadth equal to radius ⟶

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We know that

Area of circle = Area of a rectangle = Length x Breath

$= \frac{1}{2}$(circumference of the circle) × (radius of the circle)

$= \frac{1}{2}(2\pi r) \times r$

$= \pi r.r$

$= \pi r^2$

Area of a circle = πr^2 (where r is the radius)

and Area of a semicircle $= \frac{1}{2}$(Area of a circle)

$= \frac{1}{2}(\pi r^2)$

Area of semi circle $= \frac{1}{2}(\pi r^2)$

Example 1:

Radius of a circle is 28cm. find its area. $\left(\pi = \frac{22}{7}\right)$

Solution:

Radius of the circle = 28cm

∴ Area of a circle $= \pi r^2$

$= \frac{22}{7} \times (28)^2$

$= \frac{22}{7} \times 28 \times 28^4$

$= 2464cm^2$

Example 2:

Area of a circle is $154cm^2$. Find its radius.

Solution:

Area of the circle = $154cm^2$

$\pi r^2 = 154$

$\frac{22}{7} r^2 = 154$

$r^2 = {}^{7}\cancel{154} \times \frac{7}{\cancel{22}_{1}}$

$r^2 = 49$

$r^2 = (7)^2$

∴ $r = 7cm$

Hence the required radius is 7cm.

Example3:

Find the area of a semicircle having a radius of 14cm. $\left(\pi = \frac{22}{7}\right)$.

Solution:

Radius of a semicircle = 14cm

As Area of a semicircle $= \frac{1}{2}\pi r^2$

∴ Area of a semicircle $= \frac{1}{2} \times \frac{22}{7} \times (14)^2$

$= \frac{1}{\cancel{2}} \times \frac{22}{\cancel{7}} \times \cancel{14}^{2} \times 14$

$= 308cm^2$

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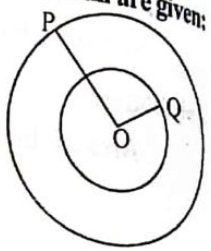
9.2.11 To find the area enclosed by two concentric circles whose radii are given:

Look at the figure:

There are two concentric circles having the same centre 'O' $\overline{OP}$ is the radial segment of the outer circle whereas $\overline{OQ}$ is the radial segment of the inner circle.



Let $\overline{OP} = R$

and $\overline{OQ} = r$

Area of the outer circle = $\pi.R^2$

and Area of the inner circle = πr^2

$\therefore$ Area enclosed by the two concentric circles

$= \pi R^2 - \pi r^2$

$= \pi\left(R^2 - r^2\right)$

Area enclosed by two concentric circles = $\pi\left(R^2 - r^2\right)$

Where R is the radius of outer circle and r is the radius of inner circle.

Example1:

Radii of two concentric circles are 10.5cm and 7 cm respectively. Find the area enclosed by these two circles. $\pi = \frac{22}{7}$.

Solution:

Radius of the outer circle = 10.5cm

Radius of the inner circle = 7cm

As Area enclosed by two concentric circles $= \pi\left(R^2 - r^2\right)$

$\therefore$ Area enclosed by two concentric circles $= \frac{22}{7}\left[(10.5)^2 - (7)^2\right]$

$= \frac{22}{7}[110.25 - 49]$

$= \frac{22}{\cancel{7}} \times \cancel{61.25}^{8.75}$

$= 22 \times 8.75$

$= 192.5cm^2$

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Example 2:

Area enclosed by two concentric circles is $66m^2$. If the radius of the outer circle is 5m, find the radius of the inner circle.

Solution:

Area enclosed by two concentric circles $= \pi\left(R^2 - r^2\right) = 66\ m^2$

$\therefore \quad \frac{22}{7} \times \left(5^2 - r^2\right) = 66$

$5^2 - r^2 = \cancel{66}^{3} \times \frac{7}{\cancel{22}_{1}}$

$25 - r^2 = 21$

$-r^2 = 21 - 25 = -4$

$r^2 = 4$

$r = \sqrt{4} = 2$

Hence, the required radius of the inner circle is 2m.

Exercise 9.5

1. Find the area of a circle for the following assumptions:
   (i) radius = 14cm (ii) radius = 21m
   (iii) radius = 3.5cm (iv) radius = 17.5m
2. Find the area of semi-circle, for the following assumptions:
   (i) radius = 9.1cm (ii) radius = 16.8m
   (iii) radius = 8.4cm (iv) radius = 17.5m
3. Find the area enclosed by concentric circles for the following assumption:
   (i) 8cm, 6cm (ii) 1.6cm, 1.2cm
   (iii) 9m, 2m (iv) 1m, 0.3m
4. Area of a circle is $616m^2$. Find its radius.
5. Area of a circle is $78.5m^2$. Find its diameter. (*Take* $\pi = 3.14$)

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6. Area of a semi-circle is 56.52cm^2. Find its radius. (*Take* $\pi = 3.14$)
7. Area enclosed by 2 concentric circles is 2200m^2. If the radius of the outer circle is 40m, find the radius of the inner circle.
8. Area enclosed by two concentric circles is 264m^2. If the radius of the inner circle is 4m, find the radius of the outer circle.

9.2.12 Word problems related with areas of triangle, rectangle, square, parallelogram and circle.

Example 1:

The sail of a boat is triangular in shape. How much nylon cloth will be required to prepare the sail if its base is 3.6m and height is 2.5m.

Solution:

Here we have to find out the area of a triangle whose base is 3.6m and height is 2.5m.

$$\text{Area of the triangle} = \frac{base \times height}{2}$$

$$= \frac{3.6 \times 2.5}{2} = 4.5 \text{ m}^2$$

$\therefore$ Size of the required nylon cloth is 4.5m^2.

Example:

Diagonal of a quadrilateral is 48m long. Perpendiculars drawn to the diagonal from its opposite vertices are 15m and 19m. Find the area of the quadrilateral.

Solution:

To find the area of the quadrilateral, we have to first find the areas of two triangles whose base is 48m and heights/altitudes are 15m and 19m respectively.

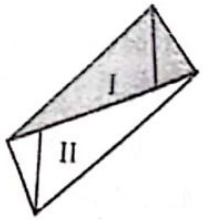
In triangle I

Base = 48m

Altitude = 15m

$\therefore$ Area of triangle-I $= \frac{^{24}\cancel{48} \times 15}{\cancel{2}_1}$

$= 360\text{m}^2$.....(i)



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In triangle II

Base = 48m

Altitude = 19m

$\therefore$ Area of triangle-II $= \frac{^{24}\cancel{48} \times 19}{\cancel{2}_1}$

$= 456\text{m}^2$.....(ii)

Area of quadrilateral = Area of triangle-I + Area of triangle-II

$= 360\text{ m}^2 + 456\text{ m}^2$

$= 816\text{ m}^2$.

Example.3:

A rectangular field is 190m long and 165m wide. What will be the total revenue if the field is based out at Rs.5 per square meter.

Solution:

Length of the field = 190m

Width of the field = 165m

Area of rectangular field = length × width

Area of rectangular field = L × W

Area of the field = 190 × 165

$= 31350\text{m}^2$

Cost of base per m^2 = Rs.5

$\therefore$ Total revenue = 5 × 31350

$\therefore$ = Rs.156750

Example.4:

Perimeter of a square shaped plot is 237.6m. Find the area of the plot.

Solution:

Perimeter of the square = 237.6m

i.e. 4 x side of the square = 237.6

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side of the square $= \frac{237.6}{4}$

$= 59.4m$

Area of the square = side x side

$= 59.4 \times 59.4$

$= 3528.36m^2.$

Example.5:

A rectangular field is 24m by 16m. how much squares of length 4m can be obtained from this field?

Solution:

Area of the rectangular field $= 24 \times 16$

$= 384m^2.$

Area of one square $= 4 \times$ side $= 4 \times 4$

$= 16m^2$

$\therefore$ No. of squares obtained $= \frac{384}{16}$

$= 24$

Example.6:

A parallelogram shape park is of base 120m long and its height/altitude is 85m. Find the cost of leveling the park at Rs.0.50 per m^2.

Solution:

Base of the parallelogram = 120m

Height of the parallelogram = 85m

$\therefore$ Area of the parallelogram $= 120 \times 85$

$= 10200m^2$

Cost of leveling per m^2 = Rs.0.50

$\therefore$ Total cost = Rs. 10200×0.50

= Rs. 5100

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Example.7:

In a circular park with a radius of 28m, there are 7 lamp poles with circular bases of 1m radius. The entire area has grass with the exception of the bases of the lamp poles. Find the area of the grassy portion of the park.

Solution:

Radius of the circular park 'r' = 28m

As Area of the circular park $= \pi r^2$

$= \frac{22}{7} \times (28)^2$

$= \frac{22}{7_1} \times 28 \times \cancel{28}^4$

$= 2464m^2$

Radius of the circular base of 1 lamp-pole 'r' = 1m

Area of the base of 1 lamp-pole $= \pi r^2$

$= \frac{22}{7} \times 1$

$= \frac{22}{7} m^2$

Area of the bases of 7 lamp-pole $= 7 \times \frac{22}{7}$

$= 22m^2.$

Area of the grassy portion of the park = (Area of circular park) – (Area of bases of lamp pole)

$= 2464\text{-}22$

$= 2442m^2$

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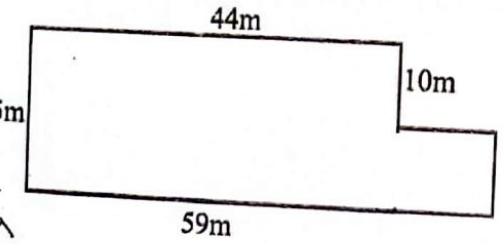
Exercise 9.6

1. A scarf is of the shape of an isoscles right angled triangle. Each of the equal side is 90cm. how much cloth will be needed to prepare 8 such scarf? Also find the cost of the cloth at the rate of Rs. 40 per m^2.
2. Sides of a triangular field are in the ratio 5:6:9. If the perimeter of the triangle is 240m, find the area of the field.
3. The front curtain of a tent is triangular in shape. How much Rexene will be required to prepare the curtain having sides 5m, 5m and 8m.
4. Diagonal of a quadrilateral plot is 65m. Its distance from the opposite vertices is 44m and 52m. Find the area of the plot. Also find the cost of its levelling at Rs.2.50 per m^2.
5. A field is in the shape of a parallelogram. Its base is 750m and height is 360m. What will be the revenue amount if it is based out at Rs.9000 per hectare. (1 Hectare = $10000m^2$).
6. The area of a courtyard, parallelogram in shape, is $39m^2$. Find its base if its height is 5.2m.
7. Find the area of the figure with given dimensions.



8. Diameter of a circular garden is 56m. A circular pavement of width 7m is constructed around it. Find the expenditure of its construction at Rs.50 per m^2.
9. Diameters of two concentric circles are 160cm and 90cm respectively. Find the area enclosed by the circles.
10. A field, rectangular in shape, is based out at the rate of Rs.7.50 per m^2. Find the total revenue, if the field is 350m long and 280m wide.

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11. Arshad covered a distance of 2800m by taking four rounds of a square park. Find the area of the park.
12. A circular fountain, having a diameter 2.8m, is situated in the middle of a square shaped park whose one side is 6m long. Find the area of the fountain. Also find the cost of turfing the park at Rs.5.50 per square meter.
13. A rectangular park is divided into square shaped flower beds having a side of 3m. How many flower beds can be made if the park is 21m long and 18m wide?

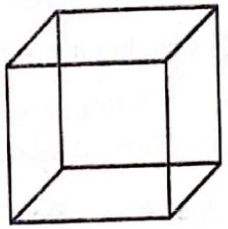
9.3.1 Volume

The amount of space occupied by a three dimensional figure is called its volume. Volume is measured in cubic units. Such as mm^3, cm^3 and m^3.

9.3.2 Cube

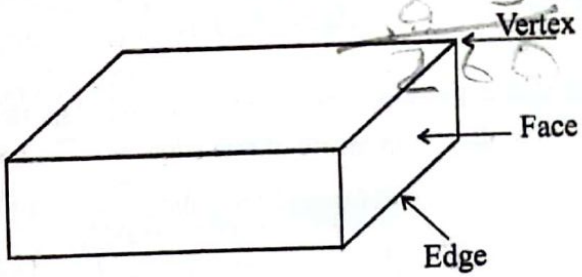
A cube is a regular solid bounded by six equal squares, called its faces. All adjacent faces of a cube are perpendicular to each other. Its dimensions i.e. length, breadth and height are equal. Each side of its square faces is called its edge.

A cube has 6 faces or sides, 12 edges and 8 vertices.



9.3.3 Cuboid

A cuboid is a box-shaped solid figure. It has six rectangular faces. Each side is called its edge. Its dimensions are its length, breadth and height. It has also six faces, twelve edges and eight vertices.



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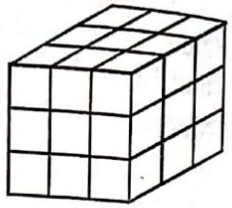
9.3.4 Finding the volume of a cube when its edge is given

If the edge of a cube is 1cm, then its volume is defined as $1cm^3$. Such a cube is called a unit cube.

Number of unit cubes needed to fill a cube is called the volume of the cube.

We see that the number of unit cubes in a cube having an edge 3cm is 27 i.e. 3 layers of 9 cubes.



We can find the volume of a cube by multiplying the area of its base by its height. As the dimensions of a cube are equal therefore.

Volume of a cube = edge x edge x edge
= $(edge)^3$

Example.1

Find the volume of a cubes having measure of one of its edges:

(i) 9cm (ii) 2.5m

Solution:

i) Measure of one edge = 9cm
Volume of the cube = $(edge)^3$
= $9^3 cm^3$
= $729cm^3$

ii) Measure of one edge = 2.5m
Volume of the cube = $(edge)^3$
= $(2.5)^3 m^3$
= $15.625m^3$

Example.2

Volume of a cube is $343cm^3$. Find the length of its side.

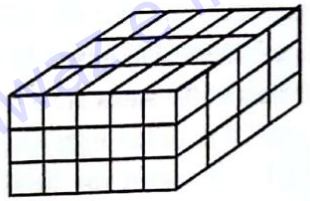
Solution:

Let the length of a side be x cm.
$\therefore$ volume of a cube = $x^3 cm^3$
But the given volume = $343cm^3$
$\therefore x^3 = 343$
$(x)^3 = (7)^3$
$x = 7$ Hence length of side = 7 cm

9.3.5 Find the volume of a cuboid when its length, breadth and height are given:

We know that the number of unit cubes needed to fill a cuboid is its volume.

For a cuboid having
Length = 5cm
Breadth = 4cm
Height = 3cm



the number of unit cubes needed to fill the space is 60 i.e. 3 layers of 5x4=20 units cubes or 60 unit cubes.

The same can be obtained by multiplying its length, breadth and height i.e.

$5 \times 4 \times 3 = 60cm^3$

Hence Volume of a cuboid = length × breadth × height
= Area of box × height

Example.1

Find the volume of a cuboid having length 15.25cm, breadth 9.4cm and height 4cm.

Solution:

Volume of cuboid = length × breath × height
= $15.25 \times 9.4 \times 4$
= $573.4\ cm^3$

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Example 2:

Volume of a cuboid is $1224m^3$. If the area of its base is $204m^2$, find its height.

Solution:

Let the height be x meter.

As Volume of a cuboid = (Area of the base) × height

$\therefore \quad 1224 = 204 \times x$

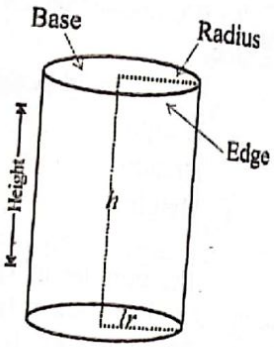
or $\quad 204x = 1224$

or $\quad x = \frac{1224}{204} = 6$

$\therefore$ required height = 6m.

9.3.6 Right Circular Cylinder

Cylinder is a three dimensional solid with two congruent and parallel bases, two edges and three faces. When the bases of a cylinder are circles and centers of the bases are directly aligned one above the other, it is called right circular cylinder. The curved surface is called the height of the cylinder. In the given figure of a cylinder, radius of each of the circular regions is 'r' and the height of the cylinder is 'h'.



9.3.7 Finding volume of right circular cylinder

We know that the base of a right circular cylinder is a circle. So, and to find the volume of a cylinder we have to multiply the area of the base with its height.

Radius of the base = r

$\therefore$ Area of the base = πr^2

Height of the cylinder = h

As volume of the cylinder = $(\pi r^2) \times (h)$

$\therefore$ Volume of a cylinder = $\pi r^2 h$

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Example.1

Find the volume of cylindrical tin with radius 3.5cm and height 12cm. $\left(Here\ \pi = \frac{22}{7}\right)$.

Solution:

Here $r = 3.5cm$

$h = 12cm$

$\therefore$ Volume of cylinder $= \pi r^2 h$

$= \frac{22}{7} \times (3.5)^2 \times 12$

$= \frac{22}{\cancel{7}} \times \cancel{12.25}^{1.75} \times 12$

$= 462cm^3$

Example-2:

Find the depth of a well of radius, 4m when the total digging is $1056m^3$. $\left(\pi = \frac{22}{7}\right)$

Solution:

We know that the shape of a well is that of a cylinder. Here depth will be taken h i.e. height in the formula for finding the volume of the cylinder.

Let the depth of the well be h and the radius is r.

$\therefore$ Radius of the top circle = 4m

Area of top circle = πr^2

= $\pi(4)^2 = 16\pi$

Volume of the well = $\pi r^2 h$

or, $1056 = 16\pi h$

$16 \times \frac{22}{7} . h = 1056$

$h = \cancel{1056}^{66} \times \frac{7}{\cancel{22}_1} \times \frac{1}{\cancel{16}_1}$

$= 21$

$\therefore$ Required Depth of the well = 21m

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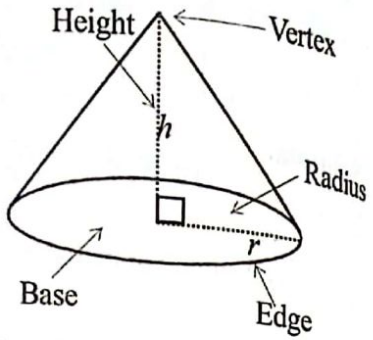
9.3.8 Right Circular Cone

Cone is a three dimensional figure that has one circular base and one vertex. If the vertex of the cone is equidistant from any point on the circle, then it is called right circular cone. A cone has two surfaces, one edge and one vertex. In the figure, radius of the circular base is r and the distance from base to the top i.e. height is h.

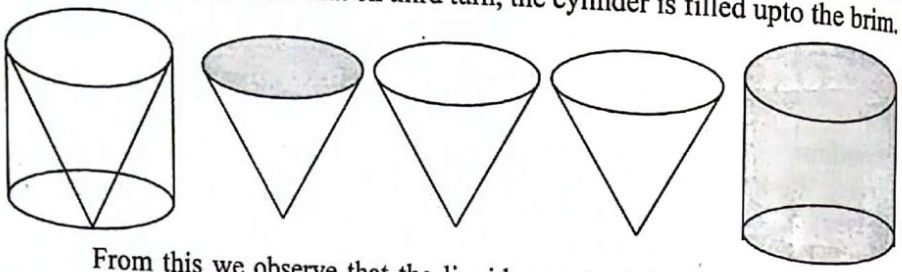


9.3.9 Finding the volume of the right Circular Cone

We find the volume of a right circular cone with the help of an activity.

Take two containers, one cylinder shaped and the other cone shaped, having the same circular diameter and the same height.

Fill the cone-shaped container with liquid and pour it into the cylinder-shaped container. We note that on third turn, the cylinder is filled upto the brim.



From this we observe that the liquid contained in the cylinder is three times the liquid contained in the cone.

In other words we can say that volume of the cylinder is three times the volume of the cone.

Since volume of the cylinder = $\pi r^2 h$

Therefore

Volume of the cone	=	$\frac{1}{3}\pi r^2 h$
	=	$\frac{1}{3}$ (Area of the base x height)

Example.1

Find the volume of a cone having radius of the base 3.5 cm and height 12cm. $\left(\pi = \frac{22}{7}\right)$.

Solution:

Here $r = 3.5cm$

and $h = 12cm$

volume of the cone $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times (3.5)^2 \times 12$

$= \frac{1}{\cancel{3}_1} \times \frac{22}{\cancel{7}_1} \times \frac{\cancel{35}^{5^1}}{\cancel{10}_{2_1}} \times \frac{\cancel{35}^{7}}{\cancel{10}_{2_1}} \cancel{12}^{4^1}$

$= 154 \text{ cm}^3$

Example-2:

The volume of a right circular cone is 1078cm^3 and its radius and height are in the ratio 1:3. Find its radius and height. $\left(\pi = \frac{22}{7}\right)$

Solution:

Volume of a cone $= \frac{1}{3}\pi r^2 h$

$1078 = \frac{1}{3}\pi r^2 h$

$\frac{1}{3}\pi r^2 h = 1078$

$r^2 h = \frac{1078 \times 3}{\pi}$

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$$r^2h = \cancel{1078}^{49} \times 3 \times \frac{7}{\cancel{22}}$$

$$r^2h = 1029 \qquad \rightarrow (1)$$

Ratio between radius and height = 1:3

i.e. $\frac{r}{h} = \frac{1}{3}$

or $\boxed{h = 3r}$ Put it in Eq. (1).

$$r^2 \times (3r) = 1029$$

or $3r^3 = 1029$

or $r^3 = \frac{\cancel{1029}^{343}}{\cancel{3}_1} = 343$

or $r^3 = (7)^3$

or $r = 7$

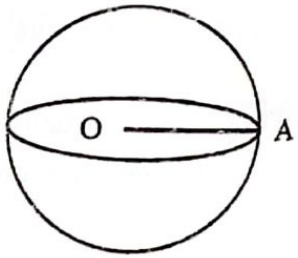
Thus radius $= 7cm$

and height $= 3 \times 7$

$= 21cm$

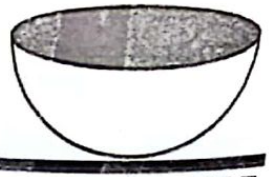
9.3.10 Sphere and Semi sphere

A sphere is a perfectly round three dimensional body, such as the shape of a cricket ball that has all its points on the surface equidistant from a fixed point called its center. This distance is known as the radius of the sphere and is denoted by r. Sphere has only one surface which is covered.



In the given figure it is a sphere, with centre 'O' and radius $\overline{OA}$.

If a sphere is divided into two equal parts, then each part is called a hemisphere.



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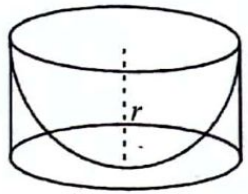
9.3.11 Volume of a hemisphere and sphere

We find the volume of a hemi-sphere with the help of an activity.

1. Take a hollow hemisphere
2. Take a cylindrical container with the same height and base area as that of the hemisphere.
3. Fill the hemisphere fully with some liquid and pour it down into the container without dropping anything.



4. You will see that the container has been filled $\frac{2}{3}$ up.

We know that

Volume of cylinder $= \pi r^2 h$

$\therefore$ volume of hemisphere $= \frac{2}{3}\pi r^2 h$

But the height of the hemisphere is its radius then the

volume of hemisphere $= \frac{2}{3}\pi r^3$ and

The volume of a sphere $= 2\left(\frac{2}{3}\pi r^3\right)$

$= \frac{4}{3}\pi r^3$

Volume of hemisphere	$= \frac{2}{3}\pi r^3$
Volume of a sphere	$= \frac{4}{3}\pi r^3$

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Example

Find the volume of a hemi-sphere if its radius is 6cm. (Take $\pi = 3.14$)

Solution:

Radius of the hemisphere $= 6cm$

Volume of a hemisphere $= \frac{2}{3}\pi r^3$

$= \frac{2}{3} \times 3.14 \times (6)^3$

$= \frac{2}{\cancel{3}_1} \times 3.14 \times \cancel{216}^{72}$

$= 2 \times 3.14 \times 72$

$= 452.16 cm^3$

Exercise 9.7

1. Find the volume of cubes having the following edges:
   (i) 6mm (ii) 8.5cm
2. Find the volume of cuboids having the following edges:
   (i) Length=6dm, Width=5dm, Height=4dm
   (ii) Length=3.5cm, Width=2.4cm, Height=2cm
3. Find the measure of an edge of cube, when its volume is:
   (i) $343cm^3$ (ii) $1.728dm^3$
4. A box is 0.9m long, 0.5m wide and 0.4m high. Find its volume.
5. Volume of a cuboid is $140m^3$. Find height if the area of its base is $35m^2$.
6. Find the width of a cuboid if its height is 5m, length is 8m and volume is $260m^3$.
7. Find the volume of a cylindrical drum having radius 0.5m and height 1.75m.
8. Find the height of a cylinder, if its volume is $6050m^3$ and radius of its base is 5m.

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9. The volume of a right circular cone is $154m^3$. Find the radius of its base if its height is 12m.
10. The volume of a right circular cone is $4312dm^3$. If the ratio between radius and height is 2:3, find radius of the base and height.
11. Find the volume of a hemisphere when its radius is:
    (i) $5\frac{1}{4}dm$ (ii) $10\frac{1}{2}cm$
12. Find the volume of a sphere to the nearest hundredths when its radius is: (*Take* $\pi = 3.14$))
    (i) 0.42m (ii) 0.6cm

Word problems relating to volume of cube, cuboid, cylinder, cone and sphere

The real life problems relating to above topics are the following:

Example-1:

A trunk dealer wants to store boxes in a store-room with a length of 7.2m, width of 3.5 cm and height of 4m. How may boxes can be stored in this space, if the dimensions of each box is 80cm long, 50cm wide and 400m high:

Solution:

Length of store-room $= 7.2m$

$= 720cm$

Width of store-room $= 3.5m$

$= 350cm$

Height of store-room $= 4m$

$= 400cm$

Volume of store-room $= 720 \times 350 \times 400 cm^3$

Length of box $= 80cm$

Width of box $= 50cm$

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10.

Height of box $= 40cm$

Volume of box $= 80\times50\times40cm^3$

$\therefore$ No. of boxes that can be stored $= \dfrac{Volume\ of\ the\ store\ room}{Volume\ of\ one\ box}$

$$= \frac{^{90}\cancel{720}\times\cancel{350}^{7}\times^{1}\cancel{400}\ cm^3}{_{1}\cancel{80}\times\cancel{50}_{1}\times\cancel{40}_{1}\ cm^3}$$

$= 630$

So 630 boxes can be stored in the room.

Example-2:

A slab of ice is cubic in shape. The length of one of its sides is 50cm. Find the weight of the slab in kgs if the weight of $1cm^3$ ice=1gram.

Solution:

Length of one side of the slab $= 50cm$

Volume of the slab $= 50\times50\times50cm^3$

Weight of $1cm^3$ ice $= 1gram$

$\therefore$ Weight of the slab of ice $= 50\times50\times50\times1\ g$

$$= \frac{50\times50\times50}{1000}\ kg$$

$= 125$ kg

$\therefore$ weight of the ice is 125 kilogram.

Example-3:

A 24m tall conical minaret is made of concrete. Find its volume when the radius of its base is 7m.

Solution:

Volume of a cone $= \dfrac{1}{3}\pi r^2h$

Here $r = 7m$

$h = 24m$

$\therefore$ volume of concrete $= \dfrac{1}{3}\times\dfrac{22}{7}\times(7)^2\times24$

$$= \frac{1}{3}\times\frac{22}{\cancel{7}_{1}}\times\cancel{49}^{7}\times24$$

$$= \frac{3696}{3}$$

So required volume $= 1232\ m^3$

Example-4:

Find the volume of children ball of radius 2.1cm.

Solution:

Here $r = 2.1cm$

volume of a sphere $= \dfrac{4}{3}\pi r^3$

$\therefore$ volume of the ball $= \dfrac{4}{3}\times\dfrac{22}{7}\times(2.1)^3$

$$= \frac{\cancel{4}}{\cancel{3}}\times\frac{22}{\cancel{7}}\times\frac{\cancel{21}^{3}}{10}\times\frac{21}{10}\times\frac{21}{10}$$

$$= \frac{38808}{1000} = 38.808$$

$= 38.8\ cm^3$ approximately

Hence volume of the ball in $38.8cm^3$

Exercise 9.8

1. A sewerage fit, having the shape of a cube, was dug with a side measuring 12m. In how many turns, three trucks will be able to take away the soil if $19.2m^3$ soil can be loaded on one truck.
2. A solid cube is made of a metal. What will be volume of the cube if one of its sides is $4\frac{1}{2}$ cm. Find its weight if $1cm^3$ of the metal weighes 16 gram.
3. Tariq bought a piece of wood 2m long, 35cm wide and 25cm thick. What will it cost him if the rate of wood is Rs.24000 per m^3?

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10.

4. Find the total expenditure on constructing a stage for a function at Rs.1500 per m^2, if the stage is 8m long, 5m wide and 1m high.
5. Two cylindrical drums of kerosene oil have the same height of 1.5m, and are made of same material. Then diameters are 1.2m and 1.6m. Find the ratio of their volumes.
6. What will be the cost of digging 21m deep well having a diameter of 2m at Rs.300 per m^2.
7. A cone shaped minaret is 12dm high and has a diameter of 14dm. find its volume.
8. Find the volume of a decorative cap, having a shape of cone, with a radius of 2.1 cm and height of 10cm.
9. Find the volume of a children's football with radius of 10.5cm.
10. Dome of a masjid is hemisphere in shape. Find its volume to the nearest tenths if its diameter is 2.1m.

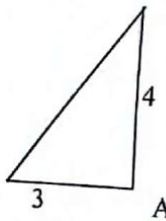
11.

11. A metallic sphere is erected in a fountain. Find its volume when its diameter is $5\frac{1}{4}$ dm.

Exercise 9.9

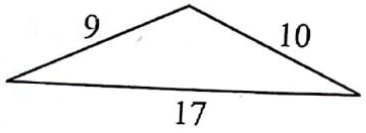
1. Find the area of the following shapes:

(i)



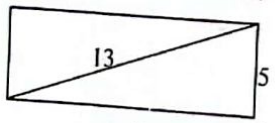
Area--cm^2

(ii)



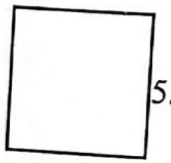
Area-----cm^2

(iii)



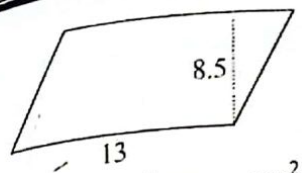
Area-----cm^2

(iv)



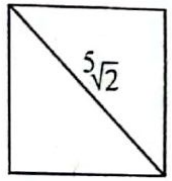
Area-----cm^2

(v)



Area-----cm^2

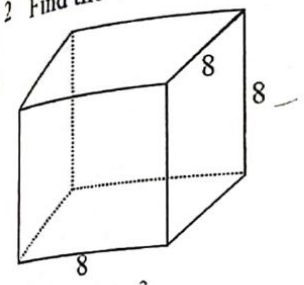
(vi)



Area-----cm^2

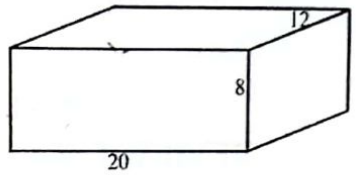
2. Find the volume of the following figures:

(i)



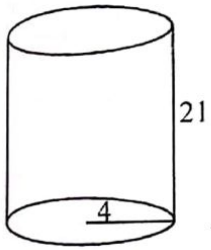
Volume--cm^3

(ii)



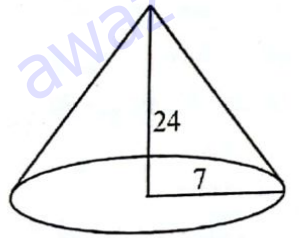
Volume-----cm^3

(iii)



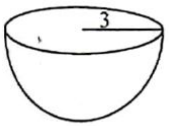
Volume----cm^3

(iv)



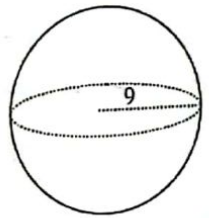
Volume-----cm^3

(v)



Volume-----cm^3

(vi)



Volume-----cm^3

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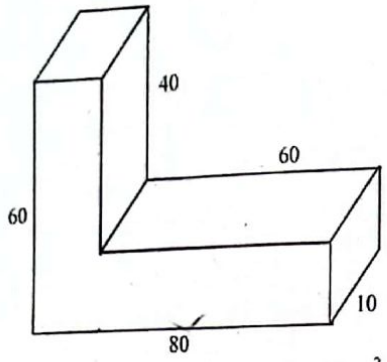
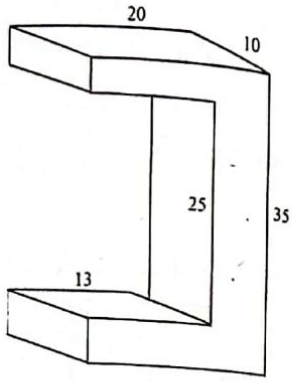
(vii)



Volume-----cm^3

(viii)



Volume-----cm^3

REVIEW EXERCISE 9 (Multiple Choice Questions)

Note: Select the correct answer from the given choices A, B, C, D in the following questions.

1. In ΔPQR, $m\angle Q = 90^o$, $m\overline{PQ} = 12cm, m\overline{QR} = 5cm$, then $m\overline{PR} =$

   A) 13cm B) 169cm
   C) 17cm D) 7cm

2. In ΔKLM, $m\angle L = 90^o$, $m\overline{KM} = 20cm, m\overline{KL} = 12cm$, then $m\overline{LM} =$

   A) 32cm B) 8cm
   C) 16cm D) 256cm

3. In ΔABC, $m\angle A = 90^o$, $m\overline{AB} = 15cm, m\overline{BC} = 25cm$, then $m\overline{CA} =$

   A) 40m B) 20m
   C) 10m D) 850m

4. In $\Delta ABC, a = 5, b = 3, c = 6$, then $S =$

   A) 14 B) 28
   C) 90 D) 7

5. If the sides of a ΔABC are a, b, c and $s = \frac{a+b+c}{2}$, then area of $\Delta ABC =$

   A) $s(s-a)(s-b)(s-c)$ B) $\sqrt{s(s+a)(s+b)(s+c)}$
   C) $\sqrt{s(s-a)(s-b)(s-c)}$ D) $\sqrt{(s-a)(s-b)(s-c)}$

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6. Area of a rectangle is $0.72m^2$. The length is 0.9m, its breadth =

   A) 8m B) 0.8m
   C) 1.6m D) 0.36m

7. Area of a square is $6.25cm^2$. its perimeter =

   A) 2.5cm B) 1.25cm
   C) 25cm D) 10cm

8. Perimeter of a square is 5.6cm. its area =

   A) $31.36cm^2$ B) 1.4cm
   C) $1.96cm^2$ D) 22.4cm

9. Base of a triangle is 8cm. its altitude is 3cm. its area =

   A) $24cm^2$ B) $12cm^2$
   C) $55cm^2$ D) $22.4cm^2$

10. Area of a triangle is $7.5cm^2$ and the altitude is 1.5cm, its base =

    A) 5cm B) 0.5cm
    C) 0.1cm D) 10cm

11. Area of a parallelogram is $5.2dm^2$ and the base is 1.3dm. its height is =

    A) 4dm B) 0.4dm
    C) 0.04dm D) 40dm

12. Area of an equilateral triangle is $4\sqrt{3}cm^2$. Its side=

    A) $\sqrt{3}cm$ B) 2cm
    C) 4cm D) 16cm

13. Diagonal of a square is $2\sqrt{2}dm$. Its area =

    A) $8dm^2$ B) $16dm^2$
    C) $8\sqrt{2}dm^2$ D) $4dm^2$

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14. Area of four walls of a room =

A) $2(l\times b)+h$ B) $2(l+b)\times h$

C) $2(l+h)\times b$ D) $2(l+b)\div h$

15. Radius of a circle is 6cm. its area =

A) $6\pi cm^2$ B) $-12\pi cm^2$

C) $18\pi cm^2$ D) $36\pi cm^2$

16. Area of a semi circle is $77cm^2$. Its radius =

A) 38.5 cm B) 7cm

C) 24.5 cm D) $\sqrt{7cm}$

17. Area enclosed by two circles having radii 7cm and 5cm, is

A) $24\pi cm^2$ B) $2\pi cm^2$

C) $74\pi cm^2$ D) $\sqrt{24}\pi cm^2$

18. Volume of a cube is $512mm^3$. Its edge =

A) $16\sqrt{2}mm$ B) 16mm

C) 8mm D) 128mm

19. Volume of a cuboid =

A) $(Area\ of\ base)\times height$ B) $(Perimeter\ of\ base)\times height$

C) $2\times(length\times breadth)\times height$ D) $\frac{1}{2}(Area\ of\ base)\times height$

20. Volume of a right circular cylinder =

A) πr^2 B) $\pi r^2 h$

C) $\frac{1}{3}\pi r^2 h$ D) $\frac{4}{3}\pi r^2$

21. Radius of the base of a cylinder is 2cm and height is 10cm. its volume =

A) $200\pi cm^3$ B) $\frac{40}{3}\pi cm^3$

C) $40\pi cm^3$ D) $20\pi cm^3$

22. Volume of a right circular cone is =

A) $Area\ of\ base\times height$ B) $\frac{4}{3}\pi r^2 h$

C) $\pi r^2 h$ D) $\frac{1}{3}(Area\ of\ base\times height)$

23. Radius of the base of a cone is 1cm and height is 3cm, its volume =

A) πcm^3 B) $3\pi cm^3$

C) $\frac{1}{3}\pi cm^3$ D) $\frac{4}{3}\pi cm^3$

24. Volume of a hemisphere =

A) πr^2 B) $\frac{4}{3}\pi r^3$

C) $2\pi r$ D) $\frac{2}{3}\pi r^3$

25. Volume of a hemisphere having radius of 3cm, is

A) $18\pi cm^3$ B) $9\pi cm^3$

C) $36\pi cm^3$ D) $6\pi cm^3$

26. Volume of a sphere =

A) $\frac{2}{3}\pi r^3$ B) $9\pi r$

C) $\frac{4}{3}\pi r^3$ D) πr^2

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27. Radius of a sphere is 3cm. its volume =

A) $18\pi cm^3$ B) $36\pi cm^3$

C) $9\pi cm^3$ D) πcm^3

28. Volume of a sphere is $288\pi cm^3$. Its radius =

A) 12 cm B) 3 cm

C) 24 cm D) 6 cm

Glossary

- Pythagoras Theorem

  Square of the hypotenuse in any right angled triangle is equal to the sum of squares of the remaining sides of the triangle.

- Hero's formula of finding the area of a triangle

  Area of triangle $= \sqrt{s(s-a)(s-b)(s-c)}$ where $s = \frac{a+b+c}{2}$ and a, b, c are the sides of the triangle.

- Area of triangle $= \frac{1}{2}(\text{measure of base} \times \text{measure of altitude})$

- Area of an equilateral triangle $= \frac{\sqrt{3}}{4}a^2$

  where "a" is the length of any one side of the triangle.

- Area of rectangle = length x breadth
- Area of a parallelogram = Base x Height
- Area of square = $(\text{side})^2$
- Area of four walls of a room = 2(length + width) x height.
- Area of circle = πr^2
- Area of a semi-circle = $\frac{1}{2}\pi r^2$

- Area enclosed by two concentric circles = $\pi(R^2 - r^2)$

  where R and r are radii of the outer and inner circles respectively.

- Volume of a cube = $(\text{edge})^3$
- Volume of cuboid = length x breadth x height

  = (Area of the base) x height

- Volume of right circular cylinder = $\pi r^2 h$
- Volume of a right circular cone = $\frac{1}{3}\pi r^2 h$
- Volume of hemisphere = $\frac{2}{3}\pi r^3$
- Volume of sphere = $\frac{4}{3}\pi r^3$

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Unit 10

INTRODUCTION TO COORDINATE GEOMETRY

In this unit, the students will be able to learn about the

- Define coordinate geometry
- Derive distance formula to find distance between two points given in Cartesian plane
- Use distance formula to find distance between two given points
- Define collinear points
- Distinguish between collinear and non- collinear points
- Use distance formula to show that given three (or more) points are collinear
- Use distance formula to show that the given three non- collinear points to form
  - An equilateral triangle
  - An isosceles triangle
  - A right angled triangle
  - A scalene triangle

10.1 Coordinate Geometry

Coordinate Geometry or Analytic Geometry is a branch of geometry in which points are represented with respect to a coordinate system and approach to view problems graphically.

Coordinate geometry was invented in 1637 by a French mathematician and philosopher Rene Descartes (1596-1650 A.D.). He was the first person to give geometrical views to algebraic equations.

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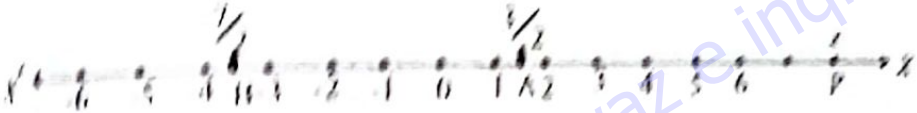
10.1.1 The coordinate lines and the Coordinate plane

The steps involved in representing real numbers on the real number line are the following. We can represent all real numbers by

i. Choosing a point on line $X'X$ as the origin
ii. Choosing a suitable unit of length
iii. Indicating a positive direction for measuring distances along the line

In fact there exists a (1-1) correspondence between the set of real numbers R and the points of the x line $X'X$ with point O as its origin.



Such a line is called a coordinate line. The real number, say x, represented by a point P of the line is called the coordinate of P.

The coordinate of a point may be positive or negative. The coordinate of a point is therefore a directed distance from the origin. Coordinate of A is $\frac{3}{2}$ and coordinate of B is $\frac{7}{2}$.

The co ordinate plane describes a two dimensional plane in terms of two directed lines $X'X$ and $Y'Y$ that are perpendicular and intersect each other at a point O. The directed horizontal line $X'OX$ is called the x axis, while the directed vertical line $Y'Y$ is called the y axis. The point of intersection O is called the origin.

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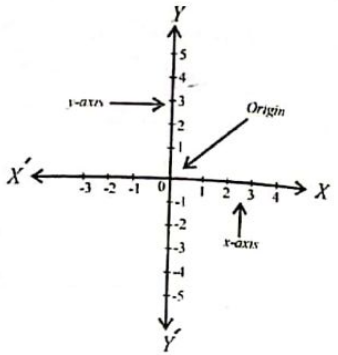
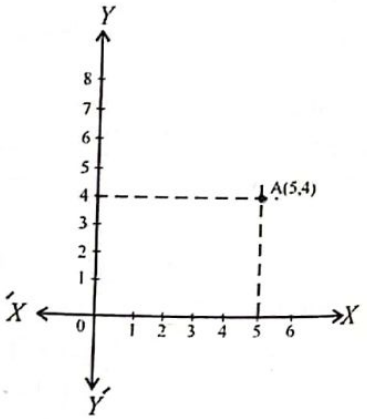
On, the point to the right of the origin O, are associated with positive real numbers and those to the left of the origin are associated with negative real numbers. Similarly on $y-axis$ the points above the origin O are associated with positive real numbers and those below the origin are associated with negative real numbers.



10.1.2 Coordinate of a point in Cartesian plane:

The position of any point on the coordinate plane is described by using two numbers, say x and y. The first number x is the horizontal position of the point from the origin. It is called the x–coordinate or abscissa. The second number is the vertical position of the point from the origin. It is called the y–coordinate or ordinate. A specific order will be used to represent the coordinates they are called ordered pairs and are written in the form (x, y).

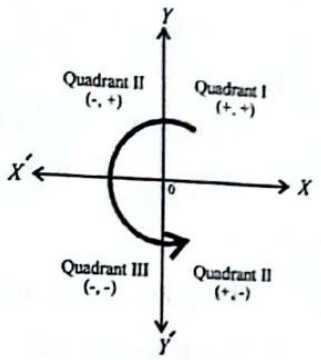
For example, the real ordered pair (5, 4) represents a point, 5 units to the right of the origin in the direction of $x-axis$ and 4 units above the origin in the direction of $y-axis$, as shown in the figure. If the point is denoted by A then we say that x–coordinate of point A is 5 and the y–coordinate of point A is 4 or simply we can say that the coordinates of A are (5,4).



10.1.3 Quadrants:

The coordinate plane or Cartesian coordinate plane is divided into four quadrants by the coordinate axes XOX′ and YOY'. These are numbered I to IV, going around counterclockwise. In Quadrant I, both x and y coordinates are positive.

In Quadrant II, the x–coordinate is negative but the y–coordinate is positive. In Quadrant III, both the coordinates are negative and in Quadrant IV, x–coordinate is positive but the y–coordinate is negative.

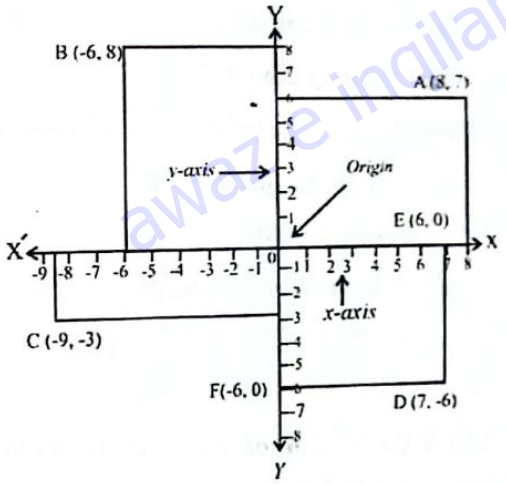


Example–1

State the coordinates of the points shown on the cartesian plane.

Solution:

i) A is 8 units to the right and 7 units above the origin. So point A is A (8, 7).



ii) B is 6 units to the left and 8 units above the origin so the coordinates of a point are B (–6, 8).

iii) C is 9 units to the left and 3 units below the origin. So the coordinates of a point C are C (–9, –3).

iv) D is 6 units to the right and 7 units below the origin. So point D is D (7, –6).

v) E is on $x-axis$ and is 6 units to the right of the origin. So point E is E (6, 0).

vi) F is on $y-axis$ and is 6 units below the origin. So point F is F(0,–6)

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Example–2:

Plot the following points and state the quadrant in which the points are lying.

K (–4, –5), L (5,8), M (–7,4),

N (6, –3), P (0,–9), Q (0, –6).

Solution:

The points are located in

i) K is in Quadrant III

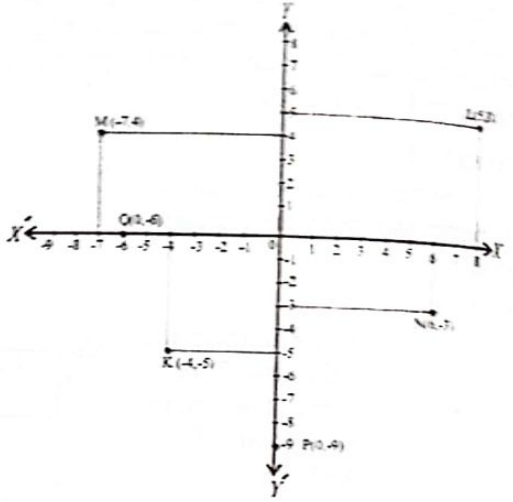
ii) L is in Quadrant I

iii) M is in Quadrant II

iv) N is in Quadrant IV

v) P is in no quadrant. It lies on $x-axis$.

vi) Q is in no quadrant. It lies on $y-axis$.



10.1.4 Derivation of Distance formula between two given points in a Cartesian plane:

In a Cartesian plane (or coordinate plane), we can use Pythagoras theorem to find the distance between any two points.

Consider the following examples:

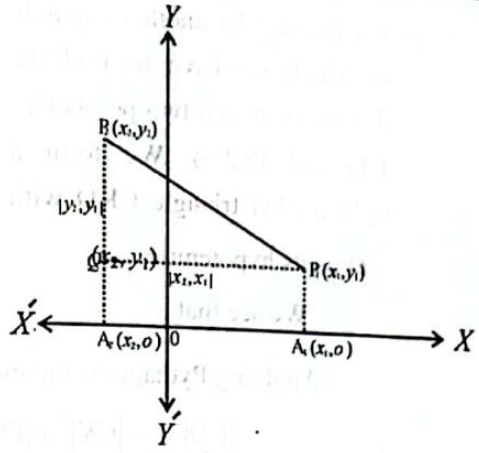
If we have to find distance between two points P_1 and P_2 where coordinates are (x_1, y_1) and (x_2, y_2) respectively then by forming a right–angled triangle P_1QP_2, we have $\overline{P_1P_2}$ as hypotenuse of the triangle.

Also $|P_1Q| = |x_2 - x_1|$

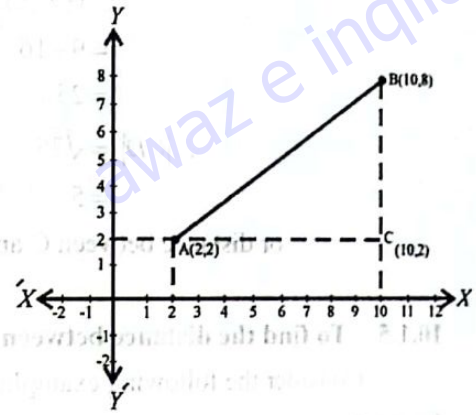
and $|P_2Q| = |y_2 - y_1|$

By Pythagoras theorem

$$|P_1P_2|^2 = |(x_2 - x_1)|^2 + |y_2 - y_1|^2$$
$$= (x_2 - x_1)^2 + (y_2 - y_1)^2$$
$$\therefore d|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



1) To find the distance between A(2,2) and B(10,8), we form a right–angled triangle ABC with $\overline{AB}$ as its hypotenuse.



$$|AC| = \sqrt{(10-2)^2 + (2-2)^2}$$
$$= \sqrt{(8)^2 + (0)^2}$$
$$= \sqrt{64+0} = \sqrt{64} = 8$$
$$|BC| = \sqrt{(10-10)2 + (2-8)^2}$$
$$= \sqrt{(0)^2 + (-6)^2} = \sqrt{36} = 6$$

Applying Pythagoras theorem

$$|AB|^2 = |AC|^2 + |BC|^2 = \left(\sqrt{64}\right)^2 + \left(\sqrt{36}\right)^2$$
$$= 64 + 36$$
$$= 100 = (10)^2$$
$$\therefore |AB| = 10$$

Or distance between A and B or d (A,B) =10.

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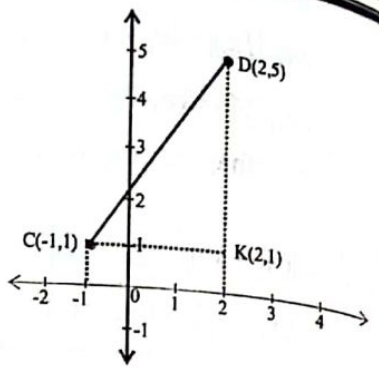
2) Let us consider another example in which we have to find the distance between two points C(–1,1) and D(2,5). We form a right–angled triangle CKD with $\overline{CD}$ as its hypotenuse.



We see that:

Applying Pythagoras theorem.

$$|CD|^2 = |CK|^2 + |DK|^2 = [2-(-1)]^2 + (5-1)^2$$
$$= (3)^2 + (4)^2$$
$$= 9+16$$
$$= 25$$
$$\therefore |CD| = \sqrt{25}$$
$$= 5$$

or distance between C and D or d (C,D)=5

10.1.5 To find the distance between two points by using distance formula:

Consider the following example.

Example

Find the distance between the points A and B when:

i) A (6, –2) , B (2, 1)

ii) A (7, –5) , B (4, 6)

Solution:

i) A (6, –2) , B (2, 1)

Here $x_1 = 6,$ $x_2 = 2,$ $y_1 = -2$ *and* $y_2 = 1$

Applying the distance formula

$$d(A,B) \text{ or } |AB| = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$
$$= \sqrt{(2-6)^2+[1-(-2)]^2}$$
$$= \sqrt{(-4)^2+(1+2)^2}$$
$$= \sqrt{16+9}$$
$$= \sqrt{25}$$
$$= 5.$$

iii) A (7, –5) , B (4, 6)

Here $x_1 = 7,$ $x_2 = 4,$ $y_1 = -5$ *and* $y_2 = 6$

Applying the distance formula

$$d(A,B) \text{ or } |AB| = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$
$$= \sqrt{(4-7)^2+[6-(-5)]^2}$$
$$= \sqrt{(-3)^2+(11)^2}$$
$$= \sqrt{9+121}$$
$$= \sqrt{130}$$

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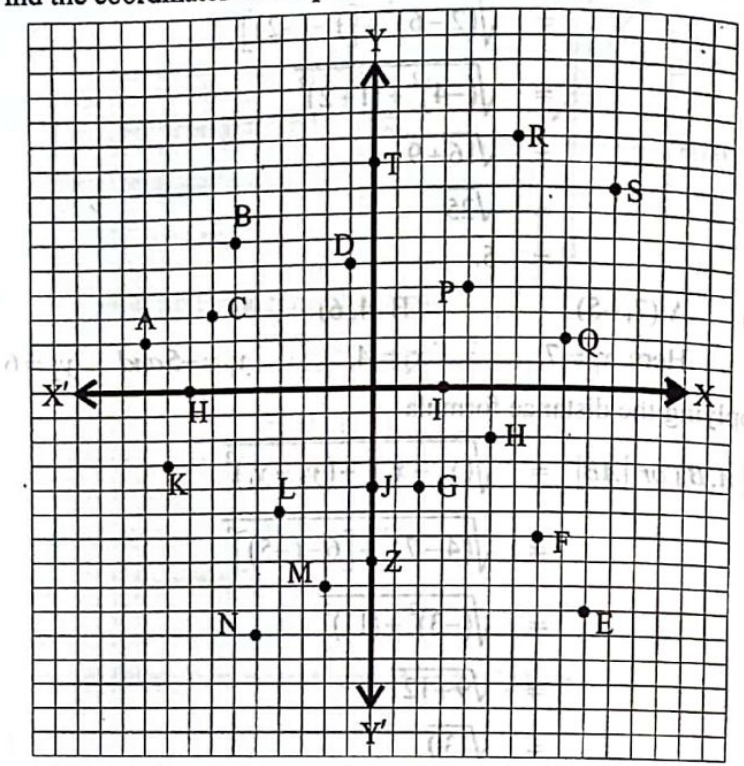
Exercise–10.1

1. Find the coordinates of the points shown in the figure.



2. Determine the x and y coordinates of the following points and then plot the points:

(i) A (–2, 5) (ii) B (3, 0) (iii) C (6, –7)
(iv) D (–4, –2) (v) E (0, –3) (vi) F (8, 4)
(vii) G (0, 6) (viii) H (–4, 0)

3. In which quadrant, the following points are lying?

(i) $(7, \sqrt{5})$ (ii) (5, 9.5)
(iii) $(\sqrt{2}, -4.1)$ (iv) (–2.9, –3.6)
(v) $\left(-\frac{3}{5}, -1\right)$ (vi) $\left(2\frac{3}{4}, -5\right)$

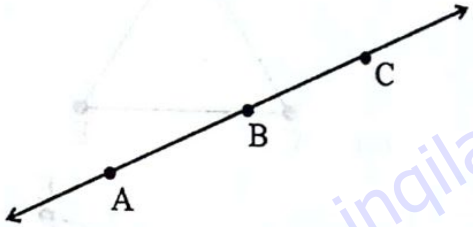
(vii) $(-6, \sqrt{3})$ (viii) (–1.49, 2.5)
(ix) (9, 0) (x) (0, –2.5)

4. Find the distance between the following points:

(i) X (–4, –5), Y (0, 2) (ii) X (3, 7), Y (–5, 1)
(iii) X (4, 0), Y (0, 3) (iv) X (–1, 9), Y (3, 4)
(v) X (–7, 2), Y (5, 7) (vi) X (–4, 12), Y (6, –12)

10.2 Collinear Points

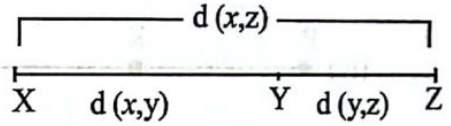
Three points A, B and C are said to be collinear if they lie on the same straight line, as shown in the figure.



If three points lie on the same straight line, the sum of the lengths of the two shorter line segments should be equal to the longest line segment, i.e. if X, Y and Z lie on the same line (as shown in the figure), then

$$d(X,Y)+d(Y,Z)=d(X,Z)$$

or $$|XY|+|YZ|=|XZ|$$

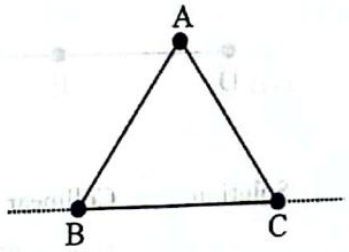


Non–collinear points do not lie on the same line. Three non–collinear points form a triangle.

If three points A, B and C do not lie on the same line (as shown in the figure), then

$$d(A,B)+d(A,C)>d(B,C)$$

or $$|AB|+|AC|>|BC|$$



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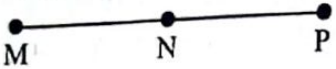
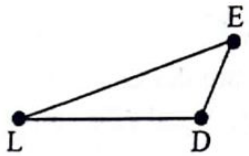
10.2.1 To distinguish between collinear and non–collinear points

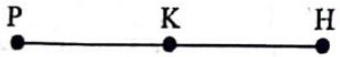
Example–1

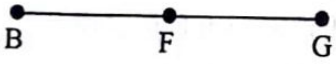
Considering the given figure, separate collinear and non–collinear points from the following:

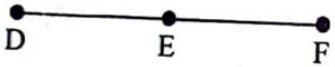
(i)


(ii)

(iii)


(iv)


(v)


(vi)


Solution: Collinear points: (i), (iv), (v), (vi),

Non–collinear points: (ii), (iii)

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10.2.2 Use distance formula to show that given three (or more) points are collinear

Example–2

Show that the points A(3,2), B(9,6) and C(12,8) are collinear.

Solution:

From A(3,2), B(9,6) and C(12,8), we have

$$d(A,B)=\sqrt{(9-3)^2+(6-2)^2}$$
$$=\sqrt{36+16}$$
$$=\sqrt{52}=2\sqrt{13}$$
$$d(B,C)=\sqrt{(12-9)^2+(8-6)^2}$$
$$=\sqrt{9+4}=\sqrt{13}$$
$$d(A,C)=\sqrt{(12-3)^2+(8-2)^2}$$
$$=\sqrt{81+36}$$
$$=\sqrt{117}=3\sqrt{13}$$

Since $d(A,B)+d(B,C)=2\sqrt{13}+\sqrt{13}$

$$=3\sqrt{13}$$
$$=d(A,C)$$

Therefore A, B and C are collinear, since the points lie on the same line.

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Exercise–10.2

1. Use the figure to indicate the collinear nd non–collinear from the following points.

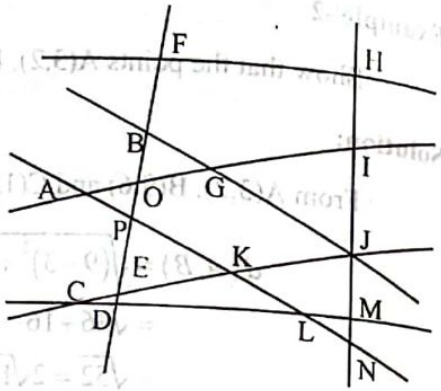
(i) (O, G, I) (ii) (E, F, G)

(iii) (M, N, O) (iv) (F, B, O)

(v) (E, K, J) (vi) (J, G, B)

(vii) (A, O, P) (viii) (C, D, L)

(ix) (H, I, J) (x) (B, C, D)



2. Show that the points p(5,–2), q(1,2) and r(–2,5) are collinear.

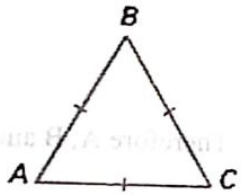
3. Show that the points A(–3,–4), B(0,2) and C(2,6) are collinear.

4. Show that the points A(–9,–5), B(–3,–2), C(3,1) and D(9,4) are collinear.

10.2.3 Use of distance formula in showing that the given three non–collinear points form an equilateral/an isosceles/a scalene/ a right angled triangle.

(i) Equilateral triangle

Three non–collinear points A,B,C form an equilateral triangle if they are equidistant from each other i.e.

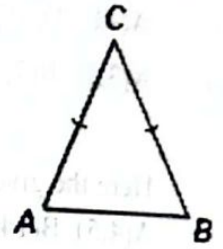


Either $d(A,B)=d(B,C)=d(C,A)$

or $|AB|=|BC|=|CA|$

(ii) Isosceles triangle

Three non–collinear points A,B,C develop an isosceles triangle if any one of the points is equidistant from the other two, points.



Either $|AB|=|AC|$

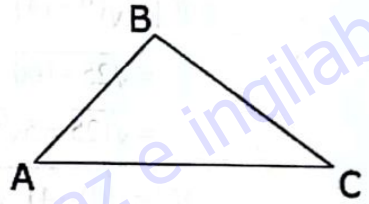
or $|BC|=|BA|$

or $|CA|=|CB|$

(iii) Scalene triangle

Three non–collinear points A,B,C develop a scalene triangle if:

$$d(A,B)\neq d(B,C)\neq d(C,A)$$

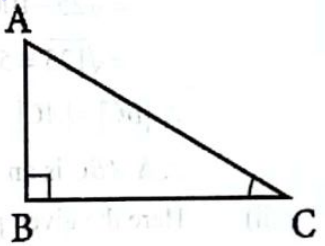


or $|AB|\neq|BC|\neq|CA|$

(iv) Right angled triangle

Three non–collinear points A,B,C develop a right–angled triangle if:

$$|AC|^2=|AB|^2+|BC|^2$$



Example

Which of the following points form:

a) an equilateral triangle
b) an isosceles triangle
c) scalene triangle
d) right angled triangle

i) A(4,5), B(14,5), C(9,15)

ii) A(3,–2), B(–2,8), C(10,–2)

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iii) $A(-1,-1), B(11,5), C(2,-7)$

iv) $A(3,2), B(7,4), C\left(5+\sqrt{3}, 3-3\sqrt{3}\right)$

Solution

i) Here the given points are:
$A(4,5), B(14,5), C(9,15)$

we have

$$|AB| = \sqrt{(14-4)^2+(5-5)^2}$$
$$= \sqrt{100} = 10$$
$$|BC| = \sqrt{(9-14)^2+(15-5)^2}$$
$$= \sqrt{25+100}$$
$$= \sqrt{125} = 5\sqrt{5}$$
$$|AC| = \sqrt{(9-4)^2+(15-5)^2}$$
$$= \sqrt{25+100}$$
$$= \sqrt{125} = 5\sqrt{5}$$
$$\therefore |BC| = |AC|$$

$\therefore \Delta ABC$ is an isosceles triangle.

ii) Here the given points are:
$A(3,-2), B(-2,8), C(10,-2)$

we have

$$|AB| = \sqrt{(-2-3)^2+[8-(-2)]^2}$$
$$= \sqrt{25+100}$$
$$= \sqrt{125} = 5\sqrt{5}$$
$$|BC| = \sqrt{[10-(-2)]^2+(-2-8)^2}$$
$$= \sqrt{144+100}$$

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$$= \sqrt{244} = 2\sqrt{61}$$
$$|AC| = \sqrt{(10-3)^2+[-2-(-2)]^2}$$
$$= \sqrt{49+0}$$
$$= 7$$
$$\therefore |AB| \neq |BC| \neq |AC|$$

$\therefore A, B, C$ are the vertices of a scalene triangle.

iii) The given points are:
$A(-1,-1), B(11,5), C(2,-7)$

we have

$$|AB| = \sqrt{[11-(-1)]^2+[5-(-1)]^2}$$
$$= \sqrt{144+36}$$
$$= \sqrt{180} = 6\sqrt{5}$$
$$|BC| = \sqrt{(2-11)^2+(-7-5)^2}$$
$$= \sqrt{81+144}$$
$$= \sqrt{225} = 15$$
$$|AC| = \sqrt{[2-(-1)]^2+[-7-(-1)]^2}$$
$$= \sqrt{9+36}$$
$$= \sqrt{45} = 3\sqrt{5}$$

We see that

$$|AB|^2+|AC|^2 = \left(6\sqrt{5}\right)^2+\left(3\sqrt{5}\right)^2$$
$$= 180+45$$
$$= 225 = (15)^2$$
$$= |BC|^2$$

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$\therefore$ A, B, C form a right angled triangle.

iv) The given points are:

$$A(3,2),\ B(7,4),\ C\left(5+\sqrt{3}, 3-3\sqrt{3}\right)$$

we have

$$|AB| = \sqrt{(7-3)^2+(4-2)^2}$$
$$= \sqrt{16+4}$$
$$= \sqrt{20} = 2\sqrt{5}$$

$$|BC| = \sqrt{\left(5+\sqrt{3}-7\right)^2+\left(3-3\sqrt{3}-4\right)^2}$$
$$= \sqrt{\left(-2+\sqrt{3}\right)^2+\left(-1-3\sqrt{3}\right)^2}$$
$$= \sqrt{4+3-4\sqrt{3}+1+6\sqrt{3}+9}$$ "
$$= \sqrt{19+2\sqrt{3}}$$

$$|AC| = \sqrt{\left(5+\sqrt{3}-3\right)^2+\left(3-2\sqrt{3}-2\right)^2}$$
$$= \sqrt{\left(2+\sqrt{3}\right)^2+\left(1-2\sqrt{3}\right)^2}$$
$$= \sqrt{4+3+4\sqrt{3}+1+12-4\sqrt{3}}$$ "
$$= \sqrt{20} = 2\sqrt{5}$$

$\therefore$ $|AB| = |BC| = |AC|$

$\therefore$ A, B, C determine an equilateral triangle.

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Exercise–10.3

1. Use distance formula to show that the following points develop an equilateral triangle.
   (i) $(5,8),(5,12),\left(5+2\sqrt{3},10\right)$ (ii) $(0,8),(0,-8),(8\sqrt{3},0)$
   (iii) $(-6,0),(10,0),\left(2,8\sqrt{3}\right)$ (iv) $(-3,-2),(-3,4),(-3+3\sqrt{3},1)$
2. Use distance formula to show that the following points are the vertices of an isosceles triangle:
   (i) (–1,6), (3,2), (6,9) (ii) (4,7), (5,0), (–3,8)
   (iii) (–4,6), (4,–2), (5,7) (iv) (0,6), (4,2), (7,9)
3. Use distance formula to show that the following points represent a scalene triangle:
   (i) (–2,5), (4,3), (7,–5) (ii) (2,1), (6,1), (5,5)
   (iii) (–5,–4), (2,3), (–2,4) (iv) (–10,4), (7,1), (–3,–3)
4. Use distance formula, to show that the following points form a right angled triangle.
   (i) (–2,3), (4,3), (–2,11) (ii) (1,4), (1,7), (8,4)
   (iii) (–6,5), (–6,–7), (–1,–7) (iv) (1,4), (3,2), (6,9)

REVISION EXERCISE–10

1. Which of the following statements are true and which are false. Encircle T in case the statement is true and F for being false.
   i) Coordinate Geometry is also called Analytic Geometry. T–F
   ii) Coordinate plane represents a two dimensional plane. T–F
   iii) On x–axis, points to the right of the origin represent positive real number. T–F
   iv) Point (4, –3) lies in quadrant III. T–F
   v) Distance between (6, 0) and (0, 8) is 10. T–F
   vi) Three collinear points form a triangle. T–F
   vii) In a triangle ABC, if $|AB| = |BC|$ then the triangle is an equilateral one. T–F
2. Fill in the blanks:
   i) (–6, –8) lies in quadrant _____.
   ii) On y–axis, the points below the origin are associated with _____ real numbers.
   iii) In quadrant II, x–coordinate is _____.
   iv) Point (0,–4) lies on _____.

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3. Four suggested answers are given for each question. Select the correct answer and write the corresponding letter (a, b, c, or d) in the box provided for each question.

i) In which quadrant, the x–coordinate is positive and the y–coordinate is negative? ☐

(a) Quadrant IV (b) Quadrant III

(c) Quadrant II (d) Quadrant I

ii) In which quadrant the point (–7,5) lies? ☐

(a) Quadrant I (b) Quadrant II

(c) Quadrant III (d) Quadrant IV

iii) Three non–collinear points A, B and C form an isosceles triangle if: ☐

(a) d (A, B) = d (A, C) (b) d (A, C) = d (B, A)

(c) d (C, A) = d (C, B) (d) All are true

iv) For a triangle ABC, with right angle at A, which of the following is true? ☐

(a) $|AB|^2 = |AC|^2 + |BC|^2$ (b) $|AC|^2 = |AB|^2 + |BC|^2$

(c) $|BC|^2 = |AB|^2 + |AC|^2$ (d) All are true.

v) For the points X, Y and Z, $|XY|+|YZ|>|XZ|$ shows that: ☐

(a) X,Y,Z are collinear (b) X,Y,Z are non collinear

(c) XYZ is a triangle (d) Both (b) and (c) are true

Glossary

- Coordinate plane describes a two–dimensional plane in terms of two perpendicular axes———x–axis and y–axis. Point of intersection of these axes is called origin'.
- In coordinate plane, points are indicated by their positions along x and y axes.
- Directed distance between two points can be positive or negative but undirected distance between two points is always positive.
- Coordinate axes divide the coordinate plane into four quadrants, numbered I–IV in counterclockwise direction.
- In quadrant I, both the coordinates are positive, while in quadrant III both the coordinates are negative.
- In quadrant II, x coordinate is negative and y coordinate is positive while in quadrant IV x–coordinate is positive and y–coordinate is negative.
- Distance between two points P_1 and P_2 having coordinates (x_1, y_1) and (x_2, y_2) respectively is calculated with the help of distance formula:

$$d(P_1, P_2) = |P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

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- Three or more points are collinear if they lie on the same line.
- If three points A, B and C are collinear then $|AB|+|BC|=|CA|$
- Three points are said to be non–collinear if they do not lie on a line. Three non–collinear points form a triangle.
- Three non–collinear points A, B and C form:
  - an equilateral triangle if $|AB|=|BC|=|CA|$
  - an isosceles triangle if any two among $|AB|,|BC|,|CA|$ are equal.
  - an scalene triangle if $|AB|\neq|BC|\neq|CA|$
  - an right–angled triangle at A if $|BC|^2=|AB|^2+|AC|^2$

ANSWERS

Exercise: 1.1

1. i , ii, iv, v are rational expression.
   iii not rational expression
2. i, iii, v not polynomials
   ii, iv polynomials
3. i, iv, v, vi rational expressions
   ii, iii, not rational expressions
4. i) 3 ii) 2 iii) 1 iv) 0 (v) 2

Exercise: 1.2

1. (i), (vi)
2. (i) $x-3$ (ii) $\frac{x^2-x+1}{x-1}$ (iii) $\frac{x+1}{y-1}$
   (iv) $\frac{x^2+1}{x^2-1}$ (v) x^2+2x+2 (vi) $z-1$

Exercise: 1.3

1. (i) $\frac{3x+3}{2x-2}$ (ii) $\frac{3y^2+2x+y}{xy}$ (iii) $\frac{2y}{y^2-1}$
   (iv) $\frac{6y^3+2y}{y^4-1}$ (v) $\frac{(xy+1)(x+y)}{x+3}$

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2. (i) $\frac{44y-93}{16y^2-9}$ (ii) $\frac{2b}{a^2-b^2}$ (iii) $\frac{y+x}{x}$
(iv) $\frac{2x}{(x+1)(x-1)}$

3. (i) $\frac{7x+2y}{x+3}$ (ii) $\frac{x-y}{x+y}$ (iii) $\frac{3(2x+3y)}{(x+1)(x-1)}$
(iv) $\frac{x^3+a^3}{x^3-a^3}$

4. (i) $\frac{2x+3}{2x+1}$ (ii) $\frac{ax^2+2}{a+2}$ (iii) $\frac{(2-x)(x-3)}{6(x+3)}$
(iv) $\frac{(x-2)(x^4+1)}{4(x^2-1)}$

5. (i) $\frac{4a}{a^2-b^2}$ (ii) $\frac{x^2+25}{x+5}$ (iii) $\frac{(2x+1)(2x+3)}{(x-3)}$

Exercise: 1.4

1. $\frac{14}{3}$, -2 2. 92 3. $\frac{8}{3}$ 4. $\frac{-17}{8}$
5. 25 6. 83 7. 2

Exercise: 1.5

1. (i) $2(x^2+y^2)$ (ii) $2(4x^4+9y^4)$ (iii) $16xy^2$
(iv) $108xy^2$ (v) $2(16x^4+y^2)$ (vi) $40xy$

2. (i) $\frac{5}{2}$ (ii) 34 (iii) 58 (iv) 450

3. (i) 561 (ii) 800 (iii) 364 (iv) 8

Exercise: 1.6

1. (i) 5 (ii) zero (iii) 26
2. $\pm 2\sqrt{3}$ (ii) $\pm 2\sqrt{3}$ (iii) $\pm 3\sqrt{2}$
3. 17 (ii) 34 (iii) -9

Exercise: 1.7

1. (i) $x^3+y^3+3xy(x+y)$ (ii) $8x^3-27y^3-36x^2y+54xy^2$
(iii) $\frac{1}{x^3}+\frac{8}{27}y^3+\frac{2y}{x}\left(\frac{1}{x}+\frac{2y}{3}\right)$

2. (i) $(x+2y)^3$ (ii) $(x-2y)^3$
3. 144 4. -80
5. (i) x^6-64y^6 (ii) $64a^6-729b^6$
6. (i) -39 (ii) -36 (iii) $\frac{-7}{2}$
7. (i) 720 (ii) 710 (iii) 2

Exercise: 1.8

1. iii, iv, vii, viii, ix, x
2. i, iii, v,
3. (i) $\frac{-\sqrt{2}+3}{7}$ (ii) $\frac{\sqrt{3}+1}{2}$ (iii) $\frac{\sqrt[3]{3}+1}{11}$ (iv) $\frac{\sqrt{5}+\sqrt{3}}{2}$ (v) $\frac{\sqrt{7}-\sqrt{3}}{4}$

Exercise: 1

1. i. T ii. T iii. F iv. F v. F
2. i. b ii. b iii. b iv. b
3. (i) $\sqrt{2}+1$ (ii) rationalization (iii) $a^3-b^3-3ab(a-b)$
(iv) $\left(\sqrt{5}-\sqrt{3}\right)\left(8+\sqrt{15}\right)$

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Exercise: 2.1

1. $2(x-4y)$
2. $3x^2(x+2)$
3. $x^2y^2(4xy-1)$
4. $2(x+y-2z)$
5. $2x^2(8x^2-2x+1)$
6. $7xy(4xy^3-5y^2+6x^3)$
7. $7abc(1-2abc)$
8. $16x(1+2y^2)$
9. $4x(x+2a^2x-a^3)$
10. $9x^3y^3(5x^7y-7x^3y^4+8)$

Exercise: 2.2

1. $(x+y)(x+z)$
2. $(a+2)(a+b)$
3. $(a+b)(a-2b^2)$
4. $(x-1)(a-1)$
5. $(3x-2)(x^2+6)$
6. $(3x+y)(1+a+b)$
7. $(2y+3)(y^4+y^2+3)$
8. $(x-y)(x+z+3)$

Exercise: 2.3

1. $(x+2)(x+2)$
2. $(y-4)(y-4)$
3. $(3a+2b)(3a+2b)$
4. $-2(x-3)(x-3)$
5. $(5a+5b-2c)(5a+5b-2c)$
6. $(7y-z)(7y-z)$
7. $(4a-3b)(4a-3b)$
8. $\left(2x+\frac{1}{x}\right)\left(2x+\frac{1}{x}\right)$

Exercise: 2.4

1. $(4x+3)(4x-3)$
2. $xy(x+y)(x-y)$
3. $2(x+12)(x-12)$
4. $2(x-y)(a+3)(a-3)$
5. $\left(\frac{x}{y}+\frac{y}{x}\right)\left(\frac{x}{y}-\frac{y}{x}\right)$
6. $\left(\frac{9}{4}x^2+25\right)\left(\frac{3x}{2}+5\right)\left(\frac{3}{2}x-5\right)$
7. $(4+5x)(4-5x)$
8. $(b+1)(b-1)(ab+2)$
9. $(3y+8x)(3y-8x)$
10. $(x^4+y^4)(x^2+y^2)(x+y)(x-y)$

Exercise: 2.5

1. $(x-y+2)(x-y-2)$
2. $(3ab-2c+5d)(3ab-2c-5d)$
3. $(1+a-b)(1-a+b)$
4. $(1+x-y)(1-x+y)$
5. $(a+b+c-d)(a+b-c+d)$
6. $(p+q+x-y)(p+q-x+y)$
7. $(b-1+a)(b-1-a)$
8. $(5ab-2c+4d)(5ab-2c-4d)$

Exercise: 2.6

1. $(x^2+2x+2)(x^2-2x+2)$
2. $(2x^2+6x+9)(2x^2-6x+9)$
3. $(2p^2+2pq+q^2)(2p^2-2pq+q^2)$
4. $(x^2+x+1)(x^2-x+1)$
5. $(x^4-x^2+1)(x^2+x+1)(x^2-x+1)$
6. $\left(x^2+\frac{1}{x^2}+3\right)\left(x^2+\frac{1}{x^2}-3\right)$
7. $(4x^2+y^2+2xy)(4x^2+y^2-2xy)$
8. $ab(2a^2+3b^2+ab)(2a^2+3b^2-ab)$

Exercise: 2.7

1. $(x+2)(x+4)$
2. $(x-4)(x+2)$
3. $(y+11)(y-5)$
4. $y(y-3)(y-5)$
5. $(z-5)(z+4)$
6. $-2x(x-1)(x-23)$
7. $-3x^2(x+6)(x-8)$
8. $(x+7)(x+3)$

Exercise: 2.8

1. $(x-3)(2x-1)$
2. $(2x+3)(2x+1)$
3. $(2y-1)(y-1)$
4. $3x(2x-3)(x-1)$
5. $2x(y+6)(y-2)$
6. $-(3y-2)(4y+1)$
7. $(x+4)(2x+5)$
8. $(2x+7)(2x+3)$
9. $-2xy(4x+3y)(2x+y)$ 10. $(x+3)(8x-1)$

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Exercise: 2.9

1. $(x+y)^3$
2. $(4x-3y)^3$
3. $(5a+b)^3$
4. $(3a+7b)^3$
5. $\left(\frac{a}{2}-\frac{b}{3}\right)^3$
6. $\left(x+\frac{1}{x}\right)^3$
7. $(2a+4b)^3$
8. $\left(2x-\frac{1}{3x}\right)^3$

Exercise: 2.10

1. $2(x+2)(x^2-2x+4)$
2. $(a-4b)(a^2+4ab+16b^2)$
3. $(x+y)(x-y)(x^2-xy+y^2)(x^2+xy+y^2)$
4. $(4y+3)(16y^2-2x+9)$
5. $3(2x+1)(4x^2-2x+1)$
6. $(a+c+d)\left[a^2-a(c+d)+(c+d)^2\right]$
7. $(x^2+y^2)(x^4-x^2y^2+y^4)(x+y)(x-y)(x^2-xy+y^2)(x^2+xy+y^2)$
8. $(7a-5b)(49a^2+35ab+25b^2+1)$
9. $(1-3y)(1+3y+9y^2)$
10. $18[x-y-2a+2b]\left[(x-y)^2+2(x-y)(a-b)+4(a-b)^2\right]$

Exercise: 2.11

1. Remainder = 4
2. Remainder = −1
3. Remainder = −4
4. $k=3$
5. $a=5$

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Exercise: 2.12

1. $(x+2)(x-2)(x-1)$
2. $(x+2)(x-2)(x+1)$
3. $(x-1)(x-3)(x-5)$
4. $(x-3)^2(x+2)$
5. $(x+1)(x+5)(x-4)$
6. $(x-2)(x-3)(x+4)$
7. $(x-1)(x-2)(x+3)$
8. $(x+2)(x-4)^2$

Exercise: 2.13

Q.1 True or false.
(i) T (ii) F (iii) T (iv) F (v) T

Q2. Fill in the blanks.
(i) $(3x+4y)(3x-4y)$ (ii) $(x-y)(x-z)$
(iii) $4a^4-12a^2b^2+9b^2$ (iv) $(x-3)(x^2+3x+9)$
(v) $(x-5)(x-6)$

Q3. Multiple choices.
(i) c (ii) a
(iii) b (iv) b (v) c

Review Exercise: 2

1. $(b+c+d)(a^2+b^2)$
2. $(a-b+3)(a-b-3)$
3. $-2x(x-23)(x-1)$
4. $(x^2+y^2)(x^4-x^2y^2+y^4)$
5. $R=28$
6. $2(x-4)(x^2+4x+16)$
7. $(y-8)(y+6)$
8. $(x-1)^2(x+5)$
9. $(3x-y)^3$
10. $(x-3)(x+1)^2$

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Exercise: 3.1

1. $x-6$
2. $a^2b^2\left(a^2+ab+b^2\right)$
3. x^2+y^2
4. $2m-1$
5. $a(x-3)$
6. $x-3$
7. y^2-2y+1
8. $x(x+1)$
9. $x+5$
10. $x(x-2)$

Exercise: 3.2

1. $(x+3)\left(x^3-8\right)$
2. $(x-3)\left(x^2-4\right)$
3. $(x+1)(x+2)(x+3)(x-4)$
4. $\left(x^4-1\right)(5x-7)\left(2x^2+1\right)$
5. $H.C.F=x-2,\quad L.C.M=\left(2x^3-5x^2+4\right)\left(x^2+2x-12\right)$
6. $H.C.F=x^2+3x+2,\quad L.C.M=(x-4)\left(x^3-2x^2-13x-10\right)$
7. $H.C.F=a^2+a+1,\quad L.C.M=\left(a^4-a^2-a+1\right)\left(a^2-a+1\right)$
8. x^2-x-2
9. $x+4$
10. $2x^2+13x+15$ and x^2+4x-5

Exercise: 3.3

1. 2
2. $\dfrac{1}{x+3}$
3. 0
4. $\dfrac{8a^7}{a^8-b^8}$
5. $\dfrac{8x^7y}{x^8-y^8}$
6. $\dfrac{2a+13}{(a+2)(a+5)(a+8)}$
7. $\dfrac{48}{(x+2)(x+3)(x+5)(x+6)}$

Exercise: 3.4

1. $x-4$
2. $\dfrac{1}{6(x-1)}$
3. $\dfrac{2\left(x^2+2x+4\right)}{x(x-3)}$
4. $\dfrac{x+y}{x^2-xz+z^2}$
5. 3
6. $\dfrac{1}{a+b}$
7. $\dfrac{x-3}{x^2+x-6}$
8. 1

Exercise:: 3.5

1. $\pm(2x-5y)$
2. $\pm(4x+9y)$
3. $\pm(x-y+3)$
4. $\pm\left(x^2+\dfrac{1}{x^2}-2\right)$
5. $\pm\left(x+\dfrac{1}{x}-4\right)$
6. $\pm\left(x^3-x^2+x\right)$
7. $\pm\left(\dfrac{x^2}{2}-2x+\dfrac{a}{3}\right)$
8. $\pm\left(x^2+\dfrac{x}{2}-4\right)$
9. $k=64$

Exercise:: 3.6

1. True and False questions.
   (i) T (ii) F (iii) F
   (iv) T (v) T
2. Fill in the blanks.
   (i) $x-2$ (ii) $(a+1)(a+2)(a+3)$
   (iii) $x-\dfrac{1}{x}$ (iv) least
3. Multiple choices
   (i) exactly
   (i) b (ii) a (c) c
   (iv) c (v) a (vi) a
   (vii) d (viii) c

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Review Exercise: 3

1. $x+3$
2. $(x-1)(x^3+6x^2+11x+6)$
3. $x^3+7x^2+8x+14$
4. $\frac{-7x+9}{(x-2)(x+3)(x-3)}$
5. $\frac{2(x^2+2x+4)}{x(x-3)}$
6. $\frac{1}{5(x-2)}$
7. $\pm(x+1)(x+5)(x+8)$

Exercise: 4.1

1. $\{4\}$
2. $\{3\}$
3. $\{4\}$
4. $\{3\}$
5. $\left\{-\frac{17}{14}\right\}$
6. $\{2\}$
7. 11 *and* 12
8. 51, 53 *and* 55
9. breadth = 4m
   Length = 6m
10. Age of daughter = 2 years
    age of mother = 13 years
11. 64
12. 7

Exercise: 4.2

1. $\{31\}$
2. $\{\ \}$
3. $\{13\}$
4. $\{1\}$
5. $\{2\}$
6. $\{23\}$
7. $\{\ \}$
8. $\{9\}$

Exercise: 4.3

1. $\{2,-2\}$
2. $\{8,-6\}$
3. $\{30,-30\}$
4. $\left\{3,-\frac{5}{3}\right\}$
5. $\left\{\frac{11}{2},\frac{-31}{2}\right\}$
6. $\left\{5,\frac{1}{5}\right\}$
7. $\left\{2,\frac{-3}{2}\right\}$
8. $\left\{-1,\frac{-13}{9}\right\}$

Exercise: 4.4

1. $x > 0$
2. $x < 0$
3. $x \le 1$
4. $x \le -5$
5. $x \ge -3$
6. $x \ge 4$
7. $-5 \le x \le 6$
8. $3 \ge x \ge -2$
9. $4 < x < 6$
10. $-3 < x < 0$

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Exercise: 3.1

1. $x-6$
2. $a^2b^2(a^2+ab+b^2)$
3. x^2+y^2
4. $2m-1$
5. $a(x-3)$
6. $x-3$
7. y^2-2y+1
8. $x(x+1)$
9. $x+5$
10. $x(x-2)$

Exercise: 3.2

1. $(x+3)(x^3-8)$
2. $(x-3)(x^2-4)$
3. $(x+1)(x+2)(x+3)(x-4)$
4. $(x^4-1)(5x-7)(2x^2+1)$
5. $H.C.F=x-2, \quad L.C.M=(2x^3-5x^2+4)(x^2+2x-12)$
6. $H.C.F=x^2+3x+2, \quad L.C.M=(x-4)(x^3-2x^2-13x-10)$
7. $H.C.F=a^2+a+1, \quad L.C.M=(a^4-a^2-a+1)(a^2-a+1)$
8. x^2-x-2
9. $x+4$
10. $2x^2+13x+15$ and x^2+4x-5

Exercise: 3.3

1. 2
2. $\frac{1}{x+3}$
3. 0
4. $\frac{8a^7}{a^8-b^8}$
5. $\frac{8x^7y}{x^8-y^8}$
6. $\frac{2a+13}{(a+2)(a+5)(a+8)}$
7. $\frac{48}{(x+2)(x+3)(x+5)(x+6)}$

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Exercise: 3.4

1. $x-4$
2. $\frac{1}{6(x-1)}$
3. $\frac{2(x^2+2x+4)}{x(x-3)}$
4. $\frac{x+y}{x^2-xz+z^2}$
5. 3
6. $\frac{1}{a+b}$
7. $\frac{x-3}{x^2+x-6}$
8. 1

Exercise:: 3.5

1. $\pm(2x-5y)$
2. $\pm(4x+9y)$
3. $\pm(x-y+3)$
4. $\pm\left(x^2+\frac{1}{x^2}-2\right)$
5. $\pm\left(x+\frac{1}{x}-4\right)$
6. $\pm(x^3-x^2+x)$
7. $\pm\left(\frac{x^2}{2}-2x+\frac{a}{3}\right)$
8. $\pm\left(x^2+\frac{x}{2}-4\right)$
9. $k=64$

Exercise:: 3.6

1. True and False questions.
   (i) T (ii) F (iii) F
   (iv) T (v) T
2. Fill in the blanks.
   (i) $x-2$ (ii) $(a+1)(a+2)(a+3)$
   (iii) $x-\frac{1}{x}$ (iv) least
   (i) exactly
3. Multiple choices
   (i) b (ii) a (c) c
   (iv) c (v) a (vi) a
   (vii) d (viii) c

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Exercise: 5.1

1. (iii), (v)
2. (i) $x^2 - 3x - 1 = 0$
   (ii) $2x^2 - 3x - 1 = 0$
   (iii) $x^2 - 0x - 3 = 0$
   (iv) $x^2 - 5x + 1 = 0$
   (v) $3x^2 - x + 0 = 0$

Exercise: 5.2

1. (i) $\{15,1\}$ (ii) $\{3,-7\}$ (iii) $\{-3,-4\}$ (iv) $\left\{1,\frac{1}{6}\right\}$ (v) $\left\{\frac{-1}{8}\right\}$
2. (i) $\left\{1+\sqrt{445}, 1-\sqrt{445}\right\}$ (ii) $\{1 \pm \sqrt{179}\}$ (iii) $\left\{2+5\sqrt{37}, 2-5\sqrt{37}\right\}$ (iv) $\{24,1\}$ (v) $\left\{5+\sqrt{15}, 5-\sqrt{15}\right\}$ (vi) $\{17,-1\}$

Exercise: 5.3

1. $\{5,3\}$ 2. $\left\{1,\frac{-2}{3}\right\}$ 3. $\left\{\frac{-3}{4}, 0\right\}$ 4. $\left\{\frac{3+\sqrt{6}}{3}, \frac{3-\sqrt{6}}{3}\right\}$
5. $\left\{\frac{-1}{3}, \frac{-1}{2}\right\}$ 6. $\{5,-2\}$ 7. $\left\{-2+2\sqrt{3}, -2-2\sqrt{3}\right\}$

Exercise: 5.4

1. Width $= -6+4\sqrt{31}$ metre and length $= 6+4\sqrt{31}$ metre.
2. 1 3. Width = 3 metre, Length = 4 metre
4. 4 5. 4.5 or −5, −4 6. −4,2 or 4, −2

Review Exercise: 5

1. (i) True (ii) False (iii) False (iv) True (v) True
2. (i) Three (ii) Two (iii) Quadratic Formula (iv) Roots
3. (i) B (ii) (B) (iii) B (iv) B (v) D

Exercise: 6.1

1. (i) Rows = 3, Columns = 3 (ii) Rows = 3, Columns = 2 (iii) Rows = 2, Columns = 3 (iv) Rows = 2, Columns = 2
2. (i) 3–by–3 (ii) 3–by–2 (iii) 2–by–3 (iv) 2–by–2
3. (i), (v)

Exercise: 6.2

1. (ii), (v)
2. (iii), (v)
3. Square matrices are:
   (ii), (v), (vii), (ix)
   Rectangular matrices are:
   (i), (iii), (iv), (vi), (viii), (x)
4. (ii), (iii), (v), (vi)
5. Scalar matrices
   (iii), (iv), (v), (vii), (viii)
   Diagonal matrices
   All
   Unit matrices
   (iii), (vii)
6. (i) $\begin{bmatrix} 1 & 5 \\ 2 & 7 \end{bmatrix}$ (ii) $\begin{bmatrix} 0 & 3 & 4 \\ 1 & 3 & 5 \\ 3 & 0 & 0 \end{bmatrix}$ (iii) $\begin{bmatrix} 3 & 1 & 5 \\ 2 & 0 & 1 \\ 1 & 4 & 7 \end{bmatrix}$ (iv) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
7. (ii), (iii), (iv)
8. (i), (iv), (v), (vi)

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Exercise: 6.3

1. (i) $\begin{bmatrix} 5 & 0 & 4 \\ 1 & 2 & -1 \end{bmatrix}$ (ii) $\begin{bmatrix} 8 \\ 9 \\ 7 \end{bmatrix}$ (iii) $[7]$

(iv) $[5 \quad 3 \quad 6]$ (v) $\begin{bmatrix} 0 & 3 & 7 \\ 7 & 1 & 9 \\ 9 & 9 & 1 \end{bmatrix}$

(vi) $\begin{bmatrix} 7 & 7 \\ 7 & 9 \end{bmatrix}$

2. (i) $\begin{bmatrix} 0 \\ 6 \\ 6 \end{bmatrix}$ (ii) $[5 \quad 5 \quad 9]$

(iii) $\begin{bmatrix} 4 & 1 & -2 \\ 0 & -11 & 0 \end{bmatrix}$ (iv) $\begin{bmatrix} -3 & 1 & 0 \\ 0 & 4 & 2 \\ -1 & 0 & 1 \end{bmatrix}$

4. (i) $[-8]$ (ii) $\begin{bmatrix} -2 & -1 \\ -3 & -3 \end{bmatrix}$

(iii) $\begin{bmatrix} -9 & 0 & 0 \\ 0 & -9 & 0 \\ 0 & 0 & -9 \end{bmatrix}$ (iv) $\begin{bmatrix} -1 & -3 & -2 \\ -5 & -8 & -9 \end{bmatrix}$

Exercise: 6.4

1. (i) $[11]$ (ii) $\begin{bmatrix} 0 & 0 \\ 6 & 6 \end{bmatrix}$ (iii) $\begin{bmatrix} 20 & 12 \\ 6 & 4 \end{bmatrix}$

(iv) $\begin{bmatrix} 14 & 14 \\ 13 & 13 \end{bmatrix}$ (v) $\begin{bmatrix} 3 & 3 \\ 2 & 2 \end{bmatrix}$ (vi) $\begin{bmatrix} -2 \\ -1 \end{bmatrix}$

4. $\begin{bmatrix} -20 & 1 \\ 1 & 11 \end{bmatrix}$

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Exercise: 6.6

(i) $\begin{bmatrix} 1 & 2 \\ 2 & -3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

(ii) $\begin{bmatrix} \frac{1}{2} & -3 \\ 1 & \frac{3}{2} \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ \frac{1}{2} \end{bmatrix}$

(iii) $\begin{bmatrix} \sqrt{2} & 1 \\ 1 & -\sqrt{3} \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \sqrt{3} \\ 1 \end{bmatrix}$

(iv) $\begin{bmatrix} \frac{4}{5} & -1 \\ \frac{1}{3} & 2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$

2. *i)* $x = \frac{5}{3}, \quad y = \frac{-1}{3}$

ii) $x = \frac{3}{11}, \quad y = \frac{-2}{11}$

iii) $x = \frac{61}{28}, \quad y = \frac{-23}{14}$

iv) $x = 3, \quad y = 1$

3. Answers same as in Q2.

4. Rs.20, Rs.60

5. 19 years, 7 years

6. 58

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5. i) $(43°, 47°)$ ii) $(69°, 21°)$ iii) $(41°, 49°)$ iv) $(79°, 11°)$

6. $87°$ 7. $45°$ 8. $40°, 60°, 80°$

9. $25°, 65°$ 10. $95°, 37°, 48°$

Exercise: 7.2

1. $b = 110°, c = 70°, d = 70°, e = 70°, f = 110°$

2. $b = 115°, c = 65°, d = 115°$

3. i) $a = 80°, b = 100°, c = 80°$ ii) $a = 92°, b = 92°$

iii) $a = 62°, b = 118°, c = 62°, d = 118°, e = 62°, f = 118°, g = 62°,$

4. $m\angle 2 = 40°, m\angle 3 = 85°, m\angle 4 = 55°, m\angle 5 = 40°, m\angle 6 = 85°, m\angle 7 = 125°,$
$m\angle 8 = 55°, m\angle 9 = 125°, m\angle 10 = 55°, m\angle 11 = 40°, m\angle 12 = 140°, m\angle 14 = 140°,$

5. $k = 120°, m = 120°, x = 70°, y = 130°, z = 130°;$ $\overline{AE} \cong \overline{CE}$

6. $a = 60°, b = 52°, c = 52°, d = 68°$

7. $\overline{BD} \cong \overline{DG}, \overline{HD} \cong \overline{DF},$
$m\angle 2 = 80°, m\angle 3 = 80°, m\angle 4 = 100°, m\angle 5 = 80°, m\angle 6 = 80°,$
$m\angle 8 = 30°, m\angle 9 = 110°,$

Exercise: 7.3

1. a) congruent figures: (i, iv), (ii, vii), (iii, v), (x, xii)
b) similar figures: (i, iv, xi), (ii, vii, ix), (vi, viii)

2. a) similar figures: ii, iii

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b) congruent figures: i

c) figures given in (iv) are neither congruent nor similar.

3. i) yes (SAS) Similarity, $x = 6$
ii) yes (SAS) Similarity, $x = 36$
iii) yes (AA) Similarity , $x = 17°$
iv) yes (AA) Similarity , $x = 36$
v) yes (SAS) Similarity , $x = 18$

Exercise: 7.4

1. i) $SSS \cong SSS$ ii) $SAS \cong SAS$
iii) $SSS \cong SSS$ iv) $RHS \cong RHS$

2. i) $\angle B \cong \angle B$ ii) $\overline{AK} \cong \overline{AK}$
iii) $\overline{RW} \cong \overline{RW}$ iv) $\overline{OK} \cong \overline{OK}$
v) $\angle 1 \cong \angle 2, \angle 3 \cong \angle 4$ vi) $\overline{AS} \cong \overline{AS}$

Exercise: 7.5

1. centre: O, chords; $\overline{CE}, \overline{DE}$
diameters: $\overline{AB}, \overline{CD}$ radial segment: $\overline{OF}$

2. Radii: $\overline{OA}, \overline{OB}, \overline{OE}$
Chords: $\overline{CF}, \overline{DF}$
Concyclic points: A, B, C, D, E, F

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3. Minor Arcs: (i) KNL (ii) ADC (iii) PSQ

Major Arcs: (i) KML (ii) ABC (iii) PRQ

4. Minor Arcs: (i) XZY (ii) ABC (iii) PQS

Major Arcs: (i) XZA (ii) ABD (iii) PQR

5. Central Angles: (i) $\angle AOB$ (ii) $\angle KOL$ (iii) $\angle GOH$

Inscribed Angles: (i) $\angle ADB$ (ii) $\angle KNL$ (iii) $\angle GPH$

6. Diameters: 4cm, 5 cm, 6cm,

7. Tangents: (i) $\overleftrightarrow{KL}$ (ii) $\overleftrightarrow{BC}$ (iii) $\overleftrightarrow{PQ}$

Segments: (i) $\overleftrightarrow{XY}$ (ii) $\overleftrightarrow{DE}$ (iii) $\overleftrightarrow{AB}$

8. $x = 18°, y = 27°$

9. $x = 18°, y = 15°$

10. 40°

11. $m\angle KNM = 45°$

Review Exercise: 7

1.	D	2.	C	3.	B
4.	A	5.	B	6.	B
7.	C	8.	D	9.	D
10.	C	11.	B	12.	A
13.	A	14.	B	15.	D

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16.	D	17.	C	18.	B
19.	D	20.	A	21.	A
22.	C	23.	B	24.	D
25.	B	26.	C	27.	B
28.	A	29.	D	30.	C
31.	A	32.	C	33.	C

Review Exercise: 8

1. i) F ii) T iii) T iv) T

2. i) median ii) perpendicular bisector iii) equidistant
iv) sides v) altitude vi) concurrent vii) 2 : 1
viii) concurrent

3. i) c ii) b iii) a iv) a
v) d vi) a vii) d viii) b

Exercise: 9.1

1. (i) 35 (ii) 5.2 (iii) 5.2
(iv) 36 (v) 7.2 (vi) 6.3

2. 8.5km 3. 0.91cm 4. 7cm

5. $13\sqrt{2}cm$ 6. 5m 7. 20m

8. 168m

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Exercise: 9.2

1. (i) $84cm^2$ (ii) $66m^2$ (iii) $84m^2$
   (iv) $420cm^2$
2. (i) $56cm^2$ (ii) $30m^2$ (iii) $9.35m^2$
   (vi) $13\frac{3}{4}m^2$
3. 16m
4. 3.6m
5. (i) $4\sqrt{3}cm^2$ (ii) $\frac{25\sqrt{3}}{2}cm^2$
6. (i) 6cm (ii) 10cm

Exercise: 9.3

1. (i) $117m^2$ (ii) $186.2m^2$ (iii) $489.5m^2$
2. (i) $499.5cm^2$ (ii) $163.2cm^2$ (iii) $892.8cm^2$
3. (i) $289cm^2$ (ii) $79.21cm^2$ (iii) $158.76m^2$
4. (i) $104.4m^2$ (ii) $70m^2$ (iii) $75.6m^2$
5. (i) 21m, 15m
6. 52m, 28m
7. $2640cm^2$
8. $420m^2$
9. 1.5m
10. 8.5cm
11. 66m
12. $182.25m^2$

Exercise: 9.4

1. Rs.1521
2. Rs.59400
3. Rs.210000
4. Rs.24977.5
5. 24400 tiles
6. 11875 tiles

Exercise: 9.5

1. (i) $616cm^2$ (ii) $1386m^2$ (iii) $38.5cm^2$
   (iv) $962.50m^2$
2. (i) $130.13cm^2$ (ii) $443.52m^2$ (iii) $110.88cm^2$ (approx.)
   (iv) $481.25m^2$
3. (i) $88cm^2$ (ii) $3.52cm^2$ (iii) $242m^2$
   (iv) $2.86m^2$
4. 14m
5. 10m
6. 6cm
7. 40m
8. 10m

Exercise: 9.6

1. $6.48m^2$; Rs.259.20
2. $1440\sqrt{2}m^2$
3. $12m^2$
4. $2120m^2$; Rs.7800
5. Rs.243000
6. 7.5m
7. $1325m^2$
8. Rs.69300
9. $13750cm^2$
10. Rs.735000
11. $30625m^2$
12. $6.16m^2$; Rs.164.12
13. 42

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Exercise: 9.7

1. (i) $216mm^3$ (ii) $614.125cm^3$
2. (i) $120dm^3$ (ii) $16.8cm^3$
3. (i) 7cm (ii) 1.2dm
4. $0.18m^3$
5. 4m
6. 6m
7. $1.375m^3$
8. 77m
9. 3.5m
10. 3.5m
11. 14dm, 21dm
12. (i) $303\frac{3}{16}dm^3$ (ii) $0.9\ cm^3$

Exercise: 9.8

1. 30 turns
2. $91.125cm^3$, 1458g
3. Rs.4200
4. Rs.60,000
5. 9:16
6. Rs.19,800
7. $616dm^3$
8. $46.2cm^3$
9. $4851cm^3$
10. $2.425m^3$
11. $606\frac{3}{8}dm^3$

Exercise: 9.9

1. (i) $6cm^2$ (ii) $36cm^2$ (iii) $60\ cm^2$
   (iv) $30.25cm^2$ (v) $110.5m^2$ (vi) $25cm^2$
2. (i) $512cm^3$ (ii) $1920cm^3$ (iii) $1056cm^3$
   (iv) $1232cm^3$ (v) $18\pi cm^3$ (vi) $972\pi cm^3$
   (vii) $24000cm^3$ (viii) $3750cm^3$

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Review Exercise: 9

(1)	A	(2)	C	(3)	B	(4)	D
(5)	C	(6)	B	(7)	D	(8)	C
(9)	B	(10)	D	(11)	A	(12)	A
(13)	D	(14)	B	(15)	D	(16)	B
(17)	A	(18)	C	(19)	A	(20)	B
(21)	C	(22)	D	(23)	A	(24)	D
(25)	A	(26)	C	(27)	B	(28)	D

Exercise: 10.1

1.
(i) A (–10, 2) (ii) B (–6, 6) (iii) C (–7, 3)
(iv) D (–1, 5) (v) E (9, –9) (vi) F (7, –6)
(vii) G (2, –4) (viii) H (5, –2) (ix) I (3, 0)
(x) (0, –4) (xi) K (–9, –3) (xii) L (–4, –5)
(xiii) M (–2, –8) (xiv) N (–6, –9) (xv) O (2, –4)
(xvi) P (4, –4) (xvii) Q (8 –6) (xviii) R(6 ,10)
(xix) S (10, 8) (xx) T (0, 9) (xxi) Z (0,–7)

2.
(*i*) $A\ (-2,\ 5)$ $x=-2, y=5$
(*ii*) $B\ (3,\ 0)$ $x=3, y=0$
(*iii*) $C\ (6,\ -7)$ $x=6, y=-7$
(*iv*) $D\ (-4,\ -2)$ $x=-4, y=-2$
(*v*) $E\ (0,\ -3)$ $x=0, y=-3$
(*vi*) $F\ (8,\ 4)$ $x=8, y=4$
(*vii*) $G\ (0,\ 6)$ $x=0, y=6$
(*viii*) $H\ (-4,\ 0)$ $x=-4, y=0$

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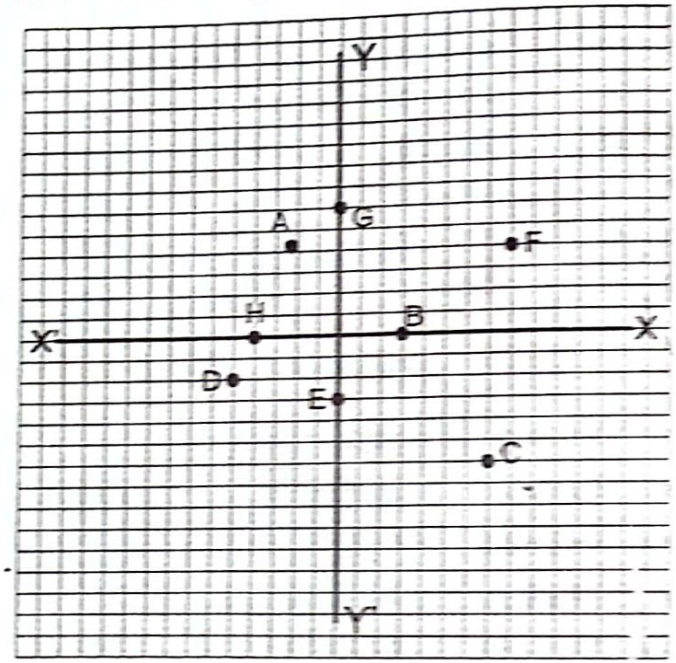
Points are shown on the graph.



3.

(i) Quadrant I
(ii) Quadrant I
(iii) Quadrant IV
(iv) Quadrant III
(v) Quadrant III
(vi) Quadrant IV
(vii) Quadrant II
(viii) Quadrant II
(ix) On x-axis
(x) On Y-axis

4.

(i) $\sqrt{65}$ (ii) 10 (iii) 5
(iv) $\sqrt{41}$ (v) 13 (vi) $\sqrt{676}$

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Collinear: (i), (iv), (vi), (viii)
Non-collinear: (ii), (iii), (vii), (ix), (x), (xi), (xii)

Review Exercise 10D

1. (i) T (ii) T (iii) T (iv) F
   (v) T (vi) F (vii) F
   (viii) negative (ix) negative

2. (i) III (ii) b (iii) d (iv) d
   (iv) y-axis

3. (i) a
   (v) d

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10.

THE AUTHORS

Mr. Abdur Rauf Jameel

A gold medalist of the University of Peshawar, Mr. Abdur Rauf Jameel did his Masters in Curriculum Studies from the University of Birmingham (U.K). He has more than 40 years of teaching experience at his credit. He served as a teacher of Mathematics at Secondary level and at Teachers' training Institute. Since 1976, he has been working as a Curriculum developer, textbook writer and a reviewer of the textbooks in Mathematics. In recognition of his services in the field of Curriculum Designing and Textbook Writing, the Pakistan Education Forum conferred upon him the National Education Award 2000. A former Principal of Islamia Collegiate School, Peshawar and a former Director of the Institute of Education and Research, University of Peshawar, Mr. Abdur Rauf Jameel has presently been working as Executive Director, Khyber Pakhtunkhwa Educational Testing and Evaluation Agency, Peshawar.

11.

1.

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مومن تو آپس میں بھائی بھائی ہیں۔
پس اپنے دو بھائیوں میں صلح کرا دیا کرو
اور اللہ سے ڈرتے رہو تا کہ تم پر رحم
کیا جائے۔

(سورۃ الحجرات: ۱۰)

حضرت محمد رسول اللہ خاتم النبیین صلی اللہ علیہ وعلیٰ آلہٖ واصحابہٖ وسلم
کا ارشاد ہے:
"میں آخری نبی ہوں۔ میرے بعد
کوئی نبی نہیں ہے"

رزق حلال کھاؤ تا کہ تمہاری عبادات قبول ہوں۔

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